aterials

esearch

ata

tandard

NCMRD SD25-02

소재 연구데이터 표준어휘 : 2025-2

2026. 1.

NCMRD 국가소재연구데이터센터
National Center for Materials Research Data

NCMRD SD25-02

소재 연구데이터 표준어휘 : 2025-2

2026. 1.

국가소재연구데이터센터
National Center for Materials Research Data

소재 연구데이터 표준어휘 제·개정 이력

문서번호	제개정 일자	제개정 내용
NCMRD SD21-01	2021.12.22.	표준어휘 597건 제정 (총 어휘 597건)
NCMRD SD22-01	2022.7.25.	표준어휘 568건 제정 및 103건 개정 (총 어휘 1,165건)
NCMRD SD22-02	2023.2.10.	표준어휘 260건 제정, 259건 개정, 111건 삭제 (총 어휘 1,314건)
NCMRD SD23-01	2023.9.15.	표준어휘 45건 제정, 31건 개정, 21건 삭제 (총 어휘 1,338건)
NCMRD SD23-02	2024.3.22.	표준어휘 408건 제정, 23건 개정, 49건 삭제 (총 어휘 1,697건)
NCMRD SD24-01	2024.10.18.	표준어휘 96건 제정, 1건 개정, 20건 삭제 (총 어휘 1,773건)
NCMRD SD24-02	2025.4.22.	표준어휘 434건 제정, 23건 개정, 15건 삭제 (총 어휘 2,192건)
NCMRD SD25-01	2025.11.26.	표준어휘 573건 제정, 126건 개정, 97건 삭제 (총 어휘 2,668건)
NCMRD SD25-02	2026.1.29.	표준어휘 93건 제정, 24건 개정, 24건 삭제 (총 어휘 2,737건)

목 차

I

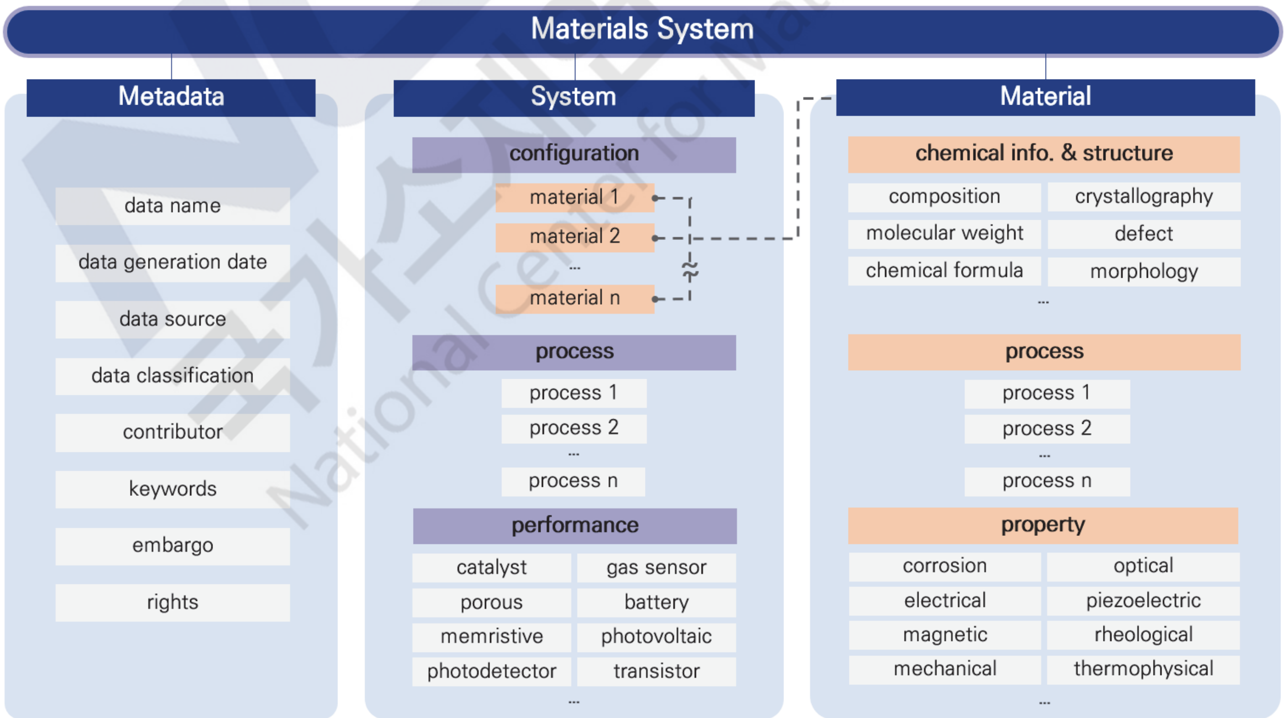
서론

1 서언

다양한 과학, 산업 응용 분야에서 개발 및 활용되는 소재에 대한 정보는 소재의 제작 공정, 구조적 특성 그리고 물성(성능) 등을 포함한 매우 광범위한 데이터 유형으로 표현될 수 있다. 이러한 데이터를 다양한 개발 목적과 활용 정도를 고려하여 연구자 및 산업 활용자들이 사용할 수 있는 체계를 구축하기 위해서는 데이터를 나타내는 어휘(Vocabulary)의 정의가 필수적이다. 본 「소재 연구데이터 표준어휘」에서는 가능한 과학적, 공학적 관점에서 공통으로 이해하고 공유할 수 있는 개념적 기반으로 체계적으로 수집한 데이터 어휘를 포함하려고 한다. 과학적, 공학적 범위에 의존적인 소재 종류에 따른 데이터 어휘는 너무 광범위하고 한 번에 정리하기 힘든 양적인 어려움을 가지고 있다. 이러한 어려움을 체계적으로 접근할 수 있도록 본 표준어휘에서는 어휘체계를 포함하여 어휘를 구성한다. 표준어휘는 고려 대상 소재의 범위에 따라 지속적인 업데이트가 진행되며, 적절한 소재 데이터 어휘 정보를 얻기 위해서는 업데이트된 표준어휘를 참고해야 한다.

2 소재 연구데이터 표준체계

본 「소재 연구데이터 표준어휘」에서는 다양한 소재 분야에서 데이터 구성 체계를 일관되게 유지하기 위해 소재 시스템(Materials System) 스키마를 기반으로 데이터를 체계화한다. 소재 시스템 스키마는 크게 세 가지 데이터군으로 구성된다. 메타데이터(Metadata) 데이터군에는 데이터명, 제공자, 키워드, 엠바고 등 데이터를 설명하는 일반정보가 포함된다. 소재(Material) 데이터군에는 단일 소재의 명칭, 구조 및 화학적 정보, 공정, 특성 데이터가 포함된다. 시스템(System) 데이터군에는 복수의 소재로 구성된 시스템 소재의 명칭, 구성, 공정, 성능 데이터가 포함된다. 예를 들어 촉매 나노입자를 평가하기 위한 촉매 시스템은 활물질인 촉매 나노입자와 촉매의 지지체 그리고 필요에 따라서는 전기를 공급하는 전극 등으로 구성된다. 이처럼 소재 데이터의 일반적인 구조는 메타데이터, 소재 시스템을 구성하는 n개 소재의 개별 데이터, 그리고 이들 소재로 구성된 소재 시스템의 데이터로 구성된다.



소재 시스템(Materials System) 스키마

소재 시스템(Materials System) 스키마를 구성하는 메타데이터, 소재 데이터, 시스템 소재 데이터군의 세부 정보 체계는 다음과 같다.

Ⅰ. 메타데이터

- Data name : 단일 소재 또는 시스템 소재의 명칭
- Data generation date : 데이터 생성 일자
- Data source : 논문, 과제정보 등 데이터 연관 정보
- Data classification : 데이터 분류체계 정보
- Contributor : 데이터 제공자 정보 (이름, 소속, 연락처, 국가연구자번호)
- Keywords : 데이터를 설명하는 주제어
- Embargo : 데이터 공개 가능 일자
- Rights : 데이터의 라이선스 정보

Ⅱ. 소재 데이터

- Name : 단일 소재의 명칭
- Physicochemical info., structure, model : 단일 소재의 물리화학적 정보, 구조 및 계산 모델 정보
- Process : 단일 소재의 공정·합성 정보
- Property : 단일 소재의 물성 정보

Ⅲ. 시스템 소재 데이터

- Description : 시스템 소재의 명칭
- Configuration : 시스템 소재의 구성
- Process : 시스템 소재의 제작 공정 정보
- Performance : 시스템 소재의 성능 정보

3 소재 연구데이터 표준어휘

소재 연구데이터 표준어휘는 소재 시스템(Materials System) 스키마의 각 데이터군을 구성하는, 단일 소재 및 시스템 소재의 구조, 공정, 특성, 물성·성능 데이터의 표준화된 명칭을 의미한다. 소재 시스템 스키마는 아래와 같이 총 5개의 표준어휘군으로 구성된다.

- **메타데이터 표준어휘군**
  데이터명, 생성 일자, 제공자, 키워드, 엠바고 등 메타데이터 관련 표준어휘로 구성
- **소재 표준어휘군**
  단일 소재의 명칭, 화학적 정보, 구조 및 계산 모델 정보, 공정, 물성 관련 표준어휘로 구성
- **시스템 소재 표준어휘군**
  시스템 소재의 명칭, 구조, 공정, 성능 관련 표준어휘로 구성
- **소재 측정·분석 공통어휘군**
  데이터의 측정 혹은 계산 조건 관련 표준어휘로 구성
- **소재 공정 공통어휘군**
  소재 공정 관련 표준어휘로 구성

본 「소재 연구데이터 표준어휘」는 각 어휘를 6개 요소로 구조화하여 정의하며, 국제표준어휘와의 연동을 위해 모두 영문으로 표기한다.

- **정의** : 표준어휘의 의미 및 설명
- **변수명** : 국가 소재 데이터 스테이션(K-MDS) 시스템에서 활용하는 표준어휘의 고유 식별자
  ※ 변수명은 K-MDS 시스템에서 자동으로 부여되는 요소로, 본 어휘집에서는 개별 어휘의 변수명 표기는 생략한다.
- **동의어** : 표준어휘와 같은 의미로 사용되는 어휘
- **유형** : 표준어휘가 정의하는 데이터의 형식 (문자열(string), 수치(numeric), 배열(numeric array) 등)
- **단위** : 측량에 해당하는 데이터의 단위
- **예시** : 표준어휘가 정의하는 데이터 예시

참고 **단위 표기 이해**

○ 표준어휘의 단위는 SI 단위계를 기반으로 하며 TeX 문법을 사용하여 표기한다.

[예시] 유도단위 표기

기호	표기
m/s	m s^{-1}
A/m^2	A m^{-2}
m^2/s^2V	m^{2} s^{-2} V^{-1}
$m^{1/3}/C$	m^{1/3} C^{-1}

[예시] 그리스문자, 단위기호 표기

기호	표기
α	\alpha
μ	\mu
ε	\epsilon
Å	\ANGSTROM

Ⅱ

표준어휘 및 체계

1 메타데이터 표준어휘

어휘번호	표준어휘 체계	표준어휘	제정	개정
E1		data name	2021-1	
E11		data classification	2022-2	2025-1
E4		data generation date	2021-1	
E12		data source	2022-2	2023-1
E2	contributor	name	2021-1	
E3		affiliation	2021-1	
E9		email address	2022-2	
E10		researcher number	2022-2	
E6		keywords	2022-1	
E7		embargo	2022-1	2022-2
E8		rights	2022-1	2022-2

2 소재 표준어휘

2-1 명칭 Name

어휘번호	표준어휘 체계	표준어휘	제정	개정
M1		name	2021-1	

2-2 물리화학적 정보 Physicochemical Information

어휘번호	표준어휘 체계				표준어휘	제정	개정
M227					doping	2025-1	2025-1
M2					composition	2021-1	
M87					density	2021-1	2022-2
M333	hybrid bonds				sp fraction	2025-1	
M334					sp2 fraction	2025-1	
M335					sp3 fraction	2025-1	
M223	molecule				CAS number	2023-2	
M224					chemical formula	2023-2	
M336					formula weight	2025-1	
M12					SMILES	2021-1	
M225					structural formula	2023-2	
M337					plasticizer	2025-1	
M338	polymer chain				classification	2025-1	
M339					degree of polymerization	2025-1	
M304		molecular weight			dispersity	2024-2	
M340					GPC (SEC) peak molar mass	2025-1	2025-2
M226					number-average molar mass	2023-2	2025-2
M341					viscosity-average molar mass	2025-1	2025-2
M303					weight-average molar mass	2024-2	2025-2
M342					z-average molar mass	2025-1	2025-2
M305		architecture			tacticity	2024-2	
M306			linearity		linearity	2024-2	
M343				branched	branch molecule	2025-1	
M344					number of branches	2025-1	
M345					degree of branching	2025-1	
M346					average branch length	2025-1	
M347					branching density	2025-1	
M348					branching index	2025-1	
M349				cross-linked	cross-linking density	2025-1	
M350					cross-linking agent	2025-1	
M351				star	core	2025-1	
M352					arm number	2025-1	
M353					arm length	2025-1	
M354				dendrimer	number of generation	2025-1	
M355					core	2025-1	
M356					branches	2025-1	
M451				ladder	degree of ladderization	2025-2	
M452					persistence length	2025-2	
M453					dihedral angle	2025-2	
M454					fractional free volume	2025-2	

어휘번호	표준어휘 체계				표준어휘	제정	개정
M308	polymer chain	architecture	homogeneity		homogeneity	2024-2	
M455	polymer chain	architecture	homogeneity	copolymer	copolymer type	2025-2	
M456	polymer chain	architecture	homogeneity	copolymer	composition	2025-2	
M357	polymer chain	architecture			end groups	2025-1	
M278					purity	2024-1	

2-3 모델 Model

어휘번호	표준어휘 체계	표준어휘	제정	개정
M15	box dimension	x	2021-1	
M16	box dimension	y	2021-1	
M17	box dimension	z	2021-1	
M122	box dimension	alpha	2022-1	
M123	box dimension	beta	2022-1	
M124	box dimension	gamma	2022-1	
M155	box dimension	cell type	2022-2	
M13		elements	2021-1	2022-2
M14		phase	2021-1	2022-1
M18		periodic boundary condition	2021-1	
M19		structure file	2021-1	

2-4 구조 Structure

어휘번호	표준어휘 체계		표준어휘	제정	개정
M309			crystallinity	2024-2	
M3	crystallography		Bravais lattice	2021-1	2022-1
M206	crystallography		ICSD collection code	2023-1	
M4	crystallography	lattice parameter	a	2021-1	2022-2
M5	crystallography	lattice parameter	b	2021-1	2022-2
M6	crystallography	lattice parameter	c	2021-1	2022-2
M7	crystallography	lattice parameter	alpha	2021-1	
M8	crystallography	lattice parameter	beta	2021-1	
M9	crystallography	lattice parameter	gamma	2021-1	
M279	crystallography	unit cell	cell volume	2024-1	
M280	crystallography	unit cell	cell formula units	2024-1	
M281	crystallography	unit cell	density of crystal structure	2024-1	
M10	crystallography		strukturbericht designation	2021-1	
M11	crystallography		space group	2021-1	2022-1
M119	crystallography		crystal system	2022-1	
M120	crystallography		Pearson symbol	2022-1	
M121	crystallography		Wyckoff symbol	2022-1	
M228	crystallography		perovskite type	2023-2	
M282	crystallography		extinction symbol	2024-1	
M310	defect	0D defect	species	2024-2	
M74	defect	0D defect	defect type	2021-1	2025-1
M75	defect	0D defect	formation energy	2021-1	
M76	defect	0D defect	density	2021-1	
M77	defect	0D defect	electric charge	2021-1	2022-2

어휘번호	표준어휘 체계			표준어휘	제정	개정
M78	defect	1D defect		defect type	2021-1	2025-1
M358				formation energy	2025-1	
M79				Burgers vector	2021-1	
M80				density	2021-1	2022-2
M81				velocity	2021-1	2022-2
M82		2D defect		defect type	2021-1	2025-1
M359				coherency	2025-1	
M83				interfacial energy	2021-1	2022-2
M360				misfit strain	2025-1	
M84				misorientation	2021-1	2022-2
M229		3D defect		defect type	2023-2	2025-1
M230				volume fraction	2023-2	
M231				size distribution	2023-2	
M86				impurity	2021-1	
M88	grain			grain size	2021-1	2022-2
M89				aspect ratio	2021-1	2025-1
M311				Lotgering factor	2024-2	
M90	morphology	0D		shape	2021-1	
M457				sphericity	2025-2	
M458				aspect ratio	2025-2	
M459				elongation	2025-2	
M91			dimension	diameter	2021-1	
M373				x	2025-1	
M374				y	2025-1	
M375				z	2025-1	
M138				facet surface	2022-1	2025-2
M460		1D		shape	2025-2	
M362				base dimension	2025-1	
M236				diameter	2023-2	
M240				inner diameter	2023-2	
M241				outer diameter	2023-2	
M237				length	2023-2	
M238				chirality	2023-2	
M235				orientation	2023-2	
M383				aspect ratio	2025-1	
M461		2D		shape	2025-2	
M283				note	2024-1	
M462				projected area	2025-2	
M463				perimeter	2025-2	
M464				feret diameter	2025-2	
M366			dimension	diameter	2025-1	
M367				thickness	2025-1	
M377				x	2025-1	
M378				y	2025-1	
M465				curvature	2025-2	
M466				roughness	2025-2	
M467				substrate	2025-2	
M468				wrinkle density	2025-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
M361	morphology	3D		shape	2025-1	
M469				volume	2025-2	
M243			dimension	dimension	2023-2	
M470				cluster size	2025-2	
M380		solution		solute	2025-1	2025-2
M381				solvent	2025-1	
M471		statistics		mean	2025-2	
M472				variance	2025-2	
M473				skewness	2025-2	
M474				kurtosis	2025-2	
M475	lattice misfit			associated phases	2025-2	
M476				lattice misfit	2025-2	
M95	phase			name	2021-1	
M184				crystal structure	2022-2	
M96				composition	2021-1	
M97				fraction	2021-1	
M384				polymer blend type	2025-1	
M477				free volume	2025-2	
M478	polymer cross-link			cross-link density	2025-2	
M479				swelling ratio	2025-2	
M480	porosity			pore connectivity	2025-2	
M141				pore diameter	2022-1	
M142				pore limiting diameter	2022-1	
M312				pore fraction	2024-2	
M313				average pore diameter	2024-2	
M98	surface	specific area		BET surface area	2021-1	2022-2
M99				Langmuir surface area	2021-1	2022-2
M139				electrochemically active surface area	2022-1	2022-2
M481				curvature	2025-2	
M385				composition	2025-1	
M386		crystallography		Miller indices	2025-1	
M387				unit cell	2025-1	
M388	texture			texture type	2025-1	
M232				texture table	2023-2	2024-2
M314				pole figure	2024-2	
M389				inverse pole figure	2025-1	
M390				ODF section plot	2025-1	
M100				total pore volume	2021-1	2022-2

2-5 공정 Process

※ '5. 소재 공정 공통어휘' 참고

2-6 물성 Property

어휘번호	표준어휘 체계		표준어휘	제정	개정
M391	biomedical property	biocompatibility	cytotoxicity	2025-1	
M392			inflammatory response	2025-1	
M393			hemocompatibility	2025-1	
M394		bioactivity	osteointegration	2025-1	
M395			surface reactivity	2025-1	
M396		biodegradability	degradation rate	2025-1	
M397			degradation product	2025-1	
M398		antibacterial property	minimum inhibitory concentration (MIC)	2025-1	
M399			zone of inhibition	2025-1	
M400			antibacterial efficiency	2025-1	
M401		cytocompatibility	cell adhesion rate	2025-1	
M402			cell proliferation rate	2025-1	
M403			cell differentiation	2025-1	
M404		immune compatibility	immune cell activation	2025-1	
M405			complement activation	2025-1	
M406		tissue compatibility	tissue integration	2025-1	
M407			tissue regeneration	2025-1	
M408		osteogenic property	alkaline phosphatase activity (ALP)	2025-1	
M409			osteocalcin expression (OCN)	2025-1	
M410			mineralized nodule formation	2025-1	
M20	corrosion property		anodic partial current	2021-1	
M21			cathodic partial current	2021-1	
M22			corrosion current	2021-1	2022-2
M23			corrosion potential	2021-1	2022-2
M24			corrosion rate	2021-1	2022-2
M25			free corrosion current	2021-1	2022-2
M26			free corrosion potential	2021-1	
M27			passivation potential	2021-1	
M28			passivation current	2021-1	
M29			pitting initiation potential	2021-1	
M30			re-activation potential	2021-1	
M31			redox potential	2021-1	2022-1
M32			transpassivation potential	2021-1	
M125			polarized potential	2022-1	
M126			standard electrode potential	2022-1	
M33	electrical property		band gap bowing parameter	2021-1	
M34			band gap energy	2021-1	2023-1
M207			band gap type	2023-1	
M244			band gap graded	2023-2	
M315			carrier type	2024-2	
M35			carrier concentration	2021-1	2022-2
M36			carrier diffusion length	2021-1	
M37			carrier lifetime	2021-1	
M245			conduction band minimum	2023-2	
M246			charge of constituents	2023-2	
M38			critical temperature	2021-1	2023-2
M411			current density	2025-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
M316	electrical property		defect energy level	2024-2	
M39			dielectric constant	2021-1	
M127			dielectric strength	2022-1	2022-2
M40			electric susceptibility	2021-1	
M41			electrical conductivity	2021-1	2022-2
M413			electron affinity	2025-1	
M42		electronic density of states	DOS data	2021-1	2023-2
M43			DOS file	2021-1	
M156			DOS type	2022-2	
M44			electron effective mass	2021-1	2022-2
M45			electron mobility	2021-1	2022-2
M46			exciton binding energy	2021-1	
M128			Fermi level	2022-1	
M129			Hall coefficient	2022-1	2022-2
M130			Hall voltage	2022-1	
M47			hole effective mass	2021-1	2022-2
M48			hole mobility	2021-1	2022-2
M414			impedance	2025-1	
M284			ionic activation energy	2024-1	
M131			ionic conductivity	2022-1	2022-2
M132			ionic mobility	2022-1	2022-2
M482			ionization energy	2025-2	
M317			loss tangent	2024-2	
M49			permittivity	2021-1	
M50			resistivity	2021-1	2022-2
M208			spin-Hall angle	2023-1	
M52			spin-orbit splitting energy	2021-1	
M415			surface resistivity	2025-1	
M318			trap density	2024-2	
M51			temperature coefficient of resistance	2021-1	
M247			valence band maximum	2023-2	
M153	electrochemical property		oxidation potential	2022-1	2022-2
M154			reduction potential	2022-1	2022-2
M203	ferroelectric property		saturation polarization	2022-2	
M204			coercive electric field	2022-2	
M205			remanent polarization	2022-2	
M416	gas transmission property		gas	2025-1	
M417			transmission rate	2025-1	
M292	magnetic property		absorption shielding effectiveness	2024-1	
M209			coercivity	2023-1	
M101			Curie temperature	2021-1	2022-2
M286			DMI energy density	2024-1	
M287			DMI field	2024-1	
M297			domain wall propagation field	2024-1	
M288			effective magnetic anisotropy energy	2024-1	
M289			interface state density	2024-1	
M210			intrinsic coercivity	2023-1	
M211			magnetic anisotropy constant	2023-1	
M290			magnetic anisotropy energy	2024-1	
M212			magnetic anisotropy field	2023-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
M248	magnetic property		magnetic field strength	2023-2	
M249			magnetic induction	2023-2	
M291			magneto-optical Kerr angle	2024-1	
M250			magnetization	2023-2	
M251			magnetostriction	2023-2	
M213			maximum BH product	2023-1	
M102			maximum permeability	2021-1	2023-1
M103			maximum susceptibility	2021-1	2023-1
M418			permeability	2025-1	
M419			Neel temperature	2025-1	
M214			recoil permeability	2023-1	
M293			reflection shielding effectiveness	2024-1	
M420			relative permeability	2025-1	
M320			remanent coercivity	2024-2	
M215			remanent flux density	2023-1	
M216			remanent magnetic polarization	2023-1	
M217			remanent magnetization	2023-1	
M252			saturation flux density	2023-2	
M218			saturation magnetic polarization	2023-1	
M219			saturation magnetization	2023-1	
M295			spin Hall angle	2024-1	
M296			spin-orbit torque efficiency	2024-1	
M220			squareness	2023-1	
M294			total shielding effectiveness	2024-1	
M298			work function	2024-1	
M53	mechanical property	compressive property	compressive yield strength	2021-1	
M54			ultimate compressive strength	2021-1	2022-2
M421			compressive modulus	2025-1	
M253			compressive fracture strength	2023-2	
M254			compressive fracture strain	2023-2	
M493		creep property	0.2% creep strain time	2025-2	
M422			creep compliance	2025-1	
M162			creep rupture strain	2022-2	
M161			creep rupture strength	2022-2	
M158			instantaneous strain	2022-2	
M56			minimum creep rate	2021-1	2022-2
M159			reduction of area	2022-2	
M55			rupture time	2021-1	
M321			stress exponent	2024-2	
M160			time to tertiary creep	2022-2	
M58		elastic property	Young's modulus	2021-1	
M60			bulk modulus	2021-1	
M61			compressibility	2021-1	2022-2
M62			Poisson’s ratio	2021-1	
M133			resilience modulus	2022-1	
M423			elastic recovery ratio	2025-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
M63	mechanical property	fatigue property	fatigue life	2021-1	
M64			fatigue limit	2021-1	
M163			fatigue type	2022-2	2025-1
M164			fatigue crack growth rate	2022-2	
M165			stress intensity factor	2022-2	
M166			stress intensity factor range	2022-2	
M167			fatigue crack growth threshold	2022-2	
M424		flexural property	flexural modulus	2025-1	
M425			flexural strength	2025-1	
M494			flexural yield strength	2025-2	
M170		fracture toughness	stress intensity factor	2022-2	
M172			plane strain fracture toughness K	2022-2	
M173			plane strain fracture toughness J	2022-2	
M174			crack tip opening displacement	2022-2	
M65			hardness	2021-1	
M175		impact toughness	impact energy	2022-2	2023-2
M495			impact strength	2025-2	
M176			upper shelf energy	2022-2	
M177			lower shelf energy	2022-2	
M178			ductile-to-brittle transition temperature	2022-2	
M426			residual stress	2025-1	
M59		shear property	shear modulus	2021-1	
M427			shear strength	2025-1	
M168		sheet metal formability	plastic strain ratio	2022-2	
M256			normal anisotropy	2023-2	
M257			planar anisotropy	2023-2	
M258			limiting drawing ratio	2023-2	
M259			bendability	2023-2	
M262			hole expansion ratio	2023-2	
M263			springback ratio	2023-2	
M221		tensile property	critical resolved shear stress	2023-1	
M323			fracture strength	2024-2	
M71			reduction of area	2021-1	
M70			strain hardening exponent	2021-1	
M322			tensile modulus	2024-2	
M222			tensile toughness	2023-1	2023-2
M69			total elongation	2021-1	
M67			ultimate tensile strength	2021-1	2022-2
M68			uniform elongation	2021-1	
M66			yield strength	2021-1	2022-2
M324	optical property		absorbance	2024-2	
M111			absorption coefficient	2021-1	
M264		electroluminescence	lifetime	2023-2	
M265			quantum efficiency	2023-2	
M267			peak position	2023-2	
M112			optical bandgap	2021-1	
M428			haze	2025-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
M266	optical property	photoluminescence	photoluminescence type	2023-2	2025-1
M113			lifetime	2021-1	
M114			quantum efficiency	2021-1	
M274			peak position	2023-2	
M325			maximum peak wavelength	2024-2	
M115			reflectance	2021-1	
M116			transmittance	2021-1	
M326			normal incidence transmittance	2024-2	
M117			color	2021-1	2022-2
M118			refractive index	2021-1	
M429		yellowness index	yellowness index	2025-1	
M497			CIE tristimulus value	2025-2	
M498		dispersion	dispersion data	2025-2	
M499			abbe number	2025-2	
M504	oxidation property		weight change	2025-2	
M505			surface morphology	2025-2	
M506			oxide layer thickness	2025-2	
M507			diffusion layer thickness	2025-2	
M508			oxidation rate	2025-2	
M509			metal loss	2025-2	
M198	piezoelectric property		piezoelectric strain coefficient	2022-2	
M199			piezoelectric voltage coefficient	2022-2	
M200			piezoelectric charge coefficient	2022-2	
M201			electromechanical coupling factor	2022-2	
M202			mechanical quality factor	2022-2	
M430	rheological property	solution viscosity	inherent viscosity	2025-1	
M431			intrinsic viscosity	2025-1	
M432			reduced viscosity	2025-1	
M433		dynamic viscosity	variability	2025-1	
M179			dynamic viscosity	2025-1	2025-1
M500			complex viscosity	2025-2	
M181		variable modulus	storage modulus	2022-2	
M182			loss modulus	2022-2	
M434			loss tangent	2025-1	
M435			stress relaxation time	2025-1	
M436	surface property		contact angle	2025-1	
M233			RMS surface roughness	2023-2	
M143	thermophysical property	activity	activity	2022-1	
M269			activity coefficient	2023-2	
M275			standard state	2023-2	
M276			temperature	2023-2	
M277			pressure	2023-2	
M268			boiling temperature	2023-2	
M144		chemical potential	chemical potential	2022-1	2022-2
M185			standard state	2022-2	
M186			temperature	2022-2	
M187			pressure	2022-2	
M145			cryoscopic constant	2022-1	2022-2
M437			decomposition temperature	2025-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
M146	thermophysical property			diffusivity	2022-1	2022-2
M188				Debye temperature	2022-2	
M147				ebullioscopic constant	2022-1	2023-1
M148		enthalpy of formation		enthalpy of formation	2022-1	2022-2
M189				standard state	2022-2	
M190				temperature	2022-2	
M191				pressure	2022-2	
M149		fugacity		fugacity	2022-1	
M192				standard state	2022-2	
M193				temperature	2022-2	
M194				pressure	2022-2	
M270				glass transition temperature	2023-2	
M150		Gibbs free energy		Gibbs free energy	2022-1	2022-2
M195				standard state	2022-2	
M196				temperature	2022-2	
M197				pressure	2022-2	
M107		heat capacity		specific heat capacity	2021-1	2025-2
M501				molar heat capacity	2025-2	
M502				volumetric heat capacity	2025-2	
M104				heat of fusion	2021-1	2022-2
M271				heat of vaporization	2023-2	
M438				heat deflection temperature (HDT)	2025-1	
M105				melting temperature	2021-1	2022-1
M106		melting range		range	2021-1	2025-1
M439				composition	2025-1	
M300				onset temperature	2024-1	
M301				softening temperature	2024-1	
M108				thermal conductivity	2021-1	2022-2
M327				thermal degradation temperature	2024-2	
M109				thermal diffusivity	2021-1	2022-2
M110				thermal expansion coefficient	2021-1	2022-2
M440				topology freezing transition temperature	2025-1	
M328				moisture absorption	2024-2	
M503				solvus temperature	2025-2	
M330	thermoelectric property			Seebeck coefficient	2024-2	
M331				power factor	2024-2	
M332				ZT	2024-2	
M441	tribological property			friction coefficient	2025-1	
M442		wear		wear type	2025-1	
M443				rate	2025-1	
M444			morphology	wear scar	2025-1	
M445				wear track	2025-1	
M446		tribochemistry		object	2025-1	
M447				composition	2025-1	
M448				structure	2025-1	
M368	triboelectric property			charge transfer	2025-1	
M370				open-circuit voltage	2025-1	
M371				short-circuit current	2025-1	
M372				triboelectric charge density	2025-1	

3 시스템 소재 표준어휘

3-1 설명 Description

어휘번호	표준어휘 체계	표준어휘	제정	개정
S1		note	2021-1	2025-2
S700		category	2025-1	

3-2 구성 Configuration

3-2-1. Configuration: Alkali-ion battery

어휘번호	표준어휘 체계		표준어휘	제정	개정
S259	cathode		material	2022-2	
S137			mass loading	2022-1	2022-2
S261			thickness	2022-2	
S262			volume	2022-2	
S138	anode		material	2022-1	2022-2
S263			mass loading	2022-2	
S265			thickness	2022-2	
S266			volume	2022-2	
S143	electrolyte		material	2022-1	2022-2
S144			mass	2022-1	2022-2
S146			thickness	2022-1	2022-2
S267			volume	2022-2	
S148	separator		material	2022-1	2022-2
S268			mass	2022-2	
S149			thickness	2022-1	
S269			porosity	2022-2	
S270			tortuosity	2022-2	
S151			cell type	2022-1	
S130	materials function list	cathode	active material	2022-1	2025-1
S131			coating material	2022-1	2025-1
S132			binder	2022-1	2025-1
S133			conducting material	2022-1	2025-1
S271			additive	2022-2	2025-1
S135			electrolyte	2022-1	2025-1
S272			current collector	2022-2	2025-1
S273		anode	active material	2022-2	2025-1
S274			coating material	2022-2	2025-1
S275			binder	2022-2	2025-1
S276			conducting material	2022-2	2025-1
S277			additive	2022-2	2025-1
S278			electrolyte	2022-2	2025-1
S279			current collector	2022-2	2025-1
S280		electrolyte	solvent	2022-2	2025-1
S281			salt	2022-2	2025-1
S282			additive	2022-2	2025-1

3-2-2. Configuration: Catalyst

어휘번호	표준어휘 체계	표준어휘	제정	개정
S2		active material	2021-1	
S3		amount of active material	2021-1	2022-2
S4		promotor	2021-1	
S5		amount of promotor	2021-1	2022-2
S64		coating material	2022-1	
S65		amount of coating material	2022-1	2022-2
S6		support materials	2021-1	

3-2-3. Configuration: Gas sensor

어휘번호	표준어휘 체계	표준어휘	제정	개정
S112	substrate	thickness	2022-1	
S113		material	2022-1	
S114	electrode	length	2022-1	2022-2
S115		width	2022-1	
S338		gap	2023-2	
S116		material	2022-1	
S117		structure type	2022-1	
S118	channel	thickness	2022-1	
S119		material	2022-1	2023-2
S120		structure type	2022-1	2023-2
S122	interlayer	material	2022-1	
S123		thickness	2022-1	
S121		structure type	2022-1	2023-2
S467	electron depletion layer	material	2024-2	
S468		thickness	2024-2	
S469	hole accumulation layer	material	2024-2	
S470		thickness	2024-2	

3-2-4. Configuration: Memcapacitive system

어휘번호	표준어휘 체계	표준어휘	제정	개정
S476		device type	2024-2	
S477	electrode_1	material	2024-2	
S478		thickness	2024-2	
S479		doping	2024-2	
S480	buffer_1	material	2024-2	
S481		thickness	2024-2	
S482		doping	2024-2	
S483	active layer	material	2024-2	
S484		thickness	2024-2	
S485		doping	2024-2	
S486	buffer_2	material	2024-2	
S487		thickness	2024-2	
S488		doping	2024-2	
S489	electrode_2	material	2024-2	
S490		thickness	2024-2	
S491		doping	2024-2	

3-2-5. Configuration: Memristive system

어휘번호	표준어휘 체계	표준어휘	제정	개정
S258		device type	2022-2	
S9	electrode_1	material	2021-1	
S10		thickness	2021-1	
S11		doping	2021-1	
S12	buffer_1	material	2021-1	
S13		thickness	2021-1	
S14		doping	2021-1	
S15	active layer	material	2021-1	
S16		thickness	2021-1	
S17		doping	2021-1	
S18	buffer_2	material	2021-1	
S19		thickness	2021-1	
S20		doping	2021-1	
S21	electrode_2	material	2021-1	
S22		thickness	2021-1	
S23		doping	2021-1	

3-2-6. Configuration: Multilayer system

어휘번호	표준어휘 체계		표준어휘	제정	개정
S706			number of repeating units	2025-1	
S326	repeating unit		material	2023-1	
S327			thickness	2023-1	
S707			orientation	2025-1	
S708		offcut	angle	2025-1	
S709			direction	2025-1	

3-2-7. Configuration: Organic thin-film transistor

어휘번호	표준어휘 체계	표준어휘	제정	개정
S429		device type	2023-2	
S430	substrate	substrate type	2023-2	
S431		material	2023-2	
S432		thickness	2023-2	
S433	electrode	electrode type	2023-2	
S434		material	2023-2	
S435		thickness	2023-2	
S436	gate dielectric	material	2023-2	
S437		thickness	2023-2	
S438	interfacial layer	interface type	2023-2	2025-1
S439		material	2023-2	
S440		thickness	2023-2	
S441	semiconductor	material	2023-2	
S442		thickness	2023-2	
S443	other structures	material	2023-2	
S444		thickness	2023-2	
S445	dimension	channel length	2023-2	
S446		channel width	2023-2	
S447		channel area	2023-2	

3-2-8. Configuration: Photoconductor

어휘번호	표준어휘 체계	표준어휘	제정	개정
S560	active layer	material	2024-2	2025-1
S712		area	2025-1	
S561		thickness	2024-2	
S555	substrate	material	2024-2	2025-1
S556		thickness	2024-2	
S557	electrode	material	2024-2	2025-1
S558		width	2024-2	
S559		gap	2024-2	
S562	interlayer	material	2024-2	2025-1
S563		size	2024-2	

3-2-9. Configuration: Photodiode

어휘번호	표준어휘 체계	표준어휘	제정	개정
S67		device type	2022-1	
S710	device size	pixel shape	2025-1	
S68		pixel size	2022-1	
S711		microjunction shape	2025-1	
S69		microjunction size	2022-1	
S512	active area	height	2024-2	
S513		width	2024-2	
S514		area	2024-2	
S111		assembly	2022-1	

3-2-10. Configuration: Phototransistor

어휘번호	표준어휘 체계		표준어휘	제정	개정
S713	active layer		area	2025-1	
S622			material	2024-2	2025-1
S623			thickness	2024-2	
S624	dielectric layer		material	2024-2	2025-1
S625			thickness	2024-2	
S616	drain electrode		material	2024-2	2025-1
S617			thickness	2024-2	
S618	gate electrode	bottom gate	material	2024-2	2025-1
S619			thickness	2024-2	
S620		top gate	material	2024-2	2025-1
S621			thickness	2024-2	
S614	source electrode		material	2024-2	2025-1
S615			thickness	2024-2	
S612	substrate		material	2024-2	2025-1
S613			thickness	2024-2	

3-2-11. Configuration: Photovoltaic system

어휘번호	표준어휘 체계	표준어휘	제정	개정
S351		architecture	2023-2	
S352	additional layer front	material	2023-2	
S353		thickness	2023-2	
S354		function	2023-2	
S355	substrate	material	2023-2	
S356		thickness	2023-2	
S357		area	2023-2	
S358	front contact	material	2023-2	
S699		thickness	2024-2	
S359	n-type layer	material	2023-2	
S360		thickness	2023-2	
S361	intrinsic layer	material	2023-2	
S362		thickness	2023-2	
S363		single crystal	2023-2	
S364	p-type layer	material	2023-2	
S365		thickness	2023-2	
S366	back contact	material	2023-2	
S367		thickness	2023-2	
S368	additional layer back	material	2023-2	
S369		thickness	2023-2	
S370		function	2023-2	
S371		encapsulation	2023-2	
S372		total area	2023-2	
S373		area measured	2023-2	
S374		number of cells per substrate	2023-2	
S375	module	number of cells in module	2023-2	
S376		total area	2023-2	
S377		active area	2023-2	2024-2
S674		designated illumination area	2024-2	
S378		JV data recalculated per cell	2023-2	

3-2-12. Configuration: Piezoelectric system

어휘번호	표준어휘 체계	표준어휘	제정	개정
S283	substrate	length	2022-2	
S284		width	2022-2	
S285		thickness	2022-2	
S286		material	2022-2	
S287	bottom electrode	length	2022-2	
S288		width	2022-2	
S289		thickness	2022-2	
S290		material	2022-2	
S291		structure type	2022-2	
S292	piezoelectric layer	length	2022-2	
S293		width	2022-2	
S294		thickness	2022-2	
S295		material	2022-2	

어휘번호	표준어휘 체계	표준어휘	제정	개정
S296	top electrode	length	2022-2	
S297		width	2022-2	
S298		thickness	2022-2	
S299		material	2022-2	
S300		structure type	2022-2	
S301	encapsulation	length	2022-2	
S302		width	2022-2	
S303		thickness	2022-2	
S304		material	2022-2	

3-2-13. Configuration: Porous materials

어휘번호	표준어휘 체계	표준어휘	제정	개정
S7		framework	2021-1	2025-1
S8		grafted molecule	2021-1	2022-2
S66		grafted molecule amount	2022-1	2022-2
S721		tortuosity	2025-2	

3-2-14. Configuration: Separation membrane

어휘번호	표준어휘 체계	표준어휘	제정	개정
S679		selective layer	2024-2	
S680		thickness of selective layer	2024-2	
S681		coating layer	2024-2	
S682		thickness of coating layer	2024-2	
S683		membrane support	2024-2	
S684		thickness of membrane support	2024-2	

3-3 공정 Process ※ '5. 소재 공정 공통어휘' 참고

3-4 성능 Performance

3-4-1. Performance: Alkali-ion battery

어휘번호	표준어휘 체계	표준어휘	제정	개정
S701		maximum cycling number	2025-1	
S702		maximum cycling time	2025-1	
S243	cycling test result	cycle number	2022-1	2025-1
S703		C-rate	2025-1	
S237		specific charge capacity (areal)	2022-1	2022-2
S238		specific charge capacity (mass)	2022-1	2022-2
S346		specific charge capacity (volume)	2023-2	
S235		specific discharge capacity (areal)	2022-1	2022-2
S236		specific discharge capacity (mass)	2022-1	2022-2
S345		specific discharge capacity (volume)	2023-2	
S309		energy density (volume)	2022-2	2025-1
S310		energy density (mass)	2022-2	2025-1

어휘번호	표준어휘 체계	표준어휘	제정	개정
S349	cycling test result	energy density (areal)	2023-2	2025-1
S311		power density (volume)	2022-2	2025-1
S312		power density (mass)	2022-2	2025-1
S350		power density (areal)	2023-2	2025-1
S719		current density (areal)	2025-1	
S720		current density (mass)	2025-1	
S245		capacity retention	2022-1	
S247		coulombic efficiency	2022-1	
S248		energy efficiency	2022-1	
S307		average voltage	2022-2	2025-1
S246		overpotential	2022-1	

3-4-2. Performance: Catalyst

어휘번호	표준어휘 체계		표준어휘	제정	개정
S329	electrochemical		reaction type	2023-1	
S24			area-specific activity	2021-1	2022-2
S25			mass-specific activity	2021-1	2022-2
S26			Faradaic efficiency	2021-1	
S152			onset potential	2022-1	
S153			half-wave potential	2022-1	
S154			overpotential	2022-1	2022-2
S155			limiting current density	2022-1	2022-2
S330			product	2023-1	
S331	photoelectrochemical		reaction type	2023-1	
S164			area-specific activity	2022-1	2022-2
S165			mass-specific activity	2022-1	2022-2
S166			Faradaic efficiency	2022-1	
S167			onset potential	2022-1	
S168			half-wave potential	2022-1	
S169			overpotential	2022-1	2022-2
S170			limiting current density	2022-1	2022-2
S33	thermal	reaction rate	reaction rate	2021-1	2025-1
S34			temperature	2021-1	
S35			reaction gas	2021-1	
S36			pressure	2021-1	2022-2
S37		conversion rate	conversion rate	2021-1	2025-1
S38			temperature	2021-1	
S39			weight	2021-1	
S332			equilibrium constant	2023-1	
S333		chemical reaction calculated	reaction step	2023-1	
S334			surface coverage	2023-1	
S335			activation energy	2023-1	
S336			reaction energy	2023-1	
S337			rate constant	2023-1	

3-4-3. Performance: Gas sensor

어휘번호	표준어휘 체계		표준어휘	제정	개정
S190	measurement condition		relative humidity	2022-1	
S191		light illumination	wavelength	2022-1	
S192			light intensity	2022-1	2022-2
S193			bias voltage	2022-1	
S194			operating temperature	2022-1	2023-2
S195		balance gas	gas type	2022-1	
S196			flow rate	2022-1	2022-2
S197		target gas	gas type	2022-1	
S198			gas concentration	2022-1	
S339			flow rate	2023-2	
S340			exposure time	2023-2	
S471			power consumption	2024-2	
S189			gas response	2022-1	2024-2
S472			baseline resistance	2024-2	
S209			recovery time	2022-1	
S219			response time	2022-1	
S229	theoretical detection limit		theoretical detection limit	2022-1	2022-2
S230			gas type	2022-1	
S207	selectivity	target gas	gas type	2022-1	2025-1
S208			concentration	2022-1	2025-1
S341			response	2023-2	2024-2
S473	sensitivity	target gas	gas type	2024-2	2025-1
S474			concentration	2024-2	2025-1
S475			sensitivity	2024-2	
S704			signal-to-noise ratio	2025-1	
S231	air stability		exposure duration	2022-1	2024-2
S342			responsivity	2023-2	
S705	humidity stability		absolute humidity	2025-1	
S343			relative humidity	2023-2	
S343			exposure duration	2022-1	2024-2
S232			responsivity	2023-2	2024-2
S344			responsivity	2023-2	

3-4-4. Performance: Memcapacitive system

어휘번호	표준어휘 체계	표준어휘	제정	개정
S492	capacitance	on capacitance	2024-2	
S493		off capacitance	2024-2	
S494		set capacitance	2024-2	
S495		on-off ratio	2024-2	
S496		number of states	2024-2	
S497	endurance	cycles	2024-2	
S498		voltage sweep	2024-2	
S499	mechanism	operating mechanism	2024-2	
S500		operating speed	2024-2	
S501	retention	time	2024-2	
S502		temperature	2024-2	
S503	voltage	set	2024-2	
S504		reset	2024-2	

3-4-5. Performance: Memristive system

어휘번호	표준어휘 체계		표준어휘	제정	개정
S46	current		Icc	2021-1	
S47			Ion	2021-1	
S48			Ioff	2021-1	
S49			Iset	2021-1	
S50	endurance		cycles	2021-1	2022-2
S51			voltage sweeps	2021-1	2022-2
S52	mechanism		operating mechanism	2021-1	
S53		conduction mechanism	LRS	2021-1	
S54			HRS	2021-1	
S55	operating speed		operating speed	2021-1	
S56	resistance		HRS	2021-1	2022-2
S57			LRS	2021-1	2022-2
S58	retention		time	2021-1	
S59			temperature	2021-1	
S60	selectivity		on-off ratio	2021-1	
S61	voltage		set	2021-1	
S62			reset	2021-1	
S63			forming	2021-1	

3-4-6. Performance: Organic thin-film transistor

어휘번호	표준어휘 체계	표준어휘	제정	개정
S448		operating voltage	2023-2	
S449		threshold voltage	2023-2	
S450		on current	2023-2	
S451		off current	2023-2	
S452		on-off ratio	2023-2	
S506		gate leakage current	2024-2	
S453		voltage gain	2023-2	
S454		current gain	2023-2	
S455		transition frequency	2023-2	
S456		carrier mobility	2023-2	
S457		capacitance	2023-2	
S507		contact resistance	2024-2	
S458		subthreshold swing	2023-2	
S459		energy consumption	2023-2	
S460	endurance	cycles	2023-2	
S461		applied voltage	2023-2	
S462		air stability	2023-2	
S463		temperature stability	2023-2	
S464		humidity stability	2023-2	
S465		bending stability	2023-2	
S466		stretching stability	2023-2	

3-4-7. Performance: Photoconductor

어휘번호	표준어휘 체계		표준어휘	제정	개정
S564	dark current		dark current	2024-2	
S565			bias voltage	2024-2	
S566			gate voltage	2024-2	
S567	dark current density		dark current density	2024-2	
S568			bias voltage	2024-2	
S569	photocurrent		photocurrent	2024-2	
S570			bias voltage	2024-2	
S571			gate voltage	2024-2	
S572		light illumination	wavelength	2024-2	
S573			light intensity	2024-2	
S574	photocurrent density		photocurrent density	2024-2	
S575			bias voltage	2024-2	
S576		light illumination	wavelength	2024-2	
S577			light intensity	2024-2	
S578	on-off ratio		on-off ratio	2024-2	
S579			bias voltage	2024-2	
S580		light illumination	wavelength	2024-2	
S581			light intensity	2024-2	
S582	photoconductive gain		photoconductive gain	2024-2	
S583			bias voltage	2024-2	
S584		light illumination	wavelength	2024-2	
S585			light intensity	2024-2	
S586	responsivity		responsivity	2024-2	
S587			bias voltage	2024-2	
S588		light illumination	wavelength	2024-2	
S589			light intensity	2024-2	
S590	specific detectivity		specific detectivity	2024-2	
S591			bias voltage	2024-2	
S592		light illumination	wavelength	2024-2	
S593			light intensity	2024-2	
S594			decay time	2024-2	
S595			rise time	2024-2	
S596	quantum efficiency		external quantum efficiency	2024-2	
S597			bias voltage	2024-2	
S598		light illumination	wavelength	2024-2	
S599			light intensity	2024-2	
S600	linear dynamic range		linear dynamic range	2024-2	
S601			bias voltage	2024-2	
S602			cutoff wavelength	2024-2	
S603			emission wavelength	2024-2	
S604			noise	2024-2	
S605			signal-to-noise	2024-2	
S606			noise equivalent power	2024-2	
S607	air stability		exposure duration	2024-2	
S608			photoresponse	2024-2	
S609	temperature stability		temperature	2024-2	
S610			operating time	2024-2	
S611			photoresponse	2024-2	

3-4-8. Performance: Photodiode

어휘번호	표준어휘 체계		표준어휘	제정	개정
S515	dark current		dark current	2024-2	
S516			bias voltage	2024-2	
S517			gate voltage	2024-2	
S180	dark current density		dark current density	2022-1	2022-2
S518			bias voltage	2024-2	
S519	photocurrent		photocurrent	2024-2	
S520			bias voltage	2024-2	
S521			gate voltage	2024-2	
S522		light illumination	wavelength	2024-2	
S523			light intensity	2024-2	
S524	photocurrent density		photocurrent density	2024-2	
S525			bias voltage	2024-2	
S526		light illumination	wavelength	2024-2	
S527			light intensity	2024-2	
S528	on-off ratio		on-off ratio	2024-2	
S529			bias voltage	2024-2	
S530		light illumination	wavelength	2024-2	
S531			light intensity	2024-2	
S532	photoconductive gain		photoconductive gain	2024-2	
S533			bias voltage	2024-2	
S534		light illumination	wavelength	2024-2	
S535			light intensity	2024-2	
S185	responsivity		responsivity	2022-1	2022-2
S536			bias voltage	2024-2	
S537		light illumination	wavelength	2024-2	
S538			light intensity	2024-2	
S539	specific detectivity		specific detectivity	2024-2	
S540			bias voltage	2024-2	
S541		light illumination	wavelength	2024-2	
S542			light intensity	2024-2	
S543			decay time	2024-2	
S544			rise time	2024-2	
S179	quantum efficiency		external quantum efficiency	2022-1	2024-2
S545			bias voltage	2024-2	
S546		light illumination	wavelength	2024-2	
S547			light intensity	2024-2	
S548	linear dynamic range		linear dynamic range	2024-2	
S549			bias voltage	2024-2	
S181			cutoff wavelength	2022-1	
S182			emission wavelength	2022-1	
S186			noise	2022-1	2022-2
S187			signal-to-noise	2022-1	2022-2
S188			noise equivalent power	2022-1	2022-2
S550	air stability		exposure duration	2024-2	
S551			photoresponse	2024-2	
S552	temperature stability		temperature	2024-2	
S553			operating time	2024-2	
S554			photoresponse	2024-2	

3-4-9. Performance: Phototransistor

어휘번호	표준어휘 체계		표준어휘	제정	개정
S626	dark current		dark current	2024-2	
S627			bias voltage	2024-2	
S628			gate voltage	2024-2	
S629	dark current density		dark current density	2024-2	
S630			bias voltage	2024-2	
S631	photocurrent		photocurrent	2024-2	
S632			bias voltage	2024-2	
S633			gate voltage	2024-2	
S634		light illumination	wavelength	2024-2	
S635			light intensity	2024-2	
S636	photocurrent density		photocurrent density	2024-2	
S637			bias voltage	2024-2	
S638		light illumination	wavelength	2024-2	
S639			light intensity	2024-2	
S640	on-off ratio		on-off ratio	2024-2	
S641			bias voltage	2024-2	
S642		light illumination	wavelength	2024-2	
S643			light intensity	2024-2	
S644	photoconductive gain		photoconductive gain	2024-2	
S645			bias voltage	2024-2	
S646		light illumination	wavelength	2024-2	
S647			light intensity	2024-2	
S648	responsivity		responsivity	2024-2	
S649			bias voltage	2024-2	
S650		light illumination	wavelength	2024-2	
S651			light intensity	2024-2	
S652	specific detectivity		specific detectivity	2024-2	
S653			bias voltage	2024-2	
S654		light illumination	wavelength	2024-2	
S655			light intensity	2024-2	
S656			decay time	2024-2	
S657			rise time	2024-2	
S658	quantum efficiency		external quantum efficiency	2024-2	
S659			bias voltage	2024-2	
S660		light illumination	wavelength	2024-2	
S661			light intensity	2024-2	
S662	linear dynamic range		linear dynamic range	2024-2	
S663			bias voltage	2024-2	
S664			cutoff wavelength	2024-2	
S665			emission wavelength	2024-2	
S666			noise	2024-2	
S667			signal-to-noise	2024-2	
S668			noise equivalent power	2024-2	
S669	air stability		exposure duration	2024-2	
S670			photoresponse	2024-2	
S671	temperature stability		temperature	2024-2	
S672			operating time	2024-2	
S673			photoresponse	2024-2	

3-4-10. Performance: Photovoltaic system

어휘번호	표준어휘 체계		표준어휘	제정	개정
S379	flexibility		minimum bending radius	2023-2	
S380	transparency		average transmittance	2023-2	
S675			color rendering index	2024-2	
S381			wavelength range	2023-2	
S382	JV property	reverse scan	Voc	2023-2	
S383			Jsc	2023-2	
S384			Pmax	2023-2	
S385			FF	2023-2	
S386			PCE	2023-2	
S387			Vmp	2023-2	
S388			Jmp	2023-2	
S389			series resistance	2023-2	
S390			shunt resistance	2023-2	
S391		forward scan	Voc	2023-2	
S392			Jsc	2023-2	
S393			Pmax	2023-2	
S394			FF	2023-2	
S395			PCE	2023-2	
S396			Vmp	2023-2	
S397			Jmp	2023-2	
S398			series resistance	2023-2	
S399			shunt resistance	2023-2	
S676			hysteresis index	2024-2	
S400	stabilized performance		PCE	2023-2	
S401			Vmp	2023-2	
S402			Jmp	2023-2	
S403	QE		integrated Jsc	2023-2	
S677			external quantum efficiency	2024-2	
S678			internal quantum efficiency	2024-2	
S404	stability	PCE	initial value	2023-2	
S405			burn in observed	2023-2	
S406			final value	2023-2	
S407			T95	2023-2	
S408			Ts95	2023-2	
S409			T80	2023-2	
S410			Ts80	2023-2	
S411			Te80	2023-2	
S412			Tse80	2023-2	
S413			after 1000h	2023-2	
S414			lifetime energy yield	2023-2	
S415	stability	bending stability	initial PCE	2023-2	
S416			final PCE	2023-2	
S417			PCE trace	2023-2	
S418	outdoor	PCE	initial value	2023-2	
S419			burn in observed	2023-2	
S420			final value	2023-2	
S421			T95	2023-2	
S422			Ts95	2023-2	
S423			T80	2023-2	
S424			Ts80	2023-2	
S425			Te80	2023-2	
S426			Tse80	2023-2	
S427			after 1000h	2023-2	
S428			power generated	2023-2	

3-4-11. Performance: Piezoelectric system

어휘번호	표준어휘 체계	표준어휘	제정	개정
S714	applied load	load type	2025-1	
S314		applied load	2022-2	
S315		load resistance	2022-2	
S313		maximum output voltage	2022-2	
S316		maximum output current	2022-2	
S319		maximum output power	2022-2	
S322		maximum power density	2022-2	

3-4-12. Performance: Porous materials

어휘번호	표준어휘 체계	표준어휘	제정	개정
S178	gas adsorption	adsorbate	2022-1	2025-1
S40		capacity	2021-1	2025-1
S41		temperature	2021-1	2022-1
S42		pressure	2021-1	2022-2
S43		binding enthalpy	2021-1	2025-2

3-4-13. Performance: Separation membrane

어휘번호	표준어휘 체계	표준어휘	제정	개정
S685		permeance	2024-2	
S686		rejection	2024-2	
S687		selectivity	2024-2	
S718		mechanical strength	2025-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A669	analysis environment			gas	2025-1	
A670				gas flow rate	2025-1	
A671				pressure	2025-1	
A672				relative humidity	2025-1	
A673				temperature	2025-1	
A674				additional condition	2025-1	
A188	Auger electron spectroscopy (AES)			instrument	2022-1	
A189		acquisition condition		electron source	2022-1	
A190				electron beam energy	2022-1	
A191				electron beam current	2022-1	
A192			sputtering condition	ion beam energy	2022-1	
A193				ion type	2022-1	
A194				ion beam current	2022-1	
A195				raster size	2022-1	
A196				sputter time	2022-1	
A197			neutralizing condition	neutralizer beam energy	2022-1	
A198				neutralizer beam current	2022-1	
A199				raw data	2022-1	
A200	atomic force microscopy (AFM)	instrument		name	2022-1	2025-2
A871				maximum scan size	2025-2	
A201		acquisition condition		mode	2022-1	
A202				cantilever type	2022-1	
A203				scan rate	2022-1	
A204				scan size	2022-1	2022-2
A205				Z servo gain	2022-1	
A206				raw data	2022-1	
A870	amperometry			instrument	2025-1	
A207		acquisition condition		upper set voltage	2022-1	
A208				lower set voltage	2022-1	
A212				temperature	2022-1	
A211				raw data	2022-1	
A213	atom probe tomography			instrument	2022-1	
A676		acquisition condition		measurement mode	2025-1	
A441				base pressure	2023-2	
A214				base temperature	2022-1	
A677				DC voltage	2025-1	
A215				voltage pulse fraction	2022-1	2025-1
A678				voltage pulse frequency	2025-1	
A216				laser pulse energy	2022-1	
A218				laser pulse frequency	2022-1	2025-1
A679				signal-to-noise ratio	2025-1	
A219				raw data	2022-1	

어휘번호	표준어휘 체계				표준어휘	제정	개정
A429	compressive test				microtruss direction	2023-1	
A430					microtruss thickness	2023-1	
A431					compressive specimen thickness	2023-1	
A432					area	2023-1	
A433					initial volume	2023-1	
A434					effective volume	2023-1	
A442					volume ratio	2023-2	
A1	creep test				instrument	2021-1	
A2					temperature	2021-1	
A3					load	2021-1	
A620					atmosphere	2024-2	
A881					creep test time	2025-2	
A680	cryoscopy				instrument	2025-1	
A681		acquisition condition			solvent	2025-1	
A682			solute		material	2025-1	
A683					mass	2025-1	
A684					cooling rate	2025-1	
A685					temperature range	2025-1	
A686					raw data	2025-1	
A621	CV measurement				instrument	2024-2	
A622		scan			frequency level	2024-2	
A623					AC drive level	2024-2	
A624					voltage step	2024-2	
A687					scan rate	2025-1	2025-2
A625					temperature	2024-2	
A626		stability test			procedure	2024-2	
A356	d33 meter				instrument	2022-2	
A357					sample type	2022-2	
A358					analysis elements	2022-2	
A359					raw data	2022-2	
A4	DFT calculation				code	2021-1	
A5					calculation mode	2021-1	
A6		basis			basis type	2021-1	2025-1
A7					basis set	2021-1	
A8					charge	2021-1	2022-2
A9					energy cutoff	2021-1	
A10					optimizer	2021-1	
A11					exchange-correlation functional	2021-1	
A12		solvent model			method	2021-1	
A13			parameters		solvent dielectric constant	2021-1	2025-1
A220					solvent	2022-1	
A221					cavity radii	2022-1	
A855					dipole moment	2025-1	
A856					closure	2025-1	
A857					box size	2025-1	
A858					grid spacing	2025-1	
A859					solvent temperature	2025-1	
A860				density	species	2025-1	
A861					density	2025-1	

어휘번호	표준어휘 체계				표준어휘	제정	개정
A862	DFT calculation	solvent model	parameters	Van der Waals parameters	species	2025-1	
A863					name	2025-1	
A864					model	2025-1	
A865					sigma	2025-1	
A866					epsilon	2025-1	
A867					mixing rule	2025-1	
A868				iterative scheme	algorithm	2025-1	
A869					convergence condition	2025-1	
A14		k-point			grid value	2021-1	
A15			k-point path for band structure		high-symmetry points	2021-1	
A16					k-point coordinates	2021-1	2022-2
A17					number of k-points	2021-1	
A18		convergence criteria			force	2021-1	2022-2
A19					energy	2021-1	
A20					scf	2021-1	
A21					symmetry	2021-1	
A22		magnetic structure			magnetic ordering	2021-1	
A360			magnetic moment		magnetic moment	2022-2	
A694					sublattice	2025-1	
A23					multiplicity	2021-1	
A24					potential	2021-1	
A25		LDA+U			scheme	2021-1	2025-1
A26					atom	2021-1	
A27					orbital	2021-1	
A28					U value	2021-1	
A29					J value	2021-1	
A30		partial occupation			method	2021-1	
A31					smearing width	2021-1	
A222					total energy	2022-1	
A32		density of states			method	2021-1	
A33					smearing width	2021-1	
A34					energy range	2021-1	
A35					number of energy point	2021-1	2022-2
A36		phonon calculations			method	2021-1	
A37					cell size	2021-1	
A38					displacement size	2021-1	2022-2
A39					number of displacement point	2021-1	2022-2
A40					q-points	2021-1	
A41		nudged elastic band			optimizer	2021-1	
A42					number of images	2021-1	2022-2
A361					images	2022-2	
A43					climbing image scheme	2021-1	
A44					force criterion	2021-1	2022-2
A427					spin orbit coupling	2023-1	
A582	differential scanning calorimetry (DSC)				instrument	2024-1	
A583		acquisition condition			start temperature	2024-1	
A584					end temperature	2024-1	
A585					temperature step	2024-1	
A586					heating rate	2024-1	
A587					raw data	2024-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A688	dilatometry			instrument	2025-1	
A689		acquisition condition		temperature range	2025-1	
A690				settling time	2025-1	
A691				pressure	2025-1	
A692				sensor	2025-1	
A693				raw data	2025-1	
A363	direct mercury analysis (DMA)			instrument	2022-2	
A364		acquisition condition		dry time	2022-2	
A365				raw data	2022-2	
A569	dynamic light scattering (DLS)			instrument	2024-1	
A570		acquisition condition		solvent	2024-1	
A571				temperature	2024-1	
A572				scattering angle	2024-1	
A573				laser wavelength	2024-1	
A574				measurement duration	2024-1	
A575				numbers of runs	2024-1	
A576		raw data		z-average size	2024-1	
A577				polydispersity index(PDI)	2024-1	
A578			size distribution	size class	2024-1	
A579				intensity percentage	2024-1	
A580			correlation function	time	2024-1	
A581				correlation	2024-1	
A695	ebullioscopy			instrument	2025-1	
A696		acquisition condition		solvent	2025-1	
A697			solute	material	2025-1	
A698				mass	2025-1	
A699				heating rate	2025-1	
A700				temperature range	2025-1	
A701				time	2025-1	
A702				raw data	2025-1	
A375	elastic recoil detection (ERD)			instrument	2022-2	
A376		acquisition condition		ion beam	2022-2	
A377				beam energy	2022-2	
A378				beam current	2022-2	
A379				beam angle	2022-2	
A380				counts	2022-2	
A381				beam size	2022-2	
A382				stopper foil	2022-2	
A383				raw data	2022-2	
A52	electrochemical activity			instrument	2021-1	2022-1
A588				geometric surface area	2024-1	
A237				reference electrode	2022-1	
A238				electrolyte composition	2022-1	2025-1
A53				temperature	2021-1	2022-1
A444				atmosphere	2023-2	
A57		raw data		data file	2021-1	2025-2

어휘번호	표준어휘 체계		표준어휘	제정	개정
A230	electrochemical impedance spectroscopy (EIS)		upper set frequency	2022-1	2024-2
A231			lower set frequency	2022-1	2024-2
A232			temperature	2022-1	
A233			resistance	2022-1	2022-2
A234			diameter	2022-1	
A235			thickness	2022-1	
A236			raw data	2022-1	
A703	electron dispersive x-ray spectroscopy (EDS)	instrument	instrument type	2025-1	
A704			microscopic model	2025-1	
A223			detector model	2022-1	2025-1
A224			mode	2022-1	
A225		acquisition condition	acceleration voltage	2022-1	
A226			beam current	2022-1	
A227			magnification	2022-1	
A705			pixel size	2025-1	
A706			field of view	2025-1	
A443			acquisition time	2023-2	
A228			frame	2022-1	2022-2
A229			raw data	2022-1	
A707	electron energy loss spectroscopy (EELS)	instrument	instrument type	2025-1	
A708			detector model	2025-1	
A709			mode	2025-1	
A710		acquisition condition	acceleration voltage	2025-1	
A711			beam current	2025-1	
A712			magnification	2025-1	
A713			pixel size	2025-1	
A714			field of view	2025-1	
A715			acquisition time	2025-1	
A716			dispersion channel	2025-1	
A717			raw data	2025-1	
A242	electron probe micro analyzer (EPMA)		instrument	2022-1	
A243		acquisition condition	sample preparation	2022-1	
A244			acceleration voltage	2022-1	
A245			current	2022-1	
A246			magnification	2022-1	
A247			spot size	2022-1	
A248			working distance	2022-1	
A249			raw data	2022-1	
A451	EQE analysis		instrument	2023-2	
A452		acquisition condition	irradiance of bias light	2023-2	
A659			type of bias light	2024-2	
A660			measurement mode	2024-2	
A661			external voltage bias	2024-2	
A662			chopping frequency	2024-2	
A453			raw data	2023-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A250	extended X-ray absorption fine structure (EXAFS)	instrument		beamline	2022-1	2025-1
A718				synchrotron facility	2025-1	
A251		acquisition condition		x-ray source	2022-1	
A719				monochromator type	2025-1	
A720				energy calibration reference	2025-1	
A721				incident angle	2025-1	
A252				beam size	2022-1	2022-2
A253				scanning step	2022-1	
A254				data collection mode	2022-1	2025-1
A722				element	2025-1	
A723				edge type	2025-1	
A255				raw data	2022-1	
A384	fatigue test			fatigue testing method	2022-2	
A97				instrument	2021-1	
A98				temperature	2021-1	
A890		temperature range		amplitude	2025-2	
A891				maximum	2025-2	
A892				minimum	2025-2	
A256				waveform	2022-1	
A105				stress ratio	2021-1	2022-2
A104				mean stress	2021-1	
A102				elastic strain amplitude	2021-1	2022-2
A103		plastic strain range		amplitude	2021-1	2025-2
A893				maximum	2025-2	
A894				minimum	2025-2	
A101		stress range		amplitude	2021-1	2025-2
A895				maximum	2025-2	
A896				minimum	2025-2	
A257				frequency	2022-1	2022-2
A897				notch	2025-2	
A898				fatigue test time	2025-2	
A589	film adhesion test			method	2024-1	
A590				equipment	2024-1	
A591				condition	2024-1	
A592				raw data	2024-1	
A258	focused ion beam (FIB)			instrument	2022-1	
A259		ion milling condition		acceleration voltage	2022-1	
A385				ion beam source type	2022-2	
A260				incident angle	2022-1	
A261				acceleration voltage	2022-1	
A262				raw data	2022-1	
A724	freezing point osmometry			instrument	2025-1	
A725		acquisition condition	solvent	material	2025-1	
A726				mass	2025-1	
A727			solute	material	2025-1	
A728				mass	2025-1	
A729				cooling rate	2025-1	
A730				temperature range	2025-1	
A731				raw data	2025-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
A263	fsLA-ICP-MS		instrument	2022-1	
A264			mode	2022-1	
A265		acquisition condition	power	2022-1	
A266			spot size	2022-1	
A267			repetition	2022-1	
A268			velocity	2022-1	2022-2
A269			analysis gas	2022-1	
A270			raw data	2022-1	
A106	gas adsorption and desorption isotherm		adsorptive	2021-1	2025-2
A107			temperature	2021-1	2025-1
A108			minimum pressure	2021-1	
A109			maximum pressure	2021-1	
A110			instrument	2021-1	
A111			image	2021-1	
A112			analysis method	2021-1	
A627	gas and liquid permeation measurement		permeating species	2024-2	
A628			temperature	2024-2	
A629			applied pressure range	2024-2	2025-1
A630			instrument	2024-2	
A631			analysis method	2024-2	
A113	gas chromatography		instrument	2021-1	
A114			measured gas	2021-1	
A115			atmosphere	2021-1	
A116			flow rate	2021-1	2022-2
A117			temperature	2021-1	
A118			voltage	2021-1	
A119			saturation time	2021-1	
A120			image	2021-1	
A121			analysis method	2021-1	
A593	gel electrophoresis		instrument	2024-1	
A594			gel type	2024-1	
A595			gel percentage	2024-1	
A596		acquisition condition	mode	2024-1	
A597			running time	2024-1	
A598			raw data	2024-1	
A732	gel permeation chromatography (GPC)		instrument	2025-1	
A733		acquisition condition	detector type	2025-1	
A734			flow rate	2025-1	
A735			solvent	2025-1	
A736			raw data	2025-1	
A122	hardness test		temperature	2021-1	
A123			test type	2021-1	2025-1
A271			test force	2022-1	
A272			dwell time	2022-1	
A273	ICP-MS		instrument	2022-1	
A274			mode	2022-1	
A275		acquisition condition	analysis gas	2022-1	
A276			raw data	2022-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
A277	ICP-OES		instrument	2022-1	
A278			mode	2022-1	
A279			acquisition condition	2022-1	
A280			raw data	2022-1	
A454	impact test		instrument	2023-2	
A124			temperature	2021-1	
A125			test type	2021-1	2025-1
A386	impedance analyzer		instrument	2022-2	
A387		acquisition condition	voltage	2022-2	
A388			current	2022-2	
A389			temperature	2022-2	
A390			frequency range	2022-2	
A391			raw data	2022-2	
A126	infrared spectroscopy		instrument	2021-1	
A127			atmosphere	2021-1	
A128			temperature	2021-1	
A129			image	2021-1	
A130			analysis method	2021-1	
A455	IV measurement		instrument	2023-2	
A632		scan	mode	2024-2	2025-1
A456			direction	2023-2	
A633			voltage range	2024-2	
A460			voltage step	2023-2	
A457			voltage sweep rate	2023-2	2024-2
A634			current range	2024-2	
A635			current step	2024-2	
A636			current sweep rate	2024-2	
A637			area of electrode	2024-2	
A458			delay time	2023-2	
A459			duration time	2023-2	
A638			sampling rate	2024-2	
A461			atmosphere	2023-2	
A462			temperature	2023-2	
A463		stability test	procedure	2023-2	
A464			measurement time	2023-2	
A465	JV measurement		equipment	2023-2	
A466		stabilized test	procedure	2023-2	
A467			metrics	2023-2	
A468			measurement time	2023-2	
A469			number of cells averaged	2023-2	
A470			certified	2023-2	
A471			certification institute	2023-2	
A472		storage before measurement	age of cell	2023-2	
A473			atmosphere	2023-2	
A474			relative humidity	2023-2	
A475			atmosphere	2023-2	
A476			relative humidity	2023-2	
A477			temperature	2023-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A478	JV measurement	light source		source type	2023-2	2025-1
A479				instrument	2023-2	
A480				simulator classification	2023-2	
A481		light		intensity	2023-2	
A482				type of spectrum	2023-2	
A483				wavelength range	2023-2	
A484				illumination direction	2023-2	
A485				masked cell	2023-2	
A486				mask area	2023-2	
A488		scan		direction	2023-2	
A489				sweep rate	2023-2	
A490				delay time	2023-2	
A491				integration time	2023-2	
A492				voltage step	2023-2	
A493				voltage settling time	2023-2	
A494		preconditioning		protocol	2023-2	
A495				time	2023-2	
A496				potential	2023-2	
A497				light intensity	2023-2	
A498				raw data	2023-2	
A599	liquid chromatography			instrument	2024-1	
A600		acquisition condition		flow rate	2024-1	
A601			column	temperature	2024-1	
A602				column type	2024-1	2025-1
A603				injection volume	2024-1	
A604				run time	2024-1	
A737			detector	detector type	2025-1	
A605				wavelength	2024-1	
A606				gas	2024-1	
A607				mobile phase solvent ratio	2024-1	
A608				raw data	2024-1	
A435	magnetic hysteresis loop			instrument	2023-1	
A436				specimen shape	2023-1	
A437				specimen length	2023-1	
A438				maximum applied field	2023-1	
A439				temperature	2023-1	
A440				magnetic hysteresis	2023-1	
A738	medium energy ion scattering (MEIS)			instrument	2025-1	
A739		acquisition condition		ion species	2025-1	
A740				ion energy	2025-1	
A741				beam current	2025-1	
A742				incident angle	2025-1	
A743				detection angle	2025-1	
A744				channeling angle	2025-1	
A745				raw data	2025-1	

어휘번호	표준어휘 체계				표준어휘	제정	개정
A746	membrane osmometry				instrument	2025-1	
A747	membrane osmometry	acquisition condition	membrane		membrane type	2025-1	
A748	membrane osmometry	acquisition condition	membrane		molecular weight cut-off (MWCO)	2025-1	
A749	membrane osmometry	acquisition condition	membrane		pore size	2025-1	
A750	membrane osmometry	acquisition condition			solvent	2025-1	
A751	membrane osmometry	acquisition condition	solute		material	2025-1	
A752	membrane osmometry	acquisition condition	solute		mass	2025-1	
A753	membrane osmometry	acquisition condition			temperature	2025-1	
A754	membrane osmometry	acquisition condition			time	2025-1	
A755	membrane osmometry				raw data	2025-1	
A131	memristive activity				instrument	2021-1	
A132	memristive activity				temperature	2021-1	
A133	memristive activity				maximum voltage	2021-1	
A134	memristive activity				minimum voltage	2021-1	
A135	memristive activity				current	2021-1	
A136	memristive activity				compliance current	2021-1	
A137	memristive activity				pulse	2021-1	
A58	molecular dynamics simulation	simulation condition	electrostatic interaction		code	2021-1	
A445	molecular dynamics simulation	simulation condition	electrostatic interaction	Ewald	fourier spacing	2023-2	
A446	molecular dynamics simulation	simulation condition	electrostatic interaction	Ewald	accuracy	2023-2	
A447	molecular dynamics simulation	simulation condition	electrostatic interaction	PME	fourier spacing	2023-2	
A448	molecular dynamics simulation	simulation condition	electrostatic interaction	PME	PME-order	2023-2	
A449	molecular dynamics simulation	simulation condition	electrostatic interaction	PPPM	accuracy	2023-2	
A450	molecular dynamics simulation	simulation condition	electrostatic interaction	MSM	accuracy	2023-2	
A59	molecular dynamics simulation	simulation condition	force field		name	2021-1	
A60	molecular dynamics simulation	simulation condition	force field		source	2021-1	
A61	molecular dynamics simulation	simulation condition	force field		parameter	2021-1	2022-1
A62	molecular dynamics simulation	simulation condition			cutoff radius	2021-1	2022-2
A63	molecular dynamics simulation	simulation condition			time step	2021-1	
A64	molecular dynamics simulation	simulation condition			integration algorithm	2021-1	
A69	molecular dynamics simulation	simulation condition			pressure	2021-1	
A70	molecular dynamics simulation	simulation condition			temperature	2021-1	
A71	molecular dynamics simulation	simulation condition			time	2021-1	
A72	molecular dynamics simulation	simulation condition			ensemble	2021-1	2025-1
A369	molecular dynamics simulation	simulation condition			thermostat	2022-2	
A370	molecular dynamics simulation	simulation condition			barostat	2022-2	
A65	molecular dynamics simulation	initialization			pressure	2021-1	
A66	molecular dynamics simulation	initialization			temperature	2021-1	
A67	molecular dynamics simulation	initialization			time	2021-1	
A366	molecular dynamics simulation	initialization			initial velocity distribution method	2022-2	
A68	molecular dynamics simulation	initialization	ensemble		scheme	2021-1	2025-1
A367	molecular dynamics simulation	initialization	ensemble		thermostat	2022-2	
A368	molecular dynamics simulation	initialization	ensemble		barostat	2022-2	
A73	molecular dynamics simulation	control	dimensional		direction	2021-1	
A74	molecular dynamics simulation	control	dimensional		strain	2021-1	
A75	molecular dynamics simulation	control	dimensional		strain rate	2021-1	2022-2

어휘번호	표준어휘 체계				표준어휘	제정	개정
A80	molecular dynamics simulation	control	thermal		initial temperature	2021-1	
A81				cycle	heating rate	2021-1	2025-1
A82					holding temperature	2021-1	2025-1
A83					holding time	2021-1	2025-1
A84					cooling rate	2021-1	2025-1
A85					final temperature	2021-1	
A90			constitutional		manipulated constituent	2021-1	2025-1
A91					energy	2021-1	
A92					direction	2021-1	
A93					interval	2021-1	
A94					total number	2021-1	2022-2
A95		output			trajectory	2021-1	
A240					result	2022-1	2025-1
A96					analysis	2021-1	
A241					input file	2022-1	
A281	nano-indentation				instrument	2022-1	
A282		acquisition condition			tip type	2022-1	
A283					test type	2022-1	
A284					time segment	2022-1	2022-2
A285					peak load	2022-1	
A286					indent pattern	2022-1	
A287					indent spacing	2022-1	
A288					raw data	2022-1	
A289	near-edge X-ray absorption fine structure (NEXAFS)	instrument			beamline	2022-1	2025-1
A756					synchrotron facility	2025-1	
A290		acquisition condition			x-ray source	2022-1	
A757					monochromator type	2025-1	
A758					energy calibration reference	2025-1	
A759					polarization	2025-1	
A760					incident angle	2025-1	
A291					beam size	2022-1	2022-2
A292					scanning step	2022-1	
A293					data collection method	2022-1	2025-1
A294					raw data	2022-1	
A761					data processing software	2025-1	
A138	nuclear magnetic resonance				instrument	2021-1	
A139					temperature	2021-1	
A140					image	2021-1	
A141					analysis method	2021-1	
A142	optical microscopy				instrument	2021-1	
A143					condition	2021-1	
A295					magnification	2022-1	2022-2
A144					image	2021-1	
A145					analysis method	2021-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A775	particle-induced X-ray emission (PIXE)			instrument	2025-1	
A776		acquisition condition		ion species	2025-1	
A777				ion energy	2025-1	
A778				beam current	2025-1	
A779				detection angle	2025-1	
A780				detector type	2025-1	
A781				filter	2025-1	
A782				quantification method	2025-1	
A783				raw data	2025-1	
A609	peel test			instrument	2024-1	
A610		acquisition condition		peel angle	2024-1	
A611				peel speed	2024-1	
A612			force range	min	2024-1	
A613				max	2024-1	
A614				raw data	2024-1	
A555	photoelectron spectroscopy (PES)	instrument		instrument name	2025-1	2025-1
A762				beamline name	2025-1	
A763				synchrotron facility	2025-1	
A764		acquisition condition		measurement mode	2025-1	
A556				beam source	2025-1	2025-1
A765				acceleration voltage	2025-1	
A766				current	2025-1	
A767				dwell time	2025-1	
A345				photon energy	2025-1	2025-1
A557				pass energy	2025-1	
A342				beamsize	2025-1	
A344				take-off angle	2025-1	2025-1
A343				charge neutralization	2025-1	2025-1
A768				sample bias voltage	2025-1	
A769				sample temperature	2025-1	
A770				element	2025-1	
A771				edge type	2025-1	
A772				scanning range	2025-1	
A558				energy step size	2025-1	2025-1
A773				surface preparation	2025-1	
A559				raw data	2025-1	
A774				data processing software	2025-1	
A392	piezoresponse force microscopy (PFM)			instrument	2022-2	
A393		acquisition condition		cantilever type	2022-2	
A394				scan rate	2022-2	
A395				scan size	2022-2	
A396				Z servo gain	2022-2	
A397				raw data	2022-2	
A527	PV outdoor test			protocol	2023-2	
A528				number of cells averaged	2023-2	
A529		location		country	2023-2	
A530				city	2023-2	
A531				coordinates	2023-2	
A532				climate zone	2023-2	

어휘번호	표준어휘 체계		표준어휘	제정	개정
A533	PV outdoor test	installation	tilt	2023-2	
A534			cardinal direction	2023-2	
A535			number of solar tracking axis	2023-2	
A536		time	season	2023-2	
A537			start	2023-2	
A538			end	2023-2	
A539			total exposure	2023-2	
A540		potential bias	load condition	2023-2	
A541			range	2023-2	
A542			passive resistance	2023-2	
A543		temperature	load condition	2023-2	
A544			range	2023-2	
A545			Tmodule	2023-2	
A546		periodic JV measurement	time between measurements	2023-2	
A547			raw data for outdoor trace	2023-2	
A548			detailed weather data	2023-2	
A549			spectral data	2023-2	
A550			irradiance data	2023-2	
A499	PV stability		protocol	2023-2	
A500			number of cells averaged	2023-2	
A501		light source	source type	2023-2	2025-1
A502			instrument	2023-2	
A503			simulator classification	2023-2	
A504		light	intensity	2023-2	
A505			type of spectrum	2023-2	
A506			wavelength range	2023-2	
A507			illumination direction	2023-2	
A508			load condition	2023-2	
A509			cycling times	2023-2	
A510			UV filter	2023-2	
A511		potential bias	load condition	2023-2	
A512			range	2023-2	
A513			passive resistance	2023-2	
A514		temperature	load condition	2023-2	
A515			range	2023-2	
A516			cycling times	2023-2	
A517			ramp speed	2023-2	
A518		atmosphere	composition	2023-2	
A519			oxygen concentration	2023-2	
A520		relative humidity	load condition	2023-2	
A521			range	2023-2	
A522			average value	2023-2	
A523			total exposure time	2023-2	
A524		periodic JV measurement	time betweeen measurements	2023-2	
A525		flexibility	number of bending cycle	2023-2	
A526			bending radius	2023-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A296	Raman spectroscopy			instrument	2022-1	
A297		acquisition condition		laser wavelength	2022-1	2023-2
A551				laser power	2023-2	
A663				laser spot diameter	2024-2	
A664				slit	2024-2	
A665				numerical aperture	2024-2	
A666				grating groove density	2024-2	
A552				spectrometer resolution	2023-2	
A553				acquisition time	2023-2	
A298				magnification	2022-1	
A299				raw data	2022-1	
A406	refractometer			instrument	2022-2	
A407				temperature	2022-2	
A408				raw data	2022-2	
A409	rheological analysis			instrument	2022-2	
A410				temperature	2022-2	
A411				humidity	2022-2	
A412				gap	2022-2	
A413				geometry	2022-2	
A414				material	2022-2	
A415				raw data	2022-2	
A398	Rutherford backscattering spectrometry (RBS)			instrument	2022-2	
A399		acquisition condition		ion beam	2022-2	
A400				beam energy	2022-2	
A401				beam current	2022-2	
A402				beam angle	2022-2	
A403				counts	2022-2	
A404				beam size	2022-2	
A405				raw data	2022-2	
A300	scanning electron microscope (SEM)			instrument	2022-1	
A416				gun type	2022-2	
A301		acquisition condition		acceleration voltage	2022-1	
A302				current (spot size)	2022-1	2022-2
A303				magnification	2022-1	
A784				pixel size	2025-1	
A785				field of view	2025-1	
A304				working distance	2022-1	
A305				raw data	2022-1	
A417				mode	2022-2	
A306	secondary ion mass spectrometry (SIMS)			instrument	2022-1	
A307		acquisition condition		primary ion	2022-1	
A308				beam energy	2022-1	
A309				beam current	2022-1	
A310			neutralizer	flood gun	2022-1	
A311				E-gun	2022-1	
A312				polarity	2022-1	
A313				sample voltage	2022-1	
A314				raster size	2022-1	
A315				sputter time	2022-1	
A316				raw data	2022-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A786	surface profilometry			instrument	2025-1	
A787	surface profilometry			scan type	2025-1	
A788	surface profilometry			raw data	2025-1	
A155	tensile test			instrument	2021-1	
A321	tensile test			specimen shape	2022-1	
A156	tensile test			specimen direction	2021-1	
A322	tensile test			gauge length	2022-1	
A157	tensile test			temperature	2021-1	
A420	tensile test			specimen width	2022-2	
A421	tensile test			specimen thickness	2022-2	
A422	tensile test			specimen diameter	2022-2	
A159	tensile test	tensile test speed		strain rate	2021-1	2022-2
A423	tensile test	tensile test speed		stress rate	2022-2	
A424	tensile test	tensile test speed		crosshead separation rate	2022-2	
A639	tensile test			test standard	2024-2	
A425	tensile test			atmosphere	2022-2	2024-2
A428	tensile test			raw data	2023-1	
A160	thermal activity			instrument	2021-1	2022-1
A161	thermal activity			flow rate	2021-1	2022-2
A162	thermal activity			temperature	2021-1	2022-1
A163	thermal activity			heating time	2021-1	2022-1
A164	thermal activity			raw data	2021-1	2022-1
A567	thermodynamic calculation			software	2024-2	
A568	thermodynamic calculation			database	2024-2	
A165	thermogravimetric analysis (TGA)			instrument	2021-1	
A791	thermogravimetric analysis (TGA)	acquisition condition		sample mass	2025-1	
A166	thermogravimetric analysis (TGA)	acquisition condition		starting temperature	2021-1	
A167	thermogravimetric analysis (TGA)	acquisition condition		final temperature	2021-1	
A168	thermogravimetric analysis (TGA)	acquisition condition		ramp rate	2021-1	2022-2
A323	thermogravimetric analysis (TGA)	acquisition condition	flow gas	gas type	2022-1	
A324	thermogravimetric analysis (TGA)	acquisition condition	flow gas	flow rate	2022-1	2022-2
A170	thermogravimetric analysis (TGA)	acquisition condition	flue gas	gas type	2021-1	
A171	thermogravimetric analysis (TGA)	acquisition condition	flue gas	flue rate	2021-1	2022-2
A667	thermogravimetric analysis (TGA)	acquisition condition		crucible	2024-2	
A792	thermogravimetric analysis (TGA)	acquisition condition		weight percentage	2025-1	
A172	thermogravimetric analysis (TGA)			image	2021-1	
A173	thermogravimetric analysis (TGA)			analysis method	2021-1	
A793	thin film residual stress measurement by curvature			curvature measurement method	2025-1	
A794	thin film residual stress measurement by curvature	substrate		thickness	2025-1	
A795	thin film residual stress measurement by curvature	substrate		biaxial elastic modulus	2025-1	
A796	thin film residual stress measurement by curvature			film thickness	2025-1	
A797	thin film residual stress measurement by curvature			curvature	2025-1	
A146	transmission electron microscope (TEM)			instrument	2021-1	
A147	transmission electron microscope (TEM)			mode	2021-1	2022-2

어휘번호	표준어휘 체계			표준어휘	제정	개정
A554	transmission electron microscope (TEM)	acquisition condition		grid	2023-2	
A148				acceleration voltage	2021-1	
A149				beam current	2021-1	
A926				beam dose	2025-2	
A151				magnification	2021-1	
A789				pixel size	2025-1	
A790				field of view	2025-1	
A317				exposure time	2022-1	
A418		analysis		zone-axis	2022-2	
A318		raw data		dimension	2022-1	2022-2
A419				scale unit	2022-2	
A319				binning	2022-1	
A320				image	2022-1	
A640	tribotest			instrument	2024-2	
A641		acquisition condition		test type	2024-2	
A642				normal load	2024-2	
A798			counterface	shape	2025-1	
A799				material	2025-1	
A643				dimension	2024-2	2025-1
A644				apparent contact area	2024-2	
A800				lubrication	2025-1	
A645				contact pressure	2024-2	
A646				sliding speed	2024-2	
A647				sliding distance	2024-2	
A648				sliding diameter	2024-2	
A649				stroke amplitude	2024-2	
A650				stroke frequency	2024-2	
A651				atmosphere	2024-2	
A652				humidity	2024-2	
A653				temperature	2024-2	
A654				time	2024-2	
A655				raw data	2024-2	
A656				analysis method	2024-2	
A325	USANS			instrument	2022-1	
A326		acquisition condition		neutron wavelength	2022-1	2022-2
A327				beamsize	2022-1	2022-2
A328				angular resolution	2022-1	2022-2
A329				scanning start angle	2022-1	
A330				scanning step	2022-1	
A331				scanning end angle	2022-1	
A332				motor velocity	2022-1	2022-2
A333				run time	2022-1	2022-2
A334				raw data	2022-1	
A560	UV-VIS-nIR spectrophotometer			instrument	2023-2	
A801		acquisition condition		wavelength selection	2025-1	
A561				wavelength range	2023-2	
A802				measurement mode	2025-1	
A805				path length	2025-1	
A806				cuvette type	2025-1	
A562				scan speed	2023-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A657	UV-VIS-nIR spectrophotometer	acquisition condition		solvent	2024-2	
A563				atmosphere	2023-2	
A668				temperature	2024-2	
A810				specimen form	2025-1	
A564				raw data	2023-2	
A811	vapor phase osmometry			instrument	2025-1	
A812		acquisition condition		temperature	2025-1	
A813			solvent	material	2025-1	
A814				mass	2025-1	
A815			solute	material	2025-1	
A816				mass	2025-1	
A817				time	2025-1	
A818				raw data	2025-1	
A819	viscometry	instrument		viscometer type	2025-1	
A933				geometry	2025-2	
A820				brand name	2025-1	
A934		acquisition condition		angular frequency	2025-2	
A935				strain amplitude	2025-2	
A936				shear stress	2025-2	
A821				shear rate	2025-1	
A822				temperature	2025-1	
A823				raw data	2025-1	
A824	voltammetry			instrument	2025-1	
A937		acquisition condition		cycle	2025-2	
A335				upper voltage	2022-1	
A336				lower voltage	2022-1	
A615				capacity cut	2024-1	
A616				voltage cut	2024-1	
A337				scan rate	2022-1	2022-2
A338				temperature	2022-1	
A426				electrode configuration	2022-2	
A339				raw data	2022-1	
A617	white light interferometry			instrument	2024-1	
A618		acquisition condition		magnification	2024-1	
A619				raw data	2024-1	
A825	X-ray absorption near edge structure (XANES)	instrument		beamline name	2025-1	
A826				synchrotron facility	2025-1	
A827		acquisition condition		x-ray source	2025-1	
A828				monochromator type	2025-1	
A829				energy calibration reference	2025-1	
A830				polarization	2025-1	
A831				incident angle	2025-1	
A832				beam size	2025-1	
A833				scanning step	2025-1	
A834				data collection mode	2025-1	
A835				element	2025-1	
A836				edge type	2025-1	
A837				pre-edge region	2025-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
A838	X-ray absorption near edge structure (XANES)	acquisition condition		edge position	2025-1	
A839				scanning range	2025-1	
A840				raw data	2025-1	
A841				data processing software	2025-1	
A174	X-ray diffraction (XRD)			instrument	2021-1	
A842				specimen form	2025-1	
A843				full width at half maximum	2025-1	
A565		acquisition condition		x-ray source	2023-2	
A175				x-ray wavelength	2021-1	2022-2
A176				acceleration voltage	2021-1	
A177				current	2021-1	2022-1
A178				scan axis	2021-1	
A179				scanning step	2021-1	2022-2
A180				incident angle	2021-1	2022-2
A181				offset angle	2021-1	2022-2
A348				2theta range	2022-1	2022-2
A185				scan rate	2021-1	2022-2
A186				sample rotation	2021-1	2022-2
A566				sample temperature	2023-2	
A187				raw data	2021-1	2022-1
A658				analysis method	2024-2	
A349	X-ray fluorescence (XRF)			instrument	2022-1	
A350		acquisition condition		x-ray target	2022-1	
A351				XRF type	2022-1	
A352				sample preparation	2022-1	
A353				analysis elements	2022-1	
A354				quantitative analysis method	2022-1	
A355				raw data	2022-1	
A844	X-ray scattering (XRS)	instrument		instrument name	2025-1	
A845				beamline name	2025-1	
A846				synchrotron facility	2025-1	
A847		acquisition condition		mode	2025-1	
A848				x-ray source	2025-1	
A938				x-ray wavelength	2025-2	
A939				incidence angle	2025-2	
A940				beam position	2025-2	
A849			detector	detector type	2025-1	
A941				pixel size	2025-2	
A942				detector matrix	2025-2	
A850				sample-to-detector distance	2025-1	
A851				beam size	2025-1	
A852				exposure time	2025-1	
A853				raw data	2025-1	
A854				data processing software	2025-1	

5 소재 공정 공통어휘

어휘번호	표준어휘 체계		표준어휘	제정	개정
P723	additive manufacturing		layer thickness	2025-1	
P724			print speed	2025-1	
P725			nozzle temperature	2025-1	
P726			bed temperature	2025-1	
P727			infusion pattern	2025-1	
P588	arc melting		hearth	2024-2	
P968			electrode	2025-2	
P589			temperature	2024-2	
P728			electric capacity	2025-1	
P590			heating rate	2024-2	
P591			holding time	2024-2	
P592			cooling rate	2024-2	2025-1
P225	atomic layer deposition (ALD)		energy type	2022-1	
P226		precursor	name	2022-1	
P227			amount	2022-1	
P228			temperature	2022-1	
P229			type of canister	2022-1	2024-2
P230		reactant	name	2022-1	
P231			amount	2022-1	
P232			temperature	2022-1	
P233			type of canister	2022-1	2024-2
P234		carrier gas	name	2022-1	
P235			flow rate	2022-1	2022-2
P236			purity	2022-1	
P237		purge gas	name	2022-1	
P238			flow rate	2022-1	2022-2
P239			purity	2022-1	
P240		pulse time	purge time	2022-1	
P241			feeding time	2022-1	
P242		cycle	cycle time	2022-1	
P243			number of cycles	2022-1	2024-2
P244			substrate temperature	2022-1	
P245			working pressure	2022-1	
P246			base pressure	2022-1	
P247			growth per cycle	2022-1	2022-2
P248	atomic layer etching (ALE)		energy type	2022-1	
P249		etching agent	name	2022-1	
P250			amount	2022-1	
P251			temperature	2022-1	
P252			type of canister	2022-1	2024-2
P253		reactant	name	2022-1	
P254			amount	2022-1	
P255			temperature	2022-1	
P256			type of canister	2022-1	2024-2
P257		carrier gas	name	2022-1	
P258			flow rate	2022-1	2022-2
P259			purity	2022-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P260	atomic layer etching (ALE)	purge gas	name	2022-1	
P261			flow rate	2022-1	2022-2
P262			purity	2022-1	
P263		pulse time	purge time	2022-1	
P264			feeding time	2022-1	
P265		cycle	cycle time	2022-1	
P266			number of cycles	2022-1	2024-2
P267			substrate temperature	2022-1	
P268			working pressure	2022-1	
P269			base pressure	2022-1	
P270			etch per cycle	2022-1	2022-2
P729	autoclave molding		pressure	2025-1	
P730			temperature cycle	2025-1	
P731			vacuum level	2025-1	
P732			curing time	2025-1	
P733			tool material	2025-1	
P734	automated fiber placement (AFP)		fiber tow width	2025-1	
P735			placement speed	2025-1	
P736			consolidation method	2025-1	
P737	automated tape placement (ATP)		tape width	2025-1	
P738			placement speed	2025-1	
P739			temperature	2025-1	
P740			consolidation pressure	2025-1	
P741			material type	2025-1	
P593	ball milling		equipment	2024-2	
P392			material	2022-2	
P271			mill type	2022-1	2025-1
P272			rotation speed	2022-1	2022-2
P273			total time	2022-1	
P274			milling time	2022-1	
P275			rest time	2022-1	
P276			ball to powder weight ratio	2022-1	2023-2
P277			material mass	2022-1	2022-2
P278			solvent	2022-1	
P393			ball material	2022-2	
P395			ball size	2022-2	
P279			jar shape	2022-1	2022-2
P280			jar volume	2022-1	
P281			jar material	2022-1	2022-2
P282			atmosphere	2022-1	
P396	blade coating		coating material	2022-2	
P397			substrate	2022-2	
P466			blade size	2023-2	
P398			gap	2022-2	
P399			speed	2022-2	2023-2
P467			feed	2023-2	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P742	blow (bottle)		materials type	2025-1	
P743			material loading	2025-1	
P744			temperature	2025-1	
P745		parison	length	2025-1	
P746			diameter	2025-1	
P747			blow pressure	2025-1	
P748			inflation rate	2025-1	
P749			mold temperature	2025-1	
P750			cooling time	2025-1	
P751			cycle time	2025-1	
P752	blown film extrusion		materials type	2025-1	
P753			melt temperature	2025-1	
P754			die temperature	2025-1	
P755			blow-up ratio	2025-1	
P756			extruder speed	2025-1	
P757			inflation pressure	2025-1	
P758			cooling method	2025-1	
P759			take-up speed	2025-1	
P760			film thickness	2025-1	
P761			additives	2025-1	
P701	Bridgman growth method		ingot diameter	2024-2	
P702			ingot length	2024-2	
P703			temperature	2024-2	
P704			holding time	2024-2	
P705			heating rate	2024-2	
P706			down rate	2024-2	
P762			reaction container	2025-1	
P283	bulk metal forming		metal forming type	2022-1	
P284			heating rate	2022-1	2022-2
P400			forming force	2022-2	
P285			forming temperature	2022-1	
P286			reduction ratio	2022-1	
P287			pass number	2022-1	2022-2
P401			cooling rate	2022-2	
P288			cooling method	2022-1	
P763	calendering		materials type	2025-1	
P764			roller temperature	2025-1	
P765			roller speed	2025-1	
P766			roller gap	2025-1	
P767			configuration	2025-1	
P768			feed rate	2025-1	
P769			cooling method	2025-1	
P770			slip agent	2025-1	
P427	casting		casting type	2022-2	2025-1
P468			casting pressure	2023-2	
P329			casting temperature	2022-1	2024-2
P594			mold material	2024-2	
P330			mold temperature	2022-1	
P331			pouring speed	2022-1	2022-2
P332			cooling rate	2022-1	2022-2
P333			solidification time	2022-1	
P974			chill	2025-2	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P1	centrifugation		rotation speed	2021-1	2025-2
P289			relative centrifugal force	2022-1	
P2			time	2021-1	
P3			temperature	2021-1	
P4		additive	name	2021-1	
P5			amount	2021-1	
P402			tube material	2022-2	
P403			tube volume	2022-2	
P469	chemical bath deposition	precursor	name	2023-2	
P470			composition	2023-2	
P471			excess	2023-2	
P472			purity	2023-2	
P473			supplier	2023-2	
P474			volume	2023-2	
P475			age	2023-2	
P476			temperature	2023-2	
P477			state	2023-2	
P478			molarity	2023-2	
P479		solvent	composition	2023-2	
P480			purity	2023-2	
P481			supplier	2023-2	
P482			temperature	2023-2	
P483		atmosphere	composition	2023-2	
P484			total pressure	2023-2	
P485			relative humidity	2023-2	
P486			time	2023-2	
P6	chemical mechanical polishing (CMP)		slurry concentration	2021-1	
P7			slurry density	2021-1	2022-2
P8			viscosity	2021-1	2022-2
P9			polishing rate	2021-1	
P10			surface roughness	2021-1	
P11		precursor	name	2021-1	
P12			amount	2021-1	
P290			pressure	2022-1	2022-2
P13			temperature	2021-1	
P14			atmosphere	2021-1	
P15			rpm	2021-1	2022-2
P16	chemical synthesis	precursor	name	2021-1	
P17			amount	2021-1	
P19			solution	2021-1	
P291			pressure	2022-1	
P20			temperature	2021-1	
P292			time	2022-1	
P771	chemical vapor deposition (CVD)		CVD type	2025-1	
P772		power supply	applied voltage	2025-1	
P773			applied current	2025-1	
P774			applied power	2025-1	
P775			pulse width	2025-1	
P776			frequency	2025-1	
P777			laser	2025-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P778	chemical vapor deposition (CVD)		microwave source	2025-1	
P779			filament	2025-1	
P293			pressure	2022-1	2024-1
P297			temperature	2022-1	
P309			time	2022-1	
P294		carrier gas	name	2022-1	
P295			flow rate	2022-1	2022-2
P303		process gas	name	2022-1	
P304			flow rate	2022-1	2022-2
P296		precursor	name	2022-1	
P301			amount	2022-1	
P568		catalyst	name	2024-1	
P569			amount	2024-1	
P570		promotor	name	2024-1	
P571			amount	2024-1	
P305		jig	name	2022-1	2025-2
P780			jig type	2025-1	
P781			rotating speed	2025-1	
P492	cleaning		method	2023-2	2025-1
P782			solvent	2025-1	
P783			time	2025-1	
P784	compression molding		materials type	2025-1	
P785			material loading	2025-1	
P786			material flow characteristics	2025-1	
P787			compression pressure	2025-1	
P788			molding temperature	2025-1	
P789			molding time	2025-1	
P790			heating method	2025-1	
P791			heating rate	2025-1	
P792			cooling method	2025-1	
P793			cooling rate	2025-1	
P794	conditioning		temperature	2025-1	
P795			humidity	2025-1	
P796			time	2025-1	
P797			airflow	2025-1	
P798			initial moisture content	2025-1	
P799			process equipment	2025-1	
P800			chemical additives	2025-1	
P801			cooling rate	2025-1	
P1027		post-conditioning	drying	2025-2	
P1028			annealing	2025-2	
P424	corona poling		corona needle	2022-2	
P425			needle to sample distance	2022-2	
P421			applied voltage	2022-2	
P423			temperature	2022-2	
P420			time	2022-2	
P422			atmosphere	2022-2	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P803	crystallization		temperature	2025-1	
P804			cooling rate	2025-1	
P805			time	2025-1	
P806			external stress	2025-1	
P807			polymer molecular weight	2025-1	
P808			additive	2025-1	
P809			shear rate	2025-1	
P810			cooling medium	2025-1	
P811			humidity	2025-1	
P812			atmosphere	2025-1	
P575	dialysis		membrane type	2024-1	
P576			membrane molecular weight cut-off	2024-1	
P577			dialysis buffer	2024-1	
P578			temperature	2024-1	
P579			time	2024-1	2025-1
P572	diamond turning (DTM)		depth	2024-1	
P573			feed rate	2024-1	
P574			rpm	2024-1	
P976	digital light processing		wavelength	2025-2	
P977			intensity	2025-2	
P978			vat temperature	2025-2	
P979			exposure time	2025-2	
P980			layer thickness	2025-2	
P981			platform temperature	2025-2	
P982			resolution	2025-2	
P983			build speed	2025-2	
P595	dip coating		coating material	2024-2	
P596			coating solvent	2024-2	
P597		substrate	material	2024-2	2025-1
P813		substrate	orientation	2025-1	
P598			speed	2024-2	
P599			temperature	2024-2	
P814			relative humidity	2025-1	
P426	direct current poling		applied electric field	2022-2	2024-2
P600			temperature	2024-2	
P601			time	2024-2	
P602			atmosphere	2024-2	
P815	direct ink writing		method	2025-1	
P493			solute	2023-2	
P494			solvent	2023-2	
P495			concentration	2023-2	
P496			temperature	2023-2	
P497			atmosphere	2023-2	
P637			traveling speed	2024-2	
P638			dispensing pressure	2024-2	
P639			nozzle diameter	2024-2	
P640			3D modelling SW	2024-2	
P328	drying		pressure	2022-1	
P31			temperature	2021-1	2022-1
P32			atmosphere	2021-1	2022-1
P33			time	2021-1	2022-1

어휘번호	표준어휘 체계			표준어휘	제정	개정
P34	e-beam lithography			acceleration voltage	2021-1	2022-1
P35		e-beam resist		name	2021-1	2022-1
P36			baking	temperature	2021-1	2022-1
P37				time	2021-1	2022-1
P38				thickness	2021-1	2022-1
P39				working distance	2021-1	2022-1
P40				beam current	2021-1	2022-1
P41				dwell time	2021-1	2022-1
P42				step size	2021-1	2022-1
P43				line spacing	2021-1	2022-1
P44				area dose	2021-1	2022-2
P45		develop		developer	2021-1	2022-1
P46				time	2021-1	2022-1
P720				temperature	2024-2	
P47	electrochemical deposition	electrolyte	precursor	name	2021-1	2022-1
P48				amount	2021-1	2022-1
P49				solvent	2021-1	2022-1
P50				concentration	2021-1	2022-1
P51				pH	2021-1	2022-1
P52				additive	2021-1	
P53				working electrode	2021-1	2022-1
P54				counter electrode	2021-1	2022-1
P487				electrode distance	2023-2	
P55				reference electrode	2021-1	2022-1
P56		deposition condition		mode	2021-1	2022-1
P57				voltage	2021-1	2022-1
P58				current density	2021-1	2022-2
P59				stirring	2021-1	2022-2
P60				temperature	2021-1	2022-1
P61				atmosphere	2021-1	
P62				time	2021-1	2022-1
P404	electrospinning			spin type	2022-2	2025-1
P405		precursor	solute	name	2022-2	
P406				amount	2022-2	
P407			solvent	name	2022-2	
P408				amount	2022-2	
P409				concentration	2022-2	
P410		tip size		inner diameter	2022-2	
P411				outer diameter	2022-2	
P412		collector		collector type	2022-2	2025-1
P413				rotating speed	2022-2	
P414				diameter	2022-2	
P721				material	2024-2	2025-1
P415				tip to collector distance	2022-2	
P416				applied voltage	2022-2	
P417				injection rate	2022-2	2023-1
P418				relative humidity	2022-2	
P419				temperature	2022-2	
P488	encapsulation			encapsulating materials	2023-2	
P489				edge sealing materials	2023-2	
P490				atmosphere	2023-2	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P816	encapsulated melting		ampoule material	2025-1	
P817			ampoule environment	2025-1	
P603			inner diameter of the ampoule	2024-2	
P818			melted material	2025-1	
P604			melting method	2024-2	
P605		thermal cycle	heating rate	2024-2	
P606			melting temperature	2024-2	
P607			holding time	2024-2	
P608			cooling rate	2024-2	
P609			cooling method	2024-2	
P610			post heat-treatment temperature	2024-2	
P611			post heat-treatment time	2024-2	
P819	evaporation deposition		atmosphere	2025-1	
P820			base pressure	2025-1	
P386			deposition rate	2022-1	
P561		evaporating material	name	2023-2	
P562			composition	2023-2	2025-1
P563			supplier	2023-2	
P564			purity	2023-2	
P565			age	2023-2	
P566			state	2023-2	
P821			evaporation method	2025-1	
P822			source power	2025-1	2025-2
P823			evaporation type	2025-1	
P824			substrate distance	2025-1	
P825			substrate rotation	2025-1	
P385			substrate temperature	2022-1	
P717			thickness	2024-2	
P387			time	2022-1	
P826	extrusion		materials type	2025-1	
P827			material feed rate	2025-1	
P828		temperature profile	T feed zone	2025-1	
P829			T compression zone	2025-1	
P830			T metering zone	2025-1	
P831			extruder speed	2025-1	
P832			line speed	2025-1	
P833			pressure	2025-1	
P834			residence time	2025-1	
P835			cooling method	2025-1	
P836	filament winding		fiber tension	2025-1	
P837			winding angle	2025-1	
P838			winding speed	2025-1	
P839			mandrel material	2025-1	
P840			number of layers	2025-1	
P841			resin injection rate	2025-1	
P842		winding atmosphere	humidity	2025-1	
P843			temperature	2025-1	
P844		curing	time	2025-1	
P845			temperature	2025-1	
P846			pressure	2025-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
P612	filtration			atmosphere	2024-2	
P613	filtration			time	2024-2	
P614	grinding			method	2024-2	
P615	grinding			speed	2024-2	
P616	grinding			time	2024-2	
P334	heat treatment			method	2022-1	2023-2
P335	heat treatment			atmosphere	2022-1	
P847	heat treatment			flow rate	2025-1	
P336	heat treatment	thermal history		temperature	2022-1	2023-2
P428	heat treatment	thermal history		heating rate	2022-2	
P337	heat treatment	thermal history		time	2022-1	2023-2
P429	heat treatment	thermal history		cooling rate	2022-2	
P984	heat treatment	thermal history		cooling method	2025-2	
P985	heat treatment	thermal history		finishing temperature	2025-2	
P722	heat treatment			crucible	2024-2	
P617	hot injection	precursor		name	2024-2	
P618	hot injection	precursor		concentration	2024-2	
P619	hot injection	precursor		amount	2024-2	
P620	hot injection	precursor		temperature	2024-2	
P621	hot injection	solvent		composition	2024-2	
P622	hot injection	solvent		volume	2024-2	
P623	hot injection			ligand composition	2024-2	2025-1
P848	hot injection			injection pressure	2025-1	
P624	hot injection			reaction temperature	2024-2	
P625	hot injection			reaction time	2024-2	
P626	hot injection			quenching temperature	2024-2	
P627	hot injection			quenching time	2024-2	
P628	hot injection	atmosphere		composition	2024-2	
P629	hot injection	atmosphere		total pressure	2024-2	
P632	hot pressing	sintering cycle		sintering pressure	2024-2	2025-1
P633	hot pressing	sintering cycle		sintering temperature	2024-2	2025-1
P634	hot pressing	sintering cycle		heating rate	2024-2	2025-1
P635	hot pressing	sintering cycle		sintering time	2024-2	2025-1
P636	hot pressing	sintering cycle		sintering atmosphere	2024-2	
P430	hydrothermal reaction	working temperature	thermal cycle	working temperature	2022-2	
P431	hydrothermal reaction	working temperature	thermal cycle	ramping rate	2022-2	
P432	hydrothermal reaction	working temperature	thermal cycle	cooling rate	2022-2	
P433	hydrothermal reaction	precursor	reactant	name	2022-2	
P434	hydrothermal reaction	precursor	reactant	amount	2022-2	
P435	hydrothermal reaction	precursor	reactant	concentration	2022-2	
P436	hydrothermal reaction	precursor	surfactant	name	2022-2	
P437	hydrothermal reaction	precursor	surfactant	amount	2022-2	
P438	hydrothermal reaction	precursor	surfactant	concentration	2022-2	
P439	hydrothermal reaction	precursor		solvent	2022-2	
P440	hydrothermal reaction			ramping rate	2022-2	
P441	hydrothermal reaction			working pressure	2022-2	
P442	hydrothermal reaction			substrate	2022-2	
P443	hydrothermal reaction			time	2022-2	
P444	hydrothermal reaction			cooling rate	2022-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
P849	injection molding			cooling time	2025-1	
P850				injection pressure	2025-1	
P851				injection speed	2025-1	
P852				instrument	2025-1	
P853				molded material	2025-1	
P854				melt temperature	2025-1	
P855				mold temperature	2025-1	
P641	interfacial reaction			reaction temperature	2024-2	
P642		composition phase 1	reactant	name	2024-2	
P643				concentration	2024-2	
P644			additive	name	2024-2	
P645				concentration	2024-2	
P646			surfactant	name	2024-2	
P647				concentration	2024-2	
P648				solvent	2024-2	
P649		composition phase 2	reactant	name	2024-2	
P650				concentration	2024-2	
P651			additive	name	2024-2	
P652				concentration	2024-2	
P653			surfactant	name	2024-2	
P654				concentration	2024-2	
P655				solvent	2024-2	
P656				substrate	2024-2	
P657				reaction time	2024-2	
P856	ion beam deposition			source material	2025-1	
P857		ion source		model	2025-1	
P858				mechanism	2025-1	
P859				power supplied	2025-1	
P860				base pressure	2025-1	
P861				working pressure	2025-1	
P862				reaction gas	2025-1	
P863				substrate distance	2025-1	
P864		substrate bias		bias type	2025-1	
P865				applied power	2025-1	
P866				bias voltage	2025-1	
P867				substrate temperature	2025-1	
P868				substrate rotation	2025-1	
P869				time	2025-1	
P580	ion intercalation	intercalation reagent		name	2024-1	
P581				amount	2024-1	
P582		matrix		name	2024-1	
P583				amount	2024-1	
P584				temperature	2024-1	
P585				time	2024-1	
P870	ion plating	source material		composition	2025-1	
P871				generation method	2025-1	
P872		ionization		ion source type	2025-1	
P873				power supply	2025-1	
P874				base pressure	2025-1	
P875				working pressure	2025-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P876	ion plating		reaction gas	2025-1	
P877			substrate distance	2025-1	
P878		substrate bias	bias type	2025-1	
P879			applied power	2025-1	
P880			bias voltage	2025-1	
P881			substrate temperature	2025-1	
P882			substrate rotation speed	2025-1	
P883			time	2025-1	
P658	mechanical alloying		milling method	2024-2	
P659			milling ball material	2024-2	
P660			milling ball diameter	2024-2	
P661			milling speed	2024-2	
P662			milling time	2024-2	
P663			milling container material	2024-2	
P664			milling container volume	2024-2	
P665			milling atmosphere	2024-2	
P666			milling agent	2024-2	
P498	metal additive manufacturing		metal AM type	2023-2	
P667			substrate	2024-2	
P499			beam wavelength	2023-2	
P500			beam power	2023-2	
P501			beam diameter	2023-2	
P502			beam current	2023-2	
P503			focus offset	2023-2	
P668			tilt angle	2024-2	
P504			build direction	2023-2	
P505			layer thickness	2023-2	
P506			scan speed	2023-2	
P507			working distance	2023-2	
P508			overlap spacing	2023-2	
P509			hatch spacing	2023-2	
P510			powder feed rate	2023-2	
P511			wire diameter	2023-2	
P512			wire feed rate	2023-2	
P513			welding speed	2023-2	
P514			welding voltage	2023-2	
P669			welding current	2024-2	
P515			gas flow rate	2023-2	
P516			substrate temperature	2023-2	
P517			preheating temperature	2023-2	
P110	microwave-assisted synthesis	precursor	name	2021-1	
P111			amount	2021-1	
P112		solution	name	2021-1	
P113			amount	2021-1	2022-2
P518			heating rate	2023-2	
P114			temperature	2021-1	
P519			cooling rate	2023-2	
P115			atmosphere	2021-1	
P116			time	2021-1	
P520			microwave power	2023-2	
P521			microwave frequency	2023-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
P670	mixing			equipment	2024-2	
P338				mixing type	2022-1	2025-2
P339				material mixed	2022-1	2023-2
P340		solvent		name	2022-1	
P341				amount	2022-1	2022-2
P342				rotation speed	2022-1	2022-2
P343				temperature	2022-1	
P344				time	2022-1	
P884				atmosphere	2025-1	
P117	molecular beam epitaxy	beam	source	name	2021-1	2022-1
P118				amount	2021-1	2022-1
P123				flux	2021-1	2025-1
P119				substrate temperature	2021-1	
P120				substrate orientation	2021-1	
P121				substrate rotation rate	2021-1	2022-2
P122				growth chamber pressure	2021-1	
P125				growth rate	2021-1	2022-2
P126				growth time	2021-1	2022-1
P127				growth interruption time	2021-1	2022-1
P671	molten metal-flux melting			metal	2024-2	
P672				composition of metals	2024-2	
P673				reaction container	2024-2	
P885				encapsulation	2025-1	
P674				heating rate	2024-2	
P675				temperature	2024-2	
P676				holding time	2024-2	
P677				cooling temperature	2024-2	
P678				cooling time	2024-2	
P679				holding time at cooled temperature	2024-2	
P680				centrifuge speed	2024-2	
P681				centrifuge time	2024-2	
P682	non-solvent induced phase separation (NIPS)	polymer solution composition	polymer	name	2024-2	
P683				concentration	2024-2	
P684			additive	name	2024-2	
P685				concentration	2024-2	
P686				solvent	2024-2	
P687				polymer solution temperature	2024-2	
P688		non-solvent composition	additive	name	2024-2	
P689				concentration	2024-2	
P690				solvent	2024-2	
P691				non-solvent temperature	2024-2	
P886				non-solvent state	2025-1	
P692				phase separation time	2024-2	
P887	out of autoclave system			pressure	2025-1	
P888				cure cycle time	2025-1	
P889				fluid type	2025-1	
P890				fluid flow rate	2025-1	
P891				fluid temperature	2025-1	
P522	perovskite postprocess			perovskite postprocess	2023-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
P892	plasma treatment			power type	2025-1	
P693				plasma gas	2024-2	
P694				plasma power	2024-2	
P695				plasma frequency	2024-2	
P696				temperature	2024-2	
P697				pressure	2024-2	
P698				time	2024-2	
P445	polishing			method	2022-2	
P446				medium	2022-2	
P128				load	2021-1	2022-2
P129				material removal rate	2021-1	2022-2
P130				contact stress	2021-1	2022-2
P131				relative velocity	2021-1	2022-2
P132				Preston's coefficient	2021-1	
P133				temperature	2021-1	
P134				atmosphere	2021-1	
P135				rpm	2021-1	2022-2
P491	pre-etching			pre-etching	2023-2	2025-2
P895	prepreg production			fiber volume fraction	2025-1	
P896				resin content	2025-1	
P897				prepreg width	2025-1	
P898				impregnation method	2025-1	
P899				curing temperature	2025-1	
P900				curing time	2025-1	
P901		storage conditions	temperature	average	2025-1	
P902				variation	2025-1	
P903				humidity	2025-1	
P904				duration	2025-1	
P905				vacuum sealed	2025-1	
P359	pressing			load type	2022-1	2025-1
P360				temperature	2022-1	
P361				pressure	2022-1	
P362				time	2022-1	
P136	pulsed laser deposition	laser		laser source	2021-1	2022-1
P137				wavelength	2021-1	2022-1
P138				frequency	2021-1	2022-1
P139				fluence	2021-1	2022-1
P350				angle	2022-1	
P351				rotation speed	2022-1	
P352				axis	2022-1	
P141				substrate temperature	2021-1	2022-1
P353				working pressure	2022-1	
P354				base pressure	2022-1	
P142				atmosphere	2021-1	
P143				time	2021-1	2022-1
P355		atmosphere		name	2022-1	
P356				pressure	2022-1	
P357				target-substrate distance	2022-1	
P358				lens-target distance	2022-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P906	pultrusion		pulling speed	2025-1	
P907			die temperature	2025-1	
P908			resin injection pressure	2025-1	
P909			fiber orientation	2025-1	
P910			profile shape	2025-1	
P523	quenching		media	2023-2	
P524			media volume	2023-2	
P525			media additive	2023-2	
P699			temperature	2024-2	
P700			time	2024-2	
P911	resin spray transfer		automation level	2025-1	
P912			cycle time	2025-1	
P913	resin transfer molding (RTM)		injection pressure	2025-1	
P914			injection time	2025-1	
P915			mold temperature	2025-1	
P916			resin viscosity	2025-1	
P917			fiber preform orientation	2025-1	
P918			post-curing conditions	2025-1	
P145	rinsing		temperature	2021-1	
P146			time	2021-1	
P586		solvent	name	2024-1	
P587			amount	2024-1	
P147		additive	name	2021-1	
P148			amount	2021-1	
P156	sintering		atmosphere	2021-1	
P157			temperature	2021-1	
P158			time	2021-1	
P159			heating rate	2021-1	2023-1
P160			cooling rate	2021-1	2023-1
P161			pressure applied	2021-1	
P455	slot coating		pump rotation	2022-2	2025-2
P456			gap	2022-2	
P457			substrate speed	2022-2	
P1018			feed rate	2025-2	
P447	slurry casting		mixer type	2022-2	2025-1
P448			material mix	2022-2	
P449		solvent	name	2022-2	
P450			amount	2022-2	
P451			rotation speed	2022-2	
P452			temperature	2022-2	
P453			humidity	2022-2	
P454			time	2022-2	
P178	sol-gel synthesis		temperature	2021-1	
P364		precursor	name	2022-1	2024-2
P713			weight	2024-2	
P365			solvent	2022-1	
P526			pH	2023-2	
P180			time	2021-1	

어휘번호	표준어휘 체계		표준어휘	제정	개정
P919	solid state reaction		materials	2025-1	
P920			milling and mixing	2025-1	
P707			temperature	2024-2	
P708			time	2024-2	
P554	solvent annealing		solvent	2023-2	
P555			temperature	2023-2	
P556			time	2023-2	
P1019			polymer concentration	2025-2	
P924	solvent casting		solvent	2025-1	
P925		mold	materials	2025-1	
P926			shape	2025-1	
P927			casting time	2025-1	
P928			temperature	2025-1	
P168	solvothermal synthesis	precursor	name	2021-1	2022-1
P169			amount	2021-1	2022-1
P170		solvent	name	2021-1	
P171			amount	2021-1	
P172		reducing agent	name	2021-1	2022-1
P173			amount	2021-1	2022-1
P174		surfactant	name	2021-1	2022-1
P175			amount	2021-1	2022-1
P363			pressure	2022-1	
P176			temperature	2021-1	2022-1
P177			time	2021-1	2022-1
P458			heating rate	2022-2	
P459			cooling rate	2022-2	
P181	sonication		composition	2021-1	2022-1
P366			power	2022-1	
P182			solvent	2021-1	
P183			temperature	2021-1	
P184			time	2021-1	
P460			sonication type	2022-2	2025-1
P1020			mode	2025-2	
P1021			frequency	2025-2	
P185	sonochemical synthesis	precursor	name	2021-1	
P186			amount	2021-1	
P187		solvent	name	2021-1	
P188			amount	2021-1	
P189			temperature	2021-1	
P190			ultrasonic frequency	2021-1	
P388			power	2022-1	
P191			time	2021-1	
P709	spark plasma sintering		temperature	2024-2	
P921			pressure	2025-1	
P922			voltage	2025-1	
P710			time	2024-2	
P711			heating rate	2024-2	
P712			average powder size	2024-2	
P923			plasma power	2025-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
P527	spin coating			coating type	2023-2	2025-1
P367		precursor		name	2022-1	
P528			composition	composition	2023-2	
P529				purity	2023-2	
P530				supplier	2023-2	
P368				volume	2022-1	2023-2
P531				age	2023-2	
P532				temperature	2023-2	
P533				state	2023-2	
P534				molarity	2023-2	
P535		solvent		composition	2023-2	
P536				purity	2023-2	
P537				supplier	2023-2	
P714		anti-solvent		chemical information	2024-2	
P715				dropping time	2024-2	
P716				dropping volume	2024-2	
P369				temperature	2022-1	
P370		atmosphere		composition	2022-1	2023-2
P538				total pressure	2023-2	
P539				relative humidity	2023-2	
P371				time	2022-1	
P372				rpm	2022-1	
P540				rpm ramping-up rate	2023-2	
P929				rpm ramping-down rate	2025-1	
P930	spinning			materials type	2025-1	
P1022		atmosphere		composition	2025-2	
P1023				total pressure	2025-2	
P1024				relative humidity	2025-2	
P931				temperature	2025-1	
P932				solution viscosity	2025-1	
P933				spinning speed	2025-1	
P934				spinneret design	2025-1	
P935				cooling method	2025-1	
P936				drawing ratio	2025-1	
P937				solvent removal ratio	2025-1	
P938				coagulation bath composition	2025-1	
P373	spray pyrolysis coating	precursor		name	2022-1	
P541			composition	composition	2023-2	
P542				purity	2023-2	
P543				supplier	2023-2	
P374				volume	2022-1	2023-2
P544				age	2023-2	
P545				temperature	2023-2	
P546				state	2023-2	
P547				molarity	2023-2	
P375		solvent		composition	2022-1	2023-2
P548				purity	2023-2	
P549				supplier	2023-2	

어휘번호	표준어휘 체계			표준어휘	제정	개정
P550	spray pyrolysis coating	atmosphere		composition	2023-2	
P551				total pressure	2023-2	
P552				relative humidity	2023-2	
P553				substrate temperature	2023-2	
P377				droplet size	2022-1	
P378				spray speed	2022-1	
P379				flame temperature	2022-1	
P380				heat treatment time	2022-1	
P939	sputter deposition			sputter type	2025-1	
P192				base pressure	2021-1	2022-1
P193				working pressure	2021-1	2022-1
P194				sputtering gas	2021-1	2022-1
P940				reaction gas	2025-1	
P195				substrate distance	2021-1	
P196				target	2021-1	2025-1
P197		sputter power		sputter power	2021-1	
P198				power source type	2021-1	2022-1
P199				time	2021-1	2022-1
P200				substrate temperature	2021-1	
P941		jig		jig type	2025-1	
P381				rotating speed	2022-1	2025-1
P942	stereolithography			layer thickness	2025-1	
P943				build speed	2025-1	
P944				exposure time	2025-1	
P945				beam power	2025-1	
P946				resin properties	2025-1	
P947				platform temperature	2025-1	
P948				post-processing	2025-1	
P557	storage before process			time	2023-2	
P558				atmosphere	2023-2	
P559				relative humidity	2023-2	
P949	stretching			stretching type	2025-1	
P950				materials type	2025-1	
P951				temperature	2025-1	
P952				stretching ratio	2025-1	
P953				stretching speed	2025-1	
P954				cooling rate	2025-1	
P955				dwell time	2025-1	
P956				humidity	2025-1	
P957				atmosphere	2025-1	
P958				additives	2025-1	
P560	surface pre-treatment			surface pre-treatment	2023-2	
P959	tailored fiber placement			base material	2025-1	
P960				stitch pattern	2025-1	
P961				stitch threads	2025-1	
P962				material placement	2025-1	

어휘번호	표준어휘 체계			표준어휘	제정	개정
P201	thermomechanical process			atmosphere	2021-1	
P202	thermomechanical process			initial temperature	2021-1	
P203	thermomechanical process	TMC cycles		heating rate	2021-1	2025-1
P204	thermomechanical process	TMC cycles		holding temperature	2021-1	2025-1
P205	thermomechanical process	TMC cycles		deformation holding	2021-1	2025-1
P206	thermomechanical process	TMC cycles		holding time	2021-1	2025-1
P207	thermomechanical process	TMC cycles		cooling rate	2021-1	2025-1
P208	thermomechanical process	TMC cycles		deformation cooling	2021-1	2025-1
P209	thermomechanical process	TMC cycles		cooling method	2021-1	2025-1
P210	thermomechanical process	TMC cycles		final temperature	2021-1	
P461	transporting			pump rotation	2022-2	2025-2
P462	transporting			pipe material	2022-2	
P463	transporting			pipe diameter	2022-2	
P464	transporting			temperature	2022-2	
P465	transporting			time	2022-2	
P567	vacuum suction			time	2023-2	
P718	washing	washing solvent		name	2024-2	2025-2
P1026	washing	washing solvent		amount	2025-2	
P719	washing			time	2024-2	
P221	wet etching			temperature	2021-1	2022-1
P222	wet etching	etching solution	etching agent	name	2021-1	2025-1
P223	wet etching	etching solution	etching agent	concentration	2021-1	2025-1
P389	wet etching	etching solution		solvent	2022-1	
P390	wet etching	etching solution		additive	2022-1	
P224	wet etching			time	2021-1	2022-1
P391	wet etching			stirring rate	2022-1	
P963	zone melting			heating method	2025-1	
P964	zone melting			temperature	2025-1	
P965	zone melting			length of heated zone	2025-1	
P966	zone melting			zone movement speed	2025-1	
P967	zone melting			atmosphere	2025-1	

Ⅲ

표준어휘 세부내용

(표준어휘 세부내용은 어휘번호 순으로 정렬되어 있음)

1 메타데이터 표준어휘 세부내용

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
E1	data name	정의	name of the data set, named by data provider; This is your own unique data or specimen identifier. With this text string alone, you should be able to identify the materials (system) in your own internal data management system.
		동의어	sample ID
		유형	string
		단위	
		예시	Zinc coated Fe-6Mn-3Al steel plate; Ti/Al dissimilar welded joint; Fe/C catalyst; Gd doped MOF 370; LLZO (Li7La3Zr2O12) Solid Electrolyte; Low Carbon TRIP Steel; 0.15C1.0Si1.5Mn1.5Al; Co single atom catalyst; Pt50Co50/TiO2; Material_1/Material_2/Material_3/Material_4;
E2	name (contributor)	정의	contributor's name in the form of "[last name], [first name]"
		동의어	
		유형	string (family name, first name)
		단위	
		예시	Jordan, Mike; Hong, Gil-Dong
E3	affiliation (contributor)	정의	contributor's affiliation (institute's name and further description of working place)
		동의어	
		유형	string
		단위	
		예시	Advanced Materials Research Center, KIST; Department of Energy, KAIST; Metallurgy Division, KIMS
E4	data generation date	정의	date when the data is added to the present database or data lake
		동의어	
		유형	string
		단위	
		예시	2021-05-24; 2008-03-01
E6	keywords	정의	keywords representing the data set or material type and description, at least 3 keywords should be provided
		동의어	
		유형	string
		단위	
		예시	[solar cell. perovskite, KIST]
E7	embargo	정의	date when allowing open access of the data
		동의어	
		유형	string
		단위	
		예시	2002-01-10 24:00;
E8	rights	정의	creative commons copyrights license
		동의어	
		유형	string
		단위	
		예시	1. CC0 2. CCBY 3. CCBY-NC 4. CCBY-ND 5. CCBY-SA 6. CCBY-NC-SA 7. CC-BY-NC-ND

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
E9	email address (contributor)	정의	contributor's email address
		동의어	
		유형	string
		단위	
		예시	Thomas.Mueller@kriss.re.kr
E10	researcher number (contributor)	정의	contributor's researcher number ID in Integrated R&D Information System of Korea (https://iris.go.kr); only applied for Korean researchers
		동의어	
		유형	string
		단위	
		예시	12345678
E11	data classification	정의	classification code for relevant fields of science and technology announced by Ministry of Science and ICT of KOREA (https://www.law.go.kr)
		동의어	
		유형	string
		단위	
		예시	EB0101; EB0102; EB0103
E12	data source	정의	source information of the data, such as lab log, source publication, related projects, DOI of data, lead author, and publication date if the data is extracted from a publication; The DOI number referring to the published paper or dataset where the data can be found. If the data is unpublished, enter "Unpublished"; lead author is the surname of the first author. If several authors, end with et al. If the DOI number is given correctly, this will be extracted automatically from www.crossref.org ; publication date will be extracted automatically from www.crossref.org if the DOI number is given correctly.
		동의어	
		유형	string
		단위	
		예시	extracted from Acta Mater vol 13, page 2089 (2000); archived on 01/01/2021; 10.1016/j.actamat.2016.08.081; generated in CNMD, POSTECH (2021)

2 소재 표준어휘 세부내용

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M1	name	정의	description of materials according to customary usage, designation system, composition, supplier, brand name, polymer name based on IUPAC recommendation (https://iupac.org/polymer-edu/polymer-naming/), etc.
		동의어	
		타입	string
		단위	
		예시	annealed INCONEL625 plate; Al7075-T6; Al10Si5Mg; Fe; CNT; TiO2; Ag; In2O2; LLZO; Aldrich CFO plate; poly(methyl methacrylate); poly[oxy(1-bromoethane-1,2-diyl)]; polypropylene; Toray T300; Owens Corning E-glass; Hyosung Carbon Fiber; Kukdo Epoxy; Hyosung Nylon; polyimide
M2	composition (physicochemical information)	정의	composition of materials
		동의어	
		타입	dictionary of {constituent : constituent quantity, … , unit : string, uncertainty : uncertainty value}
		단위	
		예시	{value : {Pt : 56.0, Ni : 34.7}, unit : at.%, uncertainty : 0.1, measurement : [{SEM : {...}}, ...]}; {value : {Fe2O3 : 97.8, Y2O3 : 2.2}, unit : wt.%, uncertainty : 0.2}
M3	Bravais lattice (structure / crystallography)	정의	name of 14 Bravais lattice
		동의어	
		타입	string
		단위	
		예시	1. simple cubic 2. body centered cubic 3. face centered cubic 4. primitive hexagonal 5. primitive trigonal 6. primitive tetragonal 7. body centered tetragonal 8. primitive orthorhombic 9. body centered orthorhombic 10. base centered orthorhombic 11. face centered orthorhombic 12. primitive monoclinic 13. body centered monoclinic 14. primitive triclinic
M4	a (structure / crystallography / lattice parameter)	정의	lattice parameter length a
		동의어	
		타입	number
		단위	nm
		예시	0.55
M5	b (structure / crystallography / lattice parameter)	정의	lattice parameter length b
		동의어	
		타입	number
		단위	nm
		예시	0.55
M6	c (structure / crystallography / lattice parameter)	정의	lattice parameter length c
		동의어	
		타입	number
		단위	nm
		예시	0.55
M7	alpha (structure / crystallography / lattice parameter)	정의	lattice parameter angle alpha
		동의어	
		타입	number
		단위	degree
		예시	92
M8	beta (structure / crystallography / lattice parameter)	정의	lattice parameter angle beta
		동의어	
		타입	number
		단위	degree
		예시	92

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M9	gamma (structure / crystallography / lattice parameter)	정의	lattice parameter angle gamma
		동의어	
		타입	number
		단위	degree
		예시	92
M10	strukturbericht designation (structure / crystallography)	정의	crystal structure classification system established by the Zeitschrift für Kristallographie
		동의어	
		타입	string
		단위	
		예시	L1_{2}; B_{k}; D0_{6}
M11	space group (structure / crystallography)	정의	space group name or number of the structure
		동의어	
		타입	string
		단위	
		예시	1; P1; 227; Fd3-m
M12	SMILES (physicochemical info. / molecule)	정의	human-readable string notation for describing the structure of molecules, simplified molecular-input line-entry system
		동의어	
		타입	string
		단위	
		예시	CH4-C; CH3CH3-CC; CH2CH2-C=C; CH3OCH3-COC
M13	elements (model)	정의	elements in the simulation model
		동의어	component; composition
		타입	dictionary of {element : number of element, ...
		단위	
		예시	{Cu : 30000, Co : 30000}
M14	phase (model)	정의	major state of the model or major crystal structure when the model is solid
		동의어	state; structure
		타입	string
		단위	
		예시	crystal; amorphous; liquid; gas
M15	x (model / box dimension)	정의	dimension in x direction
		동의어	
		타입	number
		단위	nm
		예시	10
M16	y (model / box dimension)	정의	dimension in y direction
		동의어	
		타입	number
		단위	nm
		예시	10
M17	z (model / box dimension)	정의	dimension in z direction
		동의어	
		타입	number
		단위	nm
		예시	10
M18	periodic boundary condition (model)	정의	applied periodic boundary condition
		동의어	
		타입	string
		단위	
		예시	xyz; xz
M19	structure file (model)	정의	URI of structure file used for computation or simulation. File format is determined by extension of the file name
		동의어	
		타입	string
		단위	
		예시	CuAl_surface_210721_lammps.lmp; CuAl_B2.xyz

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M20	anodic partial current (corrosion property)	정의	sum of all the currents corresponding to anodic reactions on the electrode
		동의어	
		타입	number
		단위	A
		예시	10
M21	cathodic partial current (corrosion property)	정의	sum of all the currents corresponding to cathodic electrochemical reactions on the electrode
		동의어	
		타입	number
		단위	A
		예시	10
M22	corrosion current (corrosion property)	정의	anodic partial current due to metal oxidation
		동의어	
		타입	number
		단위	A m^{-2}
		예시	10
M23	corrosion potential (corrosion property)	정의	electrode potential of a metal in a given corrosion system
		동의어	
		타입	number
		단위	mV
		예시	100
M24	corrosion rate (corrosion property)	정의	amount of corrosion loss per year in thickness
		동의어	
		타입	number
		단위	mm year^{-1}
		예시	0.1
M25	free corrosion current (corrosion property)	정의	corrosion current at the free corrosion potential
		동의어	
		타입	number
		단위	A m^{-2}
		예시	10
M26	free corrosion potential (corrosion property)	정의	corrosion potential in the absence of net (external) electrical current flowing to or from the metal surface; the potential is measured with respect to the potential of the reference electrode
		동의어	open-circuit potential
		타입	number
		단위	mV
		예시	100
M27	passivation potential (corrosion property)	정의	corrosion potential, at which the corrosion current has a peak value, and above which there is a range of potentials, where the metal is in a passive state; the potential is measured with respect to the potential of the reference electrode
		동의어	
		타입	number
		단위	mV
		예시	100
M28	passivation current (corrosion property)	정의	corrosion current at the passivation potential
		동의어	
		타입	number
		단위	A
		예시	0.1
M29	pitting initiation potential (corrosion property)	정의	the lowest value of the corrosion potential at which pit initiation is possible on a passive surface in a given corrosive environment; the potential is measured with respect to the potential of the reference electrode
		동의어	
		타입	number
		단위	mV
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M30	re-activation potential (corrosion property)	정의	corrosion potential below which reactivation takes place; the potential is measured with respect to the potential of the reference electrode
		동의어	
		타입	number
		단위	mV
		예시	100
M31	redox potential (corrosion property)	정의	potential of a reversible oxidation-reduction reaction in a given electrolyte recorded on a S.H.E. scale
		동의어	oxidation-reduction potential
		타입	number
		단위	mV
		예시	100
M32	transpassivation potential (corrosion property)	정의	corrosion potential above which the metal is in the transpassive state; the potential is measured with respect to the potential of the reference electrode
		동의어	
		타입	number
		단위	mV
		예시	100
M33	band gap bowing parameter (electrical property)	정의	a parameter that describes nonlinear variation of band gap between two materials with different lattice constants
		동의어	bowing parameter
		타입	number
		단위	eV
		예시	0.5
M34	band gap energy (electrical property)	정의	an energy range in a solid where no electronic states can exist (between valence band and conduction band)
		동의어	energy band gap; band gap energy
		타입	number
		단위	eV
		예시	10
M35	carrier concentration (electrical property)	정의	number of charge carriers per volume
		동의어	carrier density
		타입	number
		단위	m^{-3}
		예시	2100000
M36	carrier diffusion length (electrical property)	정의	an average distance that the excess carriers can cover before they recombine
		동의어	
		타입	number
		단위	m
		예시	0.0001
M37	carrier lifetime (electrical property)	정의	an average time it takes for a minority carrier to recombine
		동의어	
		타입	number
		단위	s
		예시	0.0000001
M38	critical temperature (electrical property)	정의	temperature above which certain materials lose their permanent polarization properties
		동의어	Curie temperature
		타입	number
		단위	K
		예시	273
M39	dielectric constant (electrical property)	정의	relative permittivity compared to the vacuum in induced polarization stored in a material under electric field
		동의어	relative permittivity
		타입	number
		단위	none
		예시	2.3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M40	electric susceptibility (electrical property)	정의	a dimensionless proportionality constant that indicates the degree of polarization of a material
		동의어	specific electrical resistance, volume resistivity
		타입	number
		단위	none
		예시	1.2
M41	electrical conductivity (electrical property)	정의	quantity how strongly it conducts electric current
		동의어	specific electrical conductivity, volume conductivity
		타입	number
		단위	S m^{-1}
		예시	12.3
M42	DOS data (electrical property / electronic density of states)	정의	array data of DOS
		동의어	density of states data, DOS data
		타입	numeric array [energy (eV), energy uncertainty (eV), density of state (arbitrary unit), density of state uncertainty (arbitrary unit)]
		단위	arbitrary unit
		예시	[25, 5, 500, 10], [200, 5, 300, 10]
M43	DOS file (electrical property / electronic density of states)	정의	URI of the numeral array file of the density of state
		동의어	
		타입	string
		단위	
		예시	DOS111_34.xls;
M44	electron effective mass (electrical property)	정의	a mass that a particle seems to have when responding to forces or interacting with other identical particles in a thermal distribution
		동의어	effective mass, m_e*
		타입	number
		단위	m_{e}
		예시	0.9
M45	electron mobility (electrical property)	정의	a value that represents how quickly a single electron can move through a metal or semiconductor, when pulled by an electric field
		동의어	mobility
		타입	number
		단위	cm^{2} V^{-1} s^{-1}
		예시	50
M46	exciton binding energy (electrical property)	정의	a binding energy of an exciton (an electrically neutral quasiparticle that exists in insulators or semiconductors)
		동의어	
		타입	number
		단위	eV
		예시	0.06
M47	hole effective mass (electrical property)	정의	a mass that a particle seems to have when responding to forces or interacting with other identical particles in a thermal distribution
		동의어	effective mass, m_h*
		타입	number
		단위	m_{e}
		예시	1.3
M48	hole mobility (electrical property)	정의	a value that represents how quickly a single hole can move through a metal or semiconductor, when pulled by an electric field
		동의어	mobility
		타입	number
		단위	cm^{2} V^{-1} s^{-1}
		예시	50
M49	permittivity (electrical property)	정의	measure of the electric polarizability of a material
		동의어	
		타입	number
		단위	none
		예시	2.1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M50	resistivity (electrical property)	정의	quantity how strongly it resists electric current
		동의어	specific electrical resistance, volume resistivity
		타입	number
		단위	Ohm m
		예시	12.3
M51	temperature coefficient of resistance (electrical property)	정의	a change in resistance per unit resistance per degree rise in temperature based upon the resistance of 0 degree Celcius
		동의어	temperature coefficient resistivity
		타입	number
		단위	K^{-1}
		예시	0.0004
M52	spin-orbit splitting energy (electrical property)	정의	energy level splitting energy produced by the spin-orbit interaction of a particle's spin with its motion inside a potential
		동의어	
		타입	number
		단위	meV
		예시	10
M53	compressive yield strength (mechanical property / compressive property)	정의	yield strength of material during compressive test
		동의어	
		타입	number
		단위	MPa
		예시	350
M54	ultimate compressive strength (mechanical property / compressive property)	정의	maximum compressive strength of material during compressive test
		동의어	
		타입	number
		단위	MPa
		예시	350
M55	rupture time (mechanical property / creep property)	정의	time necessary produce failure while materials is subjected to constant load at a constant temperature
		동의어	rupture life
		타입	number
		단위	h
		예시	10000
M56	minimum creep rate (mechanical property / creep property)	정의	slope of the portion of the creep vs. time diagram corresponding to secondary creep
		동의어	steady state creep rate
		타입	number
		단위	h^{-1}
		예시	0.001
M58	Young's modulus (mechanical property / elastic property)	정의	elastic properties of materials undergoing tension or compression in one direction
		동의어	
		타입	number
		단위	GPa
		예시	100
M59	shear modulus (mechanical property / shear property)	정의	the ratio of shear stress to the shear strain
		동의어	
		타입	number
		단위	GPa
		예시	5
M60	bulk modulus (mechanical property / elastic property)	정의	measure of how resistant to compression that substance is
		동의어	
		타입	number
		단위	GPa
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M61	compressibility (mechanical property / elastic property)	정의	reciprocal of bulk modulus
		동의어	
		타입	number
		단위	GPa^{-1}
		예시	100
M62	Poisson's ratio (mechanical property / elastic property)	정의	ratio of transverse strain to corresponding axial strain on a material stressed along one axis
		동의어	
		타입	number
		단위	none
		예시	0.3
M63	fatigue life (mechanical property / fatigue property)	정의	number of stress (strain) cycles required to cause failure
		동의어	number of cycles to failure, N_{f}
		타입	number
		단위	cycle
		예시	10000
M64	fatigue limit (mechanical property / fatigue property)	정의	the highest stress that materials can withstand for an infinite number of cycles without breaking
		동의어	endurance limit
		타입	number
		단위	MPa
		예시	500
M65	hardness (mechanical property)	정의	measure of the resistance to localized plastic deformation induced by either mechanical indentation or abrasion
		동의어	
		타입	number
		단위	
		예시	1. HR 2. HB 3. HV 4. HK
M66	yield strength (mechanical property / tensile property)	정의	the stress value at which the plastic deformation of a material begins to occur
		동의어	yield stress, YS, R_{p 0.2}
		타입	number
		단위	MPa
		예시	100
M67	ultimate tensile strength (mechanical property / tensile property)	정의	maximum stress that can be applied to the material before it breaks
		동의어	ultimate tensile stress, UTS, R_{m}, tensile strength
		타입	number
		단위	MPa
		예시	180
M68	uniform elongation (mechanical property / tensile property)	정의	elongation at maximum load and immediately preceding the onset of necking in a tensile test
		동의어	
		타입	number
		단위	%
		예시	12.3
M69	total elongation (mechanical property / tensile property)	정의	percentage by which the material can be stretched before it breaks
		동의어	fracture elongation, A
		타입	number
		단위	%
		예시	12.3
M70	strain hardening exponent (mechanical property / tensile property)	정의	constant used in calculations relating to stress-strain behavior in work hardening
		동의어	strain hardening index
		타입	number
		단위	none
		예시	0.23

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M71	reduction of area (mechanical property / tensile property)	정의	proportional reduction of the cross-sectional area of a tensile test piece at the plane of fracture measured after fracture
		동의어	RA
		타입	number
		단위	%
		예시	0.23
M74	defect type (structure / defect / 0D defect)	정의	type of the point defect
		동의어	
		타입	string
		단위	
		예시	1. interstitial 2. substitutional 3. vacancy 4. anti-site
M75	formation energy (structure / defect / 0D defect)	정의	defect formation energy, i.e. lattice cohesive energy in the case of vacancy
		동의어	
		타입	number
		단위	eV
		예시	2
M76	density (structure / defect / 0D defect)	정의	ratio of defected lattice sites to those containing atoms
		동의어	concentration
		타입	number
		단위	none
		예시	0.000000001
M77	electric charge (structure / defect / 0D defect)	정의	charge state of point defects in the unit of electron
		동의어	
		타입	number
		단위	e
		예시	1; 2; -1; -2
M78	defect type (structure / defect / 1D defect)	정의	type of the 1D defect
		동의어	
		타입	string
		단위	
		예시	1. edge dislocation 2. screw dislocation 3. mixed dislocation
M79	Burgers vector (structure / defect / 1D defect)	정의	magnitude and direction of the lattice distortion resulting from a dislocation
		동의어	
		타입	string
		단위	
		예시	(a/3)〈100〉
M80	density (structure / defect / 1D defect)	정의	measure of the number of dislocations in a unit volume of a crystalline material
		동의어	
		타입	number
		단위	m^{-2}
		예시	3E+20
M81	velocity (structure / defect / 1D defect)	정의	velocity of mobile dislocation upon applied stress
		동의어	
		타입	number
		단위	m s^{-1}
		예시	0.4
M82	defect type (structure / defect / 2D defect)	정의	type of the 2D defect
		동의어	
		타입	string
		단위	
		예시	1. antiphase boundary 2. grain boundary 3. interface 4. phase boundary 5. SISF (Superlattice Intrinsic Stacking Fault) 6. CSF (Complex Stacking Fault) 7. GSF (General Stacking Fault) 8. ISF (Intrinsic Stacking Fault) 9. twin boundary

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M83	interfacial energy (structure / defect / 2D defect)	정의	energy required to form a unit area of new interface
		동의어	
		타입	number
		단위	J m^{-2}
		예시	0.03
M84	misorientation (structure / defect / 2D defect)	정의	difference in crystallographic orientation between two crystallites in a polycrystalline material
		동의어	
		타입	number
		단위	degree
		예시	3.8
M86	impurity (structure / defect)	정의	impurities in the materials
		동의어	
		타입	dictionary of {composition: {constituent:concentration}, unit: unit value, uncertainty: uncertainty value}
		단위	
		예시	{composition : {B : 20}, unit : ppm, uncertainty : 0.5}
M87	density (physicochemical information)	정의	mass per unit volume of a substance
		동의어	
		타입	number
		단위	kg m^{-3}
		예시	1.45
M88	grain size (structure / grain)	정의	estimate of the average grain diameter
		동의어	
		타입	number
		단위	{\mu m}
		예시	14.5
M89	aspect ratio (structure / grain)	정의	length ratio between longest distance and shortest distance of grain
		동의어	
		타입	number
		단위	none
		예시	2; 5
M90	shape (structure / morphology / 0D)	정의	particle shape of material
		동의어	
		타입	string
		단위	
		예시	sphere; rod; plate; cube, rectangle; octahedron; truncated dodecahedron
M91	diameter (structure / morphology / 0D / dimension)	정의	average diameter of spherical or rod shaped particles
		동의어	pitch
		타입	number
		단위	nm
		예시	5
M95	name (structure / phase)	정의	name of a phase according to its thermodynamic state and/or crystal structure
		동의어	
		타입	string
		단위	
		예시	liquid; solid; ferrite; austenite; fcc1+fcc2; gamma precipitate; gamma' precipitate
M96	composition (structure / phase)	정의	composition of the phase
		동의어	
		타입	dictionary of {constituent : constituent quantity, … , unit : string, uncertainty : uncertainty value}
		단위	
		예시	{value : {Pt : 56.0, Ni : 34.7}, unit : at.%, uncertainty : 0.1, measurement : [{SEM : {...}}, ...]}; {value : {Fe2O3 : 97.8, Y2O3 : 2.2}, unit : wt.%, uncertainty : 0.2}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M97	fraction (structure / phase)	정의	relative amounts of the phase in the microstructure
		동의어	phase amounts, volume fraction of phases
		타입	number
		단위	%
		예시	34.5
M98	BET surface area (structure / surface / specific area)	정의	Brunauer-Emmett-Teller (BET) surface area of material_n
		동의어	
		타입	number
		단위	cm^{2} g^{-1}
		예시	3326
M99	Langmuir surface area (structure / surface / specific area)	정의	Langmuir surface area of material_n
		동의어	
		타입	number
		단위	cm^{2} g^{-1}
		예시	3000
M100	total pore volume (structure)	정의	total pore volume of materials
		동의어	
		타입	number
		단위	cm^{3} g^{-1}
		예시	2
M101	Curie temperature (magnetic property)	정의	temperature above which certain materials lose their permanent magnetic properties
		동의어	critical temperature
		타입	number
		단위	K
		예시	273
M102	maximum permeability (magnetic property)	정의	the maximum value of absolute permeability
		동의어	μ_{max}
		타입	number
		단위	T m A^{-1}
		예시	0.00133
M103	maximum susceptibility (magnetic property)	정의	the maximum value of absolute susceptibility
		동의어	k_{max}
		타입	number
		단위	none
		예시	0.00133
M104	heat of fusion (thermophysical property)	정의	heat absorbed by a unit mass of a solid at its melting point
		동의어	
		타입	number
		단위	J mol^{-1}
		예시	1500
M105	melting temperature (thermophysical property)	정의	temperature at which the materials melt
		동의어	melting point
		타입	number
		단위	K
		예시	2500
M106	range (thermophysical property / melting range)	정의	temperature difference between liquidus and solidus at a given composition
		동의어	
		타입	number
		단위	K
		예시	125
M107	specific heat capacity (thermophysical property / heat capacity)	정의	heat capacity of unit mass
		동의어	
		타입	number
		단위	J kg^{-1} K^{-1}
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M108	thermal conductivity (thermophysical property)	정의	quantity of heat that passes in unit time through unit area of a substance
		동의어	
		타입	number
		단위	W m^{-1} K^{-1} (watts per meter-kelvin)
		예시	12.3
M109	thermal diffusivity (thermophysical property)	정의	thermal conductivity divided by density and specific heat capacity at constant pressure
		동의어	
		타입	number
		단위	m^{2} s^{-1}
		예시	0.00000001
M110	thermal expansion coefficient (thermophysical property)	정의	fractional change in size of a material in response to a change in temperature
		동의어	
		타입	number
		단위	K^{-1}
		예시	0.00000001
M111	absorption coefficient (optical property)	정의	a value representing how far into a material light of a particular wavelength can penetrate before it is absorbed
		동의어	
		타입	number
		단위	cm^{-1}
		예시	15
M112	optical bandgap (optical property)	정의	optical band gap obtained from the absorption spectrum
		동의어	
		타입	number
		단위	eV
		예시	1.5
M113	lifetime (optical property / photoluminescence)	정의	emissive decays time from the excited state
		동의어	
		타입	number
		단위	ns
		예시	0.06
M114	quantum efficiency (optical property / photoluminescence)	정의	number of photons emitted per absorbed photons of excitation source
		동의어	
		타입	number
		단위	none
		예시	0.1
M115	reflectance (optical property)	정의	ratio of the radiation flux reflected by a sample surface to the incident radiation flux
		동의어	
		타입	number
		단위	%
		예시	95
M116	transmittance (optical property)	정의	ratio of light that is transmitted through a material to the total incident light
		동의어	
		타입	number
		단위	%
		예시	50
M117	color (optical property)	정의	color of a material
		동의어	
		타입	string
		단위	
		예시	(255, 255, 255)
M118	refractive index (optical property)	정의	the ratio of the speed of the light in the vacuum to that in a material
		동의어	
		타입	number
		단위	none
		예시	2.3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M119	crystal system (structure / crystallography)	정의	name of 7 unique crystal systems
		동의어	
		타입	string
		단위	
		예시	cubic, hexagonal; trigonal; tetragonal; orthorhombic; monoclinic; triclinic;
M120	Pearson symbol (structure / crystallography)	정의	Pearson symbol of crystal structure
		동의어	
		타입	string
		단위	
		예시	cP3; cF23; cI5; hR12; hP2; tI4; tP7; oI4; oF8; oC8; oP2; mC6; mP4; aP16
M121	Wyckoff symbol (structure / crystallography)	정의	Wyckoff description of crystal structure
		동의어	
		타입	string
		단위	
		예시	2mm; -4m2; m-3m
M122	alpha (model / box dimension)	정의	lattice parameter angle alpha
		동의어	
		타입	number
		단위	degree
		예시	92
M123	beta (model / box dimension)	정의	lattice parameter angle beta
		동의어	
		타입	number
		단위	degree
		예시	92
M124	gamma (model / box dimension)	정의	lattice parameter angle gamma
		동의어	
		타입	number
		단위	degree
		예시	92
M125	polarized potential (corrosion property)	정의	potential across the electrode/electrolyte interface that is the sum of the corrosion potential and the applied polarization
		동의어	
		타입	number
		단위	mV
		예시	10
M126	standard electrode potential (corrosion property)	정의	reversible potential for an electrode process when all products and reactants are at unit activity, recorded on a S.H.E scale.
		동의어	oxidation-reduction potential
		타입	number
		단위	V
		예시	-2.5
M127	dielectric strength (electrical property)	정의	the maximum electric field that the material can withstand under ideal conditions without undergoing electrical breakdown and becoming electrically conductive
		동의어	
		타입	number
		단위	V m^{-1}
		예시	1000
M128	Fermi level (electrical property)	정의	the thermodynamic work required to add one electron to the body
		동의어	Fermi energy
		타입	number
		단위	eV
		예시	1.2

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M129	Hall coefficient (electrical property)	정의	the ratio of the induced electric field to the product of the current density and the applied magnetic field
		동의어	Hall constant
		타입	number
		단위	m^{3} C^{-1}
		예시	-5.2 x 10^{-11}
M130	Hall voltage (electrical property)	정의	voltage difference across an electrical conductor that is transverse to an electric current in the conductor and to an applied magnetic field perpendicular to the current
		동의어	
		타입	number
		단위	V
		예시	1
M131	ionic conductivity (electrical property)	정의	quantity how strongly it conducts ionic charge
		동의어	
		타입	number
		단위	S m^{-1}
		예시	0.0001
M132	ionic mobility (electrical property)	정의	the proportionality factor between an ion's drift velocity in an electric field in a specified medium
		동의어	ion mobility
		타입	number
		단위	m^{2} s^{-1} V^{-1}
		예시	1.2
M133	resilience modulus (mechanical property / elastic property)	정의	measure of resilience modulus defined as the maximum energy that can be absorbed per unit volume without creating permanent distortion
		동의어	
		타입	number
		단위	GPa
		예시	180
M138	facet surface (structure / morphology / 0D)	정의	array of facet surface indexes of particle
		동의어	
		타입	numeric array
		단위	
		예시	[(110), (111), (110)]
M139	electrochemically active surface area (structure / surface / specific area)	정의	electrochemically active surface area (ECSA) of material_n
		동의어	
		타입	number
		단위	cm^{2} g^{-1}
		예시	4.2
M141	pore diameter (structure / porosity)	정의	size of the pores in material; in some cases, pores can have multiple number of distinguishable sizes
		동의어	pore size
		타입	number
		단위	m
		예시	2e-9; 3e-6
M142	pore limiting diameter (structure / porosity)	정의	diameter of the largest probe that can traverse through pore channels
		동의어	
		타입	number
		단위	m
		예시	2e-9; 5e-6
M143	activity (thermophysical property / activity)	정의	a measure of the effective concentration of a species under non-ideal (e.g., concentrated) conditions
		동의어	
		타입	number
		단위	none
		예시	0.1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M144	chemical potential (thermophysical property / chemical potential)	정의	rate of increase in the Gibbs free energy of the system with respect to the increase in the number of moles of species
		동의어	partial molar Gibbs free energy
		타입	number
		단위	J mol^{-1}
		예시	1000
M145	cryoscopic constant (thermophysical property)	정의	depression in freezing point produced when 1 mole of non-volatile solute is dissolved in 1 kg of solvent
		동의어	molar freezing-point depression constant, molar depression constant
		타입	number
		단위	K kg mol^{-1}
		예시	1.853
M146	diffusivity (thermophysical property)	정의	the ratio of flux density to the negative of the concentration gradient in direction of diffusion
		동의어	diffusion coefficient
		타입	number (1. Tracer (self) 2. Tracer (impurity) 3. Intrinsic (partial chemical) 4. Inter (chemical))
		단위	m^{2} s^{-1}
		예시	0.000000001
M147	ebullioscopic constant (thermophysical property)	정의	elevation in boiling point produced when 1 mole of solute is dissolved in 1 kg of solvent
		동의어	molar boiling-point elevation constant, molar elevation constant
		타입	number
		단위	K kg mol^{-1}
		예시	0.512
M148	enthalpy of formation (thermophysical property / enthalpy of formation)	정의	a measure of the energy released or consumed when one mole of a substance is created from its pure elements
		동의어	enthalpy of mixing
		타입	number
		단위	J mol^{-1}
		예시	-5000
M149	fugacity (thermophysical property / fugacity)	정의	a measure of the effective pressure of a species in a real gas
		동의어	
		타입	number
		단위	none
		예시	0.1
M150	Gibbs free energy (thermophysical property / Gibbs free energy)	정의	a measure of the potential for reversible or maximum work that may be done by a system at constant temperature and pressure
		동의어	
		타입	number
		단위	J mol^{-1}
		예시	-1000
M153	oxidation potential (electrochemical property)	정의	the oxidation potential (voltage vs. Li/Li+) of solid or liquid materials (e.g. the solvent, the conductive salt or solid electrolytes)
		동의어	
		타입	number
		단위	V
		예시	4.3
M154	reduction potential (electrochemical property)	정의	the reduction potential (voltage vs. Li/Li+) of solid or liquid materials (e.g. the solvent, the conductive salt or solid electrolytes)
		동의어	
		타입	number
		단위	V
		예시	0.5
M155	cell type (model / box dimension)	정의	the type of cell
		동의어	
		타입	string
		단위	
		예시	conventional; primitive; other

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M156	DOS type (electrical property / electronic density of states)	정의	type of dos data
		동의어	
		타입	string
		단위	
		예시	Total DOS, Mn 3px DOS
M158	instantaneous strain (mechanical property / creep property)	정의	immediate deformation of materials upon initial application of a stress
		동의어	
		타입	number
		단위	%
		예시	0.1
M159	reduction of area (mechanical property / creep property)	정의	proportional reduction of the cross-sectional area of a creep test piece at the plane of fracture measured after fracture
		동의어	RA
		타입	number
		단위	%
		예시	30
M160	time to tertiary creep (mechanical property / creep property)	정의	time necessary to reach tertiary creep stage
		동의어	
		타입	number
		단위	h
		예시	1000
M161	creep rupture strength (mechanical property / creep property)	정의	stress causing fracture in a creep test at a given time, in a specified constant environment
		동의어	stress-rupture strength, static fatigue strength
		타입	number
		단위	MPa
		예시	500
M162	creep rupture strain (mechanical property / creep property)	정의	total strain after creep rupture test
		동의어	creep elongation
		타입	number
		단위	%
		예시	20
M163	fatigue type (mechanical property / fatigue property)	정의	type of fatigue properties depending on number of cycles, applied stress and/or strain level and environment
		동의어	
		타입	string
		단위	
		예시	1. low-cycle fatigue 2. high-cycle fatigue 3. giga-cycle fatigue 4. simulated environment fatigue
M164	fatigue crack growth rate (mechanical property / fatigue property)	정의	rate of crack extension under fatigue loading, expressed in terms of crack extension per cycle, da/dN
		동의어	fatigue crack propagation rate
		타입	number
		단위	mm cycle^{-1}
		예시	1
M165	stress intensity factor (mechanical property / fatigue property)	정의	magnitude of the mathematically ideal, crack-tip stress field for a particular mode in a homogeneous, linear-elastic body
		동의어	
		타입	number
		단위	MPa m^{1/2}
		예시	500
M166	stress intensity factor range (mechanical property / fatigue property)	정의	variation in the stress intensity factor in a cycle in fatigue
		동의어	
		타입	number
		단위	MPa m^{1/2}
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M167	fatigue crack growth threshold (mechanical property / fatigue property)	정의	asymptotic value of stress intensity factor range at which crack growth rate approaches zero
		동의어	threshold stress intensity factor range
		타입	number
		단위	MPa m^{1/2}
		예시	
M168	plastic strain ratio (mechanical property / sheet metal formability)	정의	ratio of true plastic width strain to the true plastic thickness strain in a tensile test
		동의어	r_value; r
		타입	number
		단위	none
		예시	1
M170	stress intensity factor (mechanical property / fracture toughness)	정의	magnitude of the mathematically ideal, crack-tip stress field for a particular mode in a homogeneous, linear-elastic body
		동의어	
		타입	number
		단위	MPa m^{1/2}
		예시	500
M172	plane strain fracture toughness K (mechanical property / fracture toughness)	정의	crack-extension resistance under conditions of crack-tip plane strain in mode I and predominantly linear-elastic conditions
		동의어	stress intensity factor, K_{IC}
		타입	number
		단위	MPa m^{1/2}
		예시	
M173	plane strain fracture toughness J (mechanical property / fracture toughness)	정의	crack-extension resistance under conditions of crack-tip plane strain in mode I and substantial plastic deformation
		동의어	J integral, J_{IC}
		타입	number
		단위	kJ m^{-2}
		예시	
M174	crack tip opening displacement (mechanical property / fracture toughness)	정의	crack displacement resulting from the total deformation at variously defined locations near the original crack tip
		동의어	
		타입	number
		단위	mm
		예시	0.3
M175	impact energy (mechanical property / impact toughness)	정의	work done to fracture a test specimen measured by Charpy impact test
		동의어	impact toughness
		타입	number
		단위	J
		예시	
M176	upper shelf energy (mechanical property / impact toughness)	정의	high fracture energy associated with ductile behavior
		동의어	
		타입	number
		단위	J
		예시	
M177	lower shelf energy (mechanical property / impact toughness)	정의	low fracture energy associated with brittle behavior
		동의어	
		타입	number
		단위	J
		예시	
M178	ductile-to-brittle transition temperature (mechanical property / impact toughness)	정의	temperature at which there is a pronounced decrease in a material's ability to absorb force without fracturing
		동의어	DBTT
		타입	number
		단위	K
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M179	dynamic viscosity (rheological property / dynamic viscosity)	정의	resistance of a fluid (liquid or gas) to a change in shape, or movement of neighbouring portions relative to one another; in general the dynamic viscosity is dependent on the shear rate
		동의어	viscosity
		타입	dictionary of {shear rate (s^{-1}) : value, shear rate uncertainty (s^{-1}) : value, viscosity (Pa s) : value, viscosity uncertainty (Pa s) : value}
		단위	Pa s
		예시	{180, 0.5, 1.3, 0.001}, {240, 0.5, 2.2, 0.001}
M181	storage modulus (rheological property / variable modulus)	정의	storage modulus
		동의어	
		타입	numeric array : [oscillation strain (%), oscillation strain uncertainty (%), storage modulus (Pa), storage modulus uncertainty (Pa)]
		단위	Pa
		예시	[25, 1, 500, 10], [180, 5, 300, 10]
M182	loss modulus (rheological property / variable modulus)	정의	loss modulus
		동의어	
		타입	numeric array: [oscillation strain (%), oscillation strain uncertainty (%), loss modulus (Pa), loss modulus uncertainty (Pa)]
		단위	Pa
		예시	[25, 1, 500, 10], [180, 5, 300, 10]
M184	crystal structure (structure / phase)	정의	crystal structure of a phase
		동의어	
		타입	string
		단위	
		예시	fcc; bcc; A1; L12
M185	standard state (thermophysical property / chemical potential)	정의	reference state of a material (pure substance, mixture or solution) defined to calculate its properties under different conditions
		동의어	
		타입	string
		단위	
		예시	O2_Gas, Cu_FCC-A1, Fe_BCC-A2
M186	temperature (thermophysical property / chemical potential)	정의	temperature at which standard state of a material is defined
		동의어	
		타입	number
		단위	K
		예시	298.15
M187	pressure (thermophysical property / chemical potential)	정의	pressure at which standard state of a material is defined
		동의어	
		타입	number
		단위	Pa
		예시	101325
M188	Debye temperature (thermophysical property)	정의	temperature of a crystal's highest normal mode of a vibration
		동의어	
		타입	number
		단위	K
		예시	300
M189	standard state (thermophysical property / enthalpy of formation)	정의	reference state of a material (pure substance, mixture or solution) defined to calculate its properties under different conditions
		동의어	
		타입	string
		단위	
		예시	O2_Gas, Cu_FCC-A1, Fe_BCC-A2
M190	temperature (thermophysical property / enthalpy of formation)	정의	temperature at which standard state of a material is defined
		동의어	
		타입	number
		단위	K
		예시	298.15

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M191	pressure (thermophysical property / enthalpy of formation)	정의	pressure at which standard state of a material is defined
		동의어	
		타입	number
		단위	Pa
		예시	101325
M192	standard state (thermophysical property / fugacity)	정의	reference state of a material (pure substance, mixture or solution) defined to calculate its properties under different conditions
		동의어	
		타입	string
		단위	
		예시	O2, H2
M193	temperature (thermophysical property / fugacity)	정의	temperature at which standard state of a material is defined
		동의어	
		타입	number
		단위	K
		예시	298.15
M194	pressure (thermophysical property / fugacity)	정의	pressure at which standard state of a material is defined
		동의어	
		타입	number
		단위	Pa
		예시	101325
M195	standard state (thermophysical property / Gibbs free energy)	정의	reference state of a material (pure substance, mixture or solution) defined to calculate its properties under different conditions
		동의어	
		타입	string
		단위	
		예시	O2_Gas, Cu_FCC-A1, Fe_BCC-A2
M196	temperature (thermophysical property / Gibbs free energy)	정의	temperature at which standard state of a material is defined
		동의어	
		타입	number
		단위	K
		예시	298.15
M197	pressure (thermophysical property / Gibbs free energy)	정의	pressure at which standard state of a material is defined
		동의어	
		타입	number
		단위	Pa
		예시	101325
M198	piezoelectric strain coefficient (piezoelectric property)	정의	strain induced when polarized when electric field applied, presented by values assigned to symbols d*_{ij}, i.e. d*_{11}, d*_{22}, d*_{33} etc.
		동의어	Coefficient, d*
		타입	dictionary of {coefficient symbol : value, coefficient value : value}
		단위	pm V^{-1}
		예시	{coefficient symbol : d*_{33}, coefficient value : 350}, {coefficient symbol : d*_{11} : coefficient value : 320}
M199	piezoelectric voltage coefficient (piezoelectric property)	정의	voltage induced when polarized when stress applied, presented by values assigned to symbols g_{ij}, i.e. g_{11}, g_{22}, g_{33} etc.
		동의어	Coefficient, g
		타입	dictionary of {coefficient symbol : value, coefficient value : value}
		단위	V m N^{-1}
		예시	{coefficient symbol : g_{33}, coefficient value : 20}, {coefficient symbol : g_{11} : coefficient value : 10}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M200	piezoelectric charge coefficient (piezoelectric property)	정의	electrical charge induced when polarized when stress is applied, presented by values assigned to symbols d_{ij}, i.e. d_{11}, d_{22}, d_{33} etc.
		동의어	Coefficient,d
		타입	dictionary of {coefficient symbol : value, coefficient value : value}
		단위	pC N^{-1}
		예시	{coefficient symbol : d_{33}, coefficient value : 350}, {coefficient symbol : d_{11} : coefficient value : 320}
M201	electromechanical coupling factor (piezoelectric property)	정의	conversion efficiency between electrical and mechanical energy, k^{2}
		동의어	
		타입	number
		단위	none
		예시	20
M202	mechanical quality factor (piezoelectric property)	정의	variables that indicate how little loss is in piezoelectric deformation (It is the inverse of the mechanical loss tan ϕ)
		동의어	
		타입	number
		단위	none
		예시	100
M203	saturation polarization (ferroelectric property)	정의	maximum polarization when an electric field is applied to an object, presented by values assigned to symbols P_{s}
		동의어	
		타입	number
		단위	C m^{-2}
		예시	3
M204	coercive electric field (ferroelectric property)	정의	the strength of the electric field at which the macroscopic polarization of the ferroelectric capacitor disappears, presented by values assigned to symbols E_{c}
		동의어	
		타입	number
		단위	V m^{-1}
		예시	2
M205	remanent polarization (ferroelectric property)	정의	polarization that remains after removing the electric field applied to an object, presented by values assigned to symbols P_{r}
		동의어	
		타입	number
		단위	C m^{-2}
		예시	3
M206	ICSD collection code (structure / crystallography)	정의	collection code assigned by ICSD
		동의어	ICSD number
		타입	number
		단위	
		예시	238685
M207	band gap type (electrical property)	정의	band gap type (direct or indirect)
		동의어	
		타입	string array
		단위	
		예시	1. direct 2. indirect
M208	spin-Hall angle (electrical property)	정의	When a current is applied to a material with a very large spin-orbit interaction, the current does not go straight due to the strong spin-orbit interaction, and different spins diverge in the width direction on both sides of the sample, resulting in spin accumulation and spin current. The ratio of the generated spin current to the flowed current.
		동의어	SHA
		타입	number
		단위	none
		예시	0.29; -1.21

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M209	coercivity (magnetic property)	정의	the magnetic field required to reduce the magnetic flux of a material to zero
		동의어	H_c; H_cB; normal coercivity
		타입	number
		단위	A m^{-1}
		예시	13
M210	intrinsic coercivity (magnetic property)	정의	the magnetic field required to reduce the magnetization of a material to zero
		동의어	H_ci
		타입	number
		단위	A m^{-1}
		예시	13
M211	magnetic anisotropy constant (magnetic property)	정의	A measure of the energy necessary to align the magnetization from an easy axis to the hard axis of the specimen. The easy and hard axis is determined by crystal structure, interface and the shape of the specimen.
		동의어	magnetic anisotropy energy; anisotropy energy constant
		타입	number
		단위	J m^{-3}
		예시	1000000
M212	magnetic anisotropy field (magnetic property)	정의	magnetic field required for complete magnetization in the direction in which the largest magnetic field is required for magnetization in a magnetic material having magnetic anisotropy
		동의어	anisotropy field; hard-axis anisotropy field
		타입	number
		단위	A m^{-1}
		예시	1000
M213	maximum BH product (magnetic property)	정의	the product of the magnetic flux density and magnetic field strength of a permanent magnet at any point of any demagnetization curve.
		동의어	maximum energy product
		타입	number
		단위	J m^{-3}
		예시	7.958
M214	recoil permeability (magnetic property)	정의	the permeability corresponding to the slope of the recoil line
		동의어	u_rec
		타입	number
		단위	T m A^{-1}
		예시	1.1
M215	remanent flux density (magnetic property)	정의	the value of the magnetic flux density remaining in a magnetized body when, in the absence of a self-demagnetizing field, the applied magnetic field strength is brought to zero
		동의어	magnetic remanence; retentivity; residual induction
		타입	number
		단위	T
		예시	0.875
M216	remanent magnetic polarization (magnetic property)	정의	the value of the magnetic polarization remaining in a magnetized body when, in the absence of a self-demagnetizing field, the applied magnetic field strength is brought to zero
		동의어	J_r
		타입	number
		단위	T
		예시	1.2
M217	remanent magnetization (magnetic property)	정의	the value of the magnetization remaining in a magnetized body when, in the absence of a self-demagnetizing field, the applied magnetic field strength is brought to zero
		동의어	remanence; residual magnetization
		타입	number
		단위	A m^{-1}
		예시	75.047

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M218	saturation magnetic polarization (magnetic property)	정의	the maximum obtainable magnetic polarization for a given substance at a given temperature
		동의어	J_s
		타입	number
		단위	T
		예시	1.75
M219	saturation magnetization (magnetic property)	정의	the maximum obtainable magnetization for a given substance at a given temperature
		동의어	magnetization; maximum positive magnetization; Ms
		타입	number
		단위	A m^{-1}
		예시	800
M220	squareness (magnetic property)	정의	a parameter indicating how close the shape of the hysteresis curve is to a square
		동의어	magnetic hysteresis; loop squareness
		타입	number
		단위	%
		예시	97.06
M221	critical resolved shear stress (mechanical property / tensile property)	정의	component of shear stress necessary to initiate slip in a grain
		동의어	CRSS
		타입	number
		단위	GPa
		예시	3
M222	tensile toughness (mechanical property / tensile property)	정의	the amount of energy a material can absorb before it fractures characterized from the area under the stress-strain curve
		동의어	
		타입	number
		단위	J m^{-3}
		예시	94; 204e6
M223	CAS number (physicochemical info. / molecule)	정의	a numerical designation assigned to chemical substances by the U.S. Chemical Abstracts Service (CAS)
		동의어	chemical abstracts service number
		타입	string
		단위	
		예시	156074-98-5
M224	chemical formula (physicochemical info. / molecule)	정의	the molecular formula about the chemical proportions of atoms that constitute whole number ratio of atoms present in a compound or constitutional unit
		동의어	
		타입	string
		단위	
		예시	C2H6O
M225	structural formula (physicochemical info. / molecule)	정의	the molecular formula about a typographic system arose to describe organic structures in a line of functional group
		동의어	
		타입	string
		단위	
		예시	CH3CH2OH, CH3OCH3
M226	number-average molar mass (physicochemical info. / polymer chain / molecular weight)	정의	number-average molecular weights of materials; the average molecular weight of a polymer sample based on the number of molecules; M_n = sum{(N_i) x (M_i)} / sum {(N_i)}, where N_i is the number of molecules of species i and M_i the molecular weight of species i.
		동의어	M_n; number-average molecular weight
		타입	number
		단위	g mol^{-1}
		예시	1000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M227	doping (physicochemical information)	정의	dopant and its dose for modifying structure and properties of materials
		동의어	additive
		타입	dictionary of {level : {dopant : dose, … }, unit : unit value, uncertainty : value, measurement : {...}}
		단위	
		예시	{level : {B : 1e15, C : 1e10}, unit : atoms cm^{-3}, uncertainty : 1e8, measurement : {...}}
M228	perovskite type (structure / crystallography)	정의	type of perovskite
		동의어	
		타입	string
		단위	
		예시	ABC3; inspired; 2d; 3d; Dion-Jacobson; Ruddlesden-Popper; quasi-2d; quasi-3d;
M229	defect type (structure / defect / 3D defect)	정의	type of 3D defect
		동의어	
		타입	string
		단위	
		예시	pore; void; cluster, pinhole
M230	volume fraction (structure / defect / 3D defect)	정의	volume fraction of the 3D defect with respect to the volume of whole materials
		동의어	
		타입	number
		단위	none
		예시	0.04
M231	size distribution (structure / defect / 3D defect)	정의	size distribution of the 3D defects
		동의어	
		타입	dictionary of {size : {value : size value, uncertainty : size uncertainty}, fraction : {fraction : fraction value, uncertainty : uncertainty value}
		단위	nm
		예시	{size : {value : 130, uncertainty : 0.05}, fraction : {value : 0.8, uncertainty : 0.001}
M232	texture table (structure / texture)	정의	relative probability of finding the crystals comprising a polycrystalline aggregate in various orientations
		동의어	preferred orientation component fraction
		타입	dictionary of {texture component : value, fraction : value, uncertainty : value}
		단위	%
		예시	{texture component : cubic texture (Cube), fraction : 0.7, uncertainty : 0.01}
M233	RMS surface roughness (surface property)	정의	the root mean square value (RMS) of the surface roughness expressed in nm (mapping: Perovskite DB 5.4); the irregularities or deviations in the texture of a surface, typically at a micro or nanoscale level. It quantifies the variations in height of the surface features from the ideal smooth surface.
		동의어	surface roughness
		타입	number
		단위	{\mu m}
		예시	1
M235	orientation (structure / morphology / 1D)	정의	orientation in the direction of length
		동의어	
		타입	string
		단위	
		예시	〈111〉; 〈100〉
M236	diameter (structure / morphology / 1D)	정의	diameter of the wire
		동의어	
		타입	number
		단위	nm
		예시	50; 100
M237	length (structure / morphology / 1D)	정의	length of 1D tube
		동의어	
		타입	number
		단위	nm
		예시	5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M238	chirality (structure / morphology / 1D)	정의	chirality of the tube
		동의어	
		타입	string
		단위	
		예시	〈111〉; 〈100〉
M240	inner diameter (structure / morphology / 1D)	정의	inner diameter of the tube
		동의어	
		타입	number
		단위	nm
		예시	50; 100
M241	outer diameter (structure / morphology / 1D)	정의	outer diameter of the tube
		동의어	
		타입	number
		단위	nm
		예시	50; 100
M243	dimension (structure / morphology / 3D / dimension)	정의	dimension of the bulk material
		동의어	
		타입	dictionary of : {x : value, y : value, z : value, diameter : value}, uncertainty : value}
		단위	
		예시	{{x : 100, y : 800, z : 10}, uncertainty: 0.5}; {{diameter : 100}, uncertainty : 0.5}, {{diameter : 80}, uncertainty : 0.5}
M244	band gap graded (electrical property)	정의	TRUE if the band gap varies as a function of the vertical position of the materials i.e. in the perovskite layer.
		동의어	
		타입	boolean
		단위	
		예시	1. False 2. True
M245	conduction band minimum (electrical property)	정의	the lowest range of vacant electronic states
		동의어	CBM, LUMO, lowest unoccupied molecular orbital
		타입	number
		단위	eV
		예시	1.2
M246	charge of constituents (electrical property)	정의	charge of constituent in materials
		동의어	
		타입	dictionary of {value : {constituent : charge value}. … }, unit : unit value, uncertainty : uncertainty value}
		단위	e
		예시	{value : {Al : 3.2, O : -3.0}, unit : e, uncertainty : 0.1}}
M247	valence band maximum (electrical property)	정의	the highest range of electron energies in which electrons are normally present at absolute zero temperature
		동의어	VBM, HOMO, highest occupied molecular orbital
		타입	number
		단위	eV
		예시	1.2
M248	magnetic field strength (magnetic property)	정의	the externally applied magnetic field (H)
		동의어	magnetic flux density, magnetic induction, magnetic intensity
		타입	number
		단위	A m^{-1}
		예시	1
M249	magnetic induction (magnetic property)	정의	the magnitude of the internal field strength (B) within a substance that is subjected to an externally applied magnetic field (H)
		동의어	magnetic flux density
		타입	number
		단위	T
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M250	magnetization (magnetic property)	정의	In the presence of an externally applied magnetic field (H), the magnetic moments within a material tend to become aligned with the field and to reinforce it by virtue of their magnetic fields
		동의어	
		타입	number
		단위	A m^{-1}
		예시	1
M251	magnetostriction (magnetic property)	정의	the strain induced by a change in magnetization
		동의어	magnetic strain, λ
		타입	number
		단위	none
		예시	0.000001
M252	saturation flux density (magnetic property)	정의	The maximum magnetic induction (Bs) that results when all the magnetic dipoles in a solid piece are mutually aligned with the external field (H). A state reached when an increase in applied external magnetic field H cannot increase the magnetization (M) of the material further.
		동의어	maximum flux density
		타입	number
		단위	T
		예시	1
M253	compressive fracture strength (mechanical property / compressive property)	정의	maximum stress or force per unit area that a material can withstand under compression when it fails or fractures completely
		동의어	
		타입	number
		단위	MPa
		예시	300
M254	compressive fracture strain (mechanical property / compressive property)	정의	maximum deformation or strain a material can undergo in compression when it fails or fractures completely
		동의어	
		타입	number
		단위	%
		예시	5
M256	normal anisotropy (mechanical property / sheet metal formability)	정의	ratio of the logarithmic strain in the transverse direction (TD) to the logarithmic strain in the normal direction (ND)
		동의어	average r value; r_{m}
		타입	number
		단위	none
		예시	1
M257	planar anisotropy (mechanical property / sheet metal formability)	정의	variation of plastic strain ratio with direction in the plane of the sheet
		동의어	⊿r
		타입	number
		단위	none
		예시	1
M258	limiting drawing ratio (mechanical property / sheet metal formability)	정의	ratio of the maximum blank diameter that can be drawn into a cup without failure, to the diameter of the punch
		동의어	LDR
		타입	number
		단위	none
		예시	1
M259	bendability (mechanical property / sheet metal formability)	정의	ability of a material to be bent around a specified radius without fracture
		동의어	r over t ratio
		타입	number
		단위	none
		예시	3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M262	hole expansion ratio (mechanical property / sheet metal formability)	정의	ratio of the expanded hole diameter to the initial hole diameter
		동의어	
		타입	number
		단위	none
		예시	25
M263	springback ratio (mechanical property / sheet metal formability)	정의	ratio of the radius of curvature after springback to the radius at the end of forming
		동의어	
		타입	number
		단위	none
		예시	1.2
M264	lifetime (optical property / electroluminescence)	정의	emissive decays time from the excited state
		동의어	
		타입	number
		단위	ms
		예시	0.06
M265	quantum efficiency (optical property / electroluminescence)	정의	number of photons emitted per injected electrons
		동의어	
		타입	number
		단위	none
		예시	0.1
M266	photoluminescence type (optical property / photoluminescence)	정의	type of emission through photoexcitation; Luminescence is classified according to the magnitude of the delay time between absorption and reemission events. If reemission occurs for times much less than 1 s, the phenomenon is termed fluorescence. If reemission occurs for longer than 1 second, the phenomenon is termed phosphorescence.
		동의어	
		타입	string
		단위	none
		예시	1. fluorescence 2.phosphorescence
M267	peak position (optical property / electroluminescence)	정의	a wavelength where the intensity of electroluminescence reaches its maximum value; If more than one peak positions, separate those by comma
		동의어	emission peak position; EL peak
		타입	number
		단위	nm
		예시	780; 550, 770;
M268	boiling temperature (thermophysical property)	정의	temperature at which a liquid changes to a vapor
		동의어	boiling point, B.P., TB
		타입	number
		단위	K
		예시	380
M269	activity coefficient (thermophysical property / activity)	정의	activity of a species in solution divided by the actual concentration of that species
		동의어	
		타입	number
		단위	none
		예시	0.2
M270	glass transition temperature (thermophysical property)	정의	temperature at which materials transit from a hard and relatively brittle glassy state into a viscous or rubbery state
		동의어	Tg
		타입	number
		단위	K
		예시	400
M271	heat of vaporization (thermophysical property)	정의	quantity of heat required at a specific temperature to convert unit mass of liquid into vapor
		동의어	
		타입	number
		단위	J mol^{-1}
		예시	1500

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M274	peak position (optical property / photoluminescence)	정의	a wavelength where the intensity of photoluminescence reaches its maximum value; If more than one peak positions, separate those by comma
		동의어	emission peak position; EL peak
		타입	number
		단위	nm
		예시	780; 550, 770;
M275	standard state (thermophysical property / activity)	정의	reference state of a material(pure substance, mixture or solution) defined to calculate its properties under different conditions
		동의어	
		타입	string
		단위	
		예시	O2_Gas, Cu_FCC-A1, Fe_BCC-A2
M276	temperature (thermophysical property / activity)	정의	temperature at which standard state of a material is defined
		동의어	
		타입	number
		단위	K
		예시	298.15
M277	pressure (thermophysical property / activity)	정의	pressure at which standard state of a material is defined
		동의어	
		타입	number
		단위	Pa
		예시	101325
M278	purity (physicochemical information)	정의	purity of material
		동의어	
		타입	number
		단위	%
		예시	99.999; 99.3
M279	cell volume (structure / crystallography / unit cell)	정의	three-dimensional space occupied by one unit cell in a crystal lattice
		동의어	
		타입	number
		단위	\ANGSTROM^{3}
		예시	105.12
M280	cell formula units (structure / crystallography / unit cell)	정의	number of formula units of a substance contained within one unit cell of its crystal lattice; conventionally designated as Z value
		동의어	
		타입	number
		단위	none
		예시	2
M281	density of crystal structure (structure / crystallography / unit cell)	정의	mass of the atoms within a unit cell divided by the volume of the unit cell
		동의어	
		타입	number
		단위	g/\ANGSTROM^{3}
		예시	15.12
M282	extinction symbol (structure / crystallography)	정의	symbol represents specific systematic absences in a crystal's diffraction pattern
		동의어	
		타입	string
		단위	
		예시	42; 1; 101
M283	note (structure / morphology / 2D)	정의	note describing essential information of 2D structure
		동의어	
		타입	string
		단위	
		예시	graphene of two layers

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M284	ionic activation energy (electrical property)	정의	the energy barriers for ion transport
		동의어	
		타입	number
		단위	eV
		예시	1
M286	DMI energy density (magnetic property)	정의	anti-symmetric exchange interaction that favors a canted spin configuration between neighboring spins
		동의어	D, Dzyloshinskii-Moriya interaction energy density, DMI constant, antisymmetric exchange energy constant, Dzyaloshinskii-Moriya interaction constant
		타입	number
		단위	J m^{-2}
		예시	0.7
M287	DMI field (magnetic property)	정의	an effective in-plane magnetic field acting on a domain wall which tends to align the domain wall magnetization to the Neel configuration with a specific chirality
		동의어	H_{DMI}
		타입	number
		단위	T
		예시	0.4
M288	effective magnetic anisotropy energy (magnetic property)	정의	net perpendicular magnetic anisotropy which is derived from the first-order uniaxial anisotropy constant (Ku) by eliminating shape anisotropy term (2πMs2)
		동의어	
		타입	number
		단위	J m^{-3}
		예시	2.1
M289	interface state density (magnetic property)	정의	the number of states per unit area of the interface per unit energy in the semiconductor band gap
		동의어	interface trap density
		타입	number
		단위	cm^{-2} eV^{-1}
		예시	1010
M290	magnetic anisotropy energy (magnetic property)	정의	energy required to rotate the magnetization from an easy axis to a hard axis
		동의어	
		타입	number
		단위	J m^{-3}
		예시	2
M291	magneto-optical Kerr angle (magnetic property)	정의	degree of ferromagnetic ordering of materials along with applied magnetic field direction, represented by the rotation angle of reflected light polarization when the light is reflected from the surface of a magnetized material
		동의어	MOKE
		타입	number
		단위	degree
		예시	10; 45
M292	absorption shielding effectiveness (magnetic property)	정의	amount of the attenuated amplitude of electromagnetic waves in the material, compared to the amount of the transmitted electromagnetic waves into the material
		동의어	SE_{A}
		타입	array
		단위	dB
		예시	[15, 15.1, 15, 15.3, 14]
M293	reflection shielding effectiveness (magnetic property)	정의	amount of the reflected electromagnetic waves from the material
		동의어	SE_{R}
		타입	array
		단위	dB
		예시	[15, 15.1, 15, 15.3, 14]

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M294	total shielding effectiveness (magnetic property)	정의	amount of the shielded electromagnetic waves from the material
		동의어	SE_{T}
		타입	array
		단위	dB
		예시	[15, 15.1, 15, 15.3, 14]
M295	spin Hall angle (magnetic property)	정의	the conversion efficiency of the charge currents to pure spin currents
		동의어	
		타입	number
		단위	none
		예시	0.3
M296	spin-orbit torque efficiency (magnetic property)	정의	the conversion factor of the current density to the spin-orbit torque-induced effective magnetic field on a magnetization
		동의어	
		타입	number
		단위	T m^{2} A^{-1}
		예시	1.2
M297	domain wall propagation field (magnetic property)	정의	the minimum magnetic field required to move or propagate a domain wall within a ferromagnetic material
		동의어	
		타입	number
		단위	T
		예시	0.013
M298	work function (magnetic property)	정의	the minimum thermodynamic work (i.e., energy) needed to remove an electron from a solid to a point in the vacuum immediately outside the solid surface
		동의어	W
		타입	number
		단위	eV
		예시	3.8
M300	onset temperature (thermophysical property)	정의	the temperature where a significant phenomenon occurs in differential scanning calorimetry (DSC) analysis
		동의어	phase transition temperature
		타입	number
		단위	K
		예시	1200; 200
M301	softening temperature (thermophysical property)	정의	temperature at which the material becomes soft under a given load on heating
		동의어	softening point
		타입	number
		단위	K
		예시	750
M303	weight-average molar mass (physicochemical info. / polymer chain / molecular weight)	정의	weight-average molecular weights of materials; a measure of the average molecular weight of a polymer sample, taking into account the weight of each molecule; M_w = sum{(w_i) x (M_i)} / sum{(w_i)}, where w_i is the weight of species i and M_i the molecular weight of species i
		동의어	M_w; weight-average molecular weight
		타입	number
		단위	g mol^{-1}
		예시	1000
M304	dispersity (physicochemical info. / polymer chain / molecular weight)	정의	ratio of the weight-average molecular weight to the number-average molecular weight; dispersity D = (M_w) / (M_n), where M_w is the weight-average molecular weight and M_n the number-average molecular weight
		동의어	polydispersity index; PDI; molecular weight distribution
		타입	number
		단위	none
		예시	1.1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M305	tacticity (physicochemical info. / polymer chain / architecture)	정의	stereoregularity of polymer chain
		동의어	stereoisomer
		타입	string
		단위	none
		예시	1. isotactic (it-) 2. syndiotactic (st-) 3. atactic (at-)
M306	linearity (physicochemical info. / polymer chain / architecture / linearity),	정의	linearity of polymer chain
		동의어	
		타입	string
		단위	none
		예시	1. linear 2. branched 3. cross-linked 4. star 5. dendrimer
M308	homogeneity (physicochemical info. / polymer chain / architecture / homogeneity),	정의	monomer homogeneity in polymer chain
		동의어	
		타입	string
		단위	none
		예시	1. homopolymer 2. copolymer (random) 3. copolymer (block) 4. copolymer (graft)
M309	crystallinity (structure /)	정의	the fraction (percentage) of a material that is crystalline (as opposed to amorphous)
		동의어	degree of crystallinity
		타입	number
		단위	%
		예시	50
M310	species (structure / defect / 0D defect)	정의	a specific chemical species (element or molecule) constituting a point defect in the material
		동의어	
		타입	string
		단위	
		예시	iodine; Pb; bromine; In;
M311	Lotgering factor (structure /)	정의	a measurement to assess the crystallographic orientations of inorganic bulk materials. Typically calculated based on the intensities measured using X-ray diffraction pattern data. This factor serves as a convenient orientation index for determining the degree of crystallographic alignment in materials. Defined as (P-Po)/(1-Po) where P and Po are the intensity ratios for the sample and randomly oriented sample, respectively.
		동의어	
		타입	number
		단위	none
		예시	0.2
M312	pore fraction (structure / porosity)	정의	the fraction of a material's volume that is empty
		동의어	void fraction
		타입	number
		단위	%
		예시	5
M313	average pore diameter (structure / porosity)	정의	mean diameter of pores (or voids) present in a given material
		동의어	
		타입	number
		단위	m
		예시	0.000005
M314	pole figure (structure / texture)	정의	graphical representation of the orientation of texture components
		동의어	
		타입	file
		단위	
		예시	{100} pole figure.jpg
M315	carrier type (electrical property)	정의	the type of charge carrier in a material (e.g., electron or hole)
		동의어	
		타입	string
		단위	
		예시	1. electron 2. hole

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M316	defect energy level (electrical property)	정의	the energy level of the states introduced within the band gap of a semiconductor or insulator due to the presence of defects
		동의어	
		타입	number
		단위	eV
		예시	0.05; 0.3
M317	loss tangent (electrical property)	정의	signal loss due to the dielectric materials' inherent dissipation of electromagnetic energy. It is dependent on the frequency of applied electric field.
		동의어	dielectric loss tangent
		타입	number
		단위	none
		예시	0.01
M318	trap density (electrical property)	정의	the density of electronic trap states within dielectric materials responsible for carrier trapping and de-trapping
		동의어	
		타입	number
		단위	cm^{-3}
		예시	1.34e10; 1.2e13
M320	remanent coercivity (magnetic property)	정의	the magnetic field required to reduce the remanent magnetization of a material to zero, meaning that when the H field is finally returned to zero, then both B and M also fall to zero (the material reaches the origin in the hysteresis curve)
		동의어	extrinsic coercivity
		타입	number
		단위	A m^{-1}
		예시	13
M321	stress exponent (mechanical property / creep property)	정의	the value of a stress exponent, n, obtained via creep rate measurements over a range of applied stress
		동의어	
		타입	number
		단위	none
		예시	4
M322	tensile modulus (mechanical property / tensile property)	정의	stress-strain ratio in elastic region
		동의어	elastic modulus
		타입	number
		단위	GPa
		예시	100
M323	fracture strength (mechanical property / tensile property)	정의	maximum stress or force per unit area that a material can withstand under tensile condition when it fails or fractures completely
		동의어	
		타입	number
		단위	MPa
		예시	300
M324	absorbance (optical property)	정의	the quantity of light absorbed by an object, defined as the logarithm of the ratio of incident to transmitted or reflected radiation power through a material
		동의어	
		타입	number
		단위	none
		예시	15
M325	maximum peak wavelength (optical property)	정의	wavelength at which a substance absorbs the maximum amount of ultraviolet light
		동의어	maximum absorption wavelength; UV absorbance peak
		타입	number
		단위	nm
		예시	95

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M326	normal incidence transmittance (optical property)	정의	ratio of the radiation flux normally transmitted through a sample to the incident radiation flux
		동의어	
		타입	number
		단위	%
		예시	50
M327	thermal degradation temperature (thermophysical property)	정의	the onset temperature that starts the process in which a chemical species breaks down
		동의어	thermal decomposition temperature; decomposition temperature
		타입	number
		단위	K
		예시	473
M328	moisture absorption (thermophysical property)	정의	increase in weight by absorbing moisture from surrounding environment when materials are placed in a humid condition
		동의어	water absorption
		타입	number
		단위	%
		예시	1
M330	Seebeck coefficient (thermoelectric property)	정의	measure of the magnitude of an induced thermoelectric voltage in response to a temperature difference across that material
		동의어	thermoelectric power; thermopower
		타입	number
		단위	{\mu V} K^{-1}
		예시	100
M331	power factor (thermoelectric property)	정의	a measure of ability to extract electrical power from temperature difference, being calculated as the product of the square of the Seebeck coefficient and the electrical conductivity.
		동의어	
		타입	number
		단위	{\mu W} cm^{-1} K^{-2}
		예시	100
M332	ZT (thermoelectric property)	정의	thermoelectric figure of merit, a dimensionless parameter that quantifies the maximum efficiency of a thermoelectric material and device. It is defined as the product of absolute temperature and power factor divided by the total thermal conductivity of the material.
		동의어	figure of merit; thermoelectric figure of merit
		타입	number
		단위	none
		예시	1.5
M333	sp fraction (physicochemical info. / hybrid bonds)	정의	quantity of sp hybridization in percent; In chemistry, hybrid bonds are formed by hybridizing orbitals. sp hybrid bonds forms a linear geometry when one s orbital mixes with one p orbital forming two equivalent sp hybrid orbitals
		동의어	sp1
		타입	number
		단위	%
		예시	5
M334	sp2 fraction (physicochemical info. / hybrid bonds)	정의	quantity of sp2 hybridization in percent
		동의어	
		타입	number
		단위	%
		예시	35
M335	sp3 fraction (physicochemical info. / hybrid bonds)	정의	quantity of sp3 hybridization in percent
		동의어	
		타입	number
		단위	%
		예시	60

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M336	formula weight (physicochemical info. / molecule)	정의	sum of the standard atomic weights of the constituent elements, each multiplied by its stoichiometric coefficient in the formula, expressed in grams per mole (g/mol)
		동의어	molecular weight
		타입	number
		단위	g mol^{-1}
		예시	395.14; 58.08
M337	plasticizer (physicochemical info.)	정의	a substance added to a material, typically a polymer, to make it softer, more flexible, and easier to handle; plasticizers reduce intermolecular forces within the polymer matrix, decreasing the glass transition temperature and improving processability
		동의어	plasticizing agent; softener
		타입	string
		단위	
		예시	diethyl phthalates; dibutyl phthalate; citrate esters; natural oil
M338	classification (physicochemical info. / polymer chain)	정의	22 classifications based on the chemical structure of constitutional repeating unit of the polymer and related categories (Ref: https://polymer.nims.go.jp/PoLyInfo/guide/en/term_polymer.html)
		동의어	monomer types; polymer taxonomy
		타입	string
		단위	
		예시	1. polyacrylics 2. polyamides / thioamides 3. polyanhydrides / thioanhydrides 4. polycarbonates / thiocarbonates 5. polydienes 6. polyesters / thioesters 7. polyhalo-olefins 8. polyimides / thioimides 9. polyimines 10, polyketones / thioketones 11. polyolefins 12. polyoxides / ethers / acetals 13. polyphenylenes 14. polyphosphazenes 15. polysiloxanes / silanes 16. polystyrenes 17. polysulfides 18. polysulfones / sulfoxides / sulfonates / sulfoamides 19. polyurethanes / thiourethanes 20. polyureas / thioureas 21. polyvinyls 22. other polymers
M339	degree of polymerization (physicochemical info. / polymer chain)	정의	number of each monomers contained in one polymer chain or polymer molecule; it is calculated for homopolymer by dividing the molecular weight of the polymer by the molecular weight of the repeat unit
		동의어	DP
		타입	number
		단위	none
		예시	500
M340	GPC (SEC) peak molar mass (physicochemical info. / polymer chain / molecular weight)	정의	molecular weight obtained from the peak value refers to the maximum point of gel permeation chromatographic (GPC) peak or size exclusion chromatographic (SEC) peak; the peak value provides the information about the molecular weight and distribution of polymers
		동의어	M_p; GPC (SEC) molecular weight
		타입	number
		단위	g mol^{-1}
		예시	1000
M341	viscosity-average molar mass (physicochemical info. / polymer chain / molecular weight)	정의	a measure to characterize the molecular weight of a polymer based on its viscosity; particularly useful for understanding the flow behavior of polymer solutions and melts; M_v = K x [eta], where K is a constant that depends on the specific polymer and solvent used and [eta] the intrinsic viscosity of the polymer solution, which is a measure of how the viscosity of a solution changes with concentration.
		동의어	M_v; viscosity-average molecular weight
		타입	number
		단위	g mol^{-1}
		예시	1000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M342	z-average molar mass (physicochemical info. / polymer chain / molecular weight)	정의	average molecular weight that gives greater weight to larger molecules, making it particularly sensitive to high molecular weight species within the distribution; M_z= sum{(M_i)^2 x (N_i)} / sum{(M_i) x (N_i)}, where M_i is the molecular weight of species i and N_i the number of molecules of species i
		동의어	M_z; z-average molecular weight
		타입	number
		단위	g mol^{-1}
		예시	1000
M343	branch molecule (physicochemical info. / polymer chain / architecture / linearity / branched)	정의	side chain that extends from the main polymer backbone chain; it is a deviation from a perfectly linear structure
		동의어	
		타입	string
		단위	
		예시	LDPE; PP; PE; graft copolymers; PVC; PU
M344	number of branches (physicochemical info. / polymer chain / architecture / linearity / branched)	정의	average number of branches attached to a single polymer molecule
		동의어	
		타입	number
		단위	none
		예시	
M345	degree of branching (physicochemical info. / polymer chain / architecture / linearity / branched)	정의	percentage of the number of branches present in a polymer in its total number of repeating units
		동의어	
		타입	number
		단위	%
		예시	
M346	average branch length (physicochemical info. / polymer chain / architecture / linearity / branched)	정의	length of the polymer chains between branching points
		동의어	
		타입	number
		단위	nm
		예시	
M347	branching density (physicochemical info. / polymer chain / architecture / linearity / branched)	정의	number of branches per unit length of the main polymer chain
		동의어	
		타입	number
		단위	nm^{-1}
		예시	
M348	branching index (physicochemical info. / polymer chain / architecture / linearity / branched)	정의	the ratio between the radius of gyration of a branched polymer and that of a linear polymer of the same molar mass (range : 0~1)
		동의어	
		타입	number
		단위	none
		예시	
M349	cross-linking density (physicochemical info. / polymer chain / architecture / linearity / cross-linked)	정의	the number of cross-links per unit volume in a polymer network, representing the degree of interconnection between polymer chains through chemical bonds; it quantifies the three-dimensional network structure formed when polymer chains are linked together by covalent or ionic bonds
		동의어	network density; degree of cross-linking; cross-link density
		타입	number
		단위	mol cm^{-3}
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M350	cross-linking agent (physicochemical info. / polymer chain / architecture / linearity / cross-linked)	정의	a substance that promotes or regulates the formation of covalent bonds between polymer chains, resulting in a three-dimensional network structure; these agents react with functional groups on the polymer chains to create bridges or cross-links that connect separate polymer molecules
		동의어	cross-linker; curing agent; hardener; vulcanizing agent; cross-linking additive; reticulating agent
		타입	string
		단위	
		예시	Sulfur (vulcanization of rubber); Dicumyl peroxide (cross-linking of polyethylene); Diisocyanates (polyurethane network formation); Glutaraldehyde (protein cross-linking); Epoxy hardeners (amine or anhydride-based curing agents); N,N'-Methylenebisacrylamide (polyacrylamide gel crosslinking); Ethylene glycol dimethacrylate (EGDMA - crosslinking of methacrylate polymers); Trimethylolpropane triacrylate (TMPTA - radiation cross-linking); Divinylbenzene (DVB - crosslinking of styrenic polymers); Metal salts (ionic cross-linking of alginates); Boric acid (PVA cross-linking); Hexamethylene diisocyanate (HDI - polyurethane crosslinking); Glyoxal (crosslinking of cellulosic fibers); Formaldehyde (crosslinking of phenolic resins)
M351	core (physicochemical info. / polymer chain / architecture / linearity / star)	정의	central part of star polymer that acts as the central junction point from which all the polymer arms radiate outwards
		동의어	central core
		타입	string
		단위	
		예시	Glycerol; C60; Sorbitol; Divinylbenzene core
M352	arm number (physicochemical info. / polymer chain / architecture / linearity / star)	정의	number of linear polymer chains that radiate from the central core
		동의어	
		타입	number
		단위	none
		예시	
M353	arm length (physicochemical info. / polymer chain / architecture / linearity / star)	정의	length of each individual polymer arm
		동의어	
		타입	number
		단위	nm
		예시	
M354	number of generation (physicochemical info. / polymer chain / architecture / linearity / dendrimer)	정의	total number of generation to indicate the level of branching in a dendrimer; it starts from generation 0 (the core) and increases with each layer of branching
		동의어	
		타입	number
		단위	none
		예시	
M355	core (physicochemical info. / polymer chain / architecture / linearity / dendrimer)	정의	central part of star polymer that acts as the central junction point from which all the polymer arms radiate outwards
		동의어	central core
		타입	string
		단위	
		예시	Glycerol; C60; Sorbitol; Divinylbenzene core
M356	branches (physicochemical info. / polymer chain / architecture / linearity / dendrimer)	정의	average number and length of branches in each generation
		동의어	
		타입	dictionary of {generation : value, number : value, length : value}
		단위	{\mu m}
		예시	
M357	end groups (physicochemical info. / polymer chain / architecture)	정의	the chemical group or functional group present at the two ends of a polymer chain, which significantly influence the reactivity, properties and degradation of the polymer
		동의어	
		타입	string
		단위	
		예시	-H; -OH

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M358	formation energy (structure / defect / 1D defect)	정의	defect formation energy
		동의어	
		타입	number
		단위	J mol^{-1}
		예시	
M359	coherency (structure / defect / 2D defect)	정의	degree of right alignment of atoms across boundaries (2D defect)
		동의어	
		타입	string
		단위	
		예시	1. coherent 2. semi-coherent 3. incoherent
M360	misfit strain (structure / defect / 2D defect)	정의	the strain induced at the interface between two materials or phases due to a mismatch in their lattice parameters
		동의어	
		타입	number
		단위	none
		예시	
M361	shape (structure / morphology / 3D)	정의	description of the bulk shape
		동의어	
		타입	string
		단위	
		예시	rectangular, irregular, spherical
M362	base dimension (structure / morphology / 1D)	정의	dimension of rectangular or square shape bases of bar
		동의어	
		타입	dictionary of {x : value, y : value}
		단위	nm
		예시	{x : 5, y : 5}, {x : 8, y : 8}
M366	diameter (structure / morphology / 2D / dimension),	정의	diameter of disk
		동의어	
		타입	number
		단위	mm
		예시	1
M367	thickness (structure / morphology / 2D / dimension)	정의	distance between two sides of a plate-shape material
		동의어	
		타입	number
		단위	mm
		예시	20
M368	charge transfer (triboelectric property)	정의	the net electric charge that passes through the external circuit during one contact-separation cycle under specified electrical boundary conditions (e.g., short circuit), in micro coulombs {\mu C} per cycle
		동의어	
		타입	number
		단위	{\mu C}
		예시	
M370	open-circuit voltage (triboelectric property)	정의	the potential difference between the electrodes of a triboelectric device measured with no external load (zero current) during a contact-separation cycle, typically reported in volts
		동의어	
		타입	number
		단위	V
		예시	50
M371	short-circuit current (triboelectric property)	정의	current generated under short-circuit conditions
		동의어	
		타입	number
		단위	{\mu A}
		예시	4.5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M372	triboelectric charge density (triboelectric property)	정의	surface charge density generated by the triboelectric effect
		동의어	
		타입	number
		단위	{\mu C} m^{−2}
		예시	30
M373	x (structure / morphology / 0D / dimension)	정의	dimension in x direction of plate shaped particles
		동의어	
		타입	number
		단위	nm
		예시	5
M374	y (structure / morphology / 0D / dimension)	정의	dimension in y direction of plate shaped particles
		동의어	
		타입	number
		단위	nm
		예시	5
M375	z (structure / morphology / 0D / dimension)	정의	dimension in z direction of plate shaped particles
		동의어	
		타입	number
		단위	nm
		예시	5
M377	x (structure / morphology / 2D / dimension)	정의	length of the plate in x-direction
		동의어	
		타입	number
		단위	mm
		예시	100
M378	y (structure / morphology / 2D / dimension)	정의	length of the plate in y-direction
		동의어	
		타입	number
		단위	mm
		예시	800
M380	solute (structure / morphology / solution)	정의	material ID or name of solute
		동의어	dissolved molecule
		타입	string
		단위	
		예시	material_03; polyethylene (PE); polystyrene (PS); polyvinyl chloride (PVC); polyacrylamide (PAM)
M381	solvent (structure / morphology / solution)	정의	the medium in which the polymer dissolves
		동의어	
		타입	string
		단위	
		예시	hexane; toluene; ethanol; acetic acid
M383	aspect ratio (structure / morphology / 1D)	정의	the ratio of the major axis length to the minor axis length
		동의어	
		타입	number
		단위	none
		예시	
M384	polymer blend type (structure)	정의	a classification of polymer blends according to the morphology and phase distribution of the constituent polymers
		동의어	polymer blend morphology; polymer blend phase separation
		타입	string
		단위	
		예시	random; rubbery; continuous phase; bicontinuous phase; dispersed phase; spherical domain; layered; fibrous; networked; tangled; microphase separated; nanophase separated

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M385	composition (structure / surface)	정의	the quantitative description of the constituent elements, compounds, or phases present in a material or its surface layer, expressed as atomic percent (at.%)
		동의어	chemical composition; elemental composition
		타입	dictionary of {constituent : constituent quantity, ⋯ , unit : string, uncertainty : uncertainty value}
		단위	
		예시	{value : {Pt : 56.0, Ni : 34.7}, unit : at.%}}
M386	Miller indices (structure / surface / crystallography)	정의	direction of surface normal in Miller indices; Miller indices is the notation in crystallography to describe the orientation of crystal planes and directions in a crystal lattice
		동의어	
		타입	string
		단위	
		예시	[1,1,3]; [1/2, 1/2, 3]
M387	unit cell (structure / surface / crystallography)	정의	the smallest repeating unit of a crystal structure at the surface of a material
		동의어	
		타입	string
		단위	
		예시	7X7; SQR{3}xSQR{3}
M388	texture type (structure / texture)	정의	origin of texture
		동의어	
		타입	string
		단위	
		예시	fiber texture; random texture; deformation (sheet) texture; synthesis texture; recrystallzation texture
M389	inverse pole figure (structure / texture)	정의	a plot showing the distribution of sample directions in the reference frame of a crystal
		동의어	
		타입	string
		단위	
		예시	{100} inverse pole figure.jpg
M390	ODF section plot (structure / texture)	정의	URI of graphical representation of the Orientation Distribution Function (ODF)
		동의어	orientation distribution function
		타입	string
		단위	
		예시	ODF.jpg
M391	cytotoxicity (biomedical property / biocompatibility)	정의	level of toxicity on cells (measured by IC50, cell viability % etc.)
		동의어	
		타입	number
		단위	{\mu g} mL^{-1}
		예시	
M392	inflammatory response (biomedical property / biocompatibility)	정의	body's response to injury (ex: cytokine, TNF-α, IL-6 level etc.)
		동의어	
		타입	object
		단위	ng mL^{-1}
		예시	
M393	hemocompatibility (biomedical property / biocompatibility)	정의	core performance indicators of materials applied in the blood environment (platlet adhesion amount)
		동의어	
		타입	number
		단위	ng mL^{-1}
		예시	
M394	osteointegration (biomedical property / bioactivity)	정의	ability of the materials to form a direct bond with bone tissue
		동의어	
		타입	number
		단위	g cm^{-2}
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M395	surface reactivity (biomedical property / bioactivity)	정의	the ability of the material surface to interact with the cell
		동의어	
		타입	number
		단위	{\mu g} cm^{-2}
		예시	1.5
M396	degradation rate (biomedical property / biodegradability)	정의	rate of decomposition of materials in the biological environment
		동의어	
		타입	number
		단위	%
		예시	30
M397	degradation product (biomedical property / biodegradability)	정의	substances generated by the decomposition of a material, assessed in terms of their concentration and potential harmful effects on living organism
		동의어	
		타입	number
		단위	{\mu g} mL^{-1}
		예시	150
M398	minimum inhibitory concentration (MIC) (biomedical property / antibacterial property)	정의	minimum concentration to inhibit microbial growth
		동의어	
		타입	number
		단위	{\mu g} mL^{-1}
		예시	2
M399	zone of inhibition (biomedical property / antibacterial property)	정의	the area where antibacterial substances (such as antibiotics, antimicrobials) have an inhibitory effect on microbial growth
		동의어	
		타입	number
		단위	mm^{2} cm^{-2}
		예시	3.14
M400	antibacterial efficiency (biomedical property / antibacterial property)	정의	the ability of materials or antibacterial agent to inhibit the growth and multiplication of bacteria
		동의어	
		타입	number
		단위	%
		예시	99.9
M401	cell adhesion rate (biomedical property / cytocompatibility)	정의	the ability of cells to adhere to the surface of the materials
		동의어	
		타입	number
		단위	%
		예시	85
M402	cell proliferation rate (biomedical property / cytocompatibility)	정의	the rate of cell growth on the surface of the materials
		동의어	
		타입	number
		단위	%
		예시	180
M403	cell differentiation (biomedical property / cytocompatibility)	정의	whether the material supports cell differentiation to specific functional cells
		동의어	
		타입	number
		단위	%
		예시	180
M404	immune cell activation (biomedical property / immune compatibility)	정의	cells in the immune system undergo a series of physiological and functional changes after being stimulated by specific signals
		동의어	
		타입	number
		단위	%
		예시	85

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M405	complement activation (biomedical property / immune compatibility)	정의	defense mechanisms that help the body recognizes, attacks, and eliminates pathogens
		동의어	
		타입	number
		단위	ng mL^{-1}
		예시	
M406	tissue integration (biomedical property / tissue compatibility)	정의	degree of bonding of the materials to the surrounding tissue
		동의어	
		타입	number
		단위	%
		예시	70
M407	tissue regeneration (biomedical property / tissue compatibility)	정의	the process by which tissues or organs recover the structure and function after injury or damage through cell proliferation and differentiation
		동의어	
		타입	number
		단위	mm^{3}
		예시	50
M408	alkaline phosphatase activity (ALP) (biomedical property / osteogenic property)	정의	reflects early osteogenic differentiation
		동의어	
		타입	number
		단위	nmol min^{-1} mg^{-1}
		예시	
M409	osteocalcin expression (OCN) (biomedical property / osteogenic property)	정의	reflecting late osteogenic differentiation
		동의어	
		타입	number
		단위	ng mL^{-1}
		예시	
M410	mineralized nodule formation (biomedical property / osteogenic property)	정의	assessment of mineralization by alizarin red staining
		동의어	
		타입	number
		단위	%
		예시	
M411	current density (electrical property)	정의	the amount of charge per unit time that flows through a unit area of a chosen cross section at an applied voltage
		동의어	
		타입	number
		단위	A m^{-2}
		예시	
M413	electron affinity (electrical property)	정의	the amount of energy released when an atom, molecule, or material gains an electron in the gas phase
		동의어	
		타입	number
		단위	eV
		예시	
M414	impedance (electrical property)	정의	the complex, frequency-dependent opposition a circuit presents to an alternating voltage/current, defined as the ratio of the phasor voltage to phasor current; the total opposition that a circuit presents to an alternating current or voltage, including both resistance and reactance
		동의어	
		타입	number
		단위	Ohm
		예시	80
M415	surface resistivity (electrical property)	정의	electrical resistance measured across a square of a material's surface
		동의어	
		타입	number
		단위	Ohm sq^{-1}
		예시	12.3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M416	gas (gas transmission property)	정의	the chemical species of gas used for the permeability or transmission measurement
		동의어	permeant gas
		타입	string
		단위	
		예시	oxygen; air; water vapor; hydrogen; nitrogen
M417	transmission rate (gas transmission property)	정의	the rate at which a gas or vapor passes through a material, influenced by diffusion and solubility characteristics; In general, permeability of a gas is defined as the volume of the gas that passes through a materials of a unit thickness per unit area and unit time, under a unit partial-pressure difference
		동의어	permeation rate; permeability; GTR; permeability rate
		타입	number
		단위	cm^{3} m^{-2} d^{-1} Pa^{-1}
		예시	35
M418	permeability (magnetic property)	정의	the degree of magnetization of a material in response to a magnetic field
		동의어	
		타입	number
		단위	none
		예시	2.1
M419	Neel temperature (magnetic property)	정의	temperature at which an antiferromagnetic material transitions to a paramagnetic state, meaning that thermal energy overcomes the exchange interactions maintaining the antiferromagnetic ordering.
		동의어	TN
		타입	number
		단위	K
		예시	
M420	relative permeability (magnetic property)	정의	relative permeability is the ratio of the permeability of a specific medium to the permeability of vacuum
		동의어	
		타입	number
		단위	none
		예시	1.2
M421	compressive modulus (mechanical property / compressive property)	정의	the ratio of stress to strain in elastic region under compression
		동의어	elastic modulus
		타입	number
		단위	GPa
		예시	5
M422	creep compliance (mechanical property / creep property)	정의	a measure of the deformation of a material under a constant stress over time
		동의어	
		타입	number
		단위	MPa^{-1}
		예시	30
M423	elastic recovery ratio (mechanical property / elastic property)	정의	ability to recover elastically after the removal of the elastic load in nano indentation; it is defined as the ratio of elastic recovery displacement (the amount of deformation recovered after the load is removed) to the maximum indentation depth during loading
		동의어	
		타입	number
		단위	dimensionless
		예시	0.8; 0.1
M424	flexural modulus (mechanical property / flexural property)	정의	the ratio of stress to strain in elastic region in flexural deformation
		동의어	
		타입	number
		단위	GPa
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M425	flexural strength (mechanical property / flexural property)	정의	the stress in a material at the exact moment the specimen fractures (snaps)
		동의어	
		타입	number
		단위	MPa
		예시	50
M426	residual stress (mechanical property)	정의	the internal stress locked within a solid material after the original external force is removed; the stress that remains in a material or thin film after fabrication, deposition, or processing, in the absence of external forces or loads; it can be either tensile (+) or compressive (–)
		동의어	
		타입	number
		단위	MPa
		예시	700; 1.5e3
M427	shear strength (mechanical property / shear property)	정의	strength of a material against the type of yield when the material fractures in shear
		동의어	ultimate shear strength
		타입	number
		단위	MPa
		예시	100
M428	haze (optical property)	정의	amount of light that is scattered away from the direct optical path when passing through a transparent material; it is typically measured as the percentage of light that deviates by more than 2.5 degrees from the incident beam.
		동의어	
		타입	number
		단위	%
		예시	3
M429	yellowness index (optical property / yellowness index)	정의	a measure of how much a material has changed color from clear or white towards yellow. It is calculated from spectrophotometric data using the CIE Tristimulus values (ASTM E313 or ASTM D1925).
		동의어	Y.I.
		타입	number
		단위	none
		예시	2.5
M430	inherent viscosity (rheological property / solution viscosity)	정의	equal to ln{(relative viscosity)} divided by mass concentration of polymer; it is synonymous with the logarithmic viscosity number, expressed conventionally in mL g^{–1}
		동의어	
		타입	number
		단위	ml g^{–1}
		예시	208
M431	intrinsic viscosity (rheological property / solution viscosity)	정의	limiting value of the ratio of the specific viscosity to the polymer concentration as the concentration approaches zero
		동의어	limiting viscosity number
		타입	number
		단위	ml g^{–1}
		예시	208
M432	reduced viscosity (rheological property / solution viscosity)	정의	specific viscosity value adjusted to account for the effects of concentration; the ratio of the specific viscosity of a polymer solution to its concentration
		동의어	specific viscosity
		타입	number
		단위	ml g^{–1}
		예시	208
M433	variability (rheological property / dynamic viscosity)	정의	behavior of change in viscosity according to the shear rate or shear stress
		동의어	
		타입	string
		단위	
		예시	1. shear thinning 2. shear thickening 3. constant

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M434	loss tangent (rheological property / variable modulus)	정의	the ratio of loss modulus to storage modulus, indicating the viscoelastic balance of a material
		동의어	
		타입	number
		단위	none
		예시	0.65
M435	stress relaxation time (rheological property)	정의	the time it takes for a material to relax its internal stresses when a constant strain is applied; it is a measure of how quickly the material can dissipate the applied stress over time
		동의어	
		타입	number
		단위	s
		예시	180
M436	contact angle (surface property)	정의	the angle formed between a liquid droplet and a solid surface, indicating surface wettability
		동의어	
		타입	number
		단위	degree
		예시	30
M437	decomposition temperature (thermophysical property)	정의	temperature at which the polymer begins to chemically degrade
		동의어	
		타입	number
		단위	K
		예시	500; 750
M438	heat deflection temperature (HDT) (thermophysical property)	정의	the temperature at which a plastic or polymer sample deforms under a specific load
		동의어	heat distortion temperature; deflection temperature under load (DTUL); heat deflection temprature under load (HDTUL)
		타입	array
		단위	
		예시	{load : 0.46, temperature : {value : 98, uncertainty : 1}}, {load : 0.46, temperature : {value : 88, uncertainty : 1}}
M439	composition (thermophysical property / melting range)	정의	composition of materials
		동의어	
		타입	dictionary of {constituent : constituent quantity, ··· , unit : string, uncertainty : uncertainty value}
		단위	1. at.% 2. wt.% 3. mol.%
		예시	{value : {Pt : 56.0, Ni : 34.7}, unit : at.%, uncertainty : 0.1, measurement : [{SEM : {...}}, ...]}; {value : {Fe2O3 : 97.8, Y2O3 : 2.2}, unit : wt.%, uncertainty : 0.2}
M440	topology freezing transition temperature (thermophysical property)	정의	temperature at which the material transitions from a viscoelastic liquid to a viscoelastic solid state due to changes in the rate of bond exchange reactions. Below this temperature, the material's topology is effectively frozen, preventing significant rearrangement of its network structure. a characteristic temperature for vitrimers which are covalently crosslinked polymers capable of reprocessing.
		동의어	
		타입	number
		단위	K
		예시	
M441	friction coefficient (tribological property)	정의	dimensionless scalar value that quantifies the ratio of the force of friction between two surfaces to the normal force pressing them together.
		동의어	coefficient of friction
		타입	number
		단위	none
		예시	0.1; 0.001

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M442	wear type (tribological property / wear)	정의	a classification of the mode of material loss or surface damage that occurs due to mechanical action, such as sliding, rolling, or impact, under tribological conditions
		동의어	
		타입	string
		단위	
		예시	abrasive wear; adhesive wear; fatigue wear; corrosive wear; erosive wear
M443	rate (tribological property / wear)	정의	measure of the rate at which material is removed from a surface due to wear over time; the value is defined mathematically as (volume of material lost)/{(normal load)x(sliding distance)}
		동의어	
		타입	number
		단위	mm^{3} N^{-1} m^{-1}
		예시	0.2
M444	wear scar (tribological property / wear / morphology)	정의	URI of image files of wear scar after tribotest
		동의어	
		타입	string
		단위	
		예시	scar-3.jpg
M445	wear track (tribological property / wear / morphology)	정의	URI of image files of wear track after tribotest
		동의어	
		타입	string
		단위	
		예시	track-3.jpg; debris-2.jpg
M446	object (tribological property / tribochemistry)	정의	description of the objects or region in tribology system that is chemically characterized
		동의어	region of interest; region
		타입	string
		단위	
		예시	scar-3; debris-2; track-3; debris on scar 3
M447	composition (tribological property / tribochemistry)	정의	composition of materials
		동의어	
		타입	dictionary of {constituent : constituent quantity, … , unit : string, uncertainty : uncertainty value}
		단위	1. at.% 2. wt.% 3. mol.%
		예시	{value : {Pt : 56.0, Ni : 34.7}, unit : at.%, uncertainty : 0.1, measurement : [{SEM : {...}}, ...]}; {value : {Fe2O3 : 97.8, Y2O3 : 2.2}, unit : wt.%, uncertainty : 0.2}
M448	structure (tribological property / tribochemistry)	정의	chemical structure of the object in tribology system that is chemically characterized
		동의어	
		타입	string
		단위	
		예시	amorphous; 30 % sp3 and 40 % sp2; polymeric phase
M451	degree of ladderization (physicochemical info. / polymer chain / architecture / linearity / ladder)	정의	the fraction (or percentage) of the polymer backbone that has successfully formed the complete double-stranded ladder structure via cyclization reaction, without open-chain defects
		동의어	
		타입	number
		단위	%
		예시	
M452	persistence length (physicochemical info. / polymer chain / architecture / linearity / ladder)	정의	a measure of the stiffness of a polymer chain, defined as the length over which the direction of the polymer backbone segments remains correlated (i.e., the chain behaves like a rigid rod)
		동의어	
		타입	number
		단위	nm
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M453	dihedral angle (physicochemical info. / polymer chain / architecture / linearity / ladder)	정의	the angle between planes defined by the backbone atoms, quantifying the twist or contortion of the ladder structure. It determines the planarity of the ribbon-like chain
		동의어	
		타입	number
		단위	degree
		예시	
M454	fractional free volume (physicochemical info. / polymer chain / architecture / linearity / ladder)	정의	the ratio of the specific free volume (empty space not occupied by the polymer chains) to the total specific volume of the polymer material; FFV=(V_sp-1.34V_vdw)/V_sp
		동의어	FFV
		타입	number
		단위	none
		예시	
M455	copolymer type (physicochemical info. / polymer chain / architecture / homogeneity / copolymer)	정의	type of copolymer
		동의어	
		타입	string
		단위	
		예시	1. random 2. block 3. graft
M456	composition (physicochemical info. / polymer chain / architecture / homogeneity / copolymer)	정의	molar percent of each comonomer type in a copolymer chain in dictionary form {"monomer": "mol. %", ...
		동의어	comonomer ratio
		타입	object
		단위	
		예시	{"styrene": 75, "butadiene": 25}
M457	sphericity (structure / morphology / 0D)	정의	the ratio of the surface area of a sphere with the same volume as the particle to the actual surface area of the particle
		동의어	shape compactness
		타입	number
		단위	none
		예시	1.0 (perfect sphere); 0.7 (elongate particle)
M458	aspect ratio (structure / morphology / 0D)	정의	the ratio of a particle's longest dimension to its shortest dimension
		동의어	
		타입	number
		단위	none
		예시	
M459	elongation (structure / morphology / 0D)	정의	the extent to which an object is stretched along its major axis; often calculated as 1-(b/a), where a and b are major and minor axes
		동의어	
		타입	number
		단위	none
		예시	0.8 (rod-like particle); 0.0 (spherical particle)
M460	shape (structure / morphology / 1D)	정의	name of 1D materials shape in conventional way
		동의어	
		타입	string
		단위	
		예시	bar; wire; tube; cylinder
M461	shape (structure / morphology / 2D)	정의	name of 2D materials shape in conventional way
		동의어	
		타입	string
		단위	
		예시	thin film; nano sheet; disk; plate

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M462	projected area (structure / morphology / 2D)	정의	area enclosed by a 2D material's projected outline; used for estimate the effective diameter
		동의어	cross-sectional area; apparent area; TEM projected area
		타입	number
		단위	nm^{2}
		예시	
M463	perimeter (structure / morphology / 2D)	정의	the total boundary length of 2D material projection or cross-section
		동의어	circumference
		타입	number
		단위	nm
		예시	
M464	feret diameter (structure / morphology / 2D)	정의	the distance between two parallel tangents on opposite sides of a material's projection; varies with measurement direction
		동의어	caliper diameter; projected length
		타입	number
		단위	{\mu m}
		예시	
M465	curvature (structure / morphology / 2D)	정의	rate of change of the tangent angle along a curved surface
		동의어	surface curvature
		타입	number
		단위	m^{-1}
		예시	
M466	roughness (structure / morphology / 2D)	정의	the deviation of a surface from its ideal plane, expressed by average rouchness Ra
		동의어	surface roughness
		타입	number
		단위	nm
		예시	
M467	substrate (structure / morphology / 2D)	정의	materials of the substrate
		동의어	
		타입	string
		단위	
		예시	Si wafer; glass; poly(ethylene) terephtalate
M468	wrinkle density (structure / morphology / 2D)	정의	the number of wrinkles, folds, or ridge features per unit area on a surface
		동의어	fold density; ridge density
		타입	number
		단위	{\mu m}^{-2}
		예시	
M469	volume (structure / morphology / 3D)	정의	volume of 3D material
		동의어	
		타입	number
		단위	{\mu m}^3
		예시	
M470	cluster size (structure / morphology / 3D / dimension)	정의	the average or maximum dimension of aggregated particles or connected domains
		동의어	aggregate size; domain size
		타입	number
		단위	{\mu m}
		예시	
M471	mean (structure / morphology / statistics)	정의	arithmetic average of a given data values
		동의어	
		타입	number
		단위	same as the unit of value
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M472	variance (structure / morphology / statistics)	정의	average squared deviation of a set of measurements from their mean
		동의어	
		타입	number
		단위	{unit of value}^2
		예시	
M473	skewness (structure / morphology / statistics)	정의	third standardized moment describing asymmetry of a distribution; positive skewness indicates longer tail toward larger values
		동의어	
		타입	number
		단위	none
		예시	
M474	kurtosis (structure / morphology / statistics)	정의	fourth standardized moment describing the peakedness or tail heaviness of a distribution; High kurtosis (>3) means sharp peak distribution of data values
		동의어	
		타입	number
		단위	none
		예시	
M475	associated phases (structure / lattice misfit)	정의	variable names of the two phases associated with the misfit value
		동의어	
		타입	string
		단위	
		예시	phase_1; phase_2
M476	lattice misfit (structure / lattice misfit)	정의	the difference in lattice parameters between two crystalline phases
		동의어	
		타입	number
		단위	%
		예시	2
M477	free volume (structure)	정의	Unoccupied or void space within a polymer matrix that is not directly occupied by polymer chains or atoms. It represents the difference between the total volume of the polymer and the volume physically occupied by its constituent molecules. Free volume strongly affects diffusion, permeability, glass transition behavior, and mechanical relaxation in polymers.
		동의어	free space; molecular void; unoccupied volume
		타입	number
		단위	cm^{3} g^{-1}
		예시	
M478	cross-link density (structure / polymer cross-link)	정의	density value; result of Flory-Rehner equation
		동의어	cross-linking density
		타입	number
		단위	%
		예시	
M479	swelling ratio (structure / polymer cross-link)	정의	value of swelling ratio defined by the ratio of the volume or mass of swollen polymer network in equilibrium with solvent to that of dry polymer
		동의어	
		타입	number
		단위	none
		예시	
M480	pore connectivity (structure / porosity)	정의	connectivity of pores in materials
		동의어	
		타입	string
		단위	
		예시	open pore; close pore

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M481	curvature (structure / surface)	정의	rate of change of the tangent angle along a curved surface
		동의어	surface curvature
		타입	number
		단위	m^{-1}
		예시	
M482	ionization energy (electrical property)	정의	minimum energy required to remove an electron from a neutral atom or molecule in the gas phase, producing a positively charged ion
		동의어	
		타입	number
		단위	kJ mol^{-1}
		예시	
M493	0.2% creep strain time (mechanical property / creep property)	정의	the time required to reach 0.2% strain during a creep test
		동의어	
		타입	number
		단위	h
		예시	1000
M494	flexural yield strength (mechanical property / flexural property)	정의	the stress in a material just before it yields in a flexure test
		동의어	
		타입	number
		단위	MPa
		예시	50
M495	impact strength (mechanical property / impact toughness)	정의	the amount of energy absorbed per unit cross-sectional aera of a specimen during fracture by impact loading
		동의어	
		타입	number
		단위	kJ m^{-2}
		예시	
M497	CIE tristimulus value (optical property / yellowness index)	정의	adapted CIE values for yellow index
		동의어	
		타입	string
		단위	
		예시	ASTM E313; ASTM D1925
M498	dispersion data (optical property / dispersion)	정의	the ratio of the speed of the light in the vacuum to that in a material for each wavelength of incidence light
		동의어	
		타입	array
		단위	nm
		예시	
M499	abbe number (optical property / dispersion)	정의	a measure of a material's optical dispersion, defined as the ratio that indicates how much the refractive index of the material changes with wavelength
		동의어	
		타입	number
		단위	none
		예시	
M500	complex viscosity (rheological property)	정의	magnitude of the complex modulus divided by angular frequency in oscillatory shear tests
		동의어	
		타입	number
		단위	Pa s
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
M501	molar heat capacity (thermophysical property / heat capacity)	정의	heat capacity of unit mole
		동의어	
		타입	number
		단위	J mol K^{−1}
		예시	100
M502	volumetric heat capacity (thermophysical property / heat capacity)	정의	heat capacity of unit volume
		동의어	
		타입	number
		단위	J m^{−3} K^{−1}
		예시	100
M503	solvus temperature (thermophysical property)	정의	the temperature on a phase diagram above which a secondary phase is completely soluble in the primary solid solution, and below which that phase begins to precipitate as a separate solid phase
		동의어	solvus point
		타입	number
		단위	K
		예시	1500
M504	weight change (oxidation property)	정의	the variation in the weight of a specimen as a result of exposure to an oxidizing environment at elevated temperatures
		동의어	
		타입	number
		단위	mg cm^{−2}
		예시	−1.2
M505	surface morphology (oxidation property)	정의	the surface morphology after the oxidation test
		동의어	
		타입	string
		단위	
		예시	rough
M506	oxide layer thickness (oxidation property)	정의	the thickness of the continuous oxide scale layer that forms on the surface of a material as a result of exposure to an oxidizing environment at elevated temperatures
		동의어	oxide scale thickness
		타입	number
		단위	{\mu m}
		예시	11.5
M507	diffusion layer thickness (oxidation property)	정의	the thickness of the diffusion layer formed by interdiffusion between the oxide scale and the metallic matrix during the oxidation test
		동의어	
		타입	number
		단위	{\mu m}
		예시	20
M508	oxidation rate (oxidation property)	정의	the speed at which a material reacts with oxygen
		동의어	oxidation reaction rate
		타입	number
		단위	{\mu g} cm^{−2} h^{−1}
		예시	1
M509	metal loss (oxidation property)	정의	the reduction in the thickness of base metal due to its consumption during oxidation
		동의어	
		타입	number
		단위	{\mu m}
		예시	5

3 시스템 소재 표준어휘 세부내용

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S1	note	정의	conventional name of the system such as description of major research
		동의어	
		타입	string
		단위	
		예시	LIB made in UNIST Prof. Seo's lab
S2	active material (catalyst / configuration)	정의	materials ID (variable name) of active material in the catalyst system
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_2
S3	amount of active material (catalyst / configuration)	정의	amount of active materials in the catalyst system
		동의어	
		타입	number
		단위	mg cm^{-2}
		예시	0.3
S4	promotor (catalyst / configuration)	정의	additive materials which improves the catalyst performance
		동의어	
		타입	string
		단위	
		예시	Material_3; Material_4
S5	amount of promotor (catalyst / configuration)	정의	amount of promotor added in the catalyst system
		동의어	
		타입	number
		단위	%
		예시	0.1
S6	support materials (catalyst / configuration)	정의	materials ID (variable name) of support in the catalyst system
		동의어	
		타입	string
		단위	
		예시	Material_2; Material_1
S7	framework (porous materials / configuration)	정의	framework of porous material
		동의어	
		타입	string
		단위	
		예시	Materials_1
S8	grafted molecule (porous materials / configuration)	정의	chemically or physically grafted molecule inside the pore of porous material
		동의어	
		타입	string
		단위	
		예시	Materials_2
S9	material (memristive system / configuration / electrode_1)	정의	materials variable name for the bottom layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S10	thickness (memristive system / configuration / electrode_1)	정의	thickness of the bottom layer
		동의어	
		타입	number
		단위	m
		예시	1.2E-08

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S11	doping (memristive system / configuration / electrode_1)	정의	dopant and doping level of electrode_1
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S12	material (memristive system / configuration / buffer_1)	정의	materials variable name for the buffer1 layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S13	thickness (memristive system / configuration / buffer_1)	정의	thickness of the bottom layer
		동의어	
		타입	number
		단위	m
		예시	1.2E-08
S14	doping (memristive system / configuration / buffer_1)	정의	dopant and doping level of buffer_1
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S15	material (memristive system / configuration / active layer)	정의	material's variable name for the active layer
		동의어	
		타입	string
		단위	
		예시	Materials_1
S16	thickness (memristive system / configuration / active layer)	정의	thickness of the active layer
		동의어	
		타입	number
		단위	m
		예시	1.2E-07
S17	doping (memristive system / configuration / active layer)	정의	dopant and doping level of active layer
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S18	material (memristive system / configuration / buffer_2)	정의	materials variable name for the buffer_2 layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S19	thickness (memristive system / configuration / buffer_2)	정의	thickness of the bottom layer
		동의어	
		타입	number
		단위	m
		예시	0.000000012
S20	doping (memristive system / configuration / buffer_2)	정의	dopant and doping level of buffer_2
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S21	material (memristive system / configuration / electrode_2)	정의	materials variable name for the top layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S22	thickness (memristive system / configuration / electrode_2)	정의	thickness of the top layer
		동의어	
		타입	number
		단위	m
		예시	1.2E-08
S23	doping (memristive system / configuration / electrode_2)	정의	dopant and doping level of electrode_2
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S24	area-specific activity (catalyst / performance / electrochemical)	정의	activity of a catalyst per unit of its electrochemically active surface area; (area-specific activity) [mA cm^{-2}] = (current) [mA] / ((active material amount) [g] x (electrochemically active surface area) [cm^{2} g^{-1}])
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	66
S25	mass-specific activity (catalyst / performance / electrochemical)	정의	mass-specific activity of a catalyst system
		동의어	
		타입	number
		단위	A mg^{-1}
		예시	0.2
S26	Faradaic efficiency (catalyst / performance / electrochemical)	정의	faradaic efficiency of a catalyst system
		동의어	
		타입	number
		단위	%
		예시	90
S33	reaction rate (catalyst / performance / thermal / reaction rate)	정의	data value of reaction rate
		동의어	
		타입	number
		단위	mol s^{-1} g^{-1}
		예시	100000
S34	temperature (catalyst / performance / thermal / reaction rate)	정의	temperature at which reaction rate is measured
		동의어	
		타입	number
		단위	K
		예시	298
S35	reaction gas (catalyst / performance / thermal / reaction rate)	정의	reaction gas
		동의어	
		타입	string
		단위	
		예시	H2, H2+N2
S36	pressure (catalyst / performance / thermal / reaction rate)	정의	pressure of reaction gas
		동의어	
		타입	number
		단위	MPa
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S37	conversion rate (catalyst / performance / thermal / conversion rate)	정의	estimated conversion rate
		동의어	
		타입	number
		단위	%
		예시	90
S38	temperature (catalyst / performance / thermal / conversion rate)	정의	temperature at which conversion rate is measured
		동의어	
		타입	number
		단위	K
		예시	350
S39	weight (catalyst / performance / thermal / conversion rate)	정의	weight of catalyst
		동의어	
		타입	number
		단위	g
		예시	3
S40	capacity (porous materials / performance / gas adsorption)	정의	amount of adsorbed gas molecules
		동의어	
		타입	number
		단위	mol g^{-1}
		예시	0.0034
S41	temperature (porous materials / performance / gas adsorption)	정의	temperature at which adsorption capacity is measured
		동의어	
		타입	number
		단위	K
		예시	298
S42	pressure (porous materials / performance / gas adsorption)	정의	pressure at which adsorption capacity is measured
		동의어	
		타입	number
		단위	MPa
		예시	0.2
S43	binding enthalpy (porous materials / performance / gas adsorption)	정의	binding enthalpy of adsorbed gas molecules
		동의어	
		타입	number
		단위	kJ mol^{-1}
		예시	45
S46	Icc (memristive system / performance / current)	정의	current value to set not to exceed a specific current value
		동의어	compliance current
		타입	number
		단위	A
		예시	0.01
S47	Ion (memristive system / performance / current)	정의	current value when device is on state
		동의어	on current
		타입	number
		단위	A
		예시	0.0001
S48	Ioff (memristive system / performance / current)	정의	current value when device is off state
		동의어	off current
		타입	number
		단위	A
		예시	0.0000001

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S49	Iset (memristive system / performance / current)	정의	current value when device is off state
		동의어	set current
		타입	number
		단위	A
		예시	0.001
S50	cycles (memristive system / performance / endurance)	정의	a complete series of processes that voltage moves and turns back
		동의어	
		타입	number
		단위	cycle
		예시	1000000
S51	voltage sweeps (memristive system / performance / endurance)	정의	range of voltage change during endurance measurement
		동의어	
		타입	dictionary of {Vmax:V, Vmin:V}
		단위	V
		예시	{Vmax: 3, Vmin:-3.5}
S52	operating mechanism (memristive system / performance / mechanism)	정의	the operating mechanism of the device
		동의어	
		타입	string
		단위	
		예시	1. filament 2. ECM
S53	LRS (memristive system / performance / mechanism / conduction mechanism)	정의	conduction mechanism at low resistance state
		동의어	Low resistance state
		타입	string
		단위	
		예시	Ohmic
S54	HRS (memristive system / performance / mechanism / conduction mechanism)	정의	conduction mechanism at high resistance state
		동의어	High resistance state
		타입	string
		단위	
		예시	SCLC
S55	operating speed (memristive system / performance)	정의	the rate at which a device operates
		동의어	
		타입	number
		단위	ns
		예시	250
S56	HRS (memristive system / performance / resistance)	정의	high resistance state value
		동의어	high resistance state
		타입	number
		단위	Ohm
		예시	100000000
S57	LRS (memristive system / performance / resistance)	정의	low resistance state value
		동의어	low resistance state
		타입	number
		단위	Ohm
		예시	1000
S58	time (memristive system / performance / retention)	정의	the time that the device can withstand without degradation in performance
		동의어	
		타입	number
		단위	h
		예시	85440

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S59	temperature (memristive system / performance / retention)	정의	temperature measuring retention
		동의어	
		타입	number
		단위	K
		예시	273
S60	on-off ratio (memristive system / performance / selectivity)	정의	the ratio of the high resistance to the low resistance of the device
		동의어	
		타입	number
		단위	dimensionless
		예시	100000000
S61	set (memristive system / performance / voltage)	정의	voltage value that makes device on-state
		동의어	Vset, set voltage, on voltage
		타입	number
		단위	V
		예시	1.5
S62	reset (memristive system / performance / voltage)	정의	voltage value that makes devcie off-state
		동의어	Vreset, reset voltage
		타입	number
		단위	V
		예시	-2
S63	forming (memristive system / performance / voltage)	정의	voltage value that makes conductive filament
		동의어	Vforming, forming voltage
		타입	number
		단위	V
		예시	3
S64	coating material (catalyst / configuration)	정의	materials ID (variable name) of coating material in the catalyst system
		동의어	
		타입	number
		단위	
		예시	Material_1; Material_2
S65	amount of coating material (catalyst / configuration)	정의	amount of coating materials in the catalyst system
		동의어	
		타입	number
		단위	%
		예시	0.3
S66	grafted molecule amount (porous materials / configuration)	정의	loading amount of grafted molecule inside the pore of porous material
		동의어	
		타입	number
		단위	mol g^{-1}
		예시	0.002
S67	device type (photodiode / configuration)	정의	type of the device
		동의어	
		타입	string
		단위	
		예시	1. Infrared 2. Ultraviolet
S68	pixel size (photodiode / configuration / device size)	정의	pixel (mesa) size of the device
		동의어	
		타입	string ({\mu m}x{\mu m} for square, {\mu m} for circle (diameter))
		단위	
		예시	10 x 10; 20

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S69	microjunction size (photodiode / configuration / device size)	정의	microjunction size of the device
		동의어	
		타입	string (1. Circle 2. Rectangle)
		단위	um x um for square, um for circle (diameter)
		예시	10 x 10; 20
S111	assembly (photodiode / configuration)	정의	a sequential assembly of the layers to build the photodetector
		동의어	
		타입	array
		단위	
		예시	["substrate", "buffer", "bottom contact layer", "absorber", "barrier", "absorber", "barrier", "top contact layer", "top electrode", "passivation layer"]
S112	thickness (gas sensor / configuration / substrate)	정의	thickness of the substrate
		동의어	
		타입	number
		단위	m
		예시	1.2e-8; 3.3e-3
S113	material (gas sensor / configuration / substrate)	정의	material name of the substrate
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5; SiO_2; Poly(ethylene) terephthalate (PET)
S114	length (gas sensor / configuration / electrode)	정의	size of the electrode along its longest side
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-2
S115	width (gas sensor / configuration / electrode)	정의	distance across the electrode from one side to the other
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-3
S116	material (gas sensor / configuration / electrode)	정의	material name for the electrode
		동의어	
		타입	string
		단위	
		예시	Gold; Au; Platinum; Pt
S117	structure type (gas sensor / configuration / electrode)	정의	structure type of the electrode
		동의어	
		타입	string
		단위	
		예시	interdigitated electrode, bar electrode, etc.
S118	thickness (gas sensor / configuration / channel)	정의	(overall) thickness of the channel
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-6
S119	material (gas sensor / configuration / channel)	정의	material name for the channel or sensing material
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SnO_{2}; MoS_{2}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S120	structure type (gas sensor / configuration / channel)	정의	structure type of the channel
		동의어	
		타입	string
		단위	
		예시	nanosheets, nanorods, etc.
S121	structure type (gas sensor / configuration / interlayer)	정의	structure type of the interlayer
		동의어	
		타입	string
		단위	
		예시	nanosheets, nanorods, etc.
S122	material (gas sensor / configuration / interlayer)	정의	material ID or name of the interlayer
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SiO_{2}; PEDOT:PSS
S123	thickness (gas sensor / configuration / interlayer)	정의	(overall) thickness of the surface treatment layer
		동의어	
		타입	number
		단위	m
		예시	6e-7; 2e-3
S130	active material (alkali-ion battery / configuration / materials function list / cathode)	정의	active materials (contribute to the capacity of the secondary battery) variable name for cathode
		동의어	AM
		타입	string
		단위	
		예시	Materials_3
S131	coating material (alkali-ion battery / configuration / materials function list / cathode)	정의	material coated on the surface of cathode active material to improve performance, conductivity, or stability
		동의어	coating
		타입	string
		단위	
		예시	Materials_3
S132	binder (alkali-ion battery / configuration / materials function list / cathode)	정의	binding materials variable name for cathode
		동의어	BM, binding materials, binder materials
		타입	string
		단위	
		예시	Materials_3
S133	conducting material (alkali-ion battery / configuration / materials function list / cathode)	정의	conducting agent added to the cathode mixture to enhance electronic conductivity and reduce internal resistance of the electrode
		동의어	CM; conductive agent; conductive additive; conducting material
		타입	string
		단위	
		예시	Materials_3
S135	electrolyte (alkali-ion battery / configuration / materials function list / cathode)	정의	electrolyte material for cathode
		동의어	cathode electrolyte; electrolyte in cathode; catholyte
		타입	string
		단위	
		예시	Materials_3
S137	mass loading (alkali-ion battery / configuration / cathode)	정의	mass loading on current collector
		동의어	
		타입	number
		단위	mg cm^{-2}
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S138	material (alkali-ion battery / configuration / anode)	정의	materials variable name
		동의어	
		타입	string
		단위	
		예시	Materials_12
S143	material (alkali-ion battery / configuration / electrolyte)	정의	electrolyte materials variable name
		동의어	
		타입	string
		단위	
		예시	Materials_12
S144	mass (alkali-ion battery / configuration / electrolyte)	정의	mass of electrolyte
		동의어	
		타입	number
		단위	mg
		예시	30
S146	thickness (alkali-ion battery / configuration / electrolyte)	정의	thickness of electrolyte
		동의어	
		타입	number
		단위	{\mu m}
		예시	10
S148	material (alkali-ion battery / configuration / separator)	정의	separator materials variable name
		동의어	
		타입	string
		단위	
		예시	Materials_12
S149	thickness (alkali-ion battery / configuration / separator)	정의	thickness of separator
		동의어	
		타입	number
		단위	mm
		예시	0.3
S151	cell type (alkali-ion battery / configuration)	정의	shape and operation type of battery
		동의어	
		타입	string
		단위	
		예시	pouch; coin; mold cell
S152	onset potential (catalyst / performance / electrochemical)	정의	the potential in an electrochemical cell that drives the forward or reverse reaction
		동의어	
		타입	number
		단위	V
		예시	0.93
S153	half-wave potential (catalyst / performance / electrochemical)	정의	a potential at which polarographic wave current is equal to one half of diffusion current
		동의어	
		타입	number
		단위	V
		예시	0.93
S154	overpotential (catalyst / performance / electrochemical)	정의	the potential difference between a half-reaction's reduction potential which is determined thermodynamically and the potential at which the redox event is experimentally observed
		동의어	
		타입	number
		단위	V
		예시	0.93

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S155	limiting current density (catalyst / performance / electrochemical)	정의	the maximum current density required to achieve a desired electrode reaction prior to the simultaneous discharge of extraneous ions
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	2.5
S164	area-specific activity (catalyst / performance / photoelectrochemical)	정의	activity of a catalyst per unit of its electrochemically active surface area; (area-specific activity) [mA cm^{-2}] = (current) [mA] / ((active material amount) [g] x (electrochemically active surface area) [cm^{2} g^{-1}])
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	66
S165	mass-specific activity (catalyst / performance / photoelectrochemical)	정의	mass-specific activity of a catalyst system
		동의어	
		타입	number
		단위	A mg^{-1}
		예시	0.2
S166	Faradaic efficiency (catalyst / performance / photoelectrochemical)	정의	faradaic efficiency of a catalyst system
		동의어	
		타입	number
		단위	%
		예시	90
S167	onset potential (catalyst / performance / photoelectrochemical)	정의	the potential in an electrochemical cell that drives the forward or reverse reaction
		동의어	
		타입	number
		단위	V
		예시	0.93
S168	half-wave potential (catalyst / performance / photoelectrochemical)	정의	a potential at which polarographic wave current is equal to one half of diffusion current
		동의어	
		타입	number
		단위	V
		예시	0.93
S169	overpotential (catalyst / performance / photoelectrochemical)	정의	the potential difference between a half-reaction's reduction potential which is determined thermodynamically and the potential at which the redox event is experimentally observed
		동의어	
		타입	number
		단위	V
		예시	0.93
S170	limiting current density (catalyst / performance / photoelectrochemical)	정의	the maximum current density required to achieve a desired electrode reaction prior to the simultaneous discharge of extraneous ions
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	2.5
S178	adsorbate (porous materials / performance / gas adsorption)	정의	name of adsorbed gas molecules
		동의어	
		타입	string
		단위	
		예시	CO2

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S179	external quantum efficiency (photodiode / performance / quantum efficiency)	정의	the number of free electrons (which are produced by the incident photons) collected from the device to the external circuit of the device per photon incident on it
		동의어	
		타입	number
		단위	%
		예시	95.8; 150.5
S180	dark current density (photodiode / performance / dark current density)	정의	current density value measured in dark
		동의어	on current
		타입	number
		단위	A cm^{-2}
		예시	0.00001
S181	cutoff wavelength (photodiode / performance)	정의	threshold wavelength of photodetector signal detection limit
		동의어	
		타입	dictionary of {cutoff percentage : {value : cutoff percentage value (%), uncertainty : uncertainty value (%)}, temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, reverse bias voltage : {value : reverse bias voltage (mV), uncertainty : uncertainty value (mV)}, cutoff wavelength : {value : cutoff wavelength value (m), uncertainty : uncertainty value (m)}}
		단위	
		예시	{cutoff percentage: {value: 80, uncertainty: 0.5}, temperature: {value: 80, uncertainty: 0.5}, reverse bias voltage: {value: -1, uncertainty: 0.05}, cutoff wavelength: {value: 3e-4, uncertainty: 5e-6}}
S182	emission wavelength (photodiode / performance)	정의	measured emission wavelength value to estimate absorption wavelength of an absorber material
		동의어	
		타입	dictionary of {temperature: {value: temperature value (K), uncertainty: uncertainty value (K)}, emission wavelength: {value: emission wavelength value (m), uncertainty: uncertainty value (m)}}
		단위	m
		예시	{temperature: {value: 80, uncertainty: 0.5}, emission wavelength: {value: 3e-4, uncertainty: 5e-6}}
S185	responsivity (photodiode / performance / responsivity)	정의	the ratio of generated photocurrent and incident optical power (active area*light intensity)
		동의어	
		타입	number
		단위	A/W
		예시	1.28e-3; 2.3e-3
S186	noise (photodiode / performance)	정의	noise of photodetector signals
		동의어	
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, noise : {value : noise value, uncertainty : uncertainty value}}
		단위	A Hz^{1/2}
		예시	{temperature : {value : 80, uncertainty : 0.5}, noise : {value : 3e-4, uncertainty : 5e-6}}
S187	signal-to-noise (photodiode / performance)	정의	ratio between desired signal and undesired signal (noise)
		동의어	SNR; S/N
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, signal-to-noise : {value : signal-to-noise value, uncertainty : uncertainty value}}
		단위	none
		예시	{temperature: {value: 80, uncertainty: 0.5}, signal-to-noise: {value: 3e-4, uncertainty: 5e-6}}
S188	noise equivalent power (photodiode / performance)	정의	the optical input power which produces an additional output power identical to that noise power for a given bandwidth
		동의어	NEP
		타입	dictionary of {temperature: {value: temperature value (K), uncertainty: uncertainty value (K)}, noise equivalent power: {value: noise equivalent power value, uncertainty: uncertainty value}}
		단위	W Hz^{-1/2}
		예시	{temperature: {value: 80, uncertainty: 0.5}, noise equivalent power: {value: 3e-4, uncertainty: 5e-6}}
S189	gas response (gas sensor / performance)	정의	the ratio between R_0 (the resistance of the sensor in a flow of balance gas) and R_g (the steady-state resistance of the sensor in the analyte gas flow)
		동의어	sensitivity; responsivity
		타입	number
		단위	%
		예시	1.28e-3; 2.3e-3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S190	relative humidity (gas sensor / performance / measurement condition)	정의	relative humidity at which the gas response was measured
		동의어	
		타입	number
		단위	%
		예시	30; 50; 70; 90
S191	wavelength (gas sensor / performance / measurement condition / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S192	light intensity (gas sensor / performance / measurement condition / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S193	bias voltage (gas sensor / performance / measurement condition)	정의	applied bias voltage with which the responsivity was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S194	operating temperature (gas sensor / performance / measurement condition)	정의	temperature at which the responsivity was measured
		동의어	
		타입	number
		단위	K
		예시	298; 500
S195	gas type (gas sensor / performance / measurement condition / balance gas)	정의	a specific kind of balance gas
		동의어	
		타입	string
		단위	
		예시	NO_{2}
S196	flow rate (gas sensor / performance / measurement condition / balance gas)	정의	flow rate of the balance gas
		동의어	
		타입	number
		단위	standard cm^{3} min^{-1} or sccm
		예시	100
S197	gas type (gas sensor / performance / measurement condition / target gas)	정의	a specific kind of analyte gas
		동의어	
		타입	string
		단위	
		예시	NO_{2}
S198	gas concentration (gas sensor / performance / measurement condition / target gas)	정의	concentration of analyte gas
		동의어	
		타입	number
		단위	ppm
		예시	100; 50
S207	gas type (gas sensor / performance / selectivity / target gas)	정의	used target gases to measure each response values for determining selectivity
		동의어	
		타입	string
		단위	
		예시	NO_{2}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S208	concentration (gas sensor / performance / selectivity / target gas)	정의	concentration of analyte gas
		동의어	
		타입	number
		단위	ppm
		예시	100; 50
S209	recovery time (gas sensor / performance)	정의	the time required for a sensor to return to 90% of the original baseline signal upon removal of the target gas
		동의어	
		타입	number
		단위	s
		예시	32; 57
S219	response time (gas sensor / performance)	정의	the time required for a sensor to reach 90% of the total response of the signal such as resistance upon exposure to the target gas
		동의어	
		타입	number
		단위	s
		예시	32; 57
S229	theoretical detection limit (gas sensor / performance / theoretical detection limit)	정의	the minimum concentration of the target gas that can be reliably distinguished
		동의어	sensitivity; limit of detection
		타입	number
		단위	ppm
		예시	5.3e-2; 2.3e-3
S230	gas type (gas sensor / performance / theoretical detection limit)	정의	a specific kind of analyte gas
		동의어	
		타입	string
		단위	
		예시	NO_{2}
S231	exposure duration (gas sensor / performance / air repeatability)	정의	duration for exposing the sensor in ambient air
		동의어	
		타입	number
		단위	d
		예시	5, 8, 101
S232	exposure duration (gas sensor / performance / humidity repeatability)	정의	duration for exposing the sensor in the specific humidity
		동의어	
		타입	number
		단위	d
		예시	5, 8, 101
S235	specific discharge capacity (areal) (alkali-ion battery / performance / cycling test result)	정의	areal specific discharge capacity
		동의어	
		타입	number
		단위	mA h cm^{-2}
		예시	8
S236	specific discharge capacity (mass) (alkali-ion battery / performance / cycling test result)	정의	mass specific capacity
		동의어	
		타입	number
		단위	mA h g^{-1}
		예시	8
S237	specific charge capacity (areal) (alkali-ion battery / performance / cycling test result)	정의	areal specific charge capacity
		동의어	
		타입	number
		단위	mA h cm^{-2}
		예시	8

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S238	specific charge capacity (mass) (alkali-ion battery / performance / cycling test result)	정의	mass specific capacity
		동의어	
		타입	number
		단위	mA h g^{-1}
		예시	8
S243	cycle number (alkali-ion battery / performance / cycling test result)	정의	number of charge-discharge cycles
		동의어	
		타입	number
		단위	none
		예시	1
S245	capacity retention (alkali-ion battery / performance / cycling test result)	정의	capacity retention
		동의어	
		타입	number
		단위	%
		예시	99
S246	overpotential (alkali-ion battery / performance / cycling test result)	정의	overpotential
		동의어	
		타입	number
		단위	V
		예시	3.6
S247	coulombic efficiency (alkali-ion battery / performance / cycling test result)	정의	coulombic efficiency
		동의어	
		타입	number
		단위	%
		예시	99
S248	energy efficiency (alkali-ion battery / performance / cycling test result)	정의	energy efficiency
		동의어	
		타입	number
		단위	%
		예시	99
S258	device type (memristive system / configuration)	정의	type of the device
		동의어	
		타입	array
		단위	
		예시	1. NVM 2. ReRAM
S259	material (alkali-ion battery / configuration / cathode)	정의	materials variable name
		동의어	
		타입	string
		단위	
		예시	Materials_12
S261	thickness (alkali-ion battery / configuration / cathode)	정의	thickness of cathode
		동의어	
		타입	number
		단위	mm
		예시	0.3
S262	volume (alkali-ion battery / configuration / cathode)	정의	volume of cathode
		동의어	
		타입	number
		단위	mL
		예시	30

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S263	mass loading (alkali-ion battery / configuration / anode)	정의	mass loading on current collector
		동의어	
		타입	number
		단위	mg cm^{-2}
		예시	10
S265	thickness (alkali-ion battery / configuration / anode)	정의	thickness of separator
		동의어	
		타입	number
		단위	mm
		예시	0.3
S266	volume (alkali-ion battery / configuration / anode)	정의	volume of electrolyte
		동의어	
		타입	number
		단위	mL
		예시	30
S267	volume (alkali-ion battery / configuration / electrolyte)	정의	volume of electrolyte
		동의어	
		타입	number
		단위	mL
		예시	30
S268	mass (alkali-ion battery / configuration / separator)	정의	mass loading of cathode composite on current collector
		동의어	
		타입	number
		단위	mg cm^{-2}
		예시	10
S269	porosity (alkali-ion battery / configuration / separator)	정의	porosity of separator
		동의어	
		타입	number
		단위	%
		예시	75
S270	tortuosity (alkali-ion battery / configuration / separator)	정의	tortuosity of separator
		동의어	
		타입	number
		단위	none
		예시	2.3
S271	additive (alkali-ion battery / configuration / materials function list / cathode)	정의	additive materials for cathode
		동의어	
		타입	string
		단위	
		예시	Materials_3
S272	current collector (alkali-ion battery / configuration / materials function list / cathode)	정의	electrical conductor between cathode and external circuits as well as a place to support cathode materials layer (coating)
		동의어	
		타입	string
		단위	
		예시	Materials_3
S273	active material (alkali-ion battery / configuration / materials function list / anode)	정의	active materials(contribute to the capacity of the secondary battery) variable name for anode
		동의어	AM
		타입	string
		단위	
		예시	Materials_3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S274	coating material (alkali-ion battery / configuration / materials function list / anode)	정의	material coated on the surface of anode active material to improve performance, conductivity, or stability
		동의어	coating
		타입	string
		단위	
		예시	Materials_3
S275	binder (alkali-ion battery / configuration / materials function list / anode)	정의	binding materials variable name for anode
		동의어	BM, binding materials, binder materials
		타입	string
		단위	
		예시	Materials_3
S276	conducting material (alkali-ion battery / configuration / materials function list / anode)	정의	conducting agent added to the anode mixture to enhance electronic conductivity and reduce internal resistance of the electrode
		동의어	CM; conductive agent; conductive additive; conducting material
		타입	string
		단위	
		예시	Materials_3
S277	additive (alkali-ion battery / configuration / materials function list / anode)	정의	additive materials for anode
		동의어	
		타입	string
		단위	
		예시	Materials_3
S278	electrolyte (alkali-ion battery / configuration / materials function list / anode)	정의	electrolyte material for anode
		동의어	anode electrolyte; electrolyte in anode; anolyte
		타입	string
		단위	
		예시	Materials_3
S279	current collector (alkali-ion battery / configuration / materials function list / anode)	정의	electrical conductor between anode and external circuits as well as a place to support anode materials layer (coating)
		동의어	
		타입	string
		단위	
		예시	Materials_3
S280	solvent (alkali-ion battery / configuration / materials function list / electrolyte)	정의	substance that dissolves solutes, resulting in a solution
		동의어	
		타입	string
		단위	
		예시	Materials_3
S281	salt (alkali-ion battery / configuration / materials function list / electrolyte)	정의	It serves as a passage for alkali ions. So, ions should be easily dissolved or dissociated in a solvent, and those dissociated ions will move smoothly.
		동의어	solute
		타입	string
		단위	
		예시	Materials_3
S282	additive (alkali-ion battery / configuration / materials function list / electrolyte)	정의	additive materials for cathode
		동의어	
		타입	string
		단위	
		예시	Materials_3
S283	length (piezoelectric system / configuration / substrate)	정의	length of the substrate
		동의어	
		타입	number
		단위	m
		예시	0.015

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S284	width (piezoelectric system / configuration / substrate)	정의	width of the substrate
		동의어	
		타입	number
		단위	m
		예시	0.015
S285	thickness (piezoelectric system / configuration / substrate)	정의	thickness of the substrate
		동의어	
		타입	number
		단위	m
		예시	0.000075
S286	material (piezoelectric system / configuration / substrate)	정의	material name of the substrate
		동의어	
		타입	string
		단위	
		예시	Material_1; PET
S287	length (piezoelectric system / configuration / bottom electrode)	정의	length of the bottom electrode
		동의어	
		타입	number
		단위	m
		예시	0.015
S288	width (piezoelectric system / configuration / bottom electrode)	정의	width of the bottom electrode
		동의어	
		타입	number
		단위	m
		예시	0.015
S289	thickness (piezoelectric system / configuration / bottom electrode)	정의	thickness of the bottom electrode
		동의어	
		타입	number
		단위	m
		예시	0.000015
S290	material (piezoelectric system / configuration / bottom electrode)	정의	material name of the bottom electrode
		동의어	
		타입	string
		단위	
		예시	Material_1; Al
S291	structure type (piezoelectric system / configuration / bottom electrode)	정의	structure type of the electrode
		동의어	
		타입	string
		단위	
		예시	1. Vertical 2. Lateral 3. IDE 4. Others
S292	length (piezoelectric system / configuration / piezoelectric layer)	정의	length of the piezoelectric layer
		동의어	
		타입	number
		단위	m
		예시	0.015
S293	width (piezoelectric system / configuration / piezoelectric layer)	정의	width of the piezoelectric layer
		동의어	
		타입	number
		단위	m
		예시	0.015
S294	thickness (piezoelectric system / configuration / piezoelectric layer)	정의	thickness of the piezoelectric layer
		동의어	
		타입	number
		단위	m
		예시	0.00008

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S295	material (piezoelectric system / configuration / piezoelectric layer)	정의	material name of the piezoelectric layer
		동의어	
		타입	string
		단위	
		예시	Material_2; P(VDF-TrFE)
S296	length (piezoelectric system / configuration / top electrode)	정의	length of the top electrode
		동의어	
		타입	number
		단위	m
		예시	0.015
S297	width (piezoelectric system / configuration / top electrode)	정의	width of the top electrode
		동의어	
		타입	number
		단위	m
		예시	0.015
S298	thickness (piezoelectric system / configuration / top electrode)	정의	thickness of the top electrode
		동의어	
		타입	number
		단위	m
		예시	0.000015
S299	material (piezoelectric system / configuration / top electrode)	정의	material name of the top electrode
		동의어	
		타입	string
		단위	
		예시	Material_1; Al
S300	structure type (piezoelectric system / configuration / top electrode)	정의	structure type of the electrode
		동의어	
		타입	string
		단위	
		예시	1. Vertical 2. Lateral 3. IDE 4. Others
S301	length (piezoelectric system / configuration / encapsulation)	정의	length of the encapsulation layer
		동의어	
		타입	number
		단위	m
		예시	0.015
S302	width (piezoelectric system / configuration / encapsulation)	정의	width of the encapsulation layer
		동의어	
		타입	number
		단위	m
		예시	0.015
S303	thickness (piezoelectric system / configuration / encapsulation)	정의	thickness of the encapsulation layer
		동의어	
		타입	number
		단위	m
		예시	0.00008
S304	material (piezoelectric system / configuration / encapsulation)	정의	material name of the encapsulation layer
		동의어	
		타입	string
		단위	
		예시	Material_3; PDMS

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S307	average voltage (alkali-ion battery / performance / cycling test result)	정의	average values of voltage measured during cycle
		동의어	
		타입	number
		단위	V
		예시	3.6
S309	energy density (volume) (alkali-ion battery / performance / cycling test result)	정의	amount of energy stored in a given system or region of space per unit volume
		동의어	
		타입	number
		단위	W h L^{-1}
		예시	275
S310	energy density (mass) (alkali-ion battery / performance / cycling test result)	정의	amount of energy stored in a given system or region of space per unit mass
		동의어	
		타입	number
		단위	W h kg^{-1}
		예시	275
S311	power density (volume) (alkali-ion battery / performance / cycling test result)	정의	amount of power processed per unit volume
		동의어	
		타입	number
		단위	W L^{-1}
		예시	275
S312	power density (mass) (alkali-ion battery / performance / cycling test result)	정의	amount of power processed per unit mass
		동의어	
		타입	number
		단위	W kg^{-1}
		예시	275
S313	maximum output voltage (piezoelectric system / performance)	정의	measured maximum output voltage when applying the load to the piezoelectric device
		동의어	
		타입	number
		단위	V
		예시	139.5
S314	applied load (piezoelectric system / performance / applied load)	정의	applied load to measure piezoelectric performance of device
		동의어	
		타입	number
		단위	1. Bending in % 2. Tapping in N 3. Vibration in Hz
		예시	1; 5; 30
S315	load resistance (piezoelectric system / performance)	정의	loaded resistance across the device to measure voltage
		동의어	
		타입	number
		단위	Ohm
		예시	10000000
S316	maximum output current (piezoelectric system / performance)	정의	measured maximum output current when applying the load to the piezoelectric device
		동의어	
		타입	number
		단위	A
		예시	1E-09
S319	maximum output power (piezoelectric system / performance)	정의	measured maximum output power when applying the load to the piezoelectric device
		동의어	
		타입	number
		단위	W
		예시	0.0000228

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S322	maximum power density (piezoelectric system / performance)	정의	measured maximum power density when applying the load to the piezoelectric device
		동의어	
		타입	number
		단위	W cm^{-3}
		예시	0.00001
S326	material (multilayer system / configuration / repeating unit)	정의	material id (in the materials section of this schema), element name or chemical nomenclature of the layer in the repeating unit
		동의어	
		타입	string
		단위	
		예시	material_01; material_12; Au; SiO2; PET
S327	thickness (multilayer system / configuration / repeating unit)	정의	thickness of the layer in the repeating unitthickness of the layer in the repeating unit
		동의어	
		타입	number
		단위	{\mu m}
		예시	0.3
S329	reaction type (catalyst / performance / electrochemical)	정의	type of electrochemical catalytic reaction
		동의어	
		타입	string
		단위	
		예시	1. ORR (Oxygen reduction) 2. NRR (Nitrogen reduction) 3. CO2RR (CO2 reduction) 4. HER (Hydrogen evolution reaction)
S330	product (catalyst / performance / electrochemical)	정의	chemical name of product
		동의어	
		타입	string
		단위	
		예시	NH3; CO; HCOOH
S331	reaction type (catalyst / performance / photoelectrochemical)	정의	type of photoelectrochemical catalytic reaction
		동의어	
		타입	string
		단위	
		예시	1. CO2RR (CO2 reduction) 2. Water splitting
S332	equilibrium constant (catalyst / performance / thermal)	정의	the ratio of products to reactants when a chemical reaction approaches to equilibrium
		동의어	
		타입	number
		단위	none
		예시	6.63E-08
S333	reaction step (catalyst / performance / thermal / chemical reaction calculated)	정의	reactants and products of target reaction
		동의어	
		타입	string
		단위	
		예시	NH3* --> NH3(g); CO* + O* --> CO2*
S334	surface coverage (catalyst / performance / thermal / chemical reaction calculated)	정의	coverage of reactants
		동의어	
		타입	number
		단위	none
		예시	0.5; 0.75
S335	activation energy (catalyst / performance / thermal / chemical reaction calculated)	정의	calculated activation energy of target reaction
		동의어	
		타입	number
		단위	eV
		예시	1.03; 0.56

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S336	reaction energy (catalyst / performance / thermal / chemical reaction calculated)	정의	calculated reaction energy of target reaction
		동의어	enthalpy of reaction
		타입	number
		단위	eV
		예시	1.03; 0.56
S337	rate constant (catalyst / performance / thermal / chemical reaction calculated)	정의	calculated rate constant of target reaction
		동의어	
		타입	number
		단위	s^{-1}
		예시	999000
S338	gap (gas sensor / configuration / electrode)	정의	distance between electrodes
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-3
S339	flow rate (gas sensor / performance / measurement condition / target gas)	정의	flow rate of the target gas
		동의어	
		타입	number
		단위	sccm
		예시	100; 50
S340	exposure time (gas sensor / performance / measurement condition)	정의	exposure time of the target gas
		동의어	
		타입	number
		단위	s
		예시	60; 30
S341	response (gas sensor / performance / selectivity)	정의	response for each analyte gases
		동의어	sensitivity
		타입	number
		단위	%
		예시	1.28e-3; 2.3e-3
S342	responsivity (gas sensor / performance / air repeatability)	정의	peak responsivity value after specific keeping days in air condition
		동의어	sensitivity
		타입	number
		단위	%
		예시	1.28e-3; 2.3e-3
S343	relative humidity (gas sensor / performance / humidity repeatability)	정의	the specific value for humidity condition
		동의어	
		타입	number
		단위	%
		예시	10, 30, 50, 70
S344	responsivity (gas sensor / performance / humidity repeatability)	정의	peak responsivity value after specific keeping days in humidity condition
		동의어	sensitivity
		타입	number
		단위	%
		예시	1.28e-3; 2.3e-3
S345	specific discharge capacity (volume) (alkali-ion battery / performance / cycling test result)	정의	volumetric specific capacity
		동의어	
		타입	number
		단위	mA h cm^{-3}
		예시	8

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S346	specific charge capacity (volume) (alkali-ion battery / performance / cycling test result)	정의	volumetric specific capacity
		동의어	
		타입	number
		단위	mA h cm^{-3}
		예시	8
S349	energy density (areal) (alkali-ion battery / performance / cycling test result)	정의	amount of energy stored in a given system or region of space per unit area
		동의어	
		타입	number
		단위	W h m^{-2}
		예시	275
S350	power density (areal) (alkali-ion battery / performance / cycling test result)	정의	amount of power processed per unit area
		동의어	
		타입	number
		단위	W m^(-2)
		예시	275
S351	architecture (photovoltaic system / configuration)	정의	The cell architecture with respect to the direction of current flow and the order in which layers are deposited. The two most common are nip (also referred to as normal) and pin (also referred to as inverted) but there are also a few others, e.g. Back contacted • nip architecture means that the electrons are collected at the substrate side. The typical example is when a TiO2 electron selective contact is deposited between the perovskite and the substrate (e.g. SLG | FTO | TiO2-c |Perovskite | …) • pin architecture means that it instead is the holes that are collected at the substrate side. The typical example is when a PEDOT:PSS hole selective contact is deposited between the perovskite and the substrate (e.g. SLG | FTO | PEDOT:PSS |Perovskite | …)
		동의어	
		타입	string
		단위	
		예시	nip; pin;
S352	material (photovoltaic system / configuration / additional layer front)	정의	Material of the additional layer front. The layer can be a stack of layers made of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	
		타입	string
		단위	
		예시	material_3; material_1 | material_2;
S353	thickness (photovoltaic system / configuration / additional layer front)	정의	Thickness of the layers in the additional layer front. • Thickness of each layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | '). • The layers must line up with the previous material field. • Every layer in the stack have a thickness. If it is unknown, state this as 'nan' • If there are uncertainties, state the best estimate, e.g. write 100 and not 90-110
		동의어	
		타입	number
		단위	mm
		예시	3.0; 2.2 | nan;
S354	function (photovoltaic system / configuration / additional layer front)	정의	The function of the additional layers front • The function must line up with the previous material field. • Every layer should be separated by a space, a vertical bar, and a space, i.e. (' | ') • If a layer has more than one function, e.g. A and B, list the functions in order and separate them with semicolons, as in (A; B)
		동의어	
		타입	string
		단위	
		예시	1. A.R.C 2. Up conversion 3. Down conversion 4. Back reflection 5. Encapsulation 6. Light management

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S355	material (photovoltaic system / configuration / substrate)	정의	Material of the substrate. The substrate can be a stack of layers made of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	
		타입	string
		단위	
		예시	material_3; material_1 | material_2;
S356	thickness (photovoltaic system / configuration / substrate)	정의	Thickness of the layers of the substrate. • Thickness of each layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | '). • The layers must line up with the previous material field. • Every layer in the stack have a thickness. If it is unknown, state this as 'nan' • If there are uncertainties, state the best estimate, e.g write 100 and not 90-110
		동의어	
		타입	number
		단위	mm
		예시	3.0; 2.2 | nan;
S357	area (photovoltaic system / configuration / substrate)	정의	The total area in cm^{2} of the substrate over which the perovskite is deposited. This may be significantly larger than the cell area
		동의어	
		타입	number
		단위	cm^{2}
		예시	85.3
S358	material (photovoltaic system / configuration / front contact)	정의	Material of the frontcontact. The frontcontact can be a stack of layers made of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	
		타입	string
		단위	
		예시	material_3; material_1 | material_2;
S359	material (photovoltaic system / configuration / n-type layer)	정의	Material of the n-type layer. The layer can be a stack of layers made of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	
		타입	string
		단위	
		예시	material_4; material_2 | material_3;
S360	thickness (photovoltaic system / configuration / n-type layer)	정의	Thickness of the layers of the electron transport layer. • Thickness of each layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | '). • The layers must line up with the previous material field. • Every layer in the stack have a thickness. If it is unknown, state this as 'nan' • If there are uncertainties, state the best estimate, e.g. write 100 and not 90-110
		동의어	
		타입	number
		단위	nm
		예시	3.0; 2.2 | nan;
S361	material (photovoltaic system / configuration / instrinsic layer)	정의	Material of the intrinsic layer. The layer can be a stack of layers made of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	
		타입	string
		단위	
		예시	material_4; material_2 | material_3;

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S362	thickness (photovoltaic system / configuration / instrinsic layer)	정의	Thickness of the layers of the intrinsic layer • Every layer in the stack have a thickness. If it is unknown, state this as 'nan' • If there are uncertainties, state the best estimate, e.g. write 100 and not 90-110
		동의어	
		타입	number
		단위	mm
		예시	3.0; 2.2 | nan;
S363	single crystal (photovoltaic system / configuration / instrinsic layer)	정의	TRUE if the cell is based on a single crystal
		동의어	
		타입	boolean
		단위	
		예시	1. False 2. True
S364	material (photovoltaic system / configuration / p-type layer)	정의	Material of the p-type layer. The layer can be a stack of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	hole transport layer; HTL; HTM; p-type materials
		타입	string
		단위	
		예시	material_4; material_2 | material_3
S365	thickness (photovoltaic system / configuration / p-type layer)	정의	A list of thicknesses of the individual layers in the stack. • Every layer should be separated by a space, a vertical bar, and a space, i.e. (' | ') • The layers must line up with the previous filed. • State thicknesses in nm • Every layer in the stack have a thickness. If it is unknown, state this as 'nan' • If there are uncertainties, state the best estimate, e.g write 100 and not 90-110
		동의어	
		타입	number
		단위	nm
		예시	200; nan | 250; 100 | 5 | 8
S366	material (photovoltaic system / configuration / back contact)	정의	Material of the backcontact. The backcontact can be a stack of layers made of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	Stack
		타입	string
		단위	
		예시	material_4; material_2 | material_3
S367	thickness (photovoltaic system / configuration / back contact)	정의	A list of thicknesses of the individual layers in the stack. • Every layer should be separated by a space, a vertical bar, and a space, i.e. (' | ') • The layers must line up with the previous filed. • State thicknesses in nm • Every layer in the stack have a thickness. If it is unknown, state this as 'nan' • If there are uncertainties, state the best estimate, e.g. write 100 and not 90-110
		동의어	
		타입	number
		단위	nm
		예시	200; nan | 250; 100 | 5 | 8
S368	material (photovoltaic system / configuration / additional layer back)	정의	Material of the additional layer back. The layer can be a stack of layers made of material_(n) defined in the materials section. Every layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (' | ').
		동의어	
		타입	string
		단위	
		예시	material_3; material_1 | material_2;

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S369	thickness (photovoltaic system / configuration / additional layer back)	정의	Thickness of the layers in the additional layer back. • Thickness of each layer in the stack should be separated by a space, a vertical bar, and a space, i.e. (‘ | ’). • The layers must line up with the previous material field. • Every layer in the stack have a thickness. If it is unknown, state this as ‘nan’ • If there are uncertainties, state the best estimate, e.g. write 100 and not 90-110
		동의어	
		타입	number
		단위	mm
		예시	3.0; 2.2 | nan;
S370	function (photovoltaic system / configuration / additional layer back)	정의	The function of the additional layers back • The function must line up with the previous material field. • Every layer should be separated by a space, a vertical bar, and a space, i.e. (‘ | ‘) • If a layer has more than one function, e.g. A and B, list the functions in order and separate them with semicolons, as in (A; B)
		동의어	
		타입	string
		단위	
		예시	1. A.R.C 2. Up conversion 3. Down conversion 4. Back reflection 5. Encapsulation 6. Light management
S371	encapsulation (photovoltaic system / configuration)	정의	TRUE is the cell is encapsulated
		동의어	
		타입	boolean
		단위	
		예시	1. False 2. True
S372	total area (photovoltaic system / configuration)	정의	The total cell area in cm^{2}. The total area is defined as the area that would provide photovoltaic performance when illuminated without any shading, i.e. in practice the geometric overlap between the top and bottom contact.
		동의어	
		타입	number
		단위	cm^{2}
		예시	30.5
S373	area measured (photovoltaic system / configuration)	정의	The effective area of the cell during IV and stability measurements under illumination. If measured with a mask, this corresponds to the area of the hole in the mask. Otherwise this area is the same as the total cell area. (mapping: Perovskite DB 5.4)
		동의어	
		타입	number
		단위	cm^{2}
		예시	22.4
S374	number of cells per substrate (photovoltaic system / configuration)	정의	The number of individual solar cells, or pixels, on the substrate on which the reported cell is made (mapping: Perovskite DB 5.4)
		동의어	
		타입	number
		단위	none
		예시	10; 12; 1
S375	number of cells in module (photovoltaic system / configuration / module)	정의	The number of cells in the module
		동의어	
		타입	number
		단위	none
		예시	1; 6; 12
S376	total area (photovoltaic system / configuration / module)	정의	The total area of the module in cm^{2}. This includes scribes, contacts, boundaries, etc. and represent the module’s geometrical footprint.
		동의어	
		타입	number
		단위	cm^{2}
		예시	85.5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S377	active area (photovoltaic system / configuration / module)	정의	the photovoltaic active area of the module in cm^{2}
		동의어	
		타입	number
		단위	cm^{2}
		예시	72.5
S378	JV data recalculated per cell (photovoltaic system / configuration / module)	정의	The preferred way to report IV data for modules is to recalculate the IV data to average data per sub-cells in the module. That simplifies downstream comparisons, and it ensures that there is no erroneous transformation that otherwise may occur when error checking the IV data. Mark this as TRUE if the conversation is done.
		동의어	
		타입	boolean
		단위	
		예시	1. False 2. True
S379	minimum bending radius (photovoltaic system / performance / flexibility)	정의	The minimum bending radius possible without degrading the cells performance (mapping: Perovskite DB 5.4)
		동의어	
		타입	number
		단위	cm
		예시	50.7; 33.0
S380	average transmittance (photovoltaic system / performance / transparency)	정의	The average visible transmittance in the wavelength range stated in the next field. (mapping: Perovskite DB 5.4)
		동의어	
		타입	number
		단위	%
		예시	75.3; 50
S381	wavelength range (photovoltaic system / performance / transparency)	정의	the wavelength range under which the average visible transmittance is determined (mapping: Perovskite DB 5.4)
		동의어	
		타입	dictionary of {minimum wavelength : value, maximum wavelength : value}
		단위	nm
		예시	{minimum wavelength : 330, maximum wavelength : 1000}
S382	Voc (photovoltaic system / performance / JV property / reverse scan)	정의	The open circuit potential, Voc, at the reverse voltage sweep (when U scanned from Voc to 0) • Give Voc in volts [V] • If there are uncertainties, only state the best estimate, e.g. write 1.03 and not 1.01-1.05 • If unknown or not applicable, leave this field empty.
		동의어	open circuit voltage
		타입	number
		단위	V
		예시	10
S383	Jsc (photovoltaic system / performance / JV property / reverse scan)	정의	The short circuit current density, Jsc, at the reverse voltage sweep (when U scanned from Voc to 0) • Give Jsc in mA/cm2 • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	short circuit current; short circuit current density
		타입	number
		단위	mA cm^{-2}
		예시	10
S384	Pmax (photovoltaic system / performance / JV property / reverse scan)	정의	A point on the curve where the product of the current (I) and voltage (V) is maximized
		동의어	max power, maximum power
		타입	number
		단위	mW
		예시	1.8

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S385	FF (photovoltaic system / performance / JV property / reverse scan)	정의	The fill factor, FF, at the reverse voltage sweep (when U scanned from Voc to 0) • Give FF as the ratio between Vmp*Jmp/(Voc*Jsc) which gives it a value between 0 and 1 • If there are uncertainties, only state the best estimate, e.g. write 0.73 and not 0.7-0.76 • If unknown or not applicable, leave this field empty.
		동의어	fill factor
		타입	number
		단위	none
		예시	0.5
S386	PCE (photovoltaic system / performance / JV property / reverse scan)	정의	The efficiency, PCE, at the reverse voltage sweep (when U scanned from Voc to 0) • Give the efficiency in % • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	power conversion efficiency; efficiency
		타입	number
		단위	%
		예시	20.5
S387	Vmp (photovoltaic system / performance / JV property / reverse scan)	정의	The potential at the maximum power point, Vmp, at the reverse voltage sweep (when U scanned from Voc to 0) • Give Vmp in volts [V] • If there are uncertainties, only state the best estimate, e.g. write 1.03 and not 1.01-1.05 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	V
		예시	1.03
S388	Jmp (photovoltaic system / performance / JV property / reverse scan)	정의	The current density at the maximum power point, Jmp, at the reverse voltage sweep (when U scanned from Voc to 0) • Give Jmp in mA/cm2 • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	20.3
S389	series resistance (photovoltaic system / performance / JV property / reverse scan)	정의	The series resistance as extracted from the reverse voltage sweep (when U scanned from Voc to 0)
		동의어	
		타입	number
		단위	Ohm cm^{2}
		예시	105
S390	shunt resistance (photovoltaic system / performance / JV property / reverse scan)	정의	The shunt resistance as extracted from the reverse voltage sweep (when U scanned from Voc to 0)
		동의어	
		타입	number
		단위	Ohm cm^{2}
		예시	200000
S391	Voc (photovoltaic system / performance / JV property / forward scan)	정의	The open circuit potential, Voc, at the forward voltage sweep (when U scanned from 0 to Voc) • Give Voc in volts [V] • If there are uncertainties, only state the best estimate, e.g. write 1.03 and not 1.01-1.05 • If unknown or not applicable, leave this field empty.
		동의어	open circuit voltage
		타입	number
		단위	V
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S392	Jsc (photovoltaic system / performance / JV property / forward scan)	정의	The short circuit current density, Jsc, at the forward voltage sweep (when U scanned from 0 to Voc) • Give Jsc in mA/cm2 • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	short circuit current; short circuit current density
		타입	number
		단위	mA cm^{-2}
		예시	10
S393	Pmax (photovoltaic system / performance / JV property / forward scan)	정의	A point on the curve where the product of the current (I) and voltage (V) is maximized
		동의어	max power, maximum power
		타입	number
		단위	mW
		예시	1.8
S394	FF (photovoltaic system / performance / JV property / forward scan)	정의	The fill factor, FF, at the forward voltage sweep (when U scanned from 0 to Voc) • Give FF as the ratio between Vmp*Jmp/(Voc*Jsc) which gives it a value between 0 and 1 • If there are uncertainties, only state the best estimate, e.g. write 0.73 and not 0.7-0.76 • If unknown or not applicable, leave this field empty.
		동의어	fill factor
		타입	number
		단위	none
		예시	0.5
S395	PCE (photovoltaic system / performance / JV property / forward scan)	정의	The efficiency, PCE, at the forward voltage sweep (when U scanned from 0 to Voc) • Give the efficiency in % • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	power conversion efficiency; efficiency
		타입	number
		단위	%
		예시	20.5
S396	Vmp (photovoltaic system / performance / JV property / forward scan)	정의	The potential at the maximum power point, Vmp, at the forward voltage sweep (when U scanned from 0 to Voc) • Give Vmp in volts [V] • If there are uncertainties, only state the best estimate, e.g. write 1.03 and not 1.01-1.05 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	V
		예시	1.03
S397	Jmp (photovoltaic system / performance / JV property / forward scan)	정의	The current density at the maximum power point, Jmp, at the forward voltage sweep (when U scanned from 0 to Voc) • Give Jmp in mA/cm2 • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	20.3
S398	series resistance (photovoltaic system / performance / JV property / forward scan)	정의	The series resistance as extracted from the forward voltage sweep (when U scanned from 0 to Voc)
		동의어	
		타입	number
		단위	Ohm cm^{2}
		예시	105
S399	shunt resistance (photovoltaic system / performance / JV property / forward scan)	정의	The shunt resistance as extracted from the forward voltage sweep (when U scanned from 0 to Voc)
		동의어	
		타입	number
		단위	Ohm cm^{2}
		예시	200000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S400	PCE (photovoltaic system / performance / stabilized performance)	정의	The stabilized efficiency, PCE • Give the efficiency in % • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	power conversion efficiency; efficiency; steady-state PCE
		타입	number
		단위	%
		예시	15.3
S401	Vmp (photovoltaic system / performance / stabilized performance)	정의	The stabilised Vmp • Give Vmp in volts [V] • If there are uncertainties, only state the best estimate, e.g. write 1.03 and not 1.01-1.05 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	V
		예시	1.03
S402	Jmp (photovoltaic system / performance / stabilized performance)	정의	The stabilised Jmp • Give Jmp in mA cm^{-2} • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	20.5
S403	integrated Jsc (photovoltaic system / performance / QE)	정의	the integrated current density from the QE measurement
		동의어	calculated Jsc from qe; calculated Jsc from ipce
		타입	number
		단위	mA cm^{-2}
		예시	20.5
S404	initial value (photovoltaic system / performance / stability / PCE)	정의	The efficiency, PCE, of the cell before the stability measurement routine starts • Give the efficiency in % • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	12.5
S405	burn in observed (photovoltaic system / performance / stability / PCE)	정의	TRUE if the performance has a relatively fast initial decay after which the decay rate stabilises at a lower level. • There are no sharp boundary between an initial burn in phase an a catastrophic failure, but if the performance of the cell quickly degrade by more than half, it is stretching it a bit to label this as an initial burn in phase.
		동의어	
		타입	boolean
		단위	
		예시	['False', 'True']
S406	final value (photovoltaic system / performance / stability / PCE)	정의	The efficiency, PCE, of the cell at the end of the stability measurement routine • Give the efficiency in % (of the initial value of PCE) • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	75

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S407	T95 (photovoltaic system / performance / stability / PCE)	정의	The time after which the cell performance has degraded by 5 % with respect to the initial performance. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S408	Ts95 (photovoltaic system / performance / stability / PCE)	정의	The time after which the cell performance has degraded by 5 % with respect to the performance after any initial burn in phase. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S409	T80 (photovoltaic system / performance / stability / PCE)	정의	The time after which the cell performance has degraded by 20 % with respect to the initial performance. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	2000
S410	Ts80 (photovoltaic system / performance / stability / PCE)	정의	The time after which the cell performance has degraded by 20 % with respect to the performance after any initial burn in phase. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S411	Te80 (photovoltaic system / performance / stability / PCE)	정의	An estimated T80 for cells that were not measured sufficiently long for them to degrade by 20 %. with respect to the initial performance. • This value will by definition have a significant uncertainty to it, as it is not measured but extrapolated under the assumption linearity but without a detailed and stabilised extrapolation protocol. This estimate is, however, not without value as it enables a rough comparison between all cells for with the stability has been measured. • If there is an experimental T80, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S412	Tse80 (photovoltaic system / performance / stability / PCE)	정의	An estimated T80 for cells that were not measured sufficiently long for them to degrade by 20 %. with respect to the initial performance. • This value will by definition have a significant uncertainty to it, as it is not measured but extrapolated under the assumption linearity but without a detailed and stabilized extrapolation protocol. This estimate is, however, not without value as it enables a rough comparison between all cells for with the stability has been measured. • If there is an experimental T80, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S413	after 1000h (photovoltaic system / performance / stability / PCE)	정의	The efficiency, PCE, of the cell after 1000 hours • Give the efficiency in % (of initial PCE) • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	73.3
S414	lifetime energy yield (photovoltaic system / performance / stability)	정의	The lifetime energy yield • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	KW h m^{-2}
		예시	20.5
S415	initial PCE (photovoltaic system / performance / stability / bending stability)	정의	The efficiency, PCE, of the cell before the mechanical stability measurement routine starts • Give the efficiency in % • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	25.4
S416	final PCE (photovoltaic system / performance / stability / bending stability)	정의	The efficiency, PCE, of the cell after the mechanical stability measurement routine • Give the efficiency in % (of initial PCE) • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	74.3
S417	PCE trace (photovoltaic system / performance / stability / bending stability)	정의	URI of the data file for the stability data is stored • This is a beta feature. The plan is to create a file repository where the raw files for stability data can be stored and disseminated. With the link and associated protocols, it should be possible to programmatically access and analyze the raw stability data.
		동의어	
		타입	string
		단위	
		예시	*.xls; *.txt;
S418	initial value (photovoltaic system / performance / outdoor / PCE)	정의	The efficiency, PCE, of the cell before the outdoor test routine starts • Give the efficiency in % • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	12.5
S419	burn in observed (photovoltaic system / performance / outdoor / PCE)	정의	TRUE if the performance has a relatively fast initial decay after which the decay rate stabilises at a lower level. • There are no sharp boundary between an initial burn in phase an a catastrophic failure, but if the performance of the cell quickly degrade by more than half, it is stretching it a bit to label this as an initial burn in phase.
		동의어	
		타입	boolean
		단위	
		예시	['False', 'True']

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S420	final value (photovoltaic system / performance / outdoor / PCE)	정의	The efficiency, PCE, of the cell at the end of the outdoor test routine • Give the efficiency in % (of the initial value of PCE) • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	87
S421	T95 (photovoltaic system / performance / outdoor / PCE)	정의	The time after which the cell performance has degraded by 5 % with respect to the initial performance. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S422	Ts95 (photovoltaic system / performance / outdoor / PCE)	정의	The time after which the cell performance has degraded by 5 % with respect to the performance after any initial burn in phase. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S423	T80 (photovoltaic system / performance / outdoor / PCE)	정의	The time after which the cell performance has degraded by 20 % with respect to the initial performance. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	2000
S424	Ts80 (photovoltaic system / performance / outdoor / PCE)	정의	The time after which the cell performance has degraded by 20 % with respect to the performance after any initial burn in phase. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S425	Te80 (photovoltaic system / performance / outdoor / PCE)	정의	An estimated T80 for cells that were not measured sufficiently long for them to degrade by 20 %. with respect to the initial performance. • This value will by definition have a significant uncertainty to it, as it is not measured but extrapolated under the assumption linearity but without a detailed and stabilised extrapolation protocol. This estimate is, however, not without value as it enables a rough comparison between all cells for with the stability has been measured. • If there is an experimental T80, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000
S426	Tse80 (photovoltaic system / performance / outdoor / PCE)	정의	An estimated Ts80 for cells that was not measured sufficiently long for them to degrade by 20 %. with respect to the performance after any initial burn in phase. • This value will by definition have a significant uncertainty to it, as it is not measured but extrapolated under the assumption linearity but without a detailed and stabilised extrapolation protocol. This estimate is, however, not without value as it enables a ruff comparison between all cells for with the stability has been measured. • If there is an experimental T80s, leave this field empty.
		동의어	
		타입	number
		단위	h
		예시	1000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S427	after 1000h (photovoltaic system / performance / outdoor / PCE)	정의	The efficiency, PCE, of the cell after 1000 hours • Give the efficiency in % (of initial PCE) • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	73.3
S428	power generated (photovoltaic system / performance / outdoor)	정의	The yearly power generated during the measurement period • If there are uncertainties, only state the best estimate, e.g. write 20.5 and not 19-20 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	KW h m^{-2} year^{-1}
		예시	20.5
S429	device type (organic thin-film transistor / configuration)	정의	type of the device
		동의어	
		타입	string
		단위	
		예시	1. Top gate and laterally-aligned bottom contacts OTFT 2. Top gate and laterally-aligned top contacts OTFT 3. Bottom gate and laterally-aligned bottom contacts OTFT 4. Bottom gate and laterally-aligned top contacts OTFT 5. Middle gate (or base) and vertically-aligned contacts OTFT 6. Top gate (or base) and vertically-aligned contacts OTFT 7. Bottom gate (or base) and vertically-aligned contacts OTFT
S430	substrate type (organic thin-film transistor / configuration / substrate)	정의	type of the substrate
		동의어	
		타입	string
		단위	
		예시	1. Insulators 2. Source or Emitter 3. Drain or Collector 4. Gate or Base
S431	material (organic thin-film transistor / configuration / substrate)	정의	material name for the substrate
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5
S432	thickness (organic thin-film transistor / configuration / substrate)	정의	thickness of the substrate
		동의어	
		타입	number
		단위	nm
		예시	10
S433	electrode type (organic thin-film transistor / configuration / electrode)	정의	type of the electrode
		동의어	
		타입	string
		단위	
		예시	1. source or emitter 2. drain or collector 3. gate or base
S434	material (organic thin-film transistor / configuration / electrode)	정의	material name for the electrode
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5
S435	thickness (organic thin-film transistor / configuration / electrode)	정의	thickness of the electrode
		동의어	
		타입	number
		단위	nm
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S436	material (organic thin-film transistor / configuration / gate dielectric)	정의	material name for the substrate
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5
S437	thickness (organic thin-film transistor / configuration / gate dielectric)	정의	thickness of the substrate
		동의어	
		타입	number
		단위	nm
		예시	10
S438	interface type (organic thin-film transistor / configuration / interfacial layer)	정의	type of the interfacial layer
		동의어	
		타입	string
		단위	
		예시	1. between electrode and semiconductor 2. between dielectric and semiconductor
S439	material (organic thin-film transistor / configuration / interfacial layer)	정의	material name for the interfacial layer
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5
S440	thickness (organic thin-film transistor / configuration / interfacial layer)	정의	thickness of the interfacial layer
		동의어	
		타입	number
		단위	nm
		예시	10
S441	material (organic thin-film transistor / configuration / semiconductor)	정의	material name for the semiconductor
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5
S442	thickness (organic thin-film transistor / configuration / semiconductor)	정의	thickness of the semiconductor
		동의어	
		타입	number
		단위	nm
		예시	10
S443	material (organic thin-film transistor / configuration / other structures)	정의	material name for the architectural structure
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5
S444	thickness (organic thin-film transistor / configuration / other structures)	정의	thickness of the architectural structure
		동의어	
		타입	number
		단위	nm
		예시	10
S445	channel length (organic thin-film transistor / configuration / dimension)	정의	the distance between the contact electrodes (source-drain or emitter-collector)
		동의어	
		타입	number
		단위	m
		예시	0.00001

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S446	channel width (organic thin-film transistor / configuration / dimension)	정의	the width of the contact electrodes (source-drain or emitter-collector)
		동의어	
		타입	number
		단위	m
		예시	0.00001
S447	channel area (organic thin-film transistor / configuration / dimension)	정의	the area of semiconductors between source and drain electrodes
		동의어	
		타입	number
		단위	m^{2}
		예시	1E-10
S448	operating voltage (organic thin-film transistor / performance)	정의	the drain-to-source voltage (V_{DS}) that is for the saturation regime to operate OTFT, presented by values assigned to symbols V_{DD}
		동의어	on voltage, operating voltage, working voltage
		타입	number
		단위	V
		예시	3
S449	threshold voltage (organic thin-film transistor / performance)	정의	the minimum gate-to-source voltage (V_{GS}) that is needed to create a conducting path between the source and drain terminals, presented by values assigned to symbols V_{th}
		동의어	threshold voltage, pinch-off voltage
		타입	number
		단위	V
		예시	3
S450	on current (organic thin-film transistor / performance)	정의	current value when device is on state, presented by values assigned to symbols I_{on}
		동의어	on current
		타입	number
		단위	A
		예시	0.00001
S451	off current (organic thin-film transistor / performance)	정의	current value when device is off state, presented by values assigned to symbols I_{off}
		동의어	off current
		타입	number
		단위	A
		예시	1
S452	on-off ratio (organic thin-film transistor / performance)	정의	the ratio of the current in the 'on' and 'off' states, which is indicative of the switching performance of OTFTs, presented by values assigned to symbols I_{on/off}
		동의어	on/off ratio
		타입	number
		단위	none
		예시	0.2
S453	voltage gain (organic thin-film transistor / performance)	정의	the ratio between the magnitude of output and input voltage signals of V_{DS}/V_{GS}
		동의어	
		타입	number
		단위	
		예시	1,000
S454	current gain (organic thin-film transistor / performance)	정의	the ratio between the magnitude of output and input current signals of I_{DS}/I_{GS}
		동의어	
		타입	number
		단위	
		예시	1,000
S455	transition frequency (organic thin-film transistor / performance)	정의	the cut-off frequency at which a device operates with the highest frequency
		동의어	
		타입	number
		단위	Hz
		예시	1,000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S456	carrier mobility (organic thin-film transistor / performance)	정의	values how quickly a carrier can move through a semiconductor when pulled by an electric field
		동의어	
		타입	number
		단위	cm^{2} V^{-1} s^{-1}
		예시	1,000
S457	capacitance (organic thin-film transistor / performance)	정의	capacitance of gate dielectric
		동의어	
		타입	number
		단위	nF
		예시	100
S458	subthreshold swing (organic thin-film transistor / performance)	정의	Subthreshold swing is a slope of drain current versus gate voltage with drain, source, and bulk voltages fixed. It determines the gate voltage amount necessary to increase the drain-source current by an order of magnitude in the subthreshold region
		동의어	subthreshold slope
		타입	number
		단위	mV {\log mA}^{-1}
		예시	0.05
S459	energy consumption (organic thin-film transistor / performance)	정의	energy consumption
		동의어	
		타입	number
		단위	J
		예시	1E-11
S460	cycles (organic thin-film transistor / performance / endurance)	정의	a complete series of processes that voltage moves and turns back
		동의어	
		타입	number
		단위	cycle
		예시	1000000
S461	applied voltage (organic thin-film transistor / performance / endurance)	정의	the range of applied voltage
		동의어	
		타입	dictionary of {Vmin : minimum applied voltage value, Vmax : maximum applied voltage}
		단위	V
		예시	{Vmin : -3.5, Vmax : 3}
S462	air stability (organic thin-film transistor / performance / endurance)	정의	the number of days under ambient air during which the sensor maintains its performance
		동의어	
		타입	number
		단위	d
		예시	3; 60; 100
S463	temperature stability (organic thin-film transistor / performance / endurance)	정의	the number of days during which the sensor maintains its performance at the operating temperature
		동의어	
		타입	number
		단위	d
		예시	3; 60; 100
S464	humidity stability (organic thin-film transistor / performance / endurance)	정의	the number of days under humidity during which the sensor maintains its performance
		동의어	
		타입	number
		단위	d
		예시	30; 60; 100
S465	bending stability (organic thin-film transistor / performance / endurance)	정의	the number of bending cycles upto which a sensor maintains its performance
		동의어	
		타입	number
		단위	cycle
		예시	10; 50; 100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S466	stretching stability (organic thin-film transistor / performance / endurance)	정의	the number of stretching cycles upto which a sensor maintains its performance
		동의어	
		타입	number
		단위	cycle
		예시	10; 50; 100
S467	material (gas sensor / configuration / electron depletion layer)	정의	material name of the electron depletion layer
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18;
S468	thickness (gas sensor / configuration / electron depletion layer)	정의	(overall) thickness of the surface treatment layer
		동의어	
		타입	number
		단위	nm
		예시	5
S469	material (gas sensor / configuration / hole accumulation layer)	정의	material name of the hole accumulation layer
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18;
S470	thickness (gas sensor / configuration / hole accumulation layer)	정의	(overall) thickness of the surface treatment layer
		동의어	
		타입	number
		단위	nm
		예시	5
S471	power consumption (gas sensor / performance / measurement condition)	정의	power consumption for operating gas sensor
		동의어	
		타입	number
		단위	W
		예시	3
S472	baseline resistance (gas sensor / performance)	정의	the resistance of sensors when no measured variable is present
		동의어	R0, sensor initial resistance, sensor resistance
		타입	number
		단위	Ohm
		예시	1.28e3; 2.3e6
S473	gas type (gas sensor / performance / sensitivity / target gas)	정의	used target gases to measure each response values for determining selectivity
		동의어	
		타입	string
		단위	
		예시	NO_{2}
S474	concentration (gas sensor / performance / sensitivity / target gas)	정의	concentration of analyte gas
		동의어	
		타입	number
		단위	ppm
		예시	100; 50
S475	sensitivity (gas sensor / performance / sensitivity)	정의	the ability to sensitively detect the target gas; defined as the slope of the logarithmic value of response vs. the logarithmic value of target gas concentration (⊿log response/⊿log [conc.])
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S476	device type (memcapcitive system / configuration)	정의	type of the device
		동의어	
		타입	array
		단위	
		예시	1. capacitor 2. transistor
S477	material (memcapcitive system / configuration / electrode_1)	정의	materials variable name for the bottom layer
		동의어	
		타입	string
		단위	
		예시	Materials_2
S478	thickness (memcapcitive system / configuration / electrode_1)	정의	thickness of the bottom layer
		동의어	
		타입	number
		단위	m
		예시	1.2e-8
S479	doping (memcapcitive system / configuration / electrode_1)	정의	dopant and doping level of electrode_1
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S480	material (memcapcitive system / configuration / buffer_1)	정의	materials variable name for the buffer1 layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S481	thickness (memcapcitive system / configuration / buffer_1)	정의	thickness of the buffer_1 layer
		동의어	
		타입	number
		단위	m
		예시	1.2e-8
S482	doping (memcapcitive system / configuration / buffer_1)	정의	dopant and doping level of buffer_1
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S483	material (memcapcitive system / configuration / active layer)	정의	material's variable name for the active layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S484	thickness (memcapcitive system / configuration / active layer)	정의	thickness of the active layer
		동의어	
		타입	number
		단위	m
		예시	2e-7
S485	doping (memcapcitive system / configuration / active layer)	정의	dopant and doping level of active layer
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S486	material (memcapcitive system / configuration / buffer_2)	정의	materials variable name for the buffer2 layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S487	thickness (memcapcitive system / configuration / buffer_2)	정의	thickness of the buffer2 layer
		동의어	
		타입	number
		단위	m
		예시	1.2e-8
S488	doping (memcapcitive system / configuration / buffer_2)	정의	dopant and doping level of buffer_2
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S489	material (memcapcitive system / configuration / electrode_2)	정의	materials variable name for the top layer
		동의어	
		타입	string
		단위	
		예시	Materials_3
S490	thickness (memcapcitive system / configuration / electrode_2)	정의	thickness of the top layer
		동의어	
		타입	number
		단위	m
		예시	1.2e-8
S491	doping (memcapcitive system / configuration / electrode_2)	정의	dopant and doping level of electrode_2
		동의어	
		타입	dictionary of {level : {dopant : dopant amount}, unit : unit value}
		단위	atoms cm^{-3}
		예시	{level : {Nb : 0.02E10}, unit : atoms cm^{-3}}
S492	on capacitance (memcapcitive system / performance / capacitance)	정의	capacitance value when device is on state
		동의어	
		타입	number
		단위	F
		예시	1e-10
S493	off capacitance (memcapcitive system / performance / capacitance)	정의	capacitance value when device is off state
		동의어	
		타입	number
		단위	F
		예시	1e-10
S494	set capacitance (memcapcitive system / performance / capacitance)	정의	capacitance value determined by the specific past voltage history of the device
		동의어	
		타입	number
		단위	F
		예시	1e-10
S495	on-off ratio (memcapcitive system / performance / capacitance)	정의	the ratio of the on capacitance to the off capacitance of the device
		동의어	
		타입	number
		단위	none
		예시	5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S496	number of states (memcapcitive system / performance / capacitance)	정의	the number of capacitance states that can be stably maintained due to its memory effect
		동의어	
		타입	number
		단위	state
		예시	100
S497	cycles (memcapcitive system / performance / endurance)	정의	cycle number of a complete series of processes that voltage moves and turns back
		동의어	
		타입	number
		단위	none
		예시	1000000
S498	voltage sweep (memcapcitive system / performance / endurance)	정의	minimum and maximum values of the voltage gradually changed during endurance measuring process
		동의어	
		타입	dictionary
		단위	{Vmax : V, Vmin: V}
		예시	{Vmin : −3.5, Vmax : 3}
S499	operating mechanism (memcapcitive system / performance / mechanism)	정의	operating mechanism
		동의어	
		타입	string
		단위	
		예시	1. ferroelectric 2. charge shielding
S500	operating speed (memcapcitive system / performance / mechanism)	정의	the rate at which a device operates
		동의어	
		타입	number
		단위	ns
		예시	250
S501	time (memcapcitive system / performance / retention)	정의	the time that the device can withstand without degradation in retention
		동의어	
		타입	number
		단위	h
		예시	85540
S502	temperature (memcapcitive system / performance / retention)	정의	temperature measuring retention
		동의어	
		타입	number
		단위	K
		예시	298
S503	set (memcapcitive system / performance / voltage)	정의	voltage value that makes device on-state
		동의어	Vset, set voltage, on voltage
		타입	number
		단위	V
		예시	1.5
S504	reset (memcapcitive system / performance / voltage)	정의	voltage value that makes device off-state
		동의어	Vreset, reset voltage
		타입	number
		단위	V
		예시	-2
S506	gate leakage current (organic thin-film transistor / performance)	정의	the gate current that flows from gate electrode to source electrode or drain electrode at certain operating voltage, presented by values assigned to symbols I_{G}
		동의어	gate dielectric leakage
		타입	dictionary of {applied voltage (mV) : value, gate leakage current : {value : gate leakage current value, uncertainty : uncertainty value}}
		단위	{\mu A}
		예시	{applied voltage : 130, gate leakage current : {value : 0.2, uncertainty : 0.001}}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S507	contact resistance (organic thin-film transistor / performance)	정의	area normalized resistance at the interface between the metal contacts and the organic semiconductor material
		동의어	
		타입	number
		단위	Ohm cm^{-2}
		예시	100
S512	height (photodiode / configuration / active area)	정의	vertical length of active area
		동의어	
		타입	number
		단위	{\mu m}
		예시	100; 5e2
S513	width (photodiode / configuration / active area)	정의	horizontal length of active area
		동의어	
		타입	number
		단위	{\mu m}
		예시	100; 5e2
S514	area (photodiode / configuration / active area)	정의	area of active area
		동의어	
		타입	number
		단위	m^{2}
		예시	6e-7; 3e-3
S515	dark current (photodiode / performance / dark current)	정의	current value measured in dark
		동의어	
		타입	number
		단위	A
		예시	0.01
S516	bias voltage (photodiode / performance / dark current)	정의	applied voltage with which the dark current was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S517	gate voltage (photodiode / performance / dark current)	정의	applied voltage to gate with which the dark current was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S518	bias voltage (photodiode / performance / dark current density)	정의	applied bias voltage with which the dark current density was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S519	photocurrent (photodiode / performance / photocurrent)	정의	current value measured under illumination
		동의어	
		타입	number
		단위	A
		예시	0.0000001
S520	bias voltage (photodiode / performance / photocurrent)	정의	applied bias voltage with which the photocurrent was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S521	gate voltage (photodiode / performance / photocurrent)	정의	applied gate bias with which the photocurrent was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S522	wavelength (photodiode / performance / photocurrent / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S523	light intensity (photodiode / performance / photocurrent / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S524	photocurrent density (photodiode / performance / photocurrent density)	정의	current density value measured under illumination
		동의어	
		타입	number
		단위	A cm^{-2}
		예시	0.001
S525	bias voltage (photodiode / performance / photocurrent density)	정의	applied bias voltage with which the photocurrent density was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S526	wavelength (photodiode / performance / photocurrent density / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S527	light intensity (photodiode / performance / photocurrent density / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S528	on-off ratio (photodiode / performance / on-off ratio)	정의	the ratio of dark current (density) and photocurrent (density)
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S529	bias voltage (photodiode / performance / on-off ratio)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S530	wavelength (photodiode / performance / on-off ratio / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S531	light intensity (photodiode / performance / on-off ratio / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S532	photoconductive gain (photodiode / performance / photoconductive gain)	정의	the number of photogenerated charge carriers collected by the electrodes divided by the number of photons absorbed in the semiconducting photoconductor
		동의어	
		타입	number
		단위	
		예시	1.28e-3; 2.3e-3
S533	bias voltage (photodiode / performance / photoconductive gain)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S534	wavelength (photodiode / performance / photoconductive gain / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S535	light intensity (photodiode / performance / photoconductive gain / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S536	bias voltage (photodiode / performance / responsivity)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S537	wavelength (photodiode / performance / responsivity / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S538	light intensity (photodiode / performance / responsivity / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S539	specific detectivity (photodiode / performance / specific detectivity)	정의	a figure of merit used to characterize performance, equal to the reciprocal of noise-equivalent power (NEP), normalized per square root of the sensor's area and frequency bandwidth (reciprocal of twice the integration time).
		동의어	
		타입	number
		단위	Jones; cm Hz^{1/2} W^{-1}
		예시	1.28e12; 1.5e14
S540	bias voltage (photodiode / performance / specific detectivity)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S541	wavelength (photodiode / performance / specific detectivity / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S542	light intensity (photodiode / performance / specific detectivity / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S543	decay time (photodiode / performance)	정의	the time required for a sensor to return to 90% of the original baseline signal upon removal of the illumination
		동의어	
		타입	number
		단위	s
		예시	0.03; 1.5e-6
S544	rise time (photodiode / performance)	정의	the time required for a photodetector to reach 90% of the total response of the signal such as current upon exposure to illumination
		동의어	
		타입	number
		단위	s
		예시	0.03; 1.5e-6
S545	bias voltage (photodiode / performance / quantum efficiency)	정의	applied bias voltage with which the photocurrent (density) was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S546	wavelength (photodiode / performance / quantum efficiency / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S547	light intensity (photodiode / performance / quantum efficiency / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S548	linear dynamic range (photodiode / performance / linear dynamic range)	정의	a figure-of-merit for photodetectors to characterize the light intensity range in which the photodetectors have a constant responsivity
		동의어	
		타입	number
		단위	dB
		예시	100; 256
S549	bias voltage (photodiode / performance / linear dynamic range)	정의	applied bias voltage with which the photocurrent (density) was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S550	exposure duration (photodiode / performance / air stability)	정의	duration for exposing the sensor in ambient air
		동의어	
		타입	number
		단위	day
		예시	5, 8, 101

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S551	photoresponse (photodiode / performance / air stability)	정의	photocurrent, on-off ratio or responsivity value after specific keeping days in air condition
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S552	temperature (photodiode / performance / temperature stability)	정의	the specific value for high temperature condition
		동의어	
		타입	number
		단위	K
		예시	330, 350
S553	operating time (photodiode / performance / temperature stability)	정의	the specific days (or time) after keeping in the specific temperature condition
		동의어	
		타입	number
		단위	hour; day
		예시	5, 8, 101
S554	photoresponse (photodiode / performance / temperature stability)	정의	photocurrent, on-off ratio or responsivity value after specific keeping days in high temperature condition
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S555	material (photoconductor / configuration / substrate)	정의	material name of the substrate
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SiO_{2}; PEDOT:PSS
S556	thickness (photoconductor / configuration / substrate)	정의	thickness of the substrate
		동의어	
		타입	number
		단위	m
		예시	1.2e-8; 3.3e-3
S557	material (photoconductor / configuration / electrode)	정의	material name for the electrode
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SiO_{2}; PEDOT:PSS
S558	width (photoconductor / configuration / electrode)	정의	distance across the electrode from one side to the other
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-3
S559	gap (photoconductor / configuration / electrode)	정의	distance between electrodes
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-3
S560	material (photoconductor / configuration / active layer)	정의	material name for the channel or photodetection material
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SiO_{2}; PEDOT:PSS

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S561	thickness (photoconductor / configuration / active layer)	정의	(overall) thickness of the channel
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-3
S562	material (photoconductor / configuration / interlayer)	정의	material name of the interface layer
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SiO_{2}; PEDOT:PSS
S563	size (photoconductor / configuration / interlayer)	정의	(overall) thickness of the interface layer or size of the decorated nanomaterials
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-3
S564	dark current (photoconductor / performance / dark current)	정의	current value measured in dark
		동의어	
		타입	number
		단위	A
		예시	0.01
S565	bias voltage (photoconductor / performance / dark current)	정의	applied voltage with which the dark current was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S566	gate voltage (photoconductor / performance / dark current)	정의	applied voltage to gate with which the dark current was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S567	dark current density (photoconductor / performance / dark current density)	정의	current density value measured in dark
		동의어	on current
		타입	number
		단위	A cm^{-2}
		예시	0.00001
S568	bias voltage (photoconductor / performance / dark current density)	정의	applied bias voltage with which the dark current density was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S569	photocurrent (photoconductor / performance / photocurrent)	정의	current value measured under illumination
		동의어	
		타입	number
		단위	A
		예시	0.0000001
S570	bias voltage (photoconductor / performance / photocurrent)	정의	applied bias voltage with which the photocurrent was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S571	gate voltage (photoconductor / performance / photocurrent)	정의	applied gate bias with which the photocurrent was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S572	wavelength (photoconductor / performance / photocurrent / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S573	light intensity (photoconductor / performance / photocurrent / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S574	photocurrent density (photoconductor / performance / photocurrent density)	정의	current density value measured under illumination
		동의어	
		타입	number
		단위	A cm^{-2}
		예시	0.001
S575	bias voltage (photoconductor / performance / photocurrent density)	정의	applied bias voltage with which the photocurrent density was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S576	wavelength (photoconductor / performance / photocurrent density / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S577	light intensity (photoconductor / performance / photocurrent density / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S578	on-off ratio (photoconductor / performance / on-off ratio)	정의	the ratio of dark current (density) and photocurrent (density)
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S579	bias voltage (photoconductor / performance / on-off ratio)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S580	wavelength (photoconductor / performance / on-off ratio / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S581	light intensity (photoconductor / performance / on-off ratio / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S582	photoconductive gain (photoconductor / performance / photoconductive gain)	정의	the number of photogenerated charge carriers collected by the electrodes divided by the number of photons absorbed in the semiconducting photoconductor
		동의어	
		타입	number
		단위	
		예시	1.28e-3; 2.3e-3
S583	bias voltage (photoconductor / performance / photoconductive gain)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S584	wavelength (photoconductor / performance / photoconductive gain / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S585	light intensity (photoconductor / performance / photoconductive gain / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S586	responsivity (photoconductor / performance / responsivity)	정의	the ratio of generated photocurrent and incident optical power (active area*light intensity)
		동의어	
		타입	number
		단위	A/W
		예시	1.28e-3; 2.3e-3
S587	bias voltage (photoconductor / performance / responsivity)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S588	wavelength (photoconductor / performance / responsivity / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S589	light intensity (photoconductor / performance / responsivity / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S590	specific detectivity (photoconductor / performance / specific detectivity)	정의	a figure of merit used to characterize performance, equal to the reciprocal of noise-equivalent power (NEP), normalized per square root of the sensor's area and frequency bandwidth (reciprocal of twice the integration time).
		동의어	
		타입	number
		단위	Jones; cm Hz^{1/2} W^{-1}
		예시	1.28e12; 1.5e14

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S591	bias voltage (photoconductor / performance / specific detectivity)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S592	wavelength (photoconductor / performance / specific detectivity / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S593	light intensity (photoconductor / performance / specific detectivity / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S594	decay time (photoconductor / performance)	정의	the time required for a sensor to return to 90% of the original baseline signal upon removal of the illumination
		동의어	
		타입	number
		단위	s
		예시	0.03; 1.5e-6
S595	rise time (photoconductor / performance)	정의	the time required for a photodetector to reach 90% of the total response of the signal such as current upon exposure to illumination
		동의어	
		타입	number
		단위	s
		예시	0.03; 1.5e-6
S596	external quantum efficiency (photoconductor / performance / quantum efficiency)	정의	the number of free electrons (which are produced by the incident photons) collected from the device to the external circuit of the device per photon incident on it
		동의어	
		타입	number
		단위	%
		예시	95.8; 150.5
S597	bias voltage (photoconductor / performance / quantum efficiency)	정의	applied bias voltage with which the photocurrent (density) was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S598	wavelength (photoconductor / performance / quantum efficiency / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S599	light intensity (photoconductor / performance / quantum efficiency / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S600	linear dynamic range (photoconductor / performance / linear dynamic range)	정의	a figure-of-merit for photodetectors to characterize the light intensity range in which the photodetectors have a constant responsivity
		동의어	
		타입	number
		단위	dB
		예시	100; 256

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S601	bias voltage (photoconductor / performance / linear dynamic range)	정의	applied bias voltage with which the photocurrent (density) was measured
		동의어	
		타입	number
		단위	V
		예시	
S602	cutoff wavelength (photoconductor / performance)	정의	threshold wavelength of photodetector signal detection limit
		동의어	
		타입	dictionary of {cutoff percentage : {value : cutoff percentage value (%), uncertainty : uncertainty value (%)}, temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, reverse bias voltage : {value : reverse bias voltage (mV), uncertainty : uncertainty value (mV)}, cutoff wavelength : {value : cutoff wavelength value (m), uncertainty : uncertainty value (m)}}
		단위	
		예시	{cutoff percentage : {value : 80, uncertainty : 0.5}, temperature : {value : 80, uncertainty : 0.5}, reverse bias voltage : {value :-1, uncertainty : 0.05}, cutoff wavelength : {value : 3e-4, uncertainty : 5e-6}}
S603	emission wavelength (photoconductor / performance)	정의	measured emission wavelength value to estimate absorption wavelength of an absorber material
		동의어	
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, emission wavelength : {value : emission wavelength value (m), uncertainty : uncertainty value (m)}}
		단위	m
		예시	{temperature : {value : 80, uncertainty : 0.5}, emission wavelength : {value : 3e-4, uncertainty : 5e-6}}
S604	noise (photoconductor / performance)	정의	noise of photodetector signals
		동의어	
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, noise : {value : noise value, uncertainty : uncertainty value}}
		단위	A Hz^{1/2}
		예시	{temperature : {value : 80, uncertainty : 0.5}, noise : {value : 3e-4, uncertainty : 5e-6}}
S605	signal-to-noise (photoconductor / performance)	정의	ratio between desired signal and undesired signal (noise)
		동의어	SNR; S/N
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, signal-to-noise : {value : signal-to-noise value, uncertainty : uncertainty value}}
		단위	none
		예시	{temperature : {value : 80, uncertainty : 0.5}, signal-to-noise : {value : 3e-4, uncertainty : 5e-6}}
S606	noise equivalent power (photoconductor / performance)	정의	the optical input power which produces an additional output power identical to that noise power for a given bandwidth
		동의어	NEP
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, noise equivalent power : {value : noise equivalent power value, uncertainty : uncertainty value}}
		단위	W Hz^{-1/2}
		예시	{temperature : {value : 80, uncertainty : 0.5}, noise equivalent power : {value : 3e-4, uncertainty : 5e-6}}
S607	exposure duration (photoconductor / performance / air stability)	정의	duration for exposing the sensor in ambient air
		동의어	
		타입	number
		단위	day
		예시	5, 8, 101
S608	photoresponse (photoconductor / performance / air stability)	정의	photocurrent, on-off ratio or responsivity value after specific keeping days in air condition
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S609	temperature (photoconductor / performance / temperature stability)	정의	the specific value for high temperature condition
		동의어	
		타입	number
		단위	K
		예시	330, 350

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S610	operating time (photoconductor / performance / temperature stability)	정의	the specific days (or time) after keeping in the specific temperature condition
		동의어	
		타입	number
		단위	hour; day
		예시	5, 8, 101
S611	photoresponse (photoconductor / performance / temperature stability)	정의	photocurrent, on-off ratio or responsivity value after specific keeping days in high temperature condition
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S612	material (phototransistor / configuration / substrate)	정의	material name of the substrate
		동의어	
		타입	string
		단위	
		예시	Material_1; Material_5; SiO_2; Poly(ethylene) terephthalate (PET)
S613	thickness (phototransistor / configuration / substrate)	정의	thickness of the substrate
		동의어	
		타입	number
		단위	m
		예시	1.2e-8; 3.3e-3
S614	material (phototransistor / configuration / source electrode)	정의	material name for the source electrode
		동의어	
		타입	string
		단위	
		예시	Gold; Au; Platinum; Pt
S615	thickness (phototransistor / configuration / source electrode)	정의	(overall) thickness of the source electrode
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-6
S616	material (phototransistor / configuration / drain electrode)	정의	material name for the drain electrode
		동의어	
		타입	string
		단위	
		예시	Gold; Au; Platinum; Pt
S617	thickness (phototransistor / configuration / drain electrode)	정의	(overall) thickness of the drain electrode
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-6
S618	material (phototransistor / configuration / gate electrode / bottom gate)	정의	material name for the bottom gate
		동의어	
		타입	string
		단위	
		예시	Gold; Au; Platinum; Pt
S619	thickness (phototransistor / configuration / gate electrode / bottom gate)	정의	(overall) thickness of the bottom gate
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-6

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S620	material (phototransistor / configuration / gate electrode / top gate)	정의	material name for the top gate
		동의어	
		타입	string
		단위	
		예시	Gold; Au; Platinum; Pt
S621	thickness (phototransistor / configuration / gate electrode / top gate)	정의	(overall) thickness of the top gate
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-6
S622	material (phototransistor / configuration / active layer)	정의	material name for the channel or photodetection material
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SnO_{2}; MoS_{2}
S623	thickness (phototransistor / configuration / active layer)	정의	(overall) thickness of the channel
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-6
S624	material (phototransistor / configuration / dielectric layer)	정의	material name of the dielectric layer
		동의어	
		타입	string
		단위	
		예시	Material_17; Material_18; SiO_{2}; PEDOT:PSS
S625	thickness (phototransistor / configuration / dielectric layer)	정의	(overall) thickness of the dielectric layer
		동의어	
		타입	number
		단위	m
		예시	6e-7; 3e-6
S626	dark current (phototransistor / performance / dark current)	정의	current value measured in dark
		동의어	
		타입	number
		단위	A
		예시	0.01
S627	bias voltage (phototransistor / performance / dark current)	정의	applied voltage with which the dark current was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S628	gate voltage (phototransistor / performance / dark current)	정의	applied voltage to gate with which the dark current was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S629	dark current density (phototransistor / performance / dark current density)	정의	current density value measured in dark
		동의어	on current
		타입	number
		단위	A cm^{-2}
		예시	0.00001

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S630	bias voltage (phototransistor / performance / dark current density)	정의	applied bias voltage with which the dark current density was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S631	photocurrent (phototransistor / performance / photocurrent)	정의	current value measured under illumination
		동의어	
		타입	number
		단위	A
		예시	0.0000001
S632	bias voltage (phototransistor / performance / photocurrent)	정의	applied bias voltage with which the photocurrent was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S633	gate voltage (phototransistor / performance / photocurrent)	정의	applied gate bias with which the photocurrent was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S634	wavelength (phototransistor / performance / photocurrent / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S635	light intensity (phototransistor / performance / photocurrent / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S636	photocurrent density (phototransistor / performance / photocurrent density)	정의	current density value measured under illumination
		동의어	
		타입	number
		단위	A cm^{2}
		예시	0.001
S637	bias voltage (phototransistor / performance / photocurrent density)	정의	applied bias voltage with which the photocurrent density was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S638	wavelength (phototransistor / performance / photocurrent density / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S639	light intensity (phototransistor / performance / photocurrent density / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S640	on-off ratio (phototransistor / performance / on-off ratio)	정의	the ratio of dark current (density) and photocurrent (density)
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S641	bias voltage (phototransistor / performance / on-off ratio)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S642	wavelength (phototransistor / performance / on-off ratio / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S643	light intensity (phototransistor / performance / on-off ratio / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S644	photoconductive gain (phototransistor / performance / photoconductive gain)	정의	the number of photogenerated charge carriers collected by the electrodes divided by the number of photons absorbed in the semiconducting photoconductor
		동의어	
		타입	number
		단위	
		예시	1.28e-3; 2.3e-3
S645	bias voltage (phototransistor / performance / photoconductive gain)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S646	wavelength (phototransistor / performance / photoconductive gain / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S647	light intensity (phototransistor / performance / photoconductive gain / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S648	responsivity (phototransistor / performance / responsivity)	정의	the ratio of generated photocurrent and incident optical power (active area*light intensity)
		동의어	
		타입	number
		단위	A/W
		예시	1.28e-3; 2.3e-3
S649	bias voltage (phototransistor / performance / responsivity)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S650	wavelength (phototransistor / performance / responsivity / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S651	light intensity (phototransistor / performance / responsivity / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S652	specific detectivity (phototransistor / performance / specific detectivity)	정의	a figure of merit used to characterize performance, equal to the reciprocal of noise-equivalent power (NEP), normalized per square root of the sensor's area and frequency bandwidth (reciprocal of twice the integration time).
		동의어	
		타입	number
		단위	Jones; cm Hz^{1/2} W^{-1}
		예시	1.28e12; 1.5e14
S653	bias voltage (phototransistor / performance / specific detectivity)	정의	applied bias voltage to the photodetector
		동의어	
		타입	number
		단위	V
		예시	1; 5
S654	wavelength (phototransistor / performance / specific detectivity / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S655	light intensity (phototransistor / performance / specific detectivity / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S656	decay time (phototransistor / performance)	정의	the time required for a sensor to return to 90% of the original baseline signal upon removal of the illumination
		동의어	
		타입	number
		단위	s
		예시	0.03; 1.5e-6
S657	rise time (phototransistor / performance)	정의	the time required for a photodetector to reach 90% of the total response of the signal such as current upon exposure to illumination
		동의어	
		타입	number
		단위	s
		예시	0.03; 1.5e-6
S658	external quantum efficiency (phototransistor / performance / quantum efficiency)	정의	the number of free electrons (which are produced by the incident photons) collected from the device to the external circuit of the device per photon incident on it
		동의어	
		타입	number
		단위	%
		예시	95.8; 150.5
S659	bias voltage (phototransistor / performance / quantum efficiency)	정의	applied bias voltage with which the photocurrent (density) was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S660	wavelength (phototransistor / performance / quantum efficiency / light illumination)	정의	wavelength of the incident light
		동의어	
		타입	number
		단위	nm
		예시	532; 634
S661	light intensity (phototransistor / performance / quantum efficiency / light illumination)	정의	intensity of the incident light
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	10; 100
S662	linear dynamic range (phototransistor / performance / linear dynamic range)	정의	a figure-of-merit for photodetectors to characterize the light intensity range in which the photodetectors have a constant responsivity
		동의어	
		타입	number
		단위	dB
		예시	100; 256
S663	bias voltage (phototransistor / performance / linear dynamic range)	정의	applied bias voltage with which the photocurrent (density) was measured
		동의어	
		타입	number
		단위	V
		예시	1; 5
S664	cutoff wavelength (phototransistor / performance)	정의	threshold wavelength of photodetector signal detection limit
		동의어	
		타입	dictionary of {cutoff percentage : {value : cutoff percentage value (%), uncertainty : uncertainty value (%)}, temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, reverse bias voltage : {value : reverse bias voltage (mV), uncertainty : uncertainty value (mV)}, cutoff wavelength : {value : cutoff wavelength value (m), uncertainty : uncertainty value (m)}}
		단위	
		예시	{cutoff percentage : {value : 80, uncertainty : 0.5}, temperature : {value : 80, uncertainty : 0.5}, reverse bias voltage : {value :-1, uncertainty : 0.05}, cutoff wavelength : {value : 3e-4, uncertainty : 5e-6}}
S665	emission wavelength (phototransistor / performance)	정의	measured emission wavelength value to estimate absorption wavelength of an absorber material
		동의어	
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, emission wavelength : {value : emission wavelength value (m), uncertainty : uncertainty value (m)}}
		단위	m
		예시	{temperature : {value : 80, uncertainty : 0.5}, emission wavelength : {value : 3e-4, uncertainty : 5e-6}}
S666	noise (phototransistor / performance)	정의	noise of photodetector signals
		동의어	
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, noise : {value : noise value, uncertainty : uncertainty value}}
		단위	A Hz^{1/2}
		예시	{temperature : {value : 80, uncertainty : 0.5}, noise : {value : 3e-4, uncertainty : 5e-6}}
S667	signal-to-noise (phototransistor / performance)	정의	ratio between desired signal and undesired signal (noise)
		동의어	SNR; S/N
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, signal-to-noise : {value : signal-to-noise value, uncertainty : uncertainty value}}
		단위	none
		예시	{temperature : {value : 80, uncertainty : 0.5}, signal-to-noise : {value : 3e-4, uncertainty : 5e-6}}
S668	noise equivalent power (phototransistor / performance)	정의	the optical input power which produces an additional output power identical to that noise power for a given bandwidth
		동의어	NEP
		타입	dictionary of {temperature : {value : temperature value (K), uncertainty : uncertainty value (K)}, noise equivalent power : {value : noise equivalent power value, uncertainty : uncertainty value}}
		단위	W Hz^{-1/2}
		예시	{temperature : {value : 80, uncertainty : 0.5}, noise equivalent power : {value : 3e-4, uncertainty : 5e-6}}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S669	exposure duration (phototransistor / performance / air stability)	정의	duration for exposing the sensor in ambient air
		동의어	
		타입	number
		단위	day
		예시	5, 8, 101
S670	photoresponse (phototransistor / performance / air stability)	정의	photocurrent, on-off ratio or responsivity value after specific keeping days in air condition
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S671	temperature (phototransistor / performance / temperature stability)	정의	the specific value for high temperature condition
		동의어	
		타입	number
		단위	K
		예시	330, 350
S672	operating time (phototransistor / performance / temperature stability)	정의	the specific days (or time) after keeping in the specific temperature condition
		동의어	
		타입	number
		단위	hour; day
		예시	5, 8, 101
S673	photoresponse (phototransistor / performance / temperature stability)	정의	photocurrent, on-off ratio or responsivity value after specific keeping days in high temperature condition
		동의어	
		타입	number
		단위	none
		예시	1.28e-3; 2.3e-3
S674	designated illumination area (photovoltaic system / configuration / module)	정의	light illuminated area of the module in cm^{2}, including both active and interconnection areas
		동의어	
		타입	number
		단위	cm^{2}
		예시	80
S675	color rendering index (photovoltaic system / performance / transparency)	정의	a measurement of how natural colors render under an artificial white light source when compared with sunlight. The index is measured from 0-100, with a perfect 100 indicating that colors of objects under the light source appear the same as they would under natural sunlight.
		동의어	CRI
		타입	number
		단위	none
		예시	90; 76
S676	hysteresis index (photovoltaic system / performance / JV property)	정의	integral measure reflecting the difference between the forward and reverse scans defined as Solar Energy 173, 976 (2018). Also refer to http://solid.fizica.unibuc.ro/~nemnes/perovskite_solar_cells/hysteresis_index/
		동의어	J-V hysteresis
		타입	number
		단위	none
		예시	0.2
S677	external quantum efficiency (photovoltaic system / performance / QE)	정의	the integrated current density based on incident photon from the EQE measurement
		동의어	EQE; incident photon to current efficiency; ipce
		타입	dictionary of {wavelength : value (nm), intensity : value (mW cm^{-2}), quantum efficiency : {value : quantum efficiency value (%), uncertainty : uncertainty value (%)}}
		단위	%
		예시	{wavelength : 450, intensity : 100, quantum efficiency : {value : 95, uncertainty : 0.01}}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S678	internal quantum efficiency (photovoltaic system / performance / QE)	정의	the integrated current density based on absorbed photon from the EQE measurement
		동의어	IQE; absorbed photon to current efficiency
		타입	dictionary of {wavelength : value (nm), quantum efficiency : {value : quantum efficiency value (%), uncertainty : uncertainty value (%)}}
		단위	%
		예시	{wavelength : 450, quantum efficiency : {value : 95, uncertainty : 0.01}}
S679	selective layer (separation membrane / configuration)	정의	materials ID (variable name) of selective layer for controlling selective permeation of specific materials, enabling the separation function
		동의어	separation layer material; active layer material
		타입	string
		단위	
		예시	material_1
S680	thickness of selective layer (separation membrane / configuration)	정의	thickness of the selective layer material in the separation membrane system
		동의어	
		타입	number
		단위	nm
		예시	10
S681	coating layer (separation membrane / configuration)	정의	materials ID (variable name) of coating layer to improve the performance and durabillty of the separation membrane system
		동의어	
		타입	string
		단위	
		예시	material_3; polyvinyl alcohol; polyacrylic acid
S682	thickness of coating layer (separation membrane / configuration)	정의	thickness of the coating layer in the separation membrane system
		동의어	
		타입	number
		단위	nm
		예시	1
S683	membrane support (separation membrane / configuration)	정의	materials ID (variable name) of support material to mechanically reinforce the selective layer material of the separation membrane system
		동의어	
		타입	string
		단위	
		예시	material_2; porous polysulfone; porous polyethersulfone
S684	thickness of membrane support (separation membrane / configuration)	정의	thickness of the support material in the separation membrane system
		동의어	
		타입	number
		단위	{\mu m}
		예시	100
S685	permeance (separation membrane / performance)	정의	ability of a species to penetrate and permeate a membrane; (permeance) [L m^{-2} h^{-1} bar^{-1}] = (permeate volume) [L] / ((membrane area) [m^2] x (time) [h] x (applied pressure) [bar])
		동의어	
		타입	number
		단위	L m^{-2} h^{-1} bar^{-1}
		예시	10
S686	rejection (separation membrane / performance)	정의	ability of a membrane to reject a specific species; (rejection) [%] = (1 - (species concentration of the permeate) [%] / (species concentration of the feed) [%]) x 100
		동의어	
		타입	number
		단위	%
		예시	90
S687	selectivity (separation membrane / performance)	정의	ratio of the permeance of the specific species to that of the other
		동의어	permselectivity
		타입	number
		단위	none
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S699	thickness (photovoltaic system / configuration / front contact)	정의	thickness of front contact
		동의어	
		타입	number
		단위	nm
		예시	100
S700	category	정의	category of the system classified in NCMRD schema
		동의어	
		타입	string
		단위	
		예시	1. alkali ion battery 2. catalyst 3. gas sensor 4. memcapacitive 5. memristive 6. multilayer 7. organic thin film transistor 8. photodiode 9. photoconductor 10. phototransistor 11. photovoltaic 12. piezoelectric 13. porous materials 14. separation membrane 15. ferroelectric
S701	maximum cycling number (alkali-ion battery / performance)	정의	maximum cycling number
		동의어	
		타입	number
		단위	none
		예시	500
S702	maximum cycling time (alkali-ion battery / performance)	정의	maximum cycling time
		동의어	
		타입	number
		단위	h
		예시	500
S703	C-rate (alkali-ion battery / performance / cycling test result)	정의	charge or discharge rate of a cell expressed as a multiple of its nominal capacity (1 C corresponds to full charge or discharge in one hour)
		동의어	
		타입	string
		단위	
		예시	1C; 0.5C
S704	signal-to-noise ratio (gas sensor / performance)	정의	ratio of the level of a desired signal to the level of background noise
		동의어	S/N ratio
		타입	number
		단위	none
		예시	
S705	absolute humidity (gas sensor / performance / humidity repeatability)	정의	total amount of water vapor contained in a given volume of air. It measures the actual mass of water vapor present in the air, regardless of temperature
		동의어	
		타입	number
		단위	g m^{-3}
		예시	
S706	number of repeating units (multilayer system / configuration)	정의	number of repeating unit of multilayer
		동의어	
		타입	number
		단위	none
		예시	
S707	orientation (multilayer system / configuration / repeating unit)	정의	orientation of the layer in the case of single crystal layer and epitaxial film
		동의어	crystallographic orientation
		타입	string
		단위	
		예시	(111); (101)
S708	angle (multilayer system / configuration / repeating unit / offcut)	정의	the angular deviation of a substrate from a reference crystallographic plane
		동의어	
		타입	number
		단위	degree
		예시	6

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
S709	direction (multilayer system / configuration / repeating unit / offcut)	정의	direction in a plane vector form (e.g [hkl]) toward which the substrate offset is imposed
		동의어	crystallographic orientation
		타입	string
		단위	
		예시	[110]; [-100]
S710	pixel shape (photodiode / configuration / device size)	정의	shape of pixel
		동의어	
		타입	string
		단위	
		예시	1. circle 2. rectangle
S711	microjunction shape (photodiode / configuration / device size)	정의	shape of microjunction
		동의어	
		타입	string
		단위	
		예시	1. circle 2. rectangle
S712	area (photoconductor / configuration / active layer)	정의	active area in photoconductor
		동의어	
		타입	number
		단위	mm^{2}
		예시	
S713	area (phototransistor / configuration / active layer)	정의	active area in phototransistor
		동의어	
		타입	number
		단위	mm^{2}
		예시	
S714	load type (piezoelectric system / performance / applied load)	정의	type of the applied load with corresponding data unit
		동의어	
		타입	string
		단위	
		예시	1. bending 2. tapping 3. vibration
S718	mechanical strength (separation membrane / performance)	정의	the ability of a membrane to withstand stress and pressure without failure
		동의어	
		타입	number
		단위	MPa
		예시	
S719	current density (areal) (alkali-ion battery / performance / cycling test result)	정의	electric current per unit electrode area, representing the areal current load applied to the electrode
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	8
S720	current density (mass) (alkali-ion battery / performance / cycling test result)	정의	electric current per unit mass of active material, indicating the current normalized to the active material weight
		동의어	
		타입	number
		단위	mA g^{-1}
		예시	8
S721	tortuosity (porous materials / configuration)	정의	the ratio of the body dimension in a given direction to the length of the path traversed by the fluid during the transport process; a critical parameter to predict transport properties of porous media
		동의어	
		타입	number
		단위	none
		예시	2.77

4 소재 분석 공통어휘 세부내용

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A1	instrument (creep test)	정의	instrument that was used during measurement of the property
		동의어	
		타입	string
		단위	
		예시	INSTRON 5989
A2	temperature (creep test)	정의	temperature at which a given property is measured
		동의어	
		타입	number
		단위	K
		예시	800
A3	load (creep test)	정의	applied load during measurement of the property
		동의어	
		타입	number
		단위	MPa
		예시	200
A4	code (DFT calculation)	정의	DFT code used in the calculation with version information
		동의어	software
		타입	string
		단위	
		예시	VASP 5.4.4; Siesta 2.3b; Quantum Espresso; Qchem5.1; Gaussian09
A5	calculation mode (DFT calculation)	정의	calculation mode
		동의어	
		타입	string
		단위	
		예시	structural optimization; electronic optimization; phonon; density functional perturbation theory (DFPT); Ab Initio Molecular dynamics (AIMD)
A6	basis type (DFT calculation / basis)	정의	type of basis
		동의어	
		타입	string
		단위	
		예시	1. plane-wave 2. atomic orbital (e.g. Slater type orbital)
A7	basis set (DFT calculation / basis)	정의	basis set for orbital basis
		동의어	
		타입	string
		단위	
		예시	6-31G; cc-pVDZ
A8	charge (DFT calculation)	정의	total charge of the system
		동의어	
		타입	number
		단위	e
		예시	+1, 0, -1
A9	energy cutoff (DFT calculation)	정의	kinetic energy cutoff
		동의어	
		타입	number
		단위	eV
		예시	500

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A10	optimizer (DFT calculation)	정의	type of optimizer for structural optimization
		동의어	
		타입	string
		단위	
		예시	BFGS; Conjugate gradient
A11	exchange-correlation functional (DFT calculation)	정의	type of the exchange-correlation functional
		동의어	
		타입	string
		단위	
		예시	PBE; vdW-DF
A12	method (DFT calculation / solvent model)	정의	method of the solvent model
		동의어	
		타입	string
		단위	
		예시	Kirkwood-Onsager; PCM; IPCM; 3D-RISM
A13	solvent dielectric constant (DFT calculation / solvent model / parameters)	정의	dielectric constant of the solvent continuum
		동의어	
		타입	number
		단위	none
		예시	30
A14	grid value (DFT calculation / k-point)	정의	k-point grids for the conventional DFT calculations
		동의어	
		타입	string
		단위	
		예시	3x3x3
A15	high-symmetry points (DFT calculation / k-point / k-point path for band structure)	정의	high symmetry point along the k-point path
		동의어	
		타입	string
		단위	
		예시	G-X-W
A16	k-point coordinates (DFT calculation / k-point / k-point path for band structure)	정의	k-point coordinates corresponding to high-symmetry points for the band structure calculation
		동의어	
		타입	string
		단위	
		예시	[[0,0,0],[0.5,0.5,0],[0.5,0.75,0.25]]
A17	number of k-points (DFT calculation / k-point / k-point path for band structure)	정의	number of k-points between two high-symmetry points for the band structure calculation
		동의어	
		타입	number
		단위	none
		예시	20
A18	force (DFT calculation / convergence criteria)	정의	force criterion for structural optimization
		동의어	
		타입	number
		단위	eV \Angstrom^{-1}
		예시	0.01
A19	energy (DFT calculation / convergence criteria)	정의	energy criterion for structural optimization
		동의어	
		타입	number
		단위	eV
		예시	0.00001

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A20	scf (DFT calculation / convergence criteria)	정의	energy criterion for electronic optimization
		동의어	
		타입	number
		단위	eV
		예시	0.00001
A21	symmetry (DFT calculation)	정의	use of crystal symmetry in calculations
		동의어	
		타입	string
		단위	
		예시	1. on 2. off
A22	magnetic ordering (DFT calculation / magnetic structure)	정의	magnetic order configuration
		동의어	
		타입	string
		단위	
		예시	ferro; G-type antiferro; ferri; non-collinear
A23	multiplicity (DFT calculation / magnetic structure)	정의	spin multiplicity
		동의어	
		타입	string
		단위	
		예시	1. singlet 2. doublet 3. triplet
A24	potential (DFT calculation)	정의	type of pseudo potential for each element
		동의어	
		타입	string
		단위	
		예시	si_pbe_v1.uspp.F.UPF; Si_sv_GW; Si_ONCV_PBE-1.1.UPF
A25	scheme (DFT calculation / LDA+U)	정의	scheme of LDA+U method
		동의어	
		타입	string
		단위	
		예시	simplified version; rotationally invariant scheme
A26	atom (DFT calculation / LDA+U)	정의	atom to which on-site interactions are applied
		동의어	
		타입	string
		단위	
		예시	Fe; Ni; Co
A27	orbital (DFT calculation / LDA+U)	정의	orbital to which on-site interactions are applied
		동의어	
		타입	string
		단위	
		예시	d; f
A28	U value (DFT calculation / LDA+U)	정의	strength of on-site Coulomb interaction
		동의어	
		타입	number
		단위	eV
		예시	5
A29	J value (DFT calculation / LDA+U)	정의	strength of on-site exchange interaction
		동의어	
		타입	number
		단위	eV
		예시	0.7

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A30	method (DFT calculation / partial occupation)	정의	method of partial occupation
		동의어	
		타입	string
		단위	
		예시	Gaussian smearing; tetrahedra
A31	smearing width (DFT calculation / partial occupation)	정의	width of smearing
		동의어	
		타입	number
		단위	eV
		예시	0.01
A32	method (DFT calculation / density of state)	정의	method for the Brillouin zone summation
		동의어	
		타입	string
		단위	
		예시	smearing; tetrahedra
A33	smearing width (DFT calculation / density of state)	정의	width of smearing
		동의어	
		타입	number
		단위	eV
		예시	0.01
A34	energy range (DFT calculation / density of state)	정의	energy range for DOS calculation
		동의어	
		타입	string
		단위	
		예시	-10 eV:10 eV
A35	number of energy point (DFT calculation / density of state)	정의	number of energy points in given energy range
		동의어	
		타입	number
		단위	none
		예시	1000
A36	method (DFT calculation / phonon calculations)	정의	method for the phonon calculation
		동의어	
		타입	string
		단위	
		예시	Finite displacement; DFPT
A37	cell size (DFT calculation / phonon calculations)	정의	cell size for the finite displacement method
		동의어	
		타입	string
		단위	
		예시	2x2x2
A38	displacement size (DFT calculation / phonon calculations)	정의	atomic displacement size for the finite displacement method
		동의어	
		타입	number
		단위	\ANGSTROM
		예시	0.015
A39	number of displacement point (DFT calculation / phonon calculations)	정의	number of atomic displacement points for the finite displacement method
		동의어	
		타입	number
		단위	none
		예시	5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A40	q-points (DFT calculation / phonon calculations)	정의	q-point grids for the DFPT method
		동의어	
		타입	string
		단위	
		예시	2x2x2
A41	optimizer (DFT calculation / nudged elastic band)	정의	type of optimizer
		동의어	
		타입	string
		단위	
		예시	steepest descent; quasi-Newton Broyden's second method; Conjugate Gradient; Quick-Min
A42	number of images (DFT calculation / nudged elastic band)	정의	number of the images (intermediate structures) for the NEB calculation
		동의어	
		타입	number
		단위	none
		예시	5
A43	climbing image scheme (DFT calculation / nudged elastic band)	정의	use of climbing image scheme
		동의어	
		타입	string
		단위	
		예시	1. on 2. off
A44	force criterion (DFT calculation / nudged elastic band)	정의	force criterion for the NEB calculation
		동의어	
		타입	number
		단위	eV \ANGSTROM^{-1}
		예시	0.01
A52	instrument (electrochemical activity)	정의	electrochemical cell for activity measurement
		동의어	
		타입	string
		단위	
		예시	rotating rink disk electrode
A53	temperature (electrochemical activity)	정의	measurement temperature
		동의어	
		타입	number
		단위	K
		예시	298
A57	data file (electrochemical activity / raw data)	정의	URI of the raw or image data file obtained
		동의어	
		타입	string
		단위	
		예시	raw.tif
A58	code (MD simulation / simulation condition)	정의	MD code used in the simulation with version information
		동의어	
		타입	string
		단위	
		예시	VASP; LAMMPS 2.3; AMBER; GROMACS; GULP; Materials Studio
A59	name (MD simulation / simulation condition / force field)	정의	conventional force field name used in the simulation
		동의어	
		타입	string
		단위	
		예시	ab initio; LJ-12; CHARMM; ReaxFF; NEMO; Tersoff; Brenner; MLIP

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A60	source (MD simulation / simulation condition / force field)	정의	source of reference of the used force field parameters
		동의어	
		타입	string
		단위	
		예시	J. Chem. Phys. 34, 2021 (2021); DOI
A61	parameter (MD simulation / simulation condition / force field)	정의	parameter file of the potential used in the simulation
		동의어	
		타입	string
		단위	
		예시	eam-Fe-H.*; UFF.*; ReaxFF-Fe-H.*
A62	cutoff radius (MD simulation / simulation condition)	정의	cutoff radius for truncating the interatomic force
		동의어	
		타입	number
		단위	\ANGSTROM
		예시	2.5
A63	time step (MD simulation / simulation condition)	정의	time step for integrating Newton's equation of motion
		동의어	
		타입	number
		단위	ps
		예시	0.0005
A64	integration algorithm (MD simulation / simulation condition)	정의	time integration algorithm
		동의어	
		타입	string
		단위	
		예시	Velvet; Predictor-corrector;
A65	pressure (MD simulation / initialization)	정의	pressure during simulation
		동의어	
		타입	number
		단위	bar
		예시	1
A66	temperature (MD simulation / initialization)	정의	initialization temperature to relax the model system
		동의어	
		타입	number
		단위	K
		예시	1273
A67	time (MD simulation / initialization)	정의	initialization time to relax the model system
		동의어	
		타입	number
		단위	ps
		예시	1000
A68	scheme (MD simulation / initialization / ensemble)	정의	used ensemble scheme for simulation
		동의어	
		타입	string
		단위	
		예시	NVT; NPT; NVE; NPH
A69	pressure (MD simulation / simulation condition)	정의	pressure during simulation
		동의어	
		타입	number
		단위	bar
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A70	temperature (MD simulation / simulation condition)	정의	temperature during simulation
		동의어	
		타입	number
		단위	K
		예시	1273
A71	time (MD simulation / simulation condition)	정의	total simulation time
		동의어	
		타입	number
		단위	ns (nano second)
		예시	10
A72	ensemble (MD simulation / simulation condition)	정의	used ensemble for simulation
		동의어	
		타입	string
		단위	
		예시	NVT; NPT; NVE; NPH
A73	direction (MD simulation / control / dimensional)	정의	direction of dimension control in cartesian coordination [xyz]
		동의어	
		타입	string
		단위	
		예시	[001];[010];
A74	strain (MD simulation / control / dimensional)	정의	total amount of dimension change
		동의어	
		타입	number
		단위	none
		예시	0.02
A75	strain rate (MD simulation / control / dimensional)	정의	dimension change rate
		동의어	
		타입	number
		단위	ps^{-1}
		예시	0.2
A80	initial temperature (MD simulation / control / thermal)	정의	initial temperature at which the heat treatment starts
		동의어	
		타입	number
		단위	K
		예시	200
A81	heating rate (MD simulation / control / thermal / cycle)	정의	rate at which temperature is raised
		동의어	
		타입	number
		단위	K ps^{-1}
		예시	100
A82	holding temperature (MD simulation / control / thermal / cycle)	정의	soaking temperature for the heat treatment
		동의어	
		타입	number
		단위	K
		예시	403
A83	holding time (MD simulation / control / thermal / cycle)	정의	holding time for the heat treatment at holding temperature
		동의어	
		타입	number
		단위	ns
		예시	1.5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A84	cooling rate (MD simulation / control / thermal / cycle)	정의	rate at which temperature is lowered
		동의어	
		타입	number
		단위	K ps^{-1}
		예시	10
A85	final temperature (MD simulation / control / thermal)	정의	temperature when the heat treatment finished
		동의어	
		타입	number
		단위	K
		예시	200
A90	manipulated constituent (MD simulation / control / constitutional)	정의	element symbol of the added/removed atom or molecule
		동의어	
		타입	string
		단위	
		예시	Ar; Cr; C; CH4; O2
A91	energy (MD simulation / control / constitutional)	정의	energy of the added/removed atom or molecule
		동의어	
		타입	number
		단위	eV
		예시	50
A92	direction (MD simulation / control / constitutional)	정의	direction of the added/removed atom or molecule
		동의어	
		타입	string
		단위	
		예시	[001]
A93	interval (MD simulation / control / constitutional)	정의	interval between the sequential addition or removal of atoms or molecules
		동의어	
		타입	number
		단위	ps
		예시	0.1
A94	total number (MD simulation / control / constitutional)	정의	total number of added/removed atoms or molecules
		동의어	
		타입	number
		단위	none
		예시	5000
A95	trajectory (MD simulation / output)	정의	URI of trajectory file of the simulation; file format is defined by the extension of the file name
		동의어	
		타입	string
		단위	
		예시	output.xxx
A96	analysis (MD simulation)	정의	post analysis method
		동의어	
		타입	string
		단위	
		예시	RDF; bond angle distribution; potential energy surface analysis;
A97	instrument (fatigue test)	정의	instrument that was used during measurement of the property
		동의어	
		타입	string
		단위	
		예시	INSTRON 5989

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A98	temperature (fatigue test)	정의	temperature at which a given property is measured
		동의어	
		타입	number
		단위	K
		예시	800
A101	amplitude (fatigue test / stress range)	정의	one half of the stress range
		동의어	
		타입	number
		단위	MPa
		예시	1000
A102	elastic strain amplitude (fatigue test)	정의	stress amplitude divided by Young's Modulus
		동의어	
		타입	number
		단위	none
		예시	0.001
A103	amplitude (fatigue test / plastic strain range)	정의	half-width of the stress-strain hysteresis loops at zero stress
		동의어	
		타입	number
		단위	none
		예시	0.001
A104	mean stress (fatigue test)	정의	algebraic average of the maximum and minimum stresses in one cycle
		동의어	
		타입	number
		단위	MPa
		예시	1000
A105	stress ratio (fatigue test)	정의	algebraic average of two specified stress values in a stress cycle
		동의어	
		타입	number
		단위	none
		예시	1
A106	adsorptive (gas adsorption and desorption isotherm)	정의	gas molecule to be adsorbed on surface
		동의어	adsorption gas; sorptive
		타입	string
		단위	
		예시	CO_{2}
A107	temperature (gas adsorption and desorption isotherm)	정의	constant temperature in each isotherm measurement
		동의어	
		타입	number
		단위	K
		예시	298
A108	minimum pressure (gas adsorption and desorption isotherm)	정의	minimum pressure in an isotherm
		동의어	
		타입	number
		단위	bar
		예시	0
A109	maximum pressure (gas adsorption and desorption isotherm)	정의	maximum pressure in an isotherm
		동의어	
		타입	number
		단위	bar
		예시	200

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A110	instrument (gas adsorption and desorption isotherm)	정의	instrument model for isotherm measurement
		동의어	
		타입	string
		단위	
		예시	iSorb HP1 | HP2 – High Pressure Gas Sorption Analyzer
A111	image (gas adsorption and desorption isotherm)	정의	raw or image data obtained
		동의어	
		타입	file ID
		단위	
		예시	image_isotherm_co2.tif
A112	analysis method (gas adsorption and desorption isotherm)	정의	analysis method for isotherm measurement
		동의어	
		타입	string
		단위	
		예시	static; continuous; dynamic
A113	instrument (gas chromatography)	정의	gas chromatography model for faradaic efficiency
		동의어	
		타입	string
		단위	
		예시	Nexis GC-2030
A114	measured gas (gas chromatography)	정의	measured gas molecule
		동의어	
		타입	string
		단위	
		예시	CO
A115	atmosphere (gas chromatography)	정의	atmospheric condition
		동의어	
		타입	string
		단위	
		예시	Ar
A116	flow rate (gas chromatography)	정의	flow rate of gas to be measured; in standard cubic centimeter per minute
		동의어	
		타입	number
		단위	cm^{3} min^{-1}
		예시	5
A117	temperature (gas chromatography)	정의	measurement temperature
		동의어	
		타입	number
		단위	K
		예시	298
A118	voltage (gas chromatography)	정의	applied voltage
		동의어	
		타입	number
		단위	V
		예시	0.8
A119	saturation time (gas chromatography)	정의	time to reach steady-state
		동의어	
		타입	number
		단위	h
		예시	0.6

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A120	image (gas chromatography)	정의	raw or image data obtained
		동의어	
		타입	string
		단위	
		예시	image_chromato.tif
A121	analysis method (gas chromatography)	정의	analysis method for gas chromatography
		동의어	
		타입	string
		단위	
		예시	gas chromatography, gas chromatography-mass spectrometry
A122	temperature (hardness test)	정의	temperature for hardness test
		동의어	
		타입	number
		단위	K
		예시	350
A123	test type (hardness test)	정의	method to measure hardness of materials
		동의어	
		타입	string
		단위	
		예시	1. Vickers 2. Rockwell 3. Brinell 4. Knoop 5. etc.
A124	temperature (impact test)	정의	temperature for impact test
		동의어	
		타입	number
		단위	K
		예시	350
A125	test type (impact test)	정의	type of the measurement of the toughness of materials
		동의어	
		타입	string
		단위	
		예시	1. Charpy impact V-notch test 2. Izod impact test
A126	instrument (infrared spectroscopy)	정의	instrument model for IR measurement
		동의어	
		타입	string
		단위	
		예시	Nicolet iS10 FTIR | Waters
A127	atmosphere (infrared spectroscopy)	정의	atmosphere gas type
		동의어	
		타입	string
		단위	
		예시	N2
A128	temperature (infrared spectroscopy)	정의	temperature for IR measurement
		동의어	
		타입	number
		단위	K
		예시	300
A129	image (infrared spectroscopy)	정의	raw or image data obtained from IR measurement
		동의어	
		타입	string
		단위	
		예시	image_ir.tif

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A130	analysis method (infrared spectroscopy)	정의	analysis method for IR measurement
		동의어	
		타입	string
		단위	
		예시	transmittance; attenuated total reflectance
A131	instrument (memristive activity)	정의	semiconductor parameter analyzer model or pulsed generator model
		동의어	
		타입	string
		단위	
		예시	4200A-SCS Parameter Analyzer
A132	temperature (memristive activity)	정의	measurement temperature
		동의어	
		타입	number
		단위	K
		예시	300
A133	maximum voltage (memristive activity)	정의	applied maximum voltage for current voltage sweep method
		동의어	
		타입	number
		단위	V
		예시	0.9
A134	minimum voltage (memristive activity)	정의	applied minimum voltage for current voltage sweep method
		동의어	
		타입	number
		단위	V
		예시	-0.9
A135	current (memristive activity)	정의	applied current for current voltage sweep method
		동의어	
		타입	number
		단위	A
		예시	1.00E-10
A136	compliance current (memristive activity)	정의	applied compliance current for current voltage sweep method
		동의어	
		타입	number
		단위	A
		예시	1.00E-10
A137	pulse (memristive activity)	정의	applied pulse for current voltage sweep method
		동의어	
		타입	number
		단위	ns
		예시	60
A138	instrument (nuclear magnetic resonance)	정의	instrument model for NMR measurement
		동의어	
		타입	string
		단위	
		예시	AdvancedCore | Bruker
A139	temperature (nuclear magnetic resonance)	정의	temperature for NMR measurement
		동의어	
		타입	number
		단위	K
		예시	300

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A140	image (nuclear magnetic resonance)	정의	raw or image data obtained from NMR measurement
		동의어	
		타입	string
		단위	
		예시	image_nmr.tif
A141	analysis method (nuclear magnetic resonance)	정의	analysis method for NMR measurement
		동의어	
		타입	string
		단위	
		예시	liquid; solid
A142	instrument (optical microscopy)	정의	microscope model
		동의어	
		타입	string
		단위	
		예시	Zeiss 380 (2019년 모델)
A143	condition (optical microscopy)	정의	observation condition
		동의어	
		타입	string
		단위	
		예시	normal reflection
A144	image (optical microscopy)	정의	raw or image data obtained
		동의어	
		타입	string
		단위	
		예시	image_3870.tif
A145	analysis method (optical microscopy)	정의	note on microstructure analysis method
		동의어	
		타입	string
		단위	
		예시	ASTM D422
A146	instrument (TEM)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	TITAN TEM; TALOS TEM; TECNAI TEM; Cryo TEM
A147	mode (TEM)	정의	acquisition mode
		동의어	
		타입	string
		단위	
		예시	TEM BF; TEM DF; STEM HAADF; SAED; EDS; EELS
A148	acceleration voltage (TEM / acquisition condition)	정의	acceleration voltage of electron beam generation
		동의어	
		타입	number
		단위	keV
		예시	300
A149	beam current (TEM / acquisition condition)	정의	beam current
		동의어	
		타입	number
		단위	nA
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A151	magnification (TEM / acquisition condition)	정의	ratio between scan area and display size
		동의어	
		타입	number
		단위	
		예시	5000
A155	instrument (tensile test)	정의	instrument that was used during measurement of the property
		동의어	
		타입	string
		단위	
		예시	INSTRON 5989
A156	specimen direction (tensile test)	정의	direction of specimen with respect to the designated reference direction (i.e. rolling direction)
		동의어	
		타입	number
		단위	degree
		예시	45
A157	temperature (tensile test)	정의	temperature at which a given property is measured
		동의어	
		타입	number
		단위	K
		예시	300
A159	strain rate (tensile test / tensile test speed)	정의	increase of strain, measured with an extensometer, in extensometer gauge length
		동의어	
		타입	number
		단위	s^{-1}
		예시	0.1
A160	instrument (thermal activity)	정의	continuous flow reaction for thermal activity measurement
		동의어	
		타입	string
		단위	
		예시	5400 Tubular Reactor
A161	flow rate (thermal activity)	정의	flow rate of reaction gas; in standard cubic centimeter per minute
		동의어	
		타입	number
		단위	cm^{3} min^{-1}
		예시	5
A162	temperature (thermal activity)	정의	measurement temperature
		동의어	
		타입	number
		단위	K
		예시	298
A163	heating time (thermal activity)	정의	heating time before measurement
		동의어	
		타입	number
		단위	h
		예시	36
A164	raw data (thermal activity)	정의	URI of the raw or image data obtained
		동의어	
		타입	string
		단위	
		예시	image_thermal.tif

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A165	instrument (TGA)	정의	instrument model for thermogravimetric measurement
		동의어	
		타입	string
		단위	
		예시	TGA 5500
A166	starting temperature (TGA / acquisition condition)	정의	starting temperature in thermogravimetric measurement
		동의어	
		타입	number
		단위	K
		예시	298
A167	final temperature (TGA / acquisition condition)	정의	final temperature in thermogravimetric measurement
		동의어	
		타입	number
		단위	K
		예시	450
A168	ramp rate (TGA / acquisition condition)	정의	rate of temperature change in thermogravimetric measurement
		동의어	
		타입	number
		단위	K min^{-1}
		예시	20
A170	gas type (TGA / acquisition condition / flue gas)	정의	flue gas type in thermogravimetric measurement
		동의어	
		타입	string
		단위	
		예시	N2
A171	flue rate (TGA / acquisition condition / flue gas)	정의	flue rate in thermogravimetric measurement; in standard cubic centimeter per minute
		동의어	
		타입	number
		단위	cm^{3} min^{-1}
		예시	70
A172	image (TGA)	정의	raw or image data obtained from thermogravimetric measurement
		동의어	
		타입	string
		단위	
		예시	image_tga.tif
A173	analysis method (TGA)	정의	analysis method for thermogravimetric measurement
		동의어	
		타입	string
		단위	
		예시	adsorption; desorption
A174	instrument (XRD)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	PANalycal Empyrean; PANalycal X'pert Pro; Bruker D8 Advance; Bruker D8 Discover; Rigaku ATX-G; Rigaku D/MAX2500
A175	x-ray wavelength (XRD / acquisition condition)	정의	wavelength of characteristic X-ray used in measurement
		동의어	
		타입	number
		단위	\ANGSTROM
		예시	1.5406

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A176	acceleration voltage (XRD / acquisition condition)	정의	acceleration voltage of X-ray generation
		동의어	
		타입	number
		단위	kV
		예시	40
A177	current (XRD / acquisition condition)	정의	filament current
		동의어	
		타입	number
		단위	mA
		예시	250
A178	scan axis (XRD / acquisition condition)	정의	moved axes when data collected
		동의어	
		타입	string
		단위	
		예시	Theta-2theta scan; 2theta scan
A179	scanning step (XRD / acquisition condition)	정의	intervals in data collection
		동의어	
		타입	number
		단위	degree
		예시	0.02154; 0.01
A180	incident angle (XRD / acquisition condition)	정의	Incident beam angle when the beam incidence was fixed
		동의어	
		타입	number
		단위	degree
		예시	0.5
A181	offset angle (XRD / acquisition condition)	정의	angle of offset from symmetric theta-2theta configuration
		동의어	
		타입	number
		단위	degree
		예시	1
A185	scan rate (XRD / acquisition condition)	정의	how fast X-ray diffraction data collection has been done (the data can be presented with two units; deg/min or sec/step)
		동의어	
		타입	number
		단위	degree min^{-1}
		예시	2; 0.6
A186	sample rotation (XRD / acquisition condition)	정의	rotating speed when the sample is rotated
		동의어	rotation
		타입	number
		단위	revolution min^{-1}
		예시	15; 30; 60; 120
A187	raw data (XRD)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	XRD_profile.tiff; XRD_spectrum.csv
A188	instrument (AES)	정의	instrument that was used during measurement of the property
		동의어	
		타입	string
		단위	
		예시	PHI-710

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A189	electron source (AES / acquisition condition)	정의	Electron source used in measurement
		동의어	
		타입	string
		단위	
		예시	Denka TFE
A190	electron beam energy (AES / acquisition condition)	정의	energy of primary electron beam
		동의어	
		타입	number
		단위	KeV
		예시	1;3;5;10;20
A191	electron beam current (AES / acquisition condition)	정의	electron beam current on the sample
		동의어	
		타입	number
		단위	pA
		예시	0.5; 1; 5; 10
A192	ion beam energy (AES / acquisition condition / sputtering condition)	정의	accelerating sputtering ion energy
		동의어	
		타입	number
		단위	kV
		예시	0.5;1;2;3;4
A193	ion type (AES / acquisition condition / sputtering condition)	정의	ion type for sputtering
		동의어	
		타입	string
		단위	
		예시	Ar, Bi, Cs
A194	ion beam current (AES / acquisition condition / sputtering condition)	정의	ion dose for sputtering
		동의어	
		타입	number
		단위	mA
		예시	10; 0.5; 1
A195	raster size (AES / acquisition condition / sputtering condition)	정의	sputter area for measurement
		동의어	
		타입	number
		단위	mm
		예시	1;1.5;2
A196	sputter time (AES / acquisition condition / sputtering condition)	정의	sputter time for measurement
		동의어	
		타입	number
		단위	min
		예시	5;20;50
A197	neutralizer beam energy (AES / acquisition condition / neutralizing condition)	정의	positive ions (Ar) energy for neutralizing
		동의어	
		타입	number
		단위	V
		예시	10
A198	neutralizer beam current (AES / acquisition condition / neutralizing condition)	정의	ion current for neutralizing
		동의어	
		타입	number
		단위	mA
		예시	0.1, 0.5, 1, 5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A199	raw data (AES)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.tiff; data.xls
A200	name (AFM / instrument)	정의	brand and model name of AFM equipment
		동의어	
		타입	string
		단위	
		예시	Park Systems XE-100; Park Systems XE-7; Park Systems NX-10; Bruker Dimension Edge
A201	mode (AFM / acquisition condition)	정의	scanning mode used in measurement
		동의어	
		타입	string
		단위	
		예시	Contact; Non-contact; Tapping; EFM; KPFM; Conductive AFM; LFM; MFM; SSRM
A202	cantilever type (AFM / acquisition condition)	정의	type of cantilever used in measurement
		동의어	
		타입	string
		단위	
		예시	NSC36; NCHR; NSC14/Cr-Au; CONTSCPt; LFMR; MFMR
A203	scan rate (AFM / acquisition condition)	정의	the number of lines scanned per second
		동의어	
		타입	number
		단위	Hz
		예시	1
A204	scan size (AFM / acquisition condition)	정의	the size of scanning area
		동의어	
		타입	number
		단위	{\mu m}^2
		예시	10
A205	Z servo gain (AFM / acquisition condition)	정의	the sensitivity of the Z scanner feedback loop
		동의어	
		타입	number
		단위	
		예시	2
A206	raw data (AFM)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.tiff; Optic image.bmp
A207	upper set voltage (amperometry / acquisition condition)	정의	upper set voltage
		동의어	upper cutoff voltage
		타입	number
		단위	V
		예시	50
A208	lower set voltage (amperometry / acquisition condition)	정의	lower set voltage
		동의어	lower cutoff voltage
		타입	number
		단위	V
		예시	50

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A211	raw data (amperometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	spc; csv; tif; bmp; jpg; text
A212	temperature (amperometry / acquisition condition)	정의	temperature during battery operation
		동의어	
		타입	number
		단위	K
		예시	298
A213	instrument (atom probe tomography)	정의	instrument that used during measurement
		동의어	
		타입	string
		단위	
		예시	LEAP 5000 XR
A214	base temperature (atom probe tomography / acquisition condition)	정의	cryogenic base temperature used during measurement
		동의어	
		타입	number
		단위	K
		예시	100
A215	voltage pulse fraction (atom probe tomography / acquisition condition)	정의	fractional amplitude of voltage pulses applied relative to DC standing voltage during measurement
		동의어	
		타입	number
		단위	%
		예시	20
A216	laser pulse energy (atom probe tomography / acquisition condition)	정의	energy per pulse of the laser used during measurement
		동의어	
		타입	number
		단위	pJ
		예시	100
A218	laser pulse frequency (atom probe tomography / acquisition condition)	정의	frequency or repetition rate of laser pulses during laser pulse or combined mode measurement
		동의어	
		타입	number
		단위	kHz
		예시	200
A219	raw data (atom probe tomography)	정의	URI of the original measurement data consisting of mass-to-charge ratio spectra and position-sensitive detector coordinates
		동의어	
		타입	string
		단위	
		예시	Mass spectrum (*.epos), Detector hit coordinates (*.pos)
A220	solvent (DFT calculation / solvent model / parameters)	정의	solvent used in the solvent model
		동의어	
		타입	string
		단위	
		예시	Ethanol

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A221	cavity radii (DFT calculation / solvent model / parameters)	정의	the topological model and/or the set of atomic radii used for specifying molecular cavity
		동의어	
		타입	string
		단위	
		예시	Pauling, UAHF, VDW
A222	total energy (DFT calculation)	정의	calculated total energy
		동의어	
		타입	number
		단위	eV
		예시	-1234
A223	detector model (EDS / instrument)	정의	the specific model name of the EDS detector used for EDS data acquisition
		동의어	
		타입	string
		단위	
		예시	Oxford X-Max 80; Bruker XFlash 6; EDAX Elite-T; FEI Super-X
A224	mode (EDS)	정의	EDS data acquisition mode
		동의어	
		타입	string
		단위	
		예시	Spectrum; Map; Line scan; Quantitative analysis
A225	acceleration voltage (EDS / acquisition condition)	정의	acceleration voltage of electron beam generation
		동의어	
		타입	number
		단위	kV
		예시	15
A226	beam current (EDS / acquisition condition)	정의	beam current
		동의어	
		타입	number
		단위	{\mu A}
		예시	15
A227	magnification (EDS / acquisition condition)	정의	ratio between scan area and display size
		동의어	
		타입	number
		단위	
		예시	5000
A228	frame (EDS / acquisition condition)	정의	required number of scan for enough S/N
		동의어	
		타입	number
		단위	none
		예시	16
A229	raw data (EDS)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	spc; csv; tif; bmp; jpg; text
A230	upper set frequency (EIS)	정의	upper value of the frequency used in the measurement
		동의어	
		타입	number
		단위	Hz
		예시	4.5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A231	lower set frequency (EIS)	정의	lower value of the frequency used in the measurement
		동의어	
		타입	number
		단위	Hz
		예시	4.5
A232	temperature (EIS)	정의	temperature of the system during measurement
		동의어	
		타입	number
		단위	K
		예시	298
A233	resistance (EIS)	정의	resistance of EIS cell
		동의어	
		타입	number
		단위	Ohm
		예시	200
A234	diameter (EIS)	정의	diameter of EIS cell
		동의어	
		타입	number
		단위	mm
		예시	13
A235	thickness (EIS)	정의	thickness of EIS cell
		동의어	
		타입	number
		단위	mm
		예시	0.14
A236	raw data (EIS)	정의	URI of the unprocessed raw data directly obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	spc; csv; tif; bmp; jpg; text
A237	reference electrode (electrochemical activity)	정의	reference electrode of electrochemical cell
		동의어	
		타입	string
		단위	
		예시	standard hydrogen electrode
A238	electrolyte composition (electrochemical activity)	정의	chemical composition of electrolyte in electrochemical cell
		동의어	
		타입	dictionary of {value : {electrolyte : concentration, … }, unit : mol}
		단위	mol
		예시	{value : {acetone : 30, salt : 0.5}, unit : mol}
A240	result (MD simulation / output)	정의	URI of the output file from used simulation code
		동의어	
		타입	string
		단위	
		예시	output.xxx
A241	input file (MD simulation)	정의	URI of input script file for MD simulation
		동의어	
		타입	string
		단위	
		예시	difsimul.input

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A242	instrument (EPMA)	정의	instrument that was used during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Jeol JXA-8500F
A243	sample preparation (EPMA / acquisition condition)	정의	sample preparation procedure for EPMA
		동의어	
		타입	string
		단위	
		예시	Polishing; Coating
A244	acceleration voltage (EPMA / acquisition condition)	정의	acceleration voltage of electron beam generation
		동의어	
		타입	number
		단위	kV
		예시	15
A245	current (EPMA / acquisition condition)	정의	probe current
		동의어	
		타입	number
		단위	nA
		예시	10
A246	magnification (EPMA / acquisition condition)	정의	area to be analyzed
		동의어	
		타입	number
		단위	
		예시	5,000
A247	spot size (EPMA / acquisition condition)	정의	beam diameter to be analyzed
		동의어	
		타입	number
		단위	{\mu m}
		예시	1
A248	working distance (EPMA / acquisition condition)	정의	ratio between the specimen and the lower pole piece
		동의어	
		타입	number
		단위	{\mu m}
		예시	11
A249	raw data (EPMA)	정의	URI of the data file obtained after measurement and analysis
		동의어	
		타입	string
		단위	
		예시	EPMA_result.txt; EPMA_image.bmp
A250	beamline (EXAFS / instrument)	정의	name of the beamline where the measurement was performed
		동의어	
		타입	string
		단위	
		예시	1D XRS KIST-PAL
A251	x-ray source (EXAFS / acquisition condition)	정의	X-ray source used in measurement
		동의어	
		타입	string
		단위	
		예시	50keV Bending Magnet

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A252	beam size (EXAFS / acquisition condition)	정의	X-ray beam diameter at sample position
		동의어	
		타입	number
		단위	{\mu m}
		예시	100
A253	scanning step (EXAFS / acquisition condition)	정의	energy interval between data points
		동의어	
		타입	number
		단위	eV
		예시	0.4
A254	data collection mode (EXAFS / acquisition condition)	정의	detection mode for EXAFS data collection
		동의어	
		타입	string
		단위	
		예시	transmission; fluorescence
A255	raw data (EXAFS)	정의	URI of the raw absorption spectrum obtained during measurement
		동의어	
		타입	string
		단위	
		예시	EXAFS_Fe_Kedge.dat
A256	waveform (fatigue test)	정의	shape of the peak-to-peak variation of a controlled mechanical test variable(load, strain, displacement) as a function of time
		동의어	
		타입	string
		단위	
		예시	load_time.csv
A257	frequency (fatigue test)	정의	number of stress(loading) cycle per unit of time
		동의어	
		타입	number
		단위	s^{-1}
		예시	10
A258	instrument (FIB)	정의	instrument that was used during milling and observation
		동의어	
		타입	string
		단위	
		예시	Thermo Nova 600; Thermo Helios 600; Thermo Quanta 3D; Thermo Helios G4; Hitachi NX5000
A259	acceleration voltage (FIB / ion milling condition)	정의	acceleration voltage of Ion gun
		동의어	
		타입	number
		단위	kV
		예시	30
A260	incident angle (FIB / ion milling condition)	정의	the angle of the ion beam incident on the sample
		동의어	
		타입	number
		단위	
		예시	52
A261	acceleration voltage (FIB / acquisition condition)	정의	acceleration voltage of Electron gun
		동의어	
		타입	number
		단위	kV
		예시	5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A262	raw data (FIB)	정의	URI of the image obtained after ion milling
		동의어	
		타입	string
		단위	
		예시	1.tiff
A263	instrument (fsLA-ICP-MS)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	LIBS, LA-ICP-MS
A264	mode (fsLA-ICP-MS)	정의	acquisition conditions for measurement
		동의어	
		타입	string
		단위	
		예시	spot; line scanning
A265	power (fsLA-ICP-MS / acquisition condition)	정의	laser power in measurement
		동의어	
		타입	number
		단위	%
		예시	10; 20; 30; 40; 50; 60; 70; 80; 90; 100
A266	spot size (fsLA-ICP-MS / acquisition condition)	정의	spot size of laser
		동의어	numeric
		타입	
		단위	{\mu m}
		예시	2; 5; 7; 9; 25; 35; 45; 60; 75; 85; 100; 125
A267	repetition (fsLA-ICP-MS / acquisition condition)	정의	repetition of the laser ablation
		동의어	
		타입	number
		단위	Hz
		예시	1; 5; 10; 20; 50; 100; 1000
A268	velocity (fsLA-ICP-MS / acquisition condition)	정의	scan speed used for line scanning
		동의어	
		타입	string
		단위	mm s^{-1}
		예시	0.1; 1
A269	analysis gas (fsLA-ICP-MS / acquisition condition)	정의	type of gas required for analysis
		동의어	
		타입	string
		단위	
		예시	Ar; He
A270	raw data (fsLA-ICP-MS)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	ita; itm
A271	test force (hardness test)	정의	force applied to the test specimen in a direction normal to the surface
		동의어	load
		타입	number
		단위	gf
		예시	50

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A272	dwell time (hardness test)	정의	specified time during which test force is held
		동의어	duration time
		타입	number
		단위	s
		예시	10
A273	instrument (ICP-MS)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	iCAP RQ
A274	mode (ICP-MS)	정의	acquisition conditions for measurement
		동의어	
		타입	string
		단위	
		예시	Quantitative analysis
A275	analysis gas (ICP-MS / acquisition condition)	정의	type of gas required for analysis
		동의어	
		타입	string
		단위	
		예시	Ar; He
A276	raw data (ICP-MS)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	csv; pdf; jpg; txt; rtf
A277	instrument (ICP-OES)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	iCAP 6500 DUO
A278	mode (ICP-OES)	정의	acquisition conditions for measurement
		동의어	
		타입	string
		단위	
		예시	Quantitative analysis
A279	acquisition condition (ICP-OES)	정의	acquisition conditions for measurement
		동의어	
		타입	
		단위	
		예시	
A280	raw data (ICP-OES)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	csv; pdf; jpg; txt; rtf
A281	instrument (nano-indentation)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Bruker TI-950; Bruker TI-900

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A282	tip type (nano-indentation / acquisition condition)	정의	shape of probe used in measurement
		동의어	
		타입	string
		단위	
		예시	Berkovich; Cube corner; Conical; Flat ended; Vickers
A283	test type (nano-indentation / acquisition condition)	정의	type to control the indentation
		동의어	
		타입	string
		단위	
		예시	Load control; Displacement control; Partial unload
A284	time segment (nano-indentation / acquisition condition)	정의	loading, holding, and unloading time
		동의어	
		타입	number
		단위	s
		예시	5-2-5; 5-1-5; 10-10; Any value can be entered as long as the sum of the times is less than 60 seconds.
A285	peak load (nano-indentation / acquisition condition)	정의	maximum load value
		동의어	
		타입	number
		단위	{\mu N}
		예시	30~10000
A286	indent pattern (nano-indentation / acquisition condition)	정의	number of indentations
		동의어	
		타입	number
		단위	
		예시	5 x 5; 10 x 10; etc.
A287	indent spacing (nano-indentation / acquisition condition)	정의	spacing of indentations
		동의어	
		타입	number
		단위	{\mu m}
		예시	1~200
A288	raw data (nano-indentation)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Optic image.bmp; Raw data.txt; Data table.xlsx
A289	beamline (NEXAFS / instrument)	정의	name of the beamline where the measurement was performed
		동의어	
		타입	string
		단위	
		예시	1D XRS KIST-PAL
A290	x-ray source (NEXAFS / acquisition condition)	정의	X-ray source used in measurement
		동의어	
		타입	string
		단위	
		예시	3MeV Bending Magnet
A291	beam size (NEXAFS / acquisition condition)	정의	X-ray beam diameter at the sample position
		동의어	
		타입	number
		단위	{\mu m}
		예시	50

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A292	scanning step (NEXAFS / acquisition condition)	정의	energy interval between each data point
		동의어	
		타입	number
		단위	eV
		예시	0.1
A293	data collection method (NEXAFS / acquisition condition)	정의	detection method for signal collection
		동의어	
		타입	string
		단위	
		예시	total electron yield; total fluorescence yield
A294	raw data (NEXAFS)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	NEXAFS.dat
A295	magnification (optical microscopy)	정의	magnification of image
		동의어	
		타입	number
		단위	
		예시	500; 3K
A296	instrument (Raman spectroscopy)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Renishaw MicroRaman
A297	laser wavelength (Raman spectroscopy / acquisition condition)	정의	excitation wavelength
		동의어	
		타입	number
		단위	nm
		예시	532
A298	magnification (Raman spectroscopy / acquisition condition)	정의	spatial resolution by optical microscope objective lenses
		동의어	
		타입	number
		단위	
		예시	50
A299	raw data (Raman spectroscopy)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	raman_spectrum.jpg
A300	instrument (SEM)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Nova SEM; Inspect F; Inspect F50; Teneo VS; E-SEM; Regulus 8230
A301	acceleration voltage (SEM / acquisition condition)	정의	acceleration voltage of electron beam generation
		동의어	
		타입	number
		단위	kV
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A302	current (spot size) (SEM / acquisition condition)	정의	probe current (electron probe size)
		동의어	
		타입	number
		단위	{\mu A}
		예시	180 (4)
A303	magnification (SEM / acquisition condition)	정의	ratio between scan area and display size
		동의어	
		타입	number
		단위	
		예시	5000
A304	working distance (SEM / acquisition condition)	정의	distance between the specimen and the lower pole piece
		동의어	
		타입	number
		단위	cm
		예시	10
A305	raw data (SEM)	정의	URI of the spectrum data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	SEM.tif; BSE.tif; EDS.csv; EBSD.oim; EBSD.osc
A306	instrument (SIMS)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Magnetic sector SIMS; ToF-SIMS; Quadrupole SIMS
A307	primary ion (SIMS / acquisition condition)	정의	ion source used in measurement
		동의어	
		타입	string
		단위	
		예시	Bi+; Cs+; Ga+; O2+; O-; Ar cluster ions
A308	beam energy (SIMS / acquisition condition)	정의	energy of primary Ion beam
		동의어	
		타입	number
		단위	keV
		예시	0.5; 3; 6; 10
A309	beam current (SIMS / acquisition condition)	정의	current of the primary ion beam
		동의어	
		타입	number
		단위	pA
		예시	1; 5; 10; 20; 50; 100
A310	flood gun (SIMS / acquisition condition / neutralizer)	정의	used when charging occurs on the sample surface for ToF-SIMS
		동의어	
		타입	number
		단위	V
		예시	20V ;-20
A311	E-gun (SIMS / acquisition condition / neutralizer)	정의	used when charging occurs on the sample surface for D-SIMS
		동의어	
		타입	number
		단위	{\mu A}
		예시	-10; -30

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A312	polarity (SIMS / acquisition condition)	정의	polarity of secondary ions
		동의어	
		타입	string
		단위	
		예시	+; -
A313	sample voltage (SIMS / acquisition condition)	정의	applied voltages at sample position for D-SIMS
		동의어	
		타입	number
		단위	eV
		예시	5; 1.5; 2
A314	raster size (SIMS / acquisition condition)	정의	sputter area in measurement
		동의어	
		타입	number
		단위	{\mu m}
		예시	50; 100; 200; 300
A315	sputter time (SIMS / acquisition condition)	정의	sputter time in measurement
		동의어	
		타입	number
		단위	s
		예시	50; 100; 200; 300
A316	raw data (SIMS)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	ita; itm
A317	exposure time (TEM / acquisition condition)	정의	the length of time the camera collects light from sample
		동의어	
		타입	number
		단위	ms
		예시	160
A318	dimension (TEM / raw data)	정의	the length and width of image data
		동의어	
		타입	dictionary of {x : value, y : value}
		단위	none
		예시	{x : 1024, y : 1024}
A319	binning (TEM / raw data)	정의	the size combining the output of adjacent pixels on a CCD camera
		동의어	
		타입	dictionary of {x : value, y : value}
		단위	none
		예시	{x : 2, y : 2}
A320	image (TEM / raw data)	정의	URI of the raw of image data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	TEM.dm4; TEM.dm3; TEM.tif
A321	specimen shape (tensile test)	정의	shape of cross section of a tensile test specimen
		동의어	
		타입	string
		단위	
		예시	1. rectangular 2. tubular 3. round

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A322	gauge length (tensile test)	정의	original length of that portion of the specimen over which strain or change of length is determined
		동의어	
		타입	number
		단위	mm
		예시	50
A323	gas type (TGA / acquisition condition / flow gas)	정의	flow gas type in thermogravimetric measurement
		동의어	
		타입	string
		단위	
		예시	CO_{2}
A324	flow rate (TGA / acquisition condition / flow gas)	정의	flue rate in thermogravimetric measurement; in standard cubic centimeter per minute
		동의어	
		타입	number
		단위	cm^{3} min^{-1}
		예시	70
A325	instrument (USANS)	정의	Bonse-Hart-Agamalian type double crystal diffractometer to measure the very small angle scattering from the scattering vector (q) of 2x10-5 to 0.02 Å-1 using a pair of multi-bounce channel-cut perfect single crystals as a monochromator and an analyzer before and after the sample
		동의어	
		타입	string
		단위	
		예시	
A326	neutron wavelength (USANS / acquisition condition)	정의	wavelength used in ultra-small angle neutron scattering measurement
		동의어	
		타입	number
		단위	\ANGSTROM
		예시	4
A327	beamsize (USANS / acquisition condition)	정의	beam size defined by the Cd/Gd aperture at a sample position
		동의어	
		타입	number
		단위	mm
		예시	16, 76
A328	angular resolution (USANS / acquisition condition)	정의	the lowest limit of scattering angle (i.e., scattering vector). USANS has a high anglular resolution in the scattering plane while poor resolution in the perpendicular direction
		동의어	
		타입	number
		단위	\ANGSTROM^{-1}
		예시	0.00002
A329	scanning start angle (USANS / acquisition condition)	정의	initial angle at each buffer runs
		동의어	
		타입	number
		단위	mm
		예시	-0.0015; 0.00015
A330	scanning step (USANS / acquisition condition)	정의	scanning interval (i.e., interval between two data points)
		동의어	
		타입	number
		단위	mm
		예시	0.0001; 0.0002;0.001;

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A331	scanning end angle (USANS / acquisition condition)	정의	ending angle at each buffer runs
		동의어	
		타입	number
		단위	mm
		예시	0.073; 0.178
A332	motor velocity (USANS / acquisition condition)	정의	motor velocity to move from a one motor position to the next target position
		동의어	
		타입	number
		단위	mm s^{-1}
		예시	0.0001; 0.0005
A333	run time (USANS / acquisition condition)	정의	time required for measuring a one data point
		동의어	
		타입	number
		단위	s
		예시	
A334	raw data (USANS)	정의	URI of the raw data file; neutron count rate (count/sec) vs. travel distance (mm or arcsecond) of motor
		동의어	
		타입	string
		단위	
		예시	
A335	upper voltage (voltammetry / acquisition condition)	정의	data value
		동의어	
		타입	number
		단위	V
		예시	4
A336	lower voltage (voltammetry / acquisition condition)	정의	data value
		동의어	
		타입	number
		단위	V
		예시	-0.5
A337	scan rate (voltammetry / acquisition condition)	정의	the rate of voltage change over time
		동의어	
		타입	number
		단위	mV s^{-1}
		예시	0.05
A338	temperature (voltammetry / acquisition condition)	정의	temperature during battery operation
		동의어	
		타입	number
		단위	K
		예시	298
A339	raw data (voltammetry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	spc; csv; tif; bmp; jpg; text
A342	beamsize (PES / acquisition condition)	정의	X-ray beam size to be analyzed
		동의어	
		타입	number
		단위	{\mu m}
		예시	400

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A343	charge neutralization (PES / acquisition condition)	정의	whether charge compensation was used during measurement
		동의어	
		타입	string
		단위	
		예시	Electron flood gun (2 eV, 10 μA)
A344	take-off angle (PES / acquisition condition)	정의	angle between the sample surface and electron analyzer
		동의어	sample angle
		타입	number
		단위	degree
		예시	45
A345	photon energy (PES / acquisition condition)	정의	energy of the incident photon
		동의어	
		타입	number
		단위	eV
		예시	1486.6
A348	2theta range (XRD / acquisition condition)	정의	data acquistion range in case of theta-2theta and 2theta scan.
		동의어	
		타입	list of [scanning start angle, scanning finished angle]
		단위	[degree, degree]
		예시	[20.0021, 80.0011]
A349	instrument (XRF)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	PANalycal Epsilon 4; Rigaku ZSX Primus II
A350	x-ray target (XRF / acquisition condition)	정의	target material for X-ray source used in measurement
		동의어	
		타입	string
		단위	
		예시	Rh; Ag; Pd; W
A351	XRF type (XRF / acquisition condition)	정의	the type of XRF
		동의어	
		타입	string
		단위	
		예시	WD-XRF; ED-XRF
A352	sample preparation (XRF / acquisition condition)	정의	sample preparation method and sample type
		동의어	
		타입	string
		단위	
		예시	Pressed powder; Loose powder; Fused glass bead; Bulk; Filter
A353	analysis elements (XRF / acquisition condition)	정의	elements which are measured
		동의어	
		타입	string
		단위	
		예시	Na; Mg; Si; Fe; Ti; Pb; Na~U
A354	quantitative analysis method (XRF / acquisition condition)	정의	type/method of quantitative analysis
		동의어	
		타입	string
		단위	
		예시	Calibration curve method; Fundamental parameter (FP) method; Internal standard method

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A355	raw data (XRF)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	XRF_FP_result.pdf; XRF_Quant.xps; XRF_summary.xlsx
A356	instrument (d33 meter)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	PM100; YE2730; D33PZO
A357	sample type (d33 meter)	정의	the type of sample
		동의어	
		타입	string
		단위	
		예시	Bulk; Thin film
A358	analysis elements (d33 meter)	정의	elements which are measured
		동의어	
		타입	string
		단위	
		예시	PZT; BTO; PVDF
A359	raw data (d33 meter)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	*.csv; *.txt
A360	magnetic moment (DFT calculation / magnetic structure / magnetic moment)	정의	magnetic moment in bohr magneton
		동의어	
		타입	number
		단위	Bohr magneton
		예시	4
A361	images (DFT calculation / nudged elastic band)	정의	crystal structures of all images
		동의어	
		타입	string
		단위	
		예시	cif
A363	instrument (DMA)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	DMA-80
A364	dry time (DMA / acquisition condition)	정의	dry and ash time in measurement
		동의어	
		타입	number
		단위	second
		예시	30; 60; 90;
A365	raw data (DMA)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	csv

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A366	initial velocity distribution method (MD simulation / initialization)	정의	initialization velocity distribution method for temperature setting
		동의어	
		타입	string
		단위	
		예시	Random; Boltzmann distribution
A367	thermostat (MD simulation / initialization / ensemble)	정의	type of thermostat used in the ensemble
		동의어	
		타입	string
		단위	
		예시	Andersen; Nose-Hoover; Langevin; Multiple Andersen
A368	barostat (MD simulation / initialization / ensemble)	정의	type of barostat used in the ensemble
		동의어	
		타입	string
		단위	
		예시	Parrinello-Rahman
A369	thermostat (MD simulation / simulation condition)	정의	type of thermostat used in the ensemble
		동의어	
		타입	string
		단위	
		예시	Andersen; Nose-Hoover; Langevin; Multiple Andersen
A370	barostat (MD simulation / simulation condition)	정의	type of barostat used in the ensemble
		동의어	
		타입	string
		단위	
		예시	Parrinello-Rahman
A375	instrument (ERD)	정의	instruments for ERD
		동의어	
		타입	string
		단위	
		예시	Accelerator; SNICS Source; Alphatross Source; Switching magnet; Q-pol lens; PIN type Si detector
A376	ion beam (ERD / acquisition condition)	정의	the incident ion beam
		동의어	
		타입	string
		단위	
		예시	He2+;
A377	beam energy (ERD / acquisition condition)	정의	energy of the incident ion beam
		동의어	
		타입	number
		단위	MeV
		예시	2; 2.4;
A378	beam current (ERD / acquisition condition)	정의	current of the incident ion beam
		동의어	
		타입	number
		단위	nA
		예시	20; 30;
A379	beam angle (ERD / acquisition condition)	정의	angle of the incident ion beam
		동의어	
		타입	number
		단위	degree
		예시	75;

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A380	counts (ERD / acquisition condition)	정의	amount of incident ion beam
		동의어	
		타입	number
		단위	
		예시	100000; 200000;
A381	beam size (ERD / acquisition condition)	정의	size of the incident ion beam
		동의어	
		타입	string
		단위	
		예시	5mm x 5mm; 1mm x 1mm
A382	stopper foil (ERD / acquisition condition)	정의	foil to prevent He ions from reaching the detector
		동의어	stopper filter
		타입	string
		단위	
		예시	Mylar foil (10 um)
A383	raw data (ERD)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	.spe;
A384	fatigue testing method (fatigue test)	정의	types of fatigue testing method
		동의어	
		타입	string
		단위	
		예시	1. Tension-compression fatigue testing 2. Fatigue crack propagation testing 3. Plane bending fatigue testing 4. Torsional fatigue testing 5. Rotating bending fatigue testing 6. Simulated environment (corrosion) fatigue testing 7. Thermal fatigue test
A385	ion beam source type (FIB / ion milling condition)	정의	ion beam source type
		동의어	
		타입	string
		단위	
		예시	Ga+, Ar+
A386	instrument (impedance analyzer)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	DPTS-AT-600; HP-4294A
A387	voltage (impedance analyzer / acquisition condition)	정의	applied voltage for impedance measurement
		동의어	
		타입	number
		단위	mV
		예시	10
A388	current (impedance analyzer / acquisition condition)	정의	applied current for impedance measurement
		동의어	
		타입	number
		단위	mA
		예시	5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A389	temperature (impedance analyzer / acquisition condition)	정의	applied temperature for impedance measurement
		동의어	
		타입	number
		단위	K
		예시	298
A390	frequency range (impedance analyzer / acquisition condition)	정의	applied pulse frequency range for the impedance measurement
		동의어	
		타입	dictionary of {min: numeric value, max: numeric value}
		단위	Hz
		예시	{min: 100, max: 1e6}
A391	raw data (impedance analyzer)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	data221122-1.jpg
A392	instrument (PFM)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Park Systems XE-100, Park Systems XE-7, Park Systems NX-10, Bruker Dimension Edge
A393	cantilever type (PFM / acquisition condition)	정의	type of cantilever used in measurement
		동의어	
		타입	string
		단위	
		예시	NSC36; NCHR; NSC14/Cr-Au; CONTSCPt; LFMR; MFMR
A394	scan rate (PFM / acquisition condition)	정의	the number of lines scanned per second
		동의어	
		타입	number
		단위	Hz
		예시	1
A395	scan size (PFM / acquisition condition)	정의	the size of scanning area
		동의어	
		타입	number
		단위	{\mu m}
		예시	10
A396	Z servo gain (PFM / acquisition condition)	정의	the sensitivity of the Z scanner feedback loop
		동의어	
		타입	number
		단위	
		예시	2
A397	raw data (PFM)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.tiff; Optic image.bmp
A398	instrument (RBS)	정의	instruments for RBS
		동의어	
		타입	string
		단위	
		예시	Accelerator; SNICS Source; Alphatross Source; Switching magnet; Q-pol lens; PIN type Si detector

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A399	ion beam (RBS / acquisition condition)	정의	the incident ion beam
		동의어	
		타입	string
		단위	
		예시	He2+;
A400	beam energy (RBS / acquisition condition)	정의	energy of the incident ion beam
		동의어	
		타입	number
		단위	MeV
		예시	2;
A401	beam current (RBS / acquisition condition)	정의	current of the incident ion beam
		동의어	
		타입	number
		단위	nA
		예시	20; 30;
A402	beam angle (RBS / acquisition condition)	정의	angle of the incident ion beam
		동의어	
		타입	number
		단위	degree
		예시	5;
A403	counts (RBS / acquisition condition)	정의	amount of incident ion beam
		동의어	
		타입	number
		단위	
		예시	100000; 200000;
A404	beam size (RBS / acquisition condition)	정의	size of the incident ion beam
		동의어	
		타입	string
		단위	
		예시	2mm x 5mm;
A405	raw data (RBS)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	.rbs; .as1;
A406	instrument (refractometer)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	ABBEMAT 300
A407	temperature (refractometer)	정의	temperature during measurement
		동의어	
		타입	number
		단위	K
		예시	293
A408	raw data (refractometer)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	csv

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A409	instrument (rheological analysis)	정의	equipment company
		동의어	
		타입	string
		단위	
		예시	TA instrument
A410	temperature (rheological analysis)	정의	measurement temperature
		동의어	
		타입	number
		단위	K
		예시	298
A411	humidity (rheological analysis)	정의	measurement humidity
		동의어	
		타입	number
		단위	%
		예시	27
A412	gap (rheological analysis)	정의	measurement gap between geometry
		동의어	
		타입	number
		단위	mm
		예시	1
A413	geometry (rheological analysis)	정의	geometrical condition for measurement
		동의어	
		타입	string
		단위	none
		예시	plate-plate
A414	material (rheological analysis)	정의	material of measurement equipment
		동의어	
		타입	string
		단위	none
		예시	Aluminum
A415	raw data (rheological analysis)	정의	URI of the raw or image data obtained
		동의어	
		타입	string
		단위	
		예시	image_thermal.tif
A416	gun type (SEM)	정의	electron source type
		동의어	
		타입	string
		단위	
		예시	W, CeB6, LaB6
A417	mode (SEM)	정의	acquisition mode
		동의어	
		타입	string
		단위	
		예시	SEI, BSE, EBSD
A418	zone-axis (TEM / analysis)	정의	the direction to samples during the measurement
		동의어	
		타입	array
		단위	
		예시	[111]; [001];

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A419	scale unit (TEM / raw data)	정의	the scale unit of image data
		동의어	
		타입	string
		단위	
		예시	nm, um, mm
A420	specimen width (tensile test)	정의	width of tensile test specimen for rectangular and tubular products
		동의어	
		타입	number
		단위	mm
		예시	12.5
A421	specimen thickness (tensile test)	정의	thickness of tensile test specimen for rectangular and tubular products
		동의어	
		타입	number
		단위	mm
		예시	12.5
A422	specimen diameter (tensile test)	정의	diameter of tensile test specimen for round bar products
		동의어	
		타입	number
		단위	mm
		예시	12.5
A423	stress rate (tensile test / tensile test speed)	정의	increase of stress per time
		동의어	
		타입	number
		단위	MPa s^{-1}
		예시	10
A424	crosshead separation rate (tensile test / tensile test speed)	정의	displacement of the crossheads per time
		동의어	
		타입	number
		단위	mm min^{-1}
		예시	0.5
A425	atmosphere (tensile test)	정의	atmosphere to which a given testing is exposed
		동의어	testing environment
		타입	string
		단위	
		예시	Ar; N2+5H2; inert; reducing; oxidizing
A426	electrode configuration (voltammetry / acquisition condition)	정의	cell configuration to apply voltage
		동의어	
		타입	string
		단위	
		예시	SUS symmetric, Li-SUS asymmetric
A427	spin orbit coupling (DFT calculation)	정의	spin-orbit coupling
		동의어	SOC
		타입	boolean
		단위	
		예시	1. False 2. True
A428	raw data (tensile test)	정의	URI of the stress-strain curve or load-displacement curve; raw or image data
		동의어	
		타입	string
		단위	
		예시	test1.tif;

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A429	microtruss direction (compressive test)	정의	a direction of connected microtruss for variation of compressive test performance according to crystallographic direction
		동의어	
		타입	string
		단위	
		예시	[010]
A430	microtruss thickness (compressive test)	정의	thickness of truss for microstructure
		동의어	
		타입	number
		단위	um
		예시	450
A431	compressive specimen thickness (compressive test)	정의	thickness of compressive test specimen
		동의어	
		타입	number
		단위	mm
		예시	26
A432	area (compressive test)	정의	area of the substrate
		동의어	
		타입	number
		단위	mm^{2}
		예시	484.974
A433	initial volume (compressive test)	정의	initial volume of the substrate
		동의어	
		타입	number
		단위	mm^{3}
		예시	
A434	effective volume (compressive test)	정의	effective volume of the substrate
		동의어	
		타입	number
		단위	mm^{3}
		예시	2397.382
A435	instrument (magnetic hysteresis loop)	정의	the test device for magnetic material analysis of soft and hard magnetic materials
		동의어	hysteresis loop tracer
		타입	string
		단위	
		예시	
A436	specimen shape (magnetic hysteresis loop)	정의	the shape of specimen for the measurement of hysteresis loop of the soft or hard magnetic materials
		동의어	
		타입	string
		단위	
		예시	toroidal; cylinder; hexahedron
A437	specimen length (magnetic hysteresis loop)	정의	the length of specimen for measuring magnetic properties of soft and hard magnetic materials
		동의어	
		타입	number
		단위	mm
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A438	maximum applied field (magnetic hysteresis loop)	정의	the maximum field applied when the hysteresis loop is measured
		동의어	H_{max}
		타입	number
		단위	A m^{-1}
		예시	1.3e3
A439	temperature (magnetic hysteresis loop)	정의	the temperature of the test specimen during measurement
		동의어	ambient temperature
		타입	number
		단위	K
		예시	300
A440	magnetic hysteresis (magnetic hysteresis loop)	정의	Lagging of magnetization under varying external magnetic fields when the magnetic specimen is subjected to magnetic field cycle. The hysteresis appears in a M/H curve as a closed loop. This closed loop represents the matnetic hysteresis.
		동의어	magnetic hysteresis loop; hysteresis loop; magnetic hysteresis curve; hysteresis curve
		타입	numeric array: [magnetic field(A m^{-1}), magnetic field uncertainty(A m^{-1}), magnetization(A m^{-1}), magnetization uncertainty(A m^{-1})]
		단위	A m^{-1}
		예시	[130, 0.5, 0.8, 0.001], [180, 0.5, 1.3, 0.001], [240, 0.5, 2.2, 0.001]
A441	base pressure (atom probe tomography / acquisition condition)	정의	the lowest attainable pressure within a vacuum system under normal operating or idle conditions
		동의어	
		타입	number
		단위	Torr
		예시	0.0000000005
A442	volume ratio (compressive test)	정의	the ratio of the two volumes, effective volume/initial volume
		동의어	
		타입	number
		단위	none
		예시	0.190128
A443	acquisition time (EDS / acquisition condition)	정의	data collection time
		동의어	
		타입	number
		단위	s
		예시	10
A444	atmosphere (electrochemical activity)	정의	atmosphere of electrochemical cell
		동의어	
		타입	string
		단위	
		예시	Ar, N2
A445	fourier spacing (MD simulation / simulation condition / electrostatic interaction / Ewald)	정의	Fourier-space grid spacing
		동의어	
		타입	number
		단위	nm
		예시	0.12
A446	accuracy (MD simulation / simulation condition / electrostatic interaction / Ewald)	정의	relative strength of the Ewald-shifted direct potential at cutoff distance
		동의어	
		타입	number
		단위	N
		예시	0.00001

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A447	fourier spacing (MD simulation / simulation condition / electrostatic interaction / PME)	정의	Fourier-space grid spacing
		동의어	
		타입	number
		단위	nm
		예시	0.12
A448	PME-order (MD simulation / simulation condition / electrostatic interaction / PME)	정의	the number of grid points along a dimension to which a charge is mapped
		동의어	
		타입	number
		단위	none
		예시	4
A449	accuracy (MD simulation / simulation condition / electrostatic interaction / PPPM)	정의	desired relative error in forces
		동의어	
		타입	number
		단위	N
		예시	0.00001
A450	accuracy (MD simulation / simulation condition / electrostatic interaction / MSM)	정의	desired relative error in forces
		동의어	
		타입	number
		단위	N
		예시	0.00001
A451	instrument (EQE analysis)	정의	instrument that was used during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Newport QuantX-300
A452	irradiance of bias light (EQE analysis / acquisition condition)	정의	The light intensity of any bias light during the EQE measurement • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	100
A453	raw data (EQE analysis)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	EPMA_result.txt; EPMA_image.bmp
A454	instrument (impact test)	정의	name of a device to measure impact property of materials
		동의어	
		타입	string
		단위	
		예시	Pendulum impact tester 10ton; Drop weight tester; High-energy drop weight tester
A455	instrument (IV measurement)	정의	instrument that was used during IV measurement
		동의어	
		타입	string
		단위	
		예시	Keithley2450

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A456	direction (IV measurement / scan)	정의	sweep direction of the controlled parameter (voltage for potentiostatic, current for galvanostatic)
		동의어	
		타입	string
		단위	
		예시	1. forward 2. reverse
A457	voltage sweep rate (IV measurement / scan)	정의	speed of the potential sweep during the IV measurement
		동의어	speed, scan rate
		타입	number
		단위	mV s^{-1}
		예시	20; 10
A458	delay time (IV measurement / scan)	정의	delayed time between measurement in the sweep
		동의어	
		타입	number
		단위	ms
		예시	100
A459	duration time (IV measurement / scan)	정의	duration time at each value in the sweep
		동의어	measurement time; integration time
		타입	number
		단위	s
		예시	100
A460	voltage step (IV measurement / scan)	정의	voltage difference between each set values during voltage sweep in the potentiostatic measurement
		동의어	
		타입	number
		단위	mV
		예시	10
A461	atmosphere (IV measurement)	정의	atmosphere during the IV measurement
		동의어	
		타입	string
		단위	
		예시	Air; N2; Vacuum
A462	temperature (IV measurement)	정의	temperature of the device during the IV measurement
		동의어	
		타입	number
		단위	K
		예시	300; 500
A463	procedure (IV measurement / stability test)	정의	measurement condition for the stability test
		동의어	
		타입	string
		단위	
		예시	constant current; constant potential; passive resistance; short circuit; temperature; atmosphere
A464	measurement time (IV measurement / stability test)	정의	duration time of deterioration in the stability measurement
		동의어	
		타입	number
		단위	min
		예시	100
A465	equipment (JV measurement)	정의	instrument that was used during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Newport solar simulator; EQE

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A466	procedure (JV measurement / stabilized test)	정의	the potentiostatic load condition during the stabilized performance measurement
		동의어	
		타입	string
		단위	
		예시	constant current; constant potential; MPPT; passive resistance; short circuit
A467	metrics (JV measurement / stabilized test)	정의	the metrics associated to the load condition in the procedure field
		동의어	
		타입	number
		단위	• For measurement under constant current, state the current in mA cm^{-2} • For measurement under constant potential. State the potential in V • For a measurement under constant resistive load, state the resistance in Ohm
		예시	1; 23; 43
A468	measurement time (JV measurement / stabilized test)	정의	the duration of the stabilized performance measurement
		동의어	
		타입	number
		단위	min
		예시	100
A469	number of cells averaged (JV measurement)	정의	The number of cells the reported IV data is based on. • The preferred way to enter data is to give every individual cell its own entry in the data template/data base. If that is done, the data is an average over 1 cell. • If the reported IV data is not the data from one individual cell, but an average over N cells. Give the number of cells. • If the reported value is an average, but it is unknown over how many cells the value has been averaged (and no good estimate is available), state the number of cells as 2, which is the smallest number of cells that qualifies for an averaging procedure.
		동의어	
		타입	number
		단위	none
		예시	1; 8; 16;
A470	certified (JV measurement)	정의	TRUE if the IV data is measured by an independent and certification institute. If your solar simulator is calibrated by a calibrated reference diode, that does not count as a certified result.
		동의어	
		타입	boolean
		단위	
		예시	1. False 2. True
A471	certification institute (JV measurement)	정의	The name of the certification institute that has measured the certified device.
		동의어	
		타입	string
		단위	
		예시	Newport; KIER; NIM;
A472	age of cell (JV measurement / storage before measurement)	정의	The age of the cell with respect to the last deposition step was finalized
		동의어	
		타입	number
		단위	d
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A473	atmosphere (JV measurement / storage before measurement)	정의	The atmosphere in which the sample was stored between the device finalisation and the IV measurement. • If the atmosphere is a mixture of different gases, e.g. A and B, list the gases in alphabetic order and separate them with semicolons, as in (A; B) • “Dry air” represent air with low relative humidity but where the relative humidity is not known • “Ambient” represent air where the relative humidity is not known. For ambient conditions where the relative humidity is known, state this as “Air” • “Vacuum” (of unspecified pressure) is for this purpose considered as an atmospheric gas • If the atmosphere has changed during the storing time, separate the different atmospheres by a double forward angel bracket with one blank space on both sides (‘ 〉〉 ‘)
		동의어	
		타입	string
		단위	
		예시	N2; Air; N2 〉〉 Air
A474	relative humidity (JV measurement / storage before measurement)	정의	The relative humidity in the atmosphere in which the sample was stored between the device finaliastion and the IV measurement. • Relative humidity is the actual amount of water vapor in the air compared to the total amount of vapor that can exist in the air at its current temperature • If the relative humidity has changed during the storing time, separate the different relative humidity by a double forward angel bracket with one blank space on both sides (‘ 〉〉 ‘) • If the relative humidity is not known, stat that as ‘nan’ • For values with uncertainties, state the best estimate, e.g. write 35 and not 30-40.
		동의어	
		타입	number
		단위	%
		예시	35; 0; 0 〉〉 25
A475	atmosphere (JV measurement)	정의	The atmosphere in which the IV measurement is conducted • If the atmosphere is a mixture of different gases, e.g. A and B, list the gases in alphabetic order and separate them with semicolons, as in (A; B) • “Dry air” represent air with low relative humidity but where the relative humidity is not known • “Ambient” represent air where the relative humidity is not known. For ambient conditions where the relative humidity is known, state this as “Air” • “Vacuum” (of unspecified pressure) is for this purpose considered as an atmospheric gas
		동의어	
		타입	string
		단위	
		예시	Air; N2; Vacuum
A476	relative humidity (JV measurement)	정의	The relative humidity in which the IV measurement is conducted • Relative humidity is the actual amount of water vapor in the air compared to the total amount of vapor that can exist in the air at its current temperature • If there are uncertainties, only state the best estimate, e.g. write 35 and not 20-50. • If the relative humidity is not known, stat that as ‘nan’
		동의어	
		타입	number
		단위	%
		예시	50
A477	temperature (JV measurement)	정의	The relative humidity in which the IV measurement is conducted • Relative humidity is the actual amount of water vapor in the air compared to the total amount of vapor that can exist in the air at its current temperature • If there are uncertainties, only state the best estimate, e.g write 35 and not 20-50. • If the relative humidity is not known, stat that as ‘nan’
		동의어	
		타입	number
		단위	K
		예시	300; 500

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A478	source type (JV measurement / light source)	정의	The type of light source used during the IV-measurement • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template. • The category Solar simulator should only be used when you do not really know which type of light source you have in your solar simulator.
		동의어	
		타입	string
		단위	
		예시	laser; metal halide; outdoor; solar simulator; sulfur plasma; white LED; Xenon plasma
A479	instrument (JV measurement / light source)	정의	The brand name and model number of the light source/solar simulator used. • This category was included after the projects initial phase where for the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	Newport model 91192; Newport AAA; Atlas suntest
A480	simulator classification (JV measurement / light source)	정의	Classification of the light source/solar simulator used during the IV measurement according to the IEC 60904-9 Ed.3 • A three-letter code consisting of A+, A, B, and C. The order of the letters represents the quality of: spectral match, spatial non-uniformity, and temporal instability of the light source/solar simulator.
		동의어	
		타입	string
		단위	
		예시	AAA; A+AA; ABB; CAB
A481	intensity (JV measurement / light)	정의	The radiant power per unit area of light incident on a surface. In the case of the IV measurement of a photovoltaic device, it means the irradiance of the light source/solar simulator used during the measurement. • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • Standard AM 1.5 illumination correspond to 100 mW/cm2 • If you need to convert from illumination given in lux; at 550 nm, 1 mW/cm2 corresponds to 6830 lux. Be aware that the conversion change with the spectrum used. As a rule of thumb for general fluorescent/LED light sources, around 0.31mW corresponded to 1000 lux. If your light intensity is measured in lux, it probably means that your light spectra deviates quite a lot from AM 1.5, wherefore it is very important that you also specify the light spectra in the next column.
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	100
A482	type of spectrum (JV measurement / light)	정의	The spectral type of the light source/solar simulator used during the IV measurement
		동의어	
		타입	string
		단위	
		예시	AM 1.0; AM 1.5D; AM 1.5G; indoor light; monochromatic; natural sunlight; UV
A483	wavelength range (JV measurement / light)	정의	The wavelength range of the light source • Separate the lower and upper bound by a comma • For monochromatic light sources, give the same value for min and max values. • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • State unknown values as 'nan'
		동의어	
		타입	dictionary of {wavelength_min : value, wavelength_max : value]
		단위	nm
		예시	{wavelength_min : 330, wavelength_max : 1000}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A484	illumination direction (JV measurement / light)	정의	The direction of the illumination with respect to the device stack • If the cell is illuminated through the substrate, state this as 'Substrate' • If the cell is illuminated through the top contact, state this as 'Superstrate' • For back contacted cells illuminated from the non-contacted side, state this as 'Superstrate'
		동의어	
		타입	string
		단위	
		예시	Substrate; Superstrate
A485	masked cell (JV measurement / light)	정의	TRUE if the cell is illuminated through a mask with an opening that is smaller than the total cell area.
		동의어	
		타입	boolean
		단위	
		예시	1. False 2. True
A486	mask area (JV measurement / light)	정의	The area of the opening in the mask through with the cell is illuminated (if there is a mask) • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • If there is no light mask, leave this field empty.
		동의어	
		타입	number
		단위	cm^{2}
		예시	430
A488	direction (JV measurement / scan)	정의	sweep direction of voltage
		동의어	scan direction
		타입	string
		단위	
		예시	1. forward 2. reverse
A489	sweep rate (JV measurement / scan)	정의	The speed of the potential sweep during the IV measurement
		동의어	
		타입	number
		단위	mV s^{-1}
		예시	20; 10
A490	delay time (JV measurement / scan)	정의	The time at each potential value before integration in the potential sweep. • For some potentiostats you need to specify this value, whereas for others it is set automatically and is not directly accessible. • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • If unknown, leave this field empty.
		동의어	
		타입	number
		단위	ms
		예시	100
A491	integration time (JV measurement / scan)	정의	The integration time at each potential value in the potential sweep. • For some potentiostats you need to specify this value, whereas for others it is set automatically and is not directly accessible. • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • If unknown, leave this field empty.
		동의어	
		타입	number
		단위	ms
		예시	100
A492	voltage step (JV measurement / scan)	정의	The distance between the measurement point in the potential sweep • If unknown, leave this field empty.
		동의어	
		타입	number
		단위	mV
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A493	voltage settling time (JV measurement / scan)	정의	the time takes for the voltage across the solar cell to reach a stable and constant value after a change in the external circuit conditions
		동의어	stabilization time
		타입	number
		단위	ms
		예시	200
A494	protocol (JV measurement / preconditioning)	정의	Any preconditioning protocol done immediately before the IV measurement • If no preconditioning was done, state this as 'none' • If more than one preconditioning protocol was conducted in parallel, separate them with semicolons • If more than one preconditioning protocol was conducted in sequence, separate them by a double forward angel bracket (' >> ')
		동의어	
		타입	string
		단위	
		예시	cooling; heating; light soaking; light soaking >> potential biasing; potential biasing
A495	time (JV measurement / preconditioning)	정의	The duration of the preconditioning protocol • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	s
		예시	300
A496	potential (JV measurement / preconditioning)	정의	The potential at any potential biasing step • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	V
		예시	10
A497	light intensity (JV measurement / preconditioning)	정의	The light intensity at any light soaking step • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	mW cm^{-2}
		예시	85
A498	raw data (JV measurement)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	spc; csv; tif; bmp; jpg; text
A499	protocol (PV stability)	정의	The stability protocol used for the stability measurement. • For a more detailed discussion on protocols and standard nomenclature for stability measurements, please see the following paper: o Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures by: M. V. Khenkin et al. Nat. Energ. 2020. DOI: 10.1038/s41560-019-0529-5
		동의어	
		타입	string
		단위	
		예시	ISOS-D-1; ISOS-D-11; maximum power point tracing (MPPT); open-circuit condition

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A500	number of cells averaged (PV stability)	정의	The number of cells the reported stability data is based on. • The preferred way to enter data is to give every individual cell its own entry in the data template/data base. If that is done, the data is an average over 1 cell. • If the reported stability data is not the data from one individual cell, but an average over N cells. Give the number of cells. • If the reported value is an average, but it is unknown over how many cells the value has been averaged (and no good estimate is available), state the number of cells as 2, which is the smallest number of cells that qualifies for an averaging procedure.
		동의어	
		타입	number
		단위	none
		예시	18
A501	source type (PV stability / light source)	정의	The type of light source used during the stability measurement • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template. • The category Solar simulator should only be used when you do not really know which type of light source you have in your solar simulator.
		동의어	
		타입	string
		단위	
		예시	laser; metal halide; outdoor; solar simulator; sulfur plasma; white LED; Xenon plasma
A502	instrument (PV stability / light source)	정의	The brand name and model number of the light source/solar simulator used • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	Newport model 91192; MC-science, Lab100; Atlas suntest
A503	simulator classification (PV stability / light source)	정의	Classification of the light source/solar simulator used during the stability measurement according to the IEC 60904-9 Ed.3 • A three-letter code consisting of A+, A, B, and C. The order of the letters represents the quality of: spectral match, spatial non-uniformity, and temporal instability of the light source/solar simulator.
		동의어	
		타입	string
		단위	
		예시	AAA; A+AA; ABB; BAC
A504	intensity (PV stability / light)	정의	The radiant power per unit area of light incident on a surface. In the case of the stability measurement of a photovoltaic device, it means the irradiance of the light source/solar simulator used during the stability measurement. • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • Standard AM 1.5 illumination correspond to 100 mW/cm2 • If you need to convert from illumination given in lux; at 550 nm, 1 mW cm^{-2} corresponds to 6830 lux. Be aware that the conversion change with the spectrum used. As a rule of thumb for general fluorescent/LED light sources, around 0.31mW corresponded to 1000 lux. If your light intensity is measured in lux, it probably means that your light spectra deviates quite a lot from AM 1.5, wherefore it is very important that you also specify the light spectra in the next column.
		동의어	irradiance
		타입	number
		단위	mW cm^{-2}
		예시	100
A505	type of spectrum (PV stability / light)	정의	The spectral type of the light source/solar simulator used during the stability measurement
		동의어	
		타입	string
		단위	
		예시	AM1.5G; AM1.5D; AM 0; indoor light; monochromatic; natural sunlight; UV

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A506	wavelength range (PV stability / light)	정의	The wavelength range of the light source • Separate the lower and upper bound by a comma • For monochromatic light sources, give the same value for min and max values. • If there are uncertainties, only state the best estimate, e.g. write 100 and not 90-100. • State unknown values as 'nan'
		동의어	
		타입	dictionary of {wavelength_min : value, wavelength_max : value]
		단위	nm
		예시	{wavelength_min : 330, wavelength_max : 1000}
A507	illumination direction (PV stability / light)	정의	The direction of the illumination with respect to the device stack • If the cell is illuminated through the substrate, state this as 'Substrate' • If the cell is illuminated through the top contact, state this as 'Superstrate' • For back contacted cells illuminated from the non-contacted side, state this as 'Superstrate'
		동의어	
		타입	string
		단위	
		예시	Substrate; Superstrate
A508	load condition (PV stability / light)	정의	The load situation of the illumination during the stability measurement. • If the illumination is constant during the entire stability measurement, or if the cell is stored in the dark, state this as 'Constant'. • If the situation periodically is interrupted by IV-measurements, continue to consider the load condition as constant • If there is a cycling between dark and light, state this as 'Cycled' • If the illumination varies in an uncontrolled way, state this as 'Uncontrolled' • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	Constant; Cycled; Day-Night cycle; Uncontrolled
A509	cycling times (PV stability / light)	정의	If the illumination load is cycled during the stability measurement, state the time in low light followed by the time in high light for the cycling period. • If not applicable, leave blank
		동의어	
		타입	number
		단위	h
		예시	[12,12]
A510	UV filter (PV stability / light)	정의	TRUE if a UV-filter of any kind was placed between the illumination source and the device during the stability measurement.
		동의어	
		타입	boolean
		단위	
		예시	['False', 'True']
A511	load condition (PV stability / potential bias)	정의	The Potentiostatic load condition during the stability measurement • When the cell is not connected to anything, state this as 'Open circuit'
		동의어	
		타입	string
		단위	
		예시	constant current; constant potential; MMPT; open circuit; passive resistance; short circuit

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A512	range (PV stability / potential bias)	정의	The potential range during the stability measurement • The lower and upper bound recorded in numeric list format of Python • For open circuit conditions, state this as 'nan' • If there are uncertainties, only state the best estimate, e.g. write 1 and not 0.90-1.1 • State unknown values as 'nan'
		동의어	
		타입	dictionary of {Vmin : minimum potential value, Vmax : maximum potential value]
		단위	V
		예시	{Vmin : 0.9, Vmax : 1.02}
A513	passive resistance (PV stability / potential bias)	정의	The passive resistance in the measurement circuit if a resistor was used • Give the value in ohm • If there are uncertainties, only state the best estimate, e.g. write 1.03 and not 1.01-1.05 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	Ohm
		예시	1.03
A514	load condition (PV stability / temperature)	정의	The load situation of the temperature during the stability measurement. • If the temperature is constant during the entire stability measurement, state this as 'Constant'. • If there is a cycling between colder and hotter conditions, state this as 'Cycled' • If the temperature varies in an uncontrolled way, state this as 'Uncontrolled' • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	constant; uncontrolled; cycled
A515	range (PV stability / temperature)	정의	The temperature range during the stability measurement • The lower and upper bound recorded in numeric list format of Python • If there are uncertainties, only state the best estimate, e.g. write 1 and not 0.90-1.1 • State unknown values as 'nan'
		동의어	
		타입	dictionary of {Tmin : minimum temperature value, Tmax : maximum temperature value]
		단위	K
		예시	{Tmin : 289, Tmax : 373}
A516	cycling times (PV stability / temperature)	정의	If the temperature is cycled during the stability measurement, state the time in low temperature followed by the time in high temperature for the cycling period. • If not applicable, leave blank • The lower and upper bound recorded in numeric list format of Python • If there are uncertainties, only state the best estimate, e.g. write 1 and not 0.90-1.1 • State unknown values as 'nan'
		동의어	
		타입	number
		단위	h
		예시	
A517	ramp speed (PV stability / temperature)	정의	The temperature ramp speed
		동의어	
		타입	number
		단위	K min^{-1}
		예시	1.03

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A518	composition (PV stability / atmosphere)	정의	gas composition of atmosphere
		동의어	
		타입	string
		단위	
		예시	Air; N2; Vacuum
A519	oxygen concentration (PV stability / atmosphere)	정의	The oxygen concentration in the atmosphere • If unknown, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	0.4
A520	load condition (PV stability / relative humidity)	정의	The load situation of the relative humidity during the stability measurement. • If the relative humidity is constant during the entire stability measurement, state this as 'constant'. • If there is a cycling between dryer and damper conditions, state this as 'cycled' • If the relative humidity varies in an uncontrolled way, i.e. the cell is operated under ambient conditions, state this as 'ambient'
		동의어	
		타입	string
		단위	
		예시	ambient; uncontrolled; cycled
A521	range (PV stability / relative humidity)	정의	The relative humidity range during the stability measurement • The lower and upper bound recorded in numeric list format of Python • If there are uncertainties, only state the best estimate, e.g. write 1 and not 0.90-1.1 • State unknown values as 'nan'
		동의어	
		타입	dictionary of {min_RH : minimum value, max_RH : maximum value}
		단위	%
		예시	{min_RH : 30, max_RH : 65}
A522	average value (PV stability / relative humidity)	정의	The average relative humidity during the stability measurement. • If there are uncertainties, only state the best estimate, e.g. write 1 and not 0.90-1.1 • If unknown, leave this field empty.
		동의어	
		타입	number
		단위	%
		예시	48.4
A523	total exposure time (PV stability)	정의	The total duration of the stability measurement. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050
		동의어	
		타입	number
		단위	h
		예시	1000
A524	time betweeen measurements (PV stability / periodic JV measurement)	정의	The average time between JV-measurement during the stability measurement
		동의어	
		타입	number
		단위	h
		예시	8
A525	number of bending cycle (PV stability / flexibility)	정의	number of bending cycles for a flexible cell in a mechanical stability test
		동의어	
		타입	number
		단위	none
		예시	1000; 350

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A526	bending radius (PV stability / flexibility)	정의	the bending radius of the flexible cell during the mechanical stability test
		동의어	
		타입	number
		단위	mm
		예시	15
A527	protocol (PV outdoor test)	정의	The protocol used for the outdoor testing. • For a more detailed discussion on protocols and standard nomenclature for stability measurements, please see the following paper: Consensus statement for stability assessment and reporting for perovskite photovoltaics based on ISOS procedures by: M. V. Khenkin et al. Nat. Energ. 2020. DOI: 10.1038/s41560-019-0529-5
		동의어	
		타입	string
		단위	
		예시	IEC 61853-1; ISOS-O-1; ISOS-O-3;
A528	number of cells averaged (PV outdoor test)	정의	The number of cells the reported outdoor data is based on. • The preferred way to enter data is to give every individual cell its own entry in the data template/data base. If that is done, the data is an average over 1 cell. • If the reported stability data is not the data from one individual cell, but an average over N cells. Give the number of cells. • If the reported value is an average, but it is unknown over how many cells the value has been averaged (and no good estimate is available), state the number of cells as 2, which is the smallest number of cells that qualifies for an averaging procedure.
		동의어	
		타입	number
		단위	none
		예시	18
A529	country (PV outdoor test / location)	정의	The country where the outdoor testing was occurring • For measurements conducted in space, state this as 'Space International'
		동의어	
		타입	string
		단위	
		예시	Sweden; Switzerland; Space International
A530	city (PV outdoor test / location)	정의	The city where the outdoor testing was occurring
		동의어	
		타입	string
		단위	
		예시	Seoul; Sendai; Ningbo
A531	coordinates (PV outdoor test / location)	정의	The coordinates fort the places where the outdoor testing was occurring. • Use decimal degrees (DD) as the format.
		동의어	
		타입	dictionary of {latitude : value, longitude : value}
		단위	decimal degree
		예시	{latitude : 59.839116, longitude : 17.647979}
A532	climate zone (PV outdoor test / location)	정의	The climate zone for the places where the outdoor testing was occurring. Categorized 5 major zones according to Koeppen-Geiger climate classification (www.gloh2o.org/koppen)
		동의어	
		타입	string
		단위	
		예시	['Tropical', 'Dry', 'Temperate', 'Continental', 'Polar']

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A533	tilt (PV outdoor test / installation)	정의	The tilt of the installed solar cell. • A module lying flat on the ground have a tilt of 0 • A module standing straight up has a tilt of 90
		동의어	
		타입	number
		단위	degree
		예시	43
A534	cardinal direction (PV outdoor test / installation)	정의	The cardinal direction of the installed solar cell. • North is 0 • East is 90 • South is 180 • West is 270
		동의어	
		타입	number
		단위	degree
		예시	183
A535	number of solar tracking axis (PV outdoor test / installation)	정의	The number of tracking axis in the installation.
		동의어	
		타입	number
		단위	none
		예시	0; 1; 2
A536	season (PV outdoor test / time)	정의	The time of year the outdoor testing was occurring. • Order the seasons in alphabetic order and separate them with bar. • For time periods longer than a year, state all four seasons once.
		동의어	
		타입	string
		단위	
		예시	Winter; Summer; Autumn-Winter-Spring
A537	start (PV outdoor test / time)	정의	the starting time for the outdoor measurement
		동의어	
		타입	string
		단위	
		예시	2020:04:01:18:25
A538	end (PV outdoor test / time)	정의	the ending time for the outdoor measurement
		동의어	
		타입	string
		단위	
		예시	2023:04:01:18:25
A539	total exposure (PV outdoor test / time)	정의	The total duration of the outdoor measurement. • If there are uncertainties, only state the best estimate, e.g. write 1000 and not 950-1050
		동의어	
		타입	number
		단위	day
		예시	900
A540	load condition (PV outdoor test / potential bias)	정의	The Potentiostatic load condition during the outdoor measurement • When the cell is not connected to anything, state this as 'Open circuit'
		동의어	
		타입	string
		단위	
		예시	['Constant current', 'Constant potential', 'MPPT', 'Open circuit', 'Passive resistance', 'Short circuit']

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A541	range (PV outdoor test / potential bias)	정의	The potential range during the outdoor test • The lower and upper bound recorded in numeric list format of Python • For open circuit conditions, state this as 'nan' • If there are uncertainties, only state the best estimate, e.g. write 1 and not 0.90-1.1 • State unknown values as 'nan'
		동의어	
		타입	dictionary of {Vmin : minimum potential value, Vmax : maximum potential value]
		단위	V
		예시	{Vmin : 0.9, Vmax : 1.02}
A542	passive resistance (PV outdoor test / potential bias)	정의	The passive resistance in the measurement circuit if a resistor was used • Give the value in ohm • If there are uncertainties, only state the best estimate, e.g. write 1.03 and not 1.01-1.05 • If unknown or not applicable, leave this field empty.
		동의어	
		타입	number
		단위	Ohm
		예시	1.03
A543	load condition (PV outdoor test / temperature)	정의	The load situation of the temperature during the outdoor test • If the temperature is constant during the entire outdoor test, state this as 'Constant'. • If there is a cycling between colder and hotter conditions, state this as 'Cycled' • If the temperature varies in an uncontrolled way, state this as 'Uncontrolled' • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	constant; uncontrolled; cycled
A544	range (PV outdoor test / temperature)	정의	The temperature range during the outdoor test • The lower and upper bound recorded in numeric list format of Python • If there are uncertainties, only state the best estimate, e.g. write 1 and not 0.90-1.1 • State unknown values as 'nan'
		동의어	
		타입	dictionary of {Tmin : minimum temperature value, Tmax : maximum temperature value]
		단위	K
		예시	{Tmin : 289, Tmax : 373}
A545	Tmodule (PV outdoor test / temperature)	정의	The effective temperature of the module during peak hours.
		동의어	
		타입	number
		단위	K
		예시	350
A546	time between measurements (PV outdoor test / periodic JV measurement)	정의	The average time between JV-measurement during the outdoor test
		동의어	
		타입	number
		단위	h
		예시	8
A547	raw data for outdoor trace (PV outdoor test)	정의	URI of the JV measurement data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	JV.dat

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A548	detailed weather data (PV outdoor test)	정의	weather URI of data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	weather.dat
A549	spectral data (PV outdoor test)	정의	spectral URI of data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	spectra.dat
A550	irradiance data (PV outdoor test)	정의	Irradiance URI of data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	irradiance.dat
A551	laser power (Raman spectroscopy / acquisition condition)	정의	excitation laser's power
		동의어	
		타입	number
		단위	mW
		예시	60 mW
A552	spectrometer resolution (Raman spectroscopy / acquisition condition)	정의	spectrometer's instrumental resolution
		동의어	
		타입	number
		단위	nm
		예시	1 nm
A553	acquisition time (Raman spectroscopy / acquisition condition)	정의	data collection time
		동의어	
		타입	number
		단위	s
		예시	10
A554	grid (TEM / acquisition condition)	정의	a flat disc with a mesh or other shaped holes used to support thin sections of specimen for TEM measurement
		동의어	
		타입	string
		단위	None
		예시	Carbon supported 300 mesh Cu grid; Ring grid (Φ2.0 mm), Oval grid (Φ2.0x0.5 mm), Mesh grid (Cu50 mesh or Mo200 mesh)
A555	instrument name (PES / instrument)	정의	name of the instrument used for PES measurement
		동의어	
		타입	string
		단위	
		예시	PHI VersaProbe III
A556	beam source (PES / acquisition condition)	정의	beam source used in measurement
		동의어	
		타입	string
		단위	
		예시	X-ray (Al ka); UV (He I)

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A557	pass energy (PES / acquisition condition)	정의	energy resolution setting of the electron analyzer
		동의어	
		타입	number
		단위	eV
		예시	21.2
A558	energy step size (PES / acquisition condition)	정의	energy interval for scanning
		동의어	
		타입	number
		단위	eV
		예시	0.05
A559	raw data (PES)	정의	URI of the raw photoelectron spectrum obtained during measurement
		동의어	
		타입	string
		단위	
		예시	PES_spectrum_Fe_2p.dat; XPS_spectrum.spe; UPS_specrum.vgt
A560	instrument (UV-VIS-nIR spectrophotometer)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	NanoDrop One/OneC; Cary 60; Lambda 365/465; UV-2600/2700; PANalycal Epsilon 4; Rigaku ZSX Primus II
A561	wavelength range (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	the wavelength range under which the measurement was performed
		동의어	
		타입	dictionary of {wavelength_min : value, wavelength_max : value}
		단위	nm
		예시	{wavelength_min : 330, wavelength_max : 1000}
A562	scan speed (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	scan speed of the wavelength for the spectrum collection
		동의어	
		타입	string
		단위	
		예시	1. fast 2. medium 3. slow
A563	atmosphere (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	The atmosphere under which the spectrum was collected. • If the atmosphere is a mixture of different gases, e.g. A and B, list the gases in alphabetic order and separate them with semicolons, as in (A; B) • "Dry air" represent air with low relative humidity but where the relative humidity is not known • "Ambient" represent air where the relative humidity is not known. For ambient conditions where the relative humidity is known, state this as "Air" • "Vacuum" (of unspecified pressure) is for this purpose considered as an atmospheric gas
		동의어	
		타입	string
		단위	
		예시	Air; N2; Vacuum
A564	raw data (UV-VIS-nIR spectrophotometer)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	data.tiff
A565	x-ray source (XRD / acquisition condition)	정의	X-ray source used in the XRD measurement
		동의어	
		타입	string
		단위	
		예시	Cu, Cr, Mo

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A566	sample temperature (XRD / acquisition condition)	정의	sample temperature
		동의어	
		타입	number
		단위	K
		예시	300
A567	software (thermodynamic calculation)	정의	integrated computer programs designed to calculate thermodynamic properties and phase equilibria
		동의어	
		타입	string
		단위	
		예시	ThermoCalc; FactSage; PANDAT; JMatPro
A568	database (thermodynamic calculation)	정의	thermodynamic and kinetic database that were used to calculate thermodynamic properties and phase equilibria
		동의어	
		타입	string
		단위	
		예시	TCAL8; MOBAL6; FTlite; PanAluminum; *.tbd
A569	instrument (DLS)	정의	identifier for the specific DLS instrument used
		동의어	
		타입	
		단위	string
		예시	Mastersizer
A570	solvent (DLS / acquisition condition)	정의	type of solvent used for the sample
		동의어	
		타입	
		단위	string
		예시	water
A571	temperature (DLS / acquisition condition)	정의	temperature during the experiment
		동의어	
		타입	number
		단위	K
		예시	300
A572	scattering angle (DLS / acquisition condition)	정의	scattering angle in degree
		동의어	
		타입	number
		단위	degree
		예시	90
A573	laser wavelength (DLS / acquisition condition)	정의	wavelength of the laser used in nm
		동의어	
		타입	number
		단위	nm
		예시	633
A574	measurement duration (DLS / acquisition condition)	정의	duration of the measurement in seconds
		동의어	
		타입	number
		단위	s
		예시	60

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A575	numbers of runs (DLS / acquisition condition)	정의	number of runs conducted during the measurement
		동의어	
		타입	number
		단위	none
		예시	3
A576	z-average size (DLS / raw data)	정의	the average particle size obtained from DLS in nm
		동의어	
		타입	number
		단위	nm
		예시	150.5
A577	polydispersity index (PDI) (DLS / raw data)	정의	a measure of the distribution of molecular mass in a given polymer sample
		동의어	
		타입	number
		단위	
		예시	0.12
A578	size class (DLS / raw data / size distribution)	정의	particle size class in nm
		동의어	
		타입	number
		단위	nm
		예시	100
A579	intensity percentage (DLS / raw data / size distribution)	정의	intensity percentage for the particle size class
		동의어	
		타입	number
		단위	%
		예시	10
A580	time (DLS / raw data / correlation function)	정의	time in microseconds
		동의어	
		타입	number
		단위	microseconds
		예시	0.1
A581	correlation (DLS / raw data / correlation function)	정의	correlation value
		동의어	
		타입	number
		단위	
		예시	0.95
A582	instrument (DSC)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	DSC-2500
A583	start temperature (DSC / acquisition condition)	정의	starting temperature of the DSC experiment
		동의어	
		타입	number
		단위	K
		예시	300
A584	end temperature (DSC / acquisition condition)	정의	ending temperature of the DSC experiment
		동의어	
		타입	number
		단위	K
		예시	300

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A585	temperature step (DSC / acquisition condition)	정의	temperature increment per step
		동의어	
		타입	number
		단위	K
		예시	1
A586	heating rate (DSC / acquisition condition)	정의	heating rate during the experiment
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
A587	raw data (DSC)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	data.csv; data.xls; data.xlsx; data.txt; data.tds; data.dat; data.spc
A588	geometric surface area (electrochemical activity)	정의	geometric surface area of electrode
		동의어	
		타입	number
		단위	cm^2
		예시	1
A589	method (film adhesion test)	정의	method of adhesion test
		동의어	
		타입	string
		단위	
		예시	scratch test; peel test; tape test (ASTM D3359); pull-off test
A590	equipment (film adhesion test)	정의	equipment used for the adhesion test including brand name
		동의어	
		타입	string
		단위	
		예시	scratch tester TABER 550/551; 3D indentation and scratch tester SMT-5000; universal testing machine ADMET eXpert 7600; tape test kit (ASTM D3359)
A591	condition (film adhesion test)	정의	description of major test conditions
		동의어	
		타입	string
		단위	
		예시	peel speed 1mm/s; Vickers tip; scratching speed 0.1 mm/s;
A592	raw data (film adhesion test)	정의	URI of the raw data obtained by the adhesion test
		동의어	
		타입	string
		단위	
		예시	1.tiff; data.xls
A593	instrument (gel electrophoresis)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Owl Gel electrophoresis system, etc
A594	gel type (gel electrophoresis)	정의	gel types to achieve gel electrophoresis
		동의어	
		타입	string
		단위	
		예시	Agar gel, Polyacrylamide gel(PAGE), SDS polyacrylamide gel(SDS-PAGE)

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A595	gel percentage (gel electrophoresis)	정의	the percent concentration of gel
		동의어	
		타입	number
		단위	%
		예시	1, 3, 5, 10, 20
A596	mode (gel electrophoresis / acquisition condition)	정의	acquisition mode for running gel
		동의어	
		타입	string
		단위	
		예시	constant current, constant voltage, constant power
A597	running time (gel electrophoresis / acquisition condition)	정의	gel electrophoresis running time
		동의어	
		타입	number
		단위	min
		예시	10, 20, 30, 60
A598	raw data (gel electrophoresis)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	tif; jpg
A599	instrument (liquid chromatography)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Agilent Technologies 1200 Series HPLC, Agilent Technologies 1260 Series HPLC
A600	flow rate (liquid chromatography / acquisition condition)	정의	flow rate of liquid to be measured; millimeter per minute
		동의어	
		타입	number
		단위	mL min^{-1}
		예시	1
A601	temperature (liquid chromatography / acquisition condition / column)	정의	measurement of column temperature
		동의어	
		타입	number
		단위	K
		예시	305
A602	column type (liquid chromatography / acquisition condition / column)	정의	type of column used for analysis
		동의어	
		타입	string
		단위	
		예시	YMC-Pack Pro C18, 250 x 4.6 mmI.D. S-5μm, 12nm
A603	injection volume (liquid chromatography / acquisition condition)	정의	injection volume of sample
		동의어	
		타입	number
		단위	mL
		예시	0.01
A604	run time (liquid chromatography / acquisition condition)	정의	measurement time
		동의어	
		타입	number
		단위	min
		예시	30

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A605	wavelength (liquid chromatography / acquisition condition / detector)	정의	wavelength of the used light
		동의어	
		타입	number
		단위	nm
		예시	210
A606	gas (liquid chromatography / acquisition condition / detector)	정의	gas used for measurement
		동의어	
		타입	string
		단위	
		예시	N2
A607	mobile phase solvent ratio (liquid chromatography / acquisition condition)	정의	solvent composition used as mobile phase
		동의어	
		타입	dictionary of {value : {constituent : concentration, … }, unit : unit value, uncertainty : uncertainty value}
		단위	
		예시	{value : {0.1% phosphoric acid : 35.0, Acetonitrile : 65.0}, unit : vol.%, uncertainty : 1.5}
A608	raw data (liquid chromatography)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	csv; pdf;
A609	instrument (peel test)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Ametek CS2
A610	peel angle (peel test / acquisition condition)	정의	angle at which the material is peeled from the surface
		동의어	
		타입	number
		단위	degree
		예시	90; 180
A611	peel speed (peel test / acquisition condition)	정의	rate at which the material is peeled
		동의어	
		타입	number
		단위	mm min^{-1}
		예시	100; 200; 300
A612	min (peel test / acquisition condition / force range)	정의	minimum value of the force range
		동의어	
		타입	number
		단위	N
		예시	0; 10; 30; 50
A613	max (peel test / acquisition condition / force range)	정의	maximum value of the force range
		동의어	
		타입	number
		단위	N
		예시	50; 100; 500
A614	raw data (peel test)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	output.*

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A615	capacity cut (voltammetry / acquisition condition)	정의	the target capacity value to finish the galvanostatic charge-discharge test at specific cycle
		동의어	
		타입	number
		단위	mAh/g
		예시	200
A616	voltage cut (voltammetry / acquisition condition)	정의	the target voltage value to finish the galvanostatic charge-discharge test at specific cycle
		동의어	
		타입	number
		단위	V
		예시	4.3
A617	instrument (white light interferometry)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	CCI-optics; NewView; MicroXAM
A618	magnification (white light interferometry / acquisition condition)	정의	magnification power of the objective lens
		동의어	
		타입	string
		단위	
		예시	2.5x; 5x; 10x; 20x; 50x; 100x
A619	raw data (white light interferometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Surfaces.sdf; Mountains surfaces.sur; Ascii surfaces.txt; KLA Tencor surfaces.map; OpenGPS surfaces.x3p
A620	atmosphere (creep test)	정의	atmosphere in which creep test is carried out
		동의어	
		타입	string
		단위	
		예시	air; hydrogen; nitrogen
A621	instrument (CV measurement)	정의	instrument that was used during CV measurement
		동의어	
		타입	string
		단위	
		예시	Keysight E4980A
A622	frequency level (CV measurement / scan)	정의	frequency used during the measurement of capacitance
		동의어	
		타입	number
		단위	Hz
		예시	1000000
A623	AC drive level (CV measurement / scan)	정의	amplitude of the alternating current (AC) signal applied to a device during the measurement
		동의어	
		타입	number
		단위	V
		예시	0.1
A624	voltage step (CV measurement / scan)	정의	voltage difference between the measurement point in the potential sweep
		동의어	
		타입	number
		단위	mV
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A625	temperature (CV measurement)	정의	temperature of the device during the CV measurement
		동의어	
		타입	number
		단위	K
		예시	300; 500
A626	procedure (CV measurement / stability test)	정의	measurement method and conditions for the stability test
		동의어	
		타입	string
		단위	
		예시	DC drive level; AC drive level; frequency level; duration; time step; atmosphere
A627	permeating species (gas and liquid permeation measurement)	정의	species to permeate through a material
		동의어	
		타입	string
		단위	
		예시	CO_{2}; water
A628	temperature (gas and liquid permeation measurement)	정의	temperature of permeation measurement
		동의어	
		타입	number
		단위	K
		예시	298
A629	applied pressure range (gas and liquid permeation measurement)	정의	applied pressure of permeation measurement
		동의어	
		타입	dictionary of {lower : lower value of the pressure, upper : upper value of the pressure}
		단위	bar
		예시	{lower : 10, upper : 11}
A630	instrument (gas and liquid permeation measurement)	정의	instrument model for permeation measurement
		동의어	
		타입	string
		단위	
		예시	Sterlitech HP4750 – high pressure stirred cell
A631	analysis method (gas and liquid permeation measurement)	정의	analysis method for permeation measurement
		동의어	
		타입	string
		단위	
		예시	crossflow; dead-end
A632	mode (IV measurement / scan)	정의	mode of measurement · potentiostatic : measuring the current between electrodes while controlling the potential (voltage) between them · galvanostatic : monitoring the change in potential (voltage) while controlling the current between electrodes
		동의어	measurement method; characterization technique
		타입	string
		단위	
		예시	1. potentiostatic 2. galvanostatic
A633	voltage range (IV measurement / scan)	정의	voltage range applied in the potentiostatic measurement
		동의어	
		타입	dictionary {min: Vmin, max: Vmax}
		단위	V
		예시	{min:1, max:10}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A634	current range (IV measurement / scan)	정의	current range applied in the galvanostatic measurement
		동의어	
		타입	dictionary {min:Imin, max: Imax}
		단위	mA
		예시	{min:10, max:100}
A635	current step (IV measurement / scan)	정의	current difference between each values during current sweep in the galvanostatic measurement
		동의어	
		타입	number
		단위	mA
		예시	10
A636	current sweep rate (IV measurement / scan)	정의	speed of the current sweep during the galvanostatic measurement
		동의어	speed
		타입	number
		단위	mA s^{-1}
		예시	20; 10
A637	area of electrode (IV measurement / scan)	정의	area of electrode where the current is applied
		동의어	
		타입	number
		단위	cm^{2}
		예시	15
A638	sampling rate (IV measurement / scan)	정의	frequency the data (voltage or current) is recorded during the measurement
		동의어	
		타입	number
		단위	s^{-1}
		예시	10
A639	test standard (tensile test)	정의	internationally recognized standard test method for material characterization (e.g., ASTM, ISO, JIS)
		동의어	
		타입	string
		단위	
		예시	ASTM D3039; ASTM D6000; EUSTM E33498; JIS
A640	instrument (tribotest)	정의	tribotesting equipment used in the test
		동의어	
		타입	string
		단위	
		예시	Anton Paar TRB^3; Bruker UMT TriboLab; PCS Instruments MTM; Rtec Instruments MFT-5000; home-made tester
A641	test type (tribotest / acquisition condition)	정의	type of tribotest
		동의어	
		타입	string
		단위	
		예시	1. pin-on-disk 2. ball-on-disk 3. four-ball test 4. block-on-ring test 5. reciprocating sliding test 6. abrasion test 7. rolling contact fatigue test 8. (user definition)
A642	normal load (tribotest / acquisition condition)	정의	normal load applied to the counterface for tribotest
		동의어	applied load
		타입	number
		단위	N
		예시	3; 10; 100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A643	dimension (tribotest / acquisition condition / counterface)	정의	diameter for pin or ball, diameter and length for roller or cylinder
		동의어	
		타입	string
		단위	
		예시	diameter 10 mm; diameter 1 mm, length 5 mm;
A644	apparent contact area (tribotest / acquisition condition)	정의	aparent contact area; The area might be determined in the case of flat counterface materials
		동의어	
		타입	number
		단위	cm^{2}
		예시	10.3; 53.0
A645	contact pressure (tribotest / acquisition condition)	정의	contact pressure given by normal load divided by apparent contact area
		동의어	
		타입	number
		단위	MPa
		예시	10
A646	sliding speed (tribotest / acquisition condition)	정의	relative sliding speed of counterface materials against test material
		동의어	
		타입	number
		단위	m s^{-1}
		예시	10
A647	sliding distance (tribotest / acquisition condition)	정의	total sliding distance of counterface materials against test material
		동의어	stroke length
		타입	number
		단위	m
		예시	1.00E+04
A648	sliding diameter (tribotest / acquisition condition)	정의	diameter of sliding circle
		동의어	
		타입	number
		단위	cm
		예시	10; 25
A649	stroke amplitude (tribotest / acquisition condition)	정의	stroke amplitude in the reciprocating sliding test
		동의어	
		타입	number
		단위	mm
		예시	10
A650	stroke frequency (tribotest / acquisition condition)	정의	stroke frequency in the reciprocating sliding test
		동의어	
		타입	number
		단위	Hz
		예시	5; 10
A651	atmosphere (tribotest / acquisition condition)	정의	environment for tribotest
		동의어	
		타입	string
		단위	
		예시	vacuum; ambient atmosphere; N2; Ar; space simulation; deep sea simulation; NaCl; SBF; oil
A652	humidity (tribotest / acquisition condition)	정의	average humidity during tribotest
		동의어	
		타입	number
		단위	%
		예시	30; 70

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A653	temperature (tribotest / acquisition condition)	정의	temperature during tribotest
		동의어	
		타입	number
		단위	K
		예시	300
A654	time (tribotest / acquisition condition)	정의	time for tribotest
		동의어	
		타입	number
		단위	min
		예시	10; 120; 1200
A655	raw data (tribotest)	정의	URI of the tribological record during the tribotest; typically a record of friction force vs time
		동의어	
		타입	string
		단위	
		예시	test.txt; test35-1.xcl; space-tribo-1.jpg
A656	analysis method (tribotest)	정의	note on tribometer data analysis
		동의어	
		타입	string
		단위	
		예시	average friction force after initial run-in period was used for the friction coefficient calculation
A657	solvent (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	type of solvent used in the measurement
		동의어	
		타입	string
		단위	
		예시	DMSO; DCM; DI water
A658	analysis method (XRD)	정의	note on XRD data analysis
		동의어	
		타입	string
		단위	
		예시	Sin^{2}(Psi) method; Grazing incidence XRD (GXRD)
A659	type of bias light (EQE analysis / acquisition condition)	정의	type of bias light used during the EQE measurement
		동의어	
		타입	string
		단위	
		예시	Xenon lamp; Halogen lamp; 450nm LED; 800nm LED; white LED; none
A660	measurement mode (EQE analysis / acquisition condition)	정의	the measurement mode used for the EQE measurement
		동의어	measurement method; characterization technique
		타입	string
		단위	
		예시	1. DC-mode 2. AC-mode
A661	external voltage bias (EQE analysis / acquisition condition)	정의	external voltage bias applied during the EQE measurement
		동의어	
		타입	number
		단위	V
		예시	-0.6; 1.2
A662	chopping frequency (EQE analysis / acquisition condition)	정의	the chopping frequency used for the AC-mode EQE measurement
		동의어	
		타입	number
		단위	Hz
		예시	50

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A663	laser spot diameter (Raman spectroscopy / acquisition condition)	정의	diameter of laser beam
		동의어	
		타입	number
		단위	{\mu m}
		예시	50
A664	slit (Raman spectroscopy / acquisition condition)	정의	a narrow gap or opening within a wall or barrier to limit the width of the incident light
		동의어	
		타입	number
		단위	{\mu m}
		예시	30
A665	numerical aperture (Raman spectroscopy / acquisition condition)	정의	a dimensionless number measuring a system's light-gathering and resolving ability, defined as the refractive index times the sine of the half-angle of the maximum light cone, defined as (the refractive index of the medium in which the lens is working) X (sine of the half-angle of the maximum cone of lights that can enter or exit the lens)
		동의어	NA; objective NA
		타입	number
		단위	dimensionless
		예시	0.5
A666	grating groove density (Raman spectroscopy / acquisition condition)	정의	the number of grooves per unit length (typically per millimeter) on a diffraction grating
		동의어	
		타입	number
		단위	mm^{-1}
		예시	1200
A667	crucible (TGA / acquisition condition)	정의	a container used to hold the sample undergoing analysis
		동의어	dish, container, vessel
		타입	string
		단위	
		예시	alumina crucible; quartz plate
A668	temperature (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	analysis temperature
		동의어	
		타입	number
		단위	K
		예시	300
A669	gas (analysis environment)	정의	type of gas environment
		동의어	
		타입	string
		단위	
		예시	ambient air, N2, vacuum, etc.
A670	gas flow rate (analysis environment)	정의	flow rate of the gas used during measurement (if applicable)
		동의어	
		타입	number
		단위	sccm
		예시	50
A671	pressure (analysis environment)	정의	pressure or vacuum level during measurement
		동의어	
		타입	number
		단위	Torr
		예시	760

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A672	relative humidity (analysis environment)	정의	the ratio of the actual water-vapor content of air to the maximum possible at the same temperature
		동의어	
		타입	number
		단위	%
		예시	40
A673	temperature (analysis environment)	정의	temperature during measurement
		동의어	
		타입	number
		단위	K
		예시	300
A674	additional condition (analysis environment)	정의	any additional specific environmental condition applied during measurement
		동의어	
		타입	string
		단위	
		예시	UV irradiation (365 nm, 1 mW/cm^2)
A676	measurement mode (atom probe tomography / acquisition condition)	정의	mode of atom probe tomography measurement
		동의어	
		타입	string
		단위	
		예시	voltage-pulse mode, laser-pulse mode, combined-mode
A677	DC voltage (atom probe tomography / acquisition condition)	정의	the constant standing voltage applied to the sample during APT measurement
		동의어	
		타입	number
		단위	kV
		예시	10
A678	voltage pulse frequency (atom probe tomography / acquisition condition)	정의	frequency or repetition rate of voltage pulses during voltage or combined mode measurement
		동의어	
		타입	number
		단위	kHz
		예시	200
A679	signal-to-noise ratio (atom probe tomography / acquisition condition)	정의	ratio of useful signal intensity to background noise in measured data
		동의어	
		타입	number
		단위	dimensionless
		예시	100
A680	instrument (cryoscopy)	정의	brand name of instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Osmomat 3000; OM-6060 Freezing Point Osmometer; OsmoPRO® Multi-Sample Osmometer
A681	solvent (cryoscopy / acquisition condition)	정의	substance that dissolves solute, forming a solution
		동의어	
		타입	string
		단위	
		예시	water; benzene
A682	material (cryoscopy / acquisition condition / solute)	정의	solute material used
		동의어	
		타입	string
		단위	
		예시	salt; sugar

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A683	mass (cryoscopy / acquisition condition / solute)	정의	amount of matter in an object
		동의어	
		타입	number
		단위	g
		예시	
A684	cooling rate (cryoscopy / acquisition condition)	정의	rate of cooling for freezing measurement
		동의어	
		타입	number
		단위	K s^{−1}
		예시	
A685	temperature range (cryoscopy / acquisition condition)	정의	range of temperature change for the measurement
		동의어	
		타입	dictionary of {Tmin : minimum temperature value, Tmax : maximum temperature value}
		단위	K
		예시	
A686	raw data (cryoscopy)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp
A687	scan rate (CV measurement / scan)	정의	speed of the potential sweep during the CV measurement
		동의어	voltage sweep rate
		타입	number
		단위	mV s^{−1}
		예시	
A688	instrument (dilatometry)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Netzch IL 402C; TA Instrument DIL 850
A689	temperature range (dilatometry / acquisition condition)	정의	range of temperature change for the measurement
		동의어	
		타입	dictionary of {Tmin : minimum temperature value, Tmax : maximum temperature value}
		단위	K
		예시	
A690	settling time (dilatometry / acquisition condition)	정의	settling time at a specific temperature
		동의어	
		타입	number
		단위	s
		예시	
A691	pressure (dilatometry / acquisition condition)	정의	pressure applied to the sample
		동의어	
		타입	number
		단위	N m^{2}
		예시	
A692	sensor (dilatometry / acquisition condition)	정의	sensor for displacement measurement
		동의어	
		타입	string
		단위	
		예시	laser interferometer; strain gauge

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A693	raw data (dilatometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp
A694	sublattice (DFT calculation / magnetic structure / magnetic moment)	정의	sublattice associated to magnetic moment
		동의어	
		타입	string
		단위	
		예시	Mn1; O2
A695	instrument (ebullioscopy)	정의	brand name of instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	home-made; KNAUER vapor pressure and ebullioscopic system; Beckmann Ebullioscope
A696	solvent (ebullioscopy / acquisition condition)	정의	solvent for cryoscopy observation
		동의어	
		타입	string
		단위	
		예시	water; benzene
A697	material (ebullioscopy / acquisition condition / solute)	정의	solute material used
		동의어	
		타입	string
		단위	
		예시	salt; sugar
A698	mass (ebullioscopy / acquisition condition / solute)	정의	mass of solute
		동의어	
		타입	number
		단위	g
		예시	
A699	heating rate (ebullioscopy / acquisition condition)	정의	speed at which temperature increases over time
		동의어	
		타입	number
		단위	K s^{-1}
		예시	
A700	temperature range (ebullioscopy / acquisition condition)	정의	range of temperature change for the measurement
		동의어	
		타입	dictionary of {Tmin : minimum temperature value, Tmax : maximum temperature value}
		단위	K
		예시	
A701	time (ebullioscopy / acquisition condition)	정의	time of equilibration
		동의어	
		타입	number
		단위	h
		예시	
A702	raw data (ebullioscopy)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A703	instrument type (EDS / instrument)	정의	the specific equipment used to acquire EDS data
		동의어	
		타입	string
		단위	
		예시	SEM; TEM
A704	microscopic model (EDS / instrument)	정의	the specific model name of the microscope used for EDS data acquisition
		동의어	
		타입	string
		단위	
		예시	FEI Titan 300; FEI Talos G1; Hitachi High-Tech Regulus 8260
A705	pixel size (EDS / acquisition condition)	정의	physical dimension represented by a single pixel
		동의어	
		타입	number
		단위	nm
		예시	0.15
A706	field of view (EDS / acquisition condition)	정의	total observable area in the horizontal direction of the image, calculated as the product of pixel size and the number of pixels in the horizontal direction
		동의어	
		타입	number
		단위	nm
		예시	615
A707	instrument type (EELS / instrument)	정의	the specific equipment used to acquire EELS data
		동의어	
		타입	string
		단위	
		예시	STEM; TEM
A708	detector model (EELS / instrument)	정의	the specific model name of the detector used for EELS data collection
		동의어	
		타입	string
		단위	
		예시	Gatan Continuum 1069HR; Gatan Quantum 966; Thermo Scientific Zebra
A709	mode (EELS)	정의	EELS data acquisition mode
		동의어	
		타입	string
		단위	
		예시	Spectrum; Map; Line scan; Quantitative analysis
A710	acceleration voltage (EELS / acquisition condition)	정의	acceleration voltage of electron beam generation
		동의어	
		타입	number
		단위	kV
		예시	300
A711	beam current (EELS / acquisition condition)	정의	flow of electrons in the electron beam used
		동의어	
		타입	number
		단위	nA
		예시	4
A712	magnification (EELS / acquisition condition)	정의	ratio between scan area and display size
		동의어	
		타입	number
		단위	
		예시	5000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A713	pixel size (EELS / acquisition condition)	정의	physical dimension represented by a single pixel
		동의어	
		타입	number
		단위	nm
		예시	0.15
A714	field of view (EELS / acquisition condition)	정의	total observable area in the horizontal direction of the image, calculated as the product of pixel size and the number of pixels in the horizontal direction
		동의어	
		타입	number
		단위	nm
		예시	615
A715	acquisition time (EELS / acquisition condition)	정의	data collection time
		동의어	
		타입	number
		단위	s
		예시	10
A716	dispersion channel (EELS / acquisition condition)	정의	a unit of energy that's used to disperse the transmitted electron beam into an energy spectrum
		동의어	
		타입	number
		단위	eV/ch
		예시	0.1
A717	raw data (EELS)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	spc; csv; tif; bmp; jpg; text
A718	synchrotron facility (EXAFS / instrument)	정의	name of the synchrotron light source facility
		동의어	
		타입	string
		단위	
		예시	PAL
A719	monochromator type (EXAFS / acquisition condition)	정의	type of monochromator used to select X-ray energy
		동의어	
		타입	string
		단위	
		예시	Si(111) double crystal monochromator
A720	energy calibration reference (EXAFS / acquisition condition)	정의	standard used to calibrate photon energy scale
		동의어	
		타입	string
		단위	
		예시	Fe foil (K-edge at 7112 eV)
A721	incident angle (EXAFS / acquisition condition)	정의	angle between incident X-ray and sample surface
		동의어	
		타입	number
		단위	degree
		예시	45
A722	element (EXAFS / acquisition condition)	정의	target element being analyzed
		동의어	
		타입	string
		단위	
		예시	Fe

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A723	edge type (EXAFS / acquisition condition)	정의	absorption edge analyzed
		동의어	
		타입	string
		단위	
		예시	K-edge
A724	instrument (freezing point osmometry)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Multiskan FC; Multiskan SkyHigh; Varioskan ALF; Varioskan LUX
A725	material (freezing point osmometry / acquisition condition / solvent)	정의	solvent used
		동의어	
		타입	string
		단위	
		예시	water; benzene
A726	mass (freezing point osmometry / acquisition condition / solvent)	정의	mass of solvent
		동의어	
		타입	number
		단위	g
		예시	
A727	material (freezing point osmometry / acquisition condition / solute)	정의	solute material used
		동의어	
		타입	string
		단위	
		예시	salt; sugar
A728	mass (freezing point osmometry / acquisition condition / solute)	정의	mass of solute
		동의어	
		타입	number
		단위	g
		예시	
A729	cooling rate (freezing point osmometry / acquisition condition)	정의	rate to lower the temperature
		동의어	
		타입	number
		단위	K s^{-1}
		예시	
A730	temperature range (freezing point osmometry / acquisition condition)	정의	range of temperature change for the measurement
		동의어	
		타입	dictionary of {Tmin : minimum temperature value, Tmax : maximum temperature value}
		단위	K
		예시	
A731	raw data (freezing point osmometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp
A732	instrument (GPC)	정의	brand name of instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	GPC-IR; 12660 Infinity II GPC/SEC system

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A733	detector type (GPC / acquisition condition)	정의	detecting type
		동의어	
		타입	string
		단위	
		예시	Refractive Index (RI) detector; UV-VIS detector; Light Scattering detector; Viscosity detector
A734	flow rate (GPC / acquisition condition)	정의	speed at which the mobile phase moves
		동의어	
		타입	number
		단위	mL min^{-1}
		예시	0.3; 1.8
A735	solvent (GPC / acquisition condition)	정의	liquid used to dissolve the polymer sample
		동의어	
		타입	string
		단위	
		예시	DMF; DMSO
A736	raw data (GPC)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp
A737	detector type (liquid chromatography / acquisition condition / detector)	정의	type of detector
		동의어	
		타입	string
		단위	
		예시	1. DAD (Diode Array Detector) 2. ELSD (Evaporative Light Scattering Detector)
A738	instrument (MEIS)	정의	instrument used during measurement
		동의어	
		타입	string
		단위	
		예시	400 kV Ion Beam Accelerator at KIST
A739	ion species (MEIS / acquisition condition)	정의	type of ion used for scattering
		동의어	
		타입	string
		단위	
		예시	H+, He+
A740	ion energy (MEIS / acquisition condition)	정의	energy of incident ion beam
		동의어	
		타입	number
		단위	keV
		예시	100
A741	beam current (MEIS / acquisition condition)	정의	current of the ion beam
		동의어	
		타입	number
		단위	nA
		예시	10
A742	incident angle (MEIS / acquisition condition)	정의	angle between ion beam and sample surface
		동의어	
		타입	number
		단위	degree
		예시	0

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A743	detection angle (MEIS / acquisition condition)	정의	angle between sample surface and detector
		동의어	
		타입	number
		단위	degree
		예시	135
A744	channeling angle (MEIS / acquisition condition)	정의	whether ion beam is aligned to crystal orientation
		동의어	
		타입	string
		단위	
		예시	random, aligned
A745	raw data (MEIS)	정의	URI of the energy spectrum of scattered ions used for depth profile analysis
		동의어	
		타입	string
		단위	
		예시	Scattering spectrum (*.txt)
A746	instrument (membrane osmometry)	정의	brand name of instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Osmomat 090; K-7000 Vapor Pressure Osmometer; OsmoPRO® Multi-Sample Osmometer
A747	membrane type (membrane osmometry / acquisition condition / membrane)	정의	type of membrane
		동의어	
		타입	string
		단위	
		예시	polymer membrane; ceramic membrane; inorganic membrane
A748	molecular weight cut-off (MWCO) (membrane osmometry / acquisition condition / membrane)	정의	maximum molecular weight of solutes that can pass through the membrane
		동의어	
		타입	number
		단위	g mol^{-1}
		예시	
A749	pore size (membrane osmometry / acquisition condition / membrane)	정의	average pore size of the membrane
		동의어	
		타입	number
		단위	nm
		예시	
A750	solvent (membrane osmometry / acquisition condition)	정의	solvent for cryoscopy observation
		동의어	
		타입	string
		단위	
		예시	water; benzene
A751	material (membrane osmometry / acquisition condition / solute)	정의	solute material used
		동의어	
		타입	string
		단위	
		예시	salt; sugar
A752	mass (membrane osmometry / acquisition condition / solute)	정의	mass of solute
		동의어	
		타입	number
		단위	g
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A753	temperature (membrane osmometry / acquisition condition)	정의	temperature during measurement
		동의어	
		타입	number
		단위	K
		예시	300
A754	time (membrane osmometry / acquisition condition)	정의	time of equilibration
		동의어	
		타입	number
		단위	h
		예시	
A755	raw data (membrane osmometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp
A756	synchrotron facility (NEXAFS / instrument)	정의	name of the synchrotron light source facility
		동의어	
		타입	string
		단위	
		예시	PAL
A757	monochromator type (NEXAFS / acquisition condition)	정의	type of monochromator used to select X-ray energy
		동의어	
		타입	string
		단위	
		예시	VLS-PGM
A758	energy calibration reference (NEXAFS / acquisition condition)	정의	standard used to calibrate the photon energy scale
		동의어	
		타입	string
		단위	
		예시	C 1s peak at 284.5 eV from graphite
A759	polarization (NEXAFS / acquisition condition)	정의	polarization of the incident X-ray beam
		동의어	
		타입	string
		단위	
		예시	linear-horizontal
A760	incident angle (NEXAFS / acquisition condition)	정의	angle between incident X-ray and sample surface
		동의어	
		타입	number
		단위	degree
		예시	45
A761	data processing software (NEXAFS)	정의	software used for processing the raw data
		동의어	
		타입	string
		단위	
		예시	ATHENA v0.9.26
A762	beamline name (PES / instrument)	정의	name of the beamline where the measurement was performed (if applicable)
		동의어	
		타입	string
		단위	
		예시	2A MS beamline at PAL

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A763	synchrotron facility (PES / instrument)	정의	name of the synchrotron light source facility (if applicable)
		동의어	
		타입	string
		단위	
		예시	PAL
A764	measurement mode (PES / acquisition condition)	정의	mode of PES measurement depending on photon source
		동의어	
		타입	string
		단위	
		예시	
A765	acceleration voltage (PES / acquisition condition)	정의	High voltage applied to the X-ray source anode to generate characteristic X-rays for photoelectron excitation in XPS analysis. It affects the intensity and penetration of the X-rays
		동의어	
		타입	number
		단위	kV
		예시	
A766	current (PES / acquisition condition)	정의	X-ray source emission current for x-ray generation
		동의어	
		타입	number
		단위	mA
		예시	
A767	dwell time (PES / acquisition condition)	정의	time for signal accumulation
		동의어	
		타입	number
		단위	s
		예시	
A768	sample bias voltage (PES / acquisition condition)	정의	bias voltage applied to the sample holder
		동의어	
		타입	number
		단위	V
		예시	-5
A769	sample temperature (PES / acquisition condition)	정의	temperature of sample during measurement
		동의어	
		타입	number
		단위	K
		예시	
A770	element (PES / acquisition condition)	정의	target element being analyzed
		동의어	
		타입	string
		단위	
		예시	S
A771	edge type (PES / acquisition condition)	정의	absorption edge being measured
		동의어	
		타입	string
		단위	
		예시	2p
A772	scanning range (PES / acquisition condition)	정의	total energy range used for PES data acquisition
		동의어	
		타입	dictionary of {start energy : value, end energy : value}
		단위	eV
		예시	{start energy : 157, end energy : 175}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A773	surface preparation (PES / acquisition condition)	정의	method used to prepare the sample surface
		동의어	
		타입	string
		단위	
		예시	Ar+ sputtering (2 keV, 10 min)
A774	data processing software (PES)	정의	software used for processing the raw data
		동의어	
		타입	string
		단위	
		예시	CasaXPS v2.3.24
A775	instrument (PIXE)	정의	instrument used during measurement
		동의어	
		타입	string
		단위	
		예시	2 MV Pelletron accelerator at KIST
A776	ion species (PIXE / acquisition condition)	정의	type of ion used for excitation
		동의어	
		타입	string
		단위	
		예시	proton
A777	ion energy (PIXE / acquisition condition)	정의	energy of incident ion beam
		동의어	
		타입	number
		단위	MeV
		예시	2
A778	beam current (PIXE / acquisition condition)	정의	current of the ion beam
		동의어	
		타입	number
		단위	nA
		예시	5
A779	detection angle (PIXE / acquisition condition)	정의	angle between ion beam and X-ray detector
		동의어	
		타입	number
		단위	degree
		예시	135
A780	detector type (PIXE / acquisition condition)	정의	type of detector used for X-ray collection
		동의어	
		타입	string
		단위	
		예시	Si(Li), HPGe
A781	filter (PIXE / acquisition condition)	정의	absorber material used in front of the detector
		동의어	
		타입	string
		단위	
		예시	Al 50 μm
A782	quantification method (PIXE / acquisition condition)	정의	method used for quantification of elemental composition
		동의어	
		타입	string
		단위	
		예시	Fundamental Parameter, GUPIX

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A783	raw data (PIXE)	정의	URI of the X-ray spectrum obtained during measurement used for elemental analysis
		동의어	
		타입	string
		단위	
		예시	X-ray spectrum (*.mca)
A784	pixel size (SEM / acquisition condition)	정의	physical dimension represented by a single pixel
		동의어	
		타입	number
		단위	nm
		예시	0.15
A785	field of view (SEM / acquisition condition)	정의	total observable area in the horizontal direction of the image, calculated as the product of pixel size and the number of pixels in the horizontal direction
		동의어	
		타입	number
		단위	nm
		예시	615
A786	instrument (surface profilometry)	정의	Instrument used for measurement
		동의어	
		타입	string
		단위	
		예시	KLC corporation Alpha-Step D-100; Bruker Dektak XT; Taylor Hobson Talysurf i-Series; Veeco Wyko NT series; Keyence VK-X series
A787	scan type (surface profilometry)	정의	scan type for surface profiling
		동의어	
		타입	string
		단위	
		예시	1. contact stylus scanning 2. noncontact optical application 3. hybrid
A788	raw data (surface profilometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.tiff; Optic image.bmp
A789	pixel size (TEM / acquisition condition)	정의	physical dimension represented by a single pixel
		동의어	
		타입	number
		단위	nm
		예시	0.15
A790	field of view (TEM / acquisition condition)	정의	total observable area in the horizontal direction of the image, calculated as the product of pixel size and the number of pixels in the horizontal direction
		동의어	FOV
		타입	number
		단위	nm
		예시	615
A791	sample mass (TGA / acquisition condition)	정의	amount of the sample to be analyzed
		동의어	sample weight
		타입	number
		단위	g
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A792	weight percentage (TGA / acquisition condition)	정의	the percentage of the initial mass of a sample that remains at a specific temperature or time during the analysis. It is calculated by dividing the mass at a given point by the initial mass and multiplying by 100
		동의어	wt.%
		타입	number
		단위	%
		예시	
A793	curvature measurement method (thin film residual stress measurement by curvature)	정의	experimental method to measure the curvature of thin film-substrate composite
		동의어	
		타입	string
		단위	
		예시	laser scanning; stylus scanning; FIB-Imaging
A794	thickness (thin film residual stress measurement by curvature / substrate)	정의	thickness of the substrate
		동의어	
		타입	number
		단위	{\mu m}
		예시	100
A795	biaxial elastic modulus (thin film residual stress measurement by curvature / substrate)	정의	biaxial elastic modulus of the substrate = (elastic modulus of substrate)/{1-(Poisson's ratio)}
		동의어	
		타입	number
		단위	GPa
		예시	180
A796	film thickness (thin film residual stress measurement by curvature)	정의	thickness of thin coated film on the substrate
		동의어	
		타입	number
		단위	nm
		예시	10
A797	curvature (thin film residual stress measurement by curvature)	정의	measured curvature of the thin film coated substrate
		동의어	
		타입	number
		단위	m^{-1}
		예시	0.3
A798	shape (tribotest / acquisition condition / counterface)	정의	shape of the counterface
		동의어	
		타입	string
		단위	
		예시	pin; ball; roller; cylinder
A799	material (tribotest / acquisition condition / counterface)	정의	material of the counterface
		동의어	
		타입	string
		단위	
		예시	GCr15; Al2O3; Si3N4; PEEK
A800	lubrication (tribotest / acquisition condition)	정의	information of lubrication if applied
		동의어	
		타입	string
		단위	
		예시	polyvinyl alcohol; polyacrylic acid; dry
A801	wavelength selection (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	selection of the absorption wavelength suitable for the substance to be measured
		동의어	
		타입	number
		단위	nm
		예시	260, 595, 562

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A802	measurement mode (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	decide on the mode of data collection: * single wavelength : a single-beam UV-Vi-nIR spectrophotometer to measure the absorbance of a sample at a specific wavelength * dual wavelength : a technique that uses two wavelengths of light to measure the concentration of a component in a mixture, often to compensate for interfering substances * full spectrum scan : measuring a sample's absorbance or transmittance across a range of whole wavelengths (typically 190-780 nm) to generate a graph called a UV/Vis spectrum
		동의어	
		타입	string
		단위	
		예시	1. single wavelength 2. dual wavelength 3. full spectrum scan
A805	path length (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	the optical length of the cuvette
		동의어	
		타입	number
		단위	cm
		예시	1
A806	cuvette type (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	choice of material for cuvettes, affecting UV light penetration
		동의어	
		타입	number
		단위	nm
		예시	Quartz cuvette (190-2500 nm); Optical glass cuvette (350-2000 nm); Plastic cuvette (220-900 nm)
A810	specimen form (UV-VIS-nIR spectrophotometer / acquisition condition)	정의	the form of the sample used in the measurement
		동의어	
		타입	string
		단위	
		예시	thin film; powder; pellet; solution
A811	instrument (vapor phase osmometry)	정의	brand name of instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Osmomat 090; K-7000 Vapor Pressure Osmometer; OsmoPRO® Multi-Sample Osmometer
A812	temperature (vapor phase osmometry / acquisition condition)	정의	temperature during measurement
		동의어	
		타입	number
		단위	K
		예시	300
A813	material (vapor phase osmometry / acquisition condition / solvent)	정의	solvent used
		동의어	
		타입	string
		단위	
		예시	water; benzene
A814	mass (vapor phase osmometry / acquisition condition / solvent)	정의	mass of solvent
		동의어	
		타입	number
		단위	g
		예시	
A815	material (vapor phase osmometry / acquisition condition / solute)	정의	solute material used
		동의어	
		타입	string
		단위	
		예시	salt; sugar

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A816	mass (vapor phase osmometry / acquisition condition / solute)	정의	mass of solute
		동의어	
		타입	number
		단위	g
		예시	
A817	time (vapor phase osmometry / acquisition condition)	정의	time of equilibration
		동의어	
		타입	number
		단위	h
		예시	
A818	raw data (vapor phase osmometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp
A819	viscometer type (viscometry / instrument)	정의	type of viscometer
		동의어	
		타입	string
		단위	
		예시	
A820	brand name (viscometry / instrument)	정의	brand name of viscometer
		동의어	
		타입	string
		단위	
		예시	Ubbelohde Viscometer; Cannon-Fenske Viscometer; Anton Paar AMVn Automate Micro Viscometer; Malvern Kinexus Rheometer;
A821	shear rate (viscometry / acquisition condition)	정의	the rate at which a fluid is deformed under shear stress
		동의어	
		타입	number
		단위	s^{-1}
		예시	
A822	temperature (viscometry / acquisition condition)	정의	temperature during measurement
		동의어	
		타입	number
		단위	K
		예시	
A823	raw data (viscometry)	정의	URI of the data obtained during measurement and analysis
		동의어	
		타입	string
		단위	
		예시	Raw data.xls; Optic image.bmp
A824	instrument (voltammetry)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Metrohm Autolab M204; Bio-Logic VSP-300
A825	beamline name (XANES / instrument)	정의	name of the beamline where the measurement was performed
		동의어	
		타입	string
		단위	
		예시	1D XRS_KIST beamline at PAL

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A826	synchrotron facility (XANES / instrument)	정의	name of the synchrotron light source facility
		동의어	
		타입	string
		단위	
		예시	PAL
A827	x-ray source (XANES / acquisition condition)	정의	X-ray source used in measurement
		동의어	
		타입	string
		단위	
		예시	5 keV bending magnet
A828	monochromator type (XANES / acquisition condition)	정의	type of monochromator used to select X-ray energy
		동의어	
		타입	string
		단위	
		예시	Si(111) double crystal monochromator
A829	energy calibration reference (XANES / acquisition condition)	정의	standard used to calibrate the photon energy scale
		동의어	
		타입	string
		단위	
		예시	Fe foil (7112 eV)
A830	polarization (XANES / acquisition condition)	정의	polarization of the incident X-ray beam
		동의어	
		타입	string
		단위	
		예시	linear-horizontal
A831	incident angle (XANES / acquisition condition)	정의	angle between incident X-ray and sample surface
		동의어	
		타입	number
		단위	degree
		예시	45
A832	beam size (XANES / acquisition condition)	정의	X-ray beam diameter at the sample position
		동의어	
		타입	number
		단위	μm
		예시	500
A833	scanning step (XANES / acquisition condition)	정의	energy interval between each data point
		동의어	
		타입	number
		단위	eV
		예시	0.2
A834	data collection mode (XANES / acquisition condition)	정의	detection mode for XANES data collection
		동의어	
		타입	string
		단위	
		예시	transmission; fluorescence
A835	element (XANES / acquisition condition)	정의	target element being analyzed
		동의어	
		타입	string
		단위	
		예시	Fe

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A836	edge type (XANES / acquisition condition)	정의	absorption edge being measured
		동의어	
		타입	string
		단위	
		예시	K-edge
A837	pre-edge region (XANES / acquisition condition)	정의	energy range below the absorption edge analyzed for pre-edge features
		동의어	
		타입	string
		단위	
		예시	-20 to 0 eV
A838	edge position (XANES / acquisition condition)	정의	exact photon energy of absorption edge used for analysis
		동의어	
		타입	number
		단위	eV
		예시	7112
A839	scanning range (XANES / acquisition condition)	정의	total energy range used for PES data acquisition
		동의어	
		타입	dictionary of {start energy : value, end energy : value}
		단위	eV
		예시	{start energy : 157, end energy : 175}
A840	raw data (XANES)	정의	URI of the raw absorption spectrum obtained during measurement
		동의어	
		타입	string
		단위	
		예시	XANES_Fe_Kedge.dat
A841	data processing software (XANES)	정의	software used for processing the raw data
		동의어	
		타입	string
		단위	
		예시	ATHENA v0.9.26
A842	specimen form (XRD)	정의	the form of the sample used in the measurement
		동의어	type of samples, type of specimen
		타입	string
		단위	
		예시	thin film; powder; pellet
A843	full width at half maximum (XRD)	정의	the width of an X-ray diffraction peak measured at half of its maximum intensity
		동의어	FWHM
		타입	number
		단위	degree
		예시	
A844	instrument name (XRS / instrument)	정의	name of the instrument or diffractometer used for measurement
		동의어	
		타입	string
		단위	
		예시	Rigaku SmartLab
A845	beamline name (XRS / instrument)	정의	name of the beamline where the measurement was performed (if applicable)
		동의어	
		타입	string
		단위	
		예시	4C SAXS/WAXS beamline at PAL

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A846	synchrotron facility (XRS / instrument)	정의	name of the synchrotron light source facility (if applicable)
		동의어	
		타입	string
		단위	
		예시	PAL
A847	mode (XRS / acquisition condition)	정의	mode of measurement
		동의어	
		타입	string
		단위	
		예시	1. SAXS: Small-Angle X-ray Scattering mode for analyzing nano-scale structural features 2. WAXS: Wide-Angle X-ray Scattering mode for analyzing atomic-scale structural information
A848	x-ray source (XRS / acquisition condition)	정의	X-ray source used in measurement
		동의어	
		타입	string
		단위	
		예시	10 keV bending magnet
A849	detector type (XRS / acquisition condition / detector)	정의	type of detector used
		동의어	
		타입	string
		단위	
		예시	CCD
A850	sample-to-detector distance (XRS / acquisition condition)	정의	distance between sample and detector
		동의어	SDD
		타입	number
		단위	m
		예시	2.5
A851	beam size (XRS / acquisition condition)	정의	X-ray beam diameter at the sample position
		동의어	
		타입	number
		단위	\mu m
		예시	100
A852	exposure time (XRS / acquisition condition)	정의	exposure time for data collection
		동의어	
		타입	number
		단위	s
		예시	60
A853	raw data (XRS)	정의	URI of the raw scattering data obtained during SAXS measurement
		동의어	
		타입	string
		단위	
		예시	SAXS_2Dpattern.tif
A854	data processing software (XRS)	정의	software used for processing the XRS raw data
		동의어	
		타입	string
		단위	
		예시	ATSAS
A855	dipole moment (DFT calculation / solvent model / parameters)	정의	dipole moment of the solute
		동의어	
		타입	number
		단위	D
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A856	closure (DFT calculation / solvent model / parameters)	정의	a mathematical condition ensuring that the relationship between the solvent's direct and total correlation functions is satisfied; fundamental equation that ensures the consistency of the model when describing the distribution of solvent molecules around a solute; crucial for calculating the density distribution of solvent molecules around a solute
		동의어	
		타입	string
		단위	
		예시	KH (Kovalenko-Hirata); HNC (hypernetted chain); PSEn (partial series expansion upto the order n); HGK (Kobryn-Gusarobv-Kovalenko)
A857	box size (DFT calculation / solvent model / parameters)	정의	box size for solvent
		동의어	
		타입	dictionary of {Lx : value, Ly : value, Lz : value}
		단위	\ANGSTROM
		예시	
A858	grid spacing (DFT calculation / solvent model / parameters)	정의	the resolution of the spatial grid where correlation functions are calculated
		동의어	
		타입	dictionary of {dx : value, dy : value, zz : value}
		단위	\ANGSTROM
		예시	
A859	solvent temperature (DFT calculation / solvent model / parameters)	정의	temperature of solvent
		동의어	
		타입	number
		단위	K
		예시	
A860	species (DFT calculation / solvent model / parameters / density)	정의	species in solvent
		동의어	
		타입	string
		단위	
		예시	water; Na+, Cl-
A861	density (DFT calculation / solvent model / parameters / density)	정의	density of each species in solvent
		동의어	
		타입	number
		단위	\ANGSTROM^{-3}
		예시	
A862	species (DFT calculation / solvent model / parameters / Van der Waals parameters)	정의	species in solvent
		동의어	
		타입	string
		단위	
		예시	water; Na+, Cl-
A863	name (DFT calculation / solvent model / parameters / Van der Waals parameters)	정의	name of force field, if exists
		동의어	
		타입	string
		단위	
		예시	OPLS-AA; SPC
A864	model (DFT calculation / solvent model / parameters / Van der Waals parameters)	정의	VdW model
		동의어	
		타입	string
		단위	
		예시	12-6 Lennard-Jones potential; 9-6 Lennard-Jones potential

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A865	sigma (DFT calculation / solvent model / parameters / Van der Waals parameters)	정의	sigma value of the LJ potential model
		동의어	
		타입	number
		단위	none
		예시	
A866	epsilon (DFT calculation / solvent model / parameters / Van der Waals parameters)	정의	epsilon value of the LJ potential model
		동의어	
		타입	number
		단위	none
		예시	
A867	mixing rule (DFT calculation / solvent model / parameters / Van der Waals parameters)	정의	mixing rule
		동의어	
		타입	string
		단위	
		예시	arithmetic; geometric
A868	algorithm (DFT calculation / solvent model / parameters / iterative scheme)	정의	algorithm for iteration
		동의어	
		타입	string
		단위	
		예시	MDIIS
A869	convergence condition (DFT calculation / solvent model / parameters / iterative scheme)	정의	convergence criteria
		동의어	
		타입	number
		단위	eV
		예시	10^{-5}
A870	instrument (amperometry)	정의	instrument that was used during measurement
		동의어	
		타입	string
		단위	
		예시	Ultimate 3000; Mettler-Toledo PerfectION
A871	maximum scan size (AFM / instrument)	정의	the largest surface area that can be scanned by the device
		동의어	scan area limit
		타입	number
		단위	mm^{2}
		예시	100
A881	creep test time (creep test)	정의	the time at which the test is stopped before the material fractures during a creep test
		동의어	interrupt time
		타입	number
		단위	h
		예시	1000
A890	amplitude (fatigue test / temperature range)	정의	one half of the temperature range
		동의어	
		타입	number
		단위	K
		예시	100
A891	maximum (fatigue test / temperature range)	정의	the maximum temperature reached during fatigue test
		동의어	
		타입	number
		단위	K
		예시	200

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A892	minimum (fatigue test / temperature range)	정의	the minimum temperature reached during fatigue test
		동의어	
		타입	number
		단위	K
		예시	100
A893	maximum (fatigue test / plastic strain range)	정의	the maximum strain reached during a fatigue test
		동의어	max deformation
		타입	number
		단위	%
		예시	20
A894	minimum (fatigue test / plastic strain range)	정의	the minimum strain reached during a fatigue test
		동의어	min deformation
		타입	number
		단위	%
		예시	20
A895	maximum (fatigue test / stress range)	정의	the maximum stress reached during fatigue test
		동의어	max load
		타입	number
		단위	MPa
		예시	200
A896	minimum (fatigue test / stress range)	정의	the minimum stress reached during fatigue test
		동의어	min load
		타입	number
		단위	MPa
		예시	200
A897	notch (fatigue test)	정의	indicates the presence or absence of a notch in the materials or specimen
		동의어	
		타입	boolean
		단위	
		예시	true, false
A898	fatigue test time (fatigue test)	정의	time duration of the fatigue test
		동의어	
		타입	number
		단위	h
		예시	300
A926	beam dose (TEM / acquisition condition)	정의	total amount of electrons incident on a unit area of a specimen during electron beam irradiation; It quantifies the cumulative exposure of the sample to the electron beam and is a critical parameter determining image quality, signal-to-noise ratio, and radiation-induced damage in electron microscopy. Mathematically expressed as: D = It/Ae where I is the beam current (A), t is the exposure time (s), A is the irradiated area (m^2), and e is the elementary charge (1.602×10-19 C).
		동의어	electron dose; irradiation dose; dose density
		타입	number
		단위	nm^{-2}
		예시	
A933	geometry (viscometry / instrument)	정의	geometrical configuration of measuring system in viscometer (rheometer0 used to apply shear or oscillation
		동의어	
		타입	string
		단위	
		예시	cone-plate; plate-plate; concentric cylinder

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
A934	angular frequency (viscometry / acquisition condition)	정의	rate of oscillation applied during dynamic theological test
		동의어	
		타입	number
		단위	degree s^{-1}
		예시	
A935	strain amplitude (viscometry / acquisition condition)	정의	maximum strain applied in oscillatory shear experiment
		동의어	
		타입	number
		단위	%
		예시	
A936	shear stress (viscometry / acquisition condition)	정의	the stress component parallel to the material surface, resulting from applied stress
		동의어	
		타입	number
		단위	Pa
		예시	10
A937	cycle (voltammetry / acquisition condition)	정의	repeating numbers of complete potential scan from the initial potential to the designated vertex (switching) potential and then back to the initial potential, encompassing both the forward and reverse sweeps
		동의어	
		타입	number
		단위	none
		예시	
A938	x-ray wavelength (XRS / acquisition condition)	정의	wavelength of used x-ray
		동의어	
		타입	number
		단위	\ANGSTROM
		예시	
A939	incidence angle (XRS / acquisition condition)	정의	angle between the incident X-ray beam and the sample surface
		동의어	grazing angle, incident angle
		타입	number
		단위	degree
		예시	
A940	beam position (XRS / acquisition condition)	정의	beam position on 2D detector surface as [pixel position in x direction of detector matrix, pixel position in y direction of detector matrix]
		동의어	
		타입	numeric array
		단위	none
		예시	[500, 400]
A941	pixel size (XRS / acquisition condition / detector)	정의	dimensions of individual detector elements (pixels) as ["pixel size in x direction", "pixel size in y direction"]
		동의어	
		타입	numeric array
		단위	{\mu m}
		예시	[78, 78]
A942	detector matrix (XRS / acquisition condition / detector)	정의	detector dimensions in pixels as ["pixel number in x", "pixel number in y"]
		동의어	
		타입	numeric array
		단위	none
		예시	[1024, 1024]

5 소재 공정 공통어휘 세부내용

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P1	rotation speed (centrifugation)	정의	rotating speed for centrifugation process
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	6000
P2	time (centrifugation)	정의	Time of the centrifugation process
		동의어	
		타입	number
		단위	h
		예시	0.75
P3	temperature (centrifugation)	정의	Temperature of the centrifugation process
		동의어	
		타입	number
		단위	K
		예시	298
P4	name (centrifugation / additive)	정의	name of additive
		동의어	
		타입	string
		단위	
		예시	HCON(CH3)2
P5	amount (centrifugation / additive)	정의	amount of additive
		동의어	
		타입	number
		단위	L
		예시	20
P6	slurry concentration (CMP)	정의	slurry concentration by weight can be measured by evaporating a known weight of slurry – and measure the weight of dried solids
		동의어	
		타입	number
		단위	%
		예시	30
P7	slurry density (CMP)	정의	the ratio of sum of the weight of water and additives to the volume of cement slurry
		동의어	
		타입	number
		단위	kg m^{-3}
		예시	1220
P8	viscosity (CMP)	정의	how easily a liquid flows and is a measure of the fluid's resistance to gradual deformation by shear forces
		동의어	
		타입	number
		단위	N s m^{-2}
		예시	1
P9	polishing rate (CMP)	정의	percentage remaining after polishing
		동의어	
		타입	number
		단위	%
		예시	40

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P10	surface roughness (CMP)	정의	component of surface texture
		동의어	
		타입	number
		단위	{μ m}
		예시	12
P11	name (CMP / precursor)	정의	name of precursor for material_(n)
		동의어	
		타입	string
		단위	
		예시	PtCl4
P12	amount (CMP / precursor)	정의	amount of precursor for material_(n)
		동의어	
		타입	number
		단위	mg
		예시	50
P13	temperature (CMP)	정의	temperature of the process
		동의어	
		타입	number
		단위	K
		예시	298
P14	atmosphere (CMP)	정의	environment to which a given process is exposed
		동의어	process environment
		타입	string
		단위	
		예시	Ar; $N_{2}+5H_{2}$; inert; reducing; oxidizing
P15	rpm (CMP)	정의	rounds per minute for process
		동의어	
		타입	numeric
		단위	revolutions min^{-1}
		예시	0.75
P16	name (chemical synthesis / precursor)	정의	name of precursor for material_(n)
		동의어	
		타입	string
		단위	
		예시	PtCl4
P17	amount (chemical synthesis / precursor)	정의	amount of precursor
		동의어	
		타입	number
		단위	mg
		예시	64
P19	solution (chemical synthesis)	정의	solution of chemical synthesis
		동의어	
		타입	string
		단위	
		예시	CaCl2; Na2HPO4; polyelectrolyte
P20	temperature (chemical synthesis)	정의	temperature for chemical synthesis
		동의어	
		타입	number
		단위	K
		예시	673

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P31	temperature (drying)	정의	temperature of the process
		동의어	process environment
		타입	number
		단위	K
		예시	300
P32	atmosphere (drying)	정의	environment in which a process is executed
		동의어	process environment
		타입	string
		단위	
		예시	Ar; N2+5%H2; inert; reducing; oxidizing
P33	time (drying)	정의	time of the process
		동의어	drying time
		타입	number
		단위	h
		예시	0.75
P34	acceleration voltage (e-beam lithography)	정의	electric potential used to accelerate electrons
		동의어	
		타입	number
		단위	kV
		예시	200
P35	name (e-beam lithography / e-beam resist)	정의	name of e-beam resist
		동의어	
		타입	string
		단위	
		예시	PMMA A4
P36	temperature (e-beam lithography / e-beam resist / baking)	정의	baking temperature
		동의어	
		타입	number
		단위	K
		예시	393
P37	time (e-beam lithography / e-beam resist / baking)	정의	baking time
		동의어	
		타입	number
		단위	s
		예시	120
P38	thickness (e-beam lithography / e-beam resist)	정의	thickness of the e-beam resist
		동의어	
		타입	number
		단위	nm
		예시	50
P39	working distance (e-beam lithography)	정의	distance between the sample and the final pole piece
		동의어	
		타입	number
		단위	mm
		예시	6
P40	beam current (e-beam lithography)	정의	beam current used in Electron-beam lithography
		동의어	
		타입	number
		단위	pA
		예시	60

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P41	dwell time (e-beam lithography)	정의	time that electron beam remains fixed at a single point
		동의어	
		타입	number
		단위	ms
		예시	0.002
P42	step size (e-beam lithography)	정의	the distance between two adjacent points along x axis
		동의어	
		타입	number
		단위	nm
		예시	6
P43	line spacing (e-beam lithography)	정의	the distance between two adjacent points along y axis
		동의어	
		타입	number
		단위	nm
		예시	6
P44	area dose (e-beam lithography)	정의	the quantity of electric charge per square centimeters
		동의어	
		타입	number
		단위	{\mu C} cm^{-2}
		예시	350
P45	developer (e-beam lithography / develop)	정의	a chemical that removes unnecessary areas with high selectivity
		동의어	
		타입	string
		단위	
		예시	MIBK/IPA_1:3
P46	time (e-beam lithography / develop)	정의	develop time
		동의어	
		타입	number
		단위	s
		예시	120
P47	name (electrochemical deposition / electrolyte / precursor)	정의	precursor name
		동의어	
		타입	string
		단위	
		예시	CuCl_{2}
P48	amount (electrochemical deposition / electrolyte / precursor)	정의	amount of precursor for the electrochemical deposition
		동의어	
		타입	number
		단위	g
		예시	2
P49	solvent (electrochemical deposition / electrolyte)	정의	solvent of electrolyte solution
		동의어	
		타입	string
		단위	
		예시	water
P50	concentration (electrochemical deposition / electrolyte)	정의	concentration of electrolyte solution
		동의어	
		타입	number
		단위	mol
		예시	1.5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P51	pH (electrochemical deposition / electrolyte)	정의	a measure of acidity of electrolyte solution
		동의어	hydrogen concentration
		타입	number
		단위	none
		예시	2
P52	additive (electrochemical deposition)	정의	substances added for a specific purpose
		동의어	
		타입	string
		단위	
		예시	brightener; pH controller; stabilizer; complexing agent
P53	working electrode (electrochemical deposition)	정의	electrode on which reduction reactions occur
		동의어	substrate; cathode
		타입	string
		단위	
		예시	Au; Ni; Cu
P54	counter electrode (electrochemical deposition)	정의	electrode on which counter reactions occur
		동의어	anode
		타입	string
		단위	
		예시	Pt; Ni; Cu
P55	reference electrode (electrochemical deposition)	정의	electrode with reference potential
		동의어	
		타입	string
		단위	
		예시	Ag/AgCl; calomel; SHE
P56	mode (electrochemical deposition / deposition condition)	정의	method of applying power
		동의어	
		타입	array
		단위	
		예시	1. CV (constant voltage or potentiostatic) 2. CC (constant current or galvanostatic) 3. pulse (potentiodynamic)
P57	voltage (electrochemical deposition / deposition condition)	정의	voltage applied to working electrode
		동의어	reduction potential; cathodic overpotential; deposition potential
		타입	number
		단위	V
		예시	-0.7
P58	current density (electrochemical deposition / deposition condition)	정의	current density at working electrode
		동의어	
		타입	number
		단위	mA cm^{-2}
		예시	-0.1
P59	stirring (electrochemical deposition)	정의	the process of mixing an electrolyte to maintain the ionic concentration constant during the deposition process
		동의어	agitation; mixing
		타입	numeric
		단위	revolutions min^{-1}
		예시	300
P60	temperature (electrochemical deposition)	정의	process temperature
		동의어	
		타입	number
		단위	K
		예시	298

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P61	atmosphere (electrochemical deposition)	정의	atmosphere to which the electrolyte is exposed during electrochemical deposition
		동의어	process environment
		타입	string
		단위	
		예시	N_{2}; Ar; air
P62	time (electrochemical deposition)	정의	process time
		동의어	
		타입	number
		단위	s
		예시	3600
P110	name (microwave-assisted synthesis / precursor)	정의	name of precursor for the microwave-assisted process
		동의어	
		타입	string
		단위	
		예시	PtCl4
P111	amount (microwave-assisted synthesis / precursor)	정의	amount of precursor for the microwave-assisted process
		동의어	
		타입	number
		단위	mg
		예시	64
P112	name (microwave-assisted synthesis / solution)	정의	name of solution for the microwave-assisted process
		동의어	
		타입	string
		단위	
		예시	HCON(CH3)2
P113	amount (microwave-assisted synthesis / solution)	정의	amount of solution for the microwave-assisted process
		동의어	
		타입	number
		단위	mL
		예시	20
P114	temperature (microwave-assisted synthesis)	정의	temperature of the microwave-assisted process
		동의어	
		타입	number
		단위	K
		예시	298
P115	atmosphere (microwave-assisted synthesis)	정의	atmosphere environment for the microwave-assisted process
		동의어	
		타입	string
		단위	
		예시	Ar; N_{2}+5H_{2}; inert; reducing; oxidizing
P116	time (microwave-assisted synthesis)	정의	time of the microwave-assisted process
		동의어	
		타입	number
		단위	h
		예시	0.75
P117	name (molecular beam epitaxy / beam / source)	정의	source name
		동의어	
		타입	string
		단위	
		예시	Al; In; Ga; As; Sb; P; Be; Si; GaTe

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P118	amount (molecular beam epitaxy / beam / source)	정의	source amount
		동의어	
		타입	number
		단위	g
		예시	30
P119	substrate temperature (molecular beam epitaxy)	정의	substrate temperature at which epitaxial layers are grown by MBE
		동의어	
		타입	number
		단위	K
		예시	500; 600
P120	substrate orientation (molecular beam epitaxy)	정의	a crystallographic orientation of substrate that influences the condensation coefficients of adatoms on substrate
		동의어	
		타입	number
		단위	
		예시	(100); (111)
P121	substrate rotation rate (molecular beam epitaxy)	정의	substrate rotation speed to improve interface sharpness and thickness uniformity of epitaxial layers
		동의어	
		타입	number
		단위	revolution min^{-1}
		예시	10
P122	growth chamber pressure (molecular beam epitaxy)	정의	a growth chamber pressure at which epitaxial growth occurs
		동의어	
		타입	number
		단위	Pa
		예시	1.00E-08
P123	flux (molecular beam epitaxy / beam)	정의	molecular beam flux of MBE source reaching a substrate
		동의어	
		타입	number
		단위	s^{-1}
		예시	3.5e-7; 2.5e-6
P125	growth rate (molecular beam epitaxy)	정의	growth rate at which epitaxial layers are grown
		동의어	
		타입	number
		단위	\ANGSTROM s^{-1}
		예시	3
P126	growth time (molecular beam epitaxy)	정의	time spent for the entire MBE growth process
		동의어	
		타입	number
		단위	s
		예시	10800
P127	growth interruption time (molecular beam epitaxy)	정의	time interval before a heterointerface started to grow
		동의어	
		타입	number
		단위	s
		예시	2; 5
P128	load (polishing)	정의	the applied load in polishing
		동의어	
		타입	number
		단위	N
		예시	50

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P129	material removal rate (polishing)	정의	the amount of material removed per time unit
		동의어	MRR
		타입	number
		단위	cm^{3} s^{-1}
		예시	3600
P130	contact stress (polishing)	정의	localized stresses that develop as two curved surfaces come in contact and deform slightly under the imposed loads
		동의어	
		타입	number
		단위	N m^{-2}
		예시	15
P131	relative velocity (polishing)	정의	relative velocity between the wafer and polishing pad
		동의어	
		타입	number
		단위	m min^{-1}
		예시	10
P132	Preston's coefficient (polishing)	정의	the dependence of the removal rate with sample composition
		동의어	
		타입	string
		단위	
		예시	K
P133	temperature (polishing)	정의	temperature of the process
		동의어	
		타입	number
		단위	K
		예시	298
P134	atmosphere (polishing)	정의	environment to which a given process is exposed
		동의어	process environment
		타입	string
		단위	
		예시	Ar; N_{2}+5H_{2}; inert; reducing; oxidizing
P135	rpm (polishing)	정의	rounds per minute for process
		동의어	
		타입	numeric
		단위	revolutions min^{-1}
		예시	500
P136	laser source (pulsed laser deposition / laser)	정의	laser source used for a PLD process
		동의어	
		타입	string
		단위	
		예시	Nd:YAG, KrF
P137	wavelength (pulsed laser deposition / laser)	정의	wavelength of a pulsed laser used for a PLD process
		동의어	
		타입	number
		단위	nm
		예시	256
P138	frequency (pulsed laser deposition / laser)	정의	frequency of a pulsed laser used for a PLD process
		동의어	
		타입	number
		단위	Hz
		예시	30

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P139	fluence (pulsed laser deposition / laser)	정의	energy of a pulsed laser divided by its illumination area
		동의어	energy intensity
		타입	number
		단위	J cm^{2}
		예시	5
P141	substrate temperature (pulsed laser deposition)	정의	temperature of the substrate
		동의어	
		타입	number
		단위	K
		예시	300
P142	atmosphere (pulsed laser deposition)	정의	environment to which a given process is exposed
		동의어	process environment
		타입	string
		단위	
		예시	Ar; vacuum; inert; reducing; oxidizing
P143	time (pulsed laser deposition)	정의	duration of the PLD process
		동의어	
		타입	number
		단위	s
		예시	75
P145	temperature (rinsing)	정의	Temperature of the rinsing process
		동의어	
		타입	number
		단위	K
		예시	300
P146	time (rinsing)	정의	time of the rinsing process
		동의어	
		타입	number
		단위	h
		예시	0.75
P147	name (rinsing / additive)	정의	name of additive
		동의어	
		타입	string
		단위	
		예시	ethanol
P148	amount (rinsing / additive)	정의	amount of additive
		동의어	
		타입	number
		단위	L
		예시	20
P156	atmosphere (sintering)	정의	environment to which a given process is exposed
		동의어	process environment
		타입	string
		단위	
		예시	vacuum; H2
P157	temperature (sintering)	정의	selected temperature to which temperature is raised, lowered or isothermally maintained
		동의어	
		타입	number
		단위	K
		예시	2000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P158	time (sintering)	정의	selected time to which temperature is raised, lowered or isothermally maintained
		동의어	
		타입	number
		단위	min
		예시	1000
P159	heating rate (sintering)	정의	rate at which temperature is raised
		동의어	
		타입	number
		단위	K s^{-1}
		예시	10
P160	cooling rate (sintering)	정의	rate at which temperature is lowered
		동의어	
		타입	number
		단위	K s^{-1}
		예시	10
P161	pressure applied (sintering)	정의	amount of pressure applied to materials at constant temperature or during cooling
		동의어	
		타입	number
		단위	MPa
		예시	50
P168	name (solvothermal synthesis / precursor)	정의	name of the precursor
		동의어	
		타입	string
		단위	
		예시	PtCl4
P169	amount (solvothermal synthesis / precursor)	정의	amount of the precursor
		동의어	
		타입	number
		단위	mg
		예시	64
P170	name (solvothermal synthesis / solvent)	정의	name of the solvent
		동의어	
		타입	string
		단위	
		예시	HCON(CH3)2
P171	amount (solvothermal synthesis / solvent)	정의	amount of the solvent
		동의어	
		타입	number
		단위	mL
		예시	20
P172	name (solvothermal synthesis / reducing agent)	정의	name of the reducing agent
		동의어	
		타입	string
		단위	
		예시	NaBH4
P173	amount (solvothermal synthesis / reducing agent)	정의	amount of the reducing agent
		동의어	
		타입	number
		단위	mg
		예시	20

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P174	name (solvothermal synthesis / surfactant)	정의	name of the surfactant
		동의어	
		타입	string
		단위	
		예시	CH3(CH2)7CH=CH(CH2)7CH2NH2
P175	amount (solvothermal synthesis / surfactant)	정의	amount of the surfactant
		동의어	
		타입	number
		단위	mg
		예시	20
P176	temperature (solvothermal synthesis)	정의	temperature for material synthesis
		동의어	
		타입	number
		단위	K
		예시	160
P177	time (solvothermal synthesis)	정의	time for material synthesis
		동의어	heat time
		타입	number
		단위	h
		예시	12
P178	temperature (sol-gel synthesis)	정의	temperature for sol-gel synthesis
		동의어	
		타입	number
		단위	K
		예시	1173
P180	time (sol-gel synthesis)	정의	time of the sol-gel process
		동의어	
		타입	number
		단위	h
		예시	512
P181	composition (sonication)	정의	materials composition for sonication
		동의어	
		타입	dictionary of {value : {constituent : concentration, …}, unit : unit value}
		단위	mL
		예시	{value : {material_01 : 20, material_02 : 30}, unit : mL}
P182	solvent (sonication)	정의	name of solvent for sonication
		동의어	
		타입	string
		단위	
		예시	toluene
P183	temperature (sonication)	정의	sonication temperature
		동의어	
		타입	number
		단위	K
		예시	300
P184	time (sonication)	정의	sonication time
		동의어	
		타입	number
		단위	h
		예시	2.4

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P185	name (sonochemical synthesis / precursor)	정의	name of precursor for the sonochemical process
		동의어	
		타입	string
		단위	
		예시	PtCl4
P186	amount (sonochemical synthesis / precursor)	정의	amount of precursor for the sonochemical process
		동의어	
		타입	number
		단위	mg
		예시	64
P187	name (sonochemical synthesis / solvent)	정의	name of solution for the sonochemical process
		동의어	
		타입	string
		단위	
		예시	HCON(CH3)2
P188	amount (sonochemical synthesis / solvent)	정의	amount of solution for the sonochemical process
		동의어	
		타입	number
		단위	mL
		예시	20
P189	temperature (sonochemical synthesis)	정의	temperature of the sonochemical process
		동의어	
		타입	number
		단위	K
		예시	298
P190	ultrasonic frequency (sonochemical synthesis)	정의	ultrasonic frequency for the sonochemical method
		동의어	
		타입	number
		단위	kHz
		예시	37
P191	time (sonochemical synthesis)	정의	time of sonication for the sonochemical method
		동의어	
		타입	number
		단위	h
		예시	2
P192	base pressure (sputter deposition)	정의	the lowest pressure of the processing chamber to pump down without any gas flows
		동의어	
		타입	number
		단위	Torr
		예시	0.000001
P193	working pressure (sputter deposition)	정의	the pressure in the processing chamber during the sputter deposition process
		동의어	processing pressure; sputtering pressure
		타입	number
		단위	Torr
		예시	0.001
P194	sputtering gas (sputter deposition)	정의	flow rate of sputter gas that form ionized particles in an electric field for sputtering process
		동의어	processing pressure; sputtering pressure
		타입	dictionary of {composition : {constituent : concentration, … }, unit : unit value}
		단위	cm^{3} min^{-1}
		예시	{composition : {He: 50, Ar : 100}, unit : SCCM}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P195	substrate distance (sputter deposition)	정의	the distance between the target and the substrate
		동의어	working distance
		타입	number
		단위	mm
		예시	60
P196	target (sputter deposition)	정의	sputter target materials
		동의어	
		타입	string
		단위	
		예시	Au, Pt, Ti, Cr, Cu
P197	sputter power (sputter deposition / sputter power)	정의	electric power for the sputtering process
		동의어	
		타입	number
		단위	W
		예시	60
P198	power source type (sputter deposition / sputter power)	정의	type of power source
		동의어	
		타입	string
		단위	
		예시	1. DC (direct current) 2. RF (radio frequency) 3. pulsed DC 4. high power impulse
P199	time (sputter deposition)	정의	time of sputter deposition process
		동의어	coating time; sputtering time
		타입	number
		단위	s
		예시	120
P200	substrate temperature (sputter deposition)	정의	temperature of the substrate
		동의어	
		타입	number
		단위	K
		예시	300
P201	atmosphere (thermomechanical process)	정의	environment to which a given process is exposed
		동의어	process environment
		타입	string
		단위	
		예시	Ar, N_{2}+5H_{2}, inert, reducing, oxidizing
P202	initial temperature (thermomechanical process)	정의	initial temperature at which the TMP process starts
		동의어	
		타입	number
		단위	K
		예시	403
P203	heating rate (thermomechanical process / TMC cycles)	정의	rate at which temperature is raised to holding(or target) temperature
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P204	holding temperature (thermomechanical process / TMC cycles)	정의	soaking temperature maintained during the isothermal heat treatment
		동의어	
		타입	number
		단위	K
		예시	403

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P205	deformation holding (thermomechanical process / TMC cycles)	정의	amount of plastic deformation applied to materials at the holding temperature
		동의어	
		타입	number
		단위	%
		예시	50
P206	holding time (thermomechanical process / TMC cycles)	정의	holding time for the heat treatment at the holding temperature
		동의어	
		타입	number
		단위	h
		예시	0.5
P207	cooling rate (thermomechanical process / TMC cycles)	정의	rate at which temperature is lowered to the final temperature
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P208	deformation cooling (thermomechanical process / TMC cycles)	정의	amount of plastic deformation applied to materials during cooling to the final temperature
		동의어	
		타입	number
		단위	%
		예시	50
P209	cooling method (thermomechanical process / TMC cycles)	정의	method of cooling without specified cooling rate
		동의어	
		타입	string
		단위	
		예시	1. air cooling (AC) 2. furnace cooling (FC) 3. oil quenching (OQ) 4. water quenching (WQ)
P210	final temperature (thermomechanical process / TMC cycles)	정의	temperature at which the TMP cycle finished
		동의어	
		타입	number
		단위	K
		예시	298
P221	temperature (wet etching)	정의	temperature of an etching solution
		동의어	
		타입	number
		단위	K
		예시	350
P222	name (wet etching / etching solution / etching agent)	정의	name of the etching agent (chemical)
		동의어	
		타입	string
		단위	
		예시	KOH, hydrazine
P223	concentration (wet etching / etching solution / etching agent)	정의	concentration of etchant
		동의어	
		타입	number
		단위	mol
		예시	0.1
P224	time (wet etching)	정의	etching time
		동의어	
		타입	number
		단위	h
		예시	0.75

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P225	energy type (ALD)	정의	energy type used for chemical reactions
		동의어	
		타입	string
		단위	
		예시	thermal; plasma enhanced
P226	name (ALD / precursor)	정의	name of a precursor material
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P227	amount (ALD / precursor)	정의	amount of a precursor material
		동의어	
		타입	number
		단위	mg
		예시	64
P228	temperature (ALD / precursor)	정의	precursor temperature
		동의어	
		타입	number
		단위	K
		예시	400
P229	type of canister (ALD / precursor)	정의	type of the device used to convert metalorganic compounds from a liquid or solid phase into a usable vapor phase
		동의어	
		타입	string
		단위	
		예시	1. bubbler 2. non-bubbler
P230	name (ALD / reactant)	정의	name of a reactant material
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P231	amount (ALD / reactant)	정의	amount of a reactant material
		동의어	
		타입	number
		단위	mg
		예시	64
P232	temperature (ALD / reactant)	정의	reactant temperature
		동의어	
		타입	number
		단위	K
		예시	400
P233	type of canister (ALD / reactant)	정의	a device used to convert metalorganic compounds from a liquid or solid phase into a usable vapor phase
		동의어	bubbler cylinder
		타입	string
		단위	
		예시	1. bubbler 2. non-bubbler
P234	name (ALD / carrier gas)	정의	name of a carrier gas
		동의어	
		타입	string
		단위	
		예시	Ar

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P235	flow rate (ALD / carrier gas)	정의	volumetric flow rate of the carrier gas; in standard cubic centimeter per minute.
		동의어	flow gas rate; volumetric flow rate
		타입	number
		단위	cm^{3} min^{-1}
		예시	200
P236	purity (ALD / carrier gas)	정의	purity of a carrier gas
		동의어	
		타입	number
		단위	%
		예시	99.999
P237	name (ALD / purge gas)	정의	name of a purge gas
		동의어	
		타입	string
		단위	
		예시	Ar
P238	flow rate (ALD / purge gas)	정의	volumetric flow rate of the carrier gas; in standard cubic centimeter per minute.
		동의어	flow gas rate; volumetric flow rate
		타입	number
		단위	cm^{3} min^{-1}
		예시	200
P239	purity (ALD / purge gas)	정의	purity of a carrier gas
		동의어	
		타입	number
		단위	%
		예시	99.999
P240	purge time (ALD / pulse time)	정의	time imposed to remove residual precursors or reactants from the processing chamber
		동의어	
		타입	number
		단위	s
		예시	0.3
P241	feeding time (ALD / pulse time)	정의	time imposed to insert precursors or reactants into the processing chamber
		동의어	
		타입	number
		단위	s
		예시	0.3
P242	cycle time (ALD / cycle)	정의	time spent for each atomic layer deposition cycle
		동의어	
		타입	number
		단위	s
		예시	3
P243	number of cycles (ALD / cycle)	정의	a total number of atomic layer deposition cycles
		동의어	
		타입	number
		단위	none
		예시	20
P244	substrate temperature (ALD)	정의	substrate temperature
		동의어	
		타입	number
		단위	K
		예시	350

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P245	working pressure (ALD)	정의	pressure in a processing chamber during an ALD process
		동의어	
		타입	number
		단위	Torr
		예시	0.03
P246	base pressure (ALD)	정의	the lowest pressure of the processing chamber to pump down without any gas flows
		동의어	
		타입	number
		단위	Torr
		예시	0.000015
P247	growth per cycle (ALD)	정의	deposited film thickness per cycle
		동의어	
		타입	number
		단위	nm cycle^{-1}
		예시	0.1
P248	energy type (ALE)	정의	energy type used for chemical reactions
		동의어	
		타입	string
		단위	
		예시	thermal; plasma enhanced
P249	name (ALE / etching agent)	정의	name of a precursor material
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P250	amount (ALE / etching agent)	정의	amount of a precursor material
		동의어	
		타입	number
		단위	mg
		예시	64
P251	temperature (ALE / etching agent)	정의	precursor temperature
		동의어	
		타입	number
		단위	K
		예시	400
P252	type of canister (ALE / etching agent)	정의	type of the device used to convert metalorganic compounds from a liquid or solid phase into a usable vapor phase
		동의어	
		타입	string
		단위	
		예시	1. bubbler 2. non-bubbler
P253	name (ALE / reactant)	정의	name of a reactant material
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P254	amount (ALE / reactant)	정의	amount of a reactant material
		동의어	
		타입	number
		단위	mg
		예시	64

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P255	temperature (ALE / reactant)	정의	reactant temperature
		동의어	
		타입	number
		단위	K
		예시	400
P256	type of canister (ALE / reactant)	정의	type of the device used to convert metalorganic compounds from a liquid or solid phase into a usable vapor phase
		동의어	
		타입	string
		단위	
		예시	1. bubbler 2. non-bubbler
P257	name (ALE / carrier gas)	정의	name of a carrier gas
		동의어	
		타입	string
		단위	
		예시	Ar
P258	flow rate (ALE / carrier gas)	정의	volumetric flow rate of the carrier gas; in standard cubic centimeter per minute.
		동의어	flow gas rate; volumetric flow rate
		타입	number
		단위	cm^{3} min^{-1}
		예시	200
P259	purity (ALE / carrier gas)	정의	purity of a carrier gas
		동의어	
		타입	number
		단위	%
		예시	99.999
P260	name (ALE / purge gas)	정의	name of a purge gas
		동의어	
		타입	string
		단위	
		예시	Ar
P261	flow rate (ALE / purge gas)	정의	volumetric flow rate of the carrier gas; in standard cubic centimeter per minute.
		동의어	flow gas rate; volumetric flow rate
		타입	number
		단위	cm^{3} min^{-1}
		예시	200
P262	purity (ALE / purge gas)	정의	purity of a carrier gas
		동의어	
		타입	number
		단위	%
		예시	99.999
P263	purge time (ALE / pulse time)	정의	time imposed to remove residual precursors or reactants from the processing chamber
		동의어	
		타입	number
		단위	s
		예시	0.3
P264	feeding time (ALE / pulse time)	정의	time imposed to insert precursors or reactants into the processing chamber
		동의어	
		타입	number
		단위	s
		예시	0.3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P265	cycle time (ALE / cycle)	정의	time spent for each atomic layer etching cycle
		동의어	
		타입	number
		단위	s
		예시	3
P266	number of cycles (ALE / cycle)	정의	a total number of atomic layer etching cycles
		동의어	
		타입	number
		단위	none
		예시	20
P267	substrate temperature (ALE)	정의	substrate temperature
		동의어	
		타입	number
		단위	K
		예시	350
P268	working pressure (ALE)	정의	pressure in a processing chamber during an ALE process
		동의어	
		타입	number
		단위	Torr
		예시	0.03
P269	base pressure (ALE)	정의	the lowest pressure of the processing chamber to pump down without any gas flows
		동의어	
		타입	number
		단위	Torr
		예시	0.000015
P270	etch per cycle (ALE)	정의	etching thickness per cycle
		동의어	
		타입	number
		단위	nm cycle^{-1}
		예시	0.1
P271	mill type (ball milling)	정의	ball milling type
		동의어	
		타입	string
		단위	
		예시	planetary; impact; shake
P272	rotation speed (ball milling)	정의	number of revolution per unit time
		동의어	rotation rate
		타입	numeric
		단위	revolutions min^{-1}
		예시	30
P273	total time (ball milling)	정의	total milling time
		동의어	
		타입	number
		단위	h
		예시	100
P274	milling time (ball milling)	정의	time spent for milling
		동의어	
		타입	number
		단위	h
		예시	80

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P275	rest time (ball milling)	정의	time for resting between milling sequence
		동의어	
		타입	number
		단위	h
		예시	20
P276	ball to powder weight ratio (ball milling)	정의	mass of ball over mass of material (precursor)
		동의어	
		타입	number
		단위	none
		예시	40
P277	material mass (ball milling)	정의	mass of material (precursor) to be milled
		동의어	
		타입	number
		단위	g
		예시	1
P278	solvent (ball milling)	정의	materials of solvent
		동의어	
		타입	string
		단위	
		예시	NMP
P279	jar shape (ball milling)	정의	shape of jar
		동의어	
		타입	string
		단위	
		예시	cylinder; sphere
P280	jar volume (ball milling)	정의	volume of jar
		동의어	
		타입	number
		단위	mL
		예시	45, 80
P281	jar material (ball milling)	정의	materials of jar
		동의어	
		타입	string
		단위	
		예시	Stainless steel; Zirconium oxide; Silicon nitride
P282	atmosphere (ball milling)	정의	environment to which a given process is exposed
		동의어	process environment
		타입	string
		단위	
		예시	Ar; Air; inert; reducing; oxidizing
P283	metal forming type (bulk metal forming)	정의	type of bulk metal forming where materials are subjected to plastic deformation to obtain the required size and shape
		동의어	
		타입	string
		단위	
		예시	1. rolling 2. forging 3. extrusion 4. drawing
P284	heating rate (bulk metal forming)	정의	rate being raised to metal forming (working) temperature
		동의어	
		타입	number
		단위	K s^{-1}
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P285	forming temperature (bulk metal forming)	정의	selected temperature at which bulk metal forming process is performed
		동의어	
		타입	number
		단위	K
		예시	1000
P286	reduction ratio (bulk metal forming)	정의	ratio of the feed size to the product size in any metal forming operation
		동의어	
		타입	number
		단위	%
		예시	50
P287	pass number (bulk metal forming)	정의	repeating number of metal working process applied to materials
		동의어	
		타입	number
		단위	none
		예시	5
P288	cooling method (bulk metal forming)	정의	cooling method of a workpiece to obtain certain material properties
		동의어	
		타입	string
		단위	
		예시	1. air cooling (AC) 2. oil quenching (OQ) 3. water quenching (WQ)
P289	relative centrifugal force (centrifugation)	정의	applied force for centrifugation
		동의어	
		타입	number
		단위	N
		예시	12298
P290	pressure (CMP)	정의	applied pressure for polishing
		동의어	
		타입	number
		단위	kgf mm^{-2}
		예시	10; 30
P291	pressure (chemical synthesis)	정의	reaction pressure for synthesis
		동의어	
		타입	number
		단위	atm
		예시	1; 0.5
P292	time (chemical synthesis)	정의	time of the synthesis process
		동의어	
		타입	numeric
		단위	h
		예시	0.75
P293	pressure (CVD)	정의	reaction pressure in a processing chamber
		동의어	
		타입	number
		단위	Torr
		예시	1e-3; 2.4e0
P294	name (CVD / carrier gas)	정의	name of carrier gas
		동의어	
		타입	string
		단위	
		예시	Ar; N_{2}

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P295	flow rate (CVD / carrier gas)	정의	flow rate of carrier gas
		동의어	
		타입	
		단위	sccm
		예시	100; 200; 250
P296	name (CVD / precursor)	정의	name of a precursor
		동의어	
		타입	string
		단위	
		예시	$[(NH_{4})]_{2}MoS_{4}$; S; P; SnO_{2}
P297	temperature (CVD)	정의	reaction temperature in a processing chamber
		동의어	
		타입	number
		단위	K
		예시	750; 1000
P301	amount (CVD / precursor)	정의	amount of a precursor
		동의어	
		타입	number
		단위	mg
		예시	100; 200; 500
P303	name (CVD / process gas)	정의	name of reactant gas
		동의어	
		타입	string
		단위	
		예시	H_{2}; NH_{3}
P304	flow rate (CVD / process gas)	정의	the flow rate of reactant gas
		동의어	
		타입	number
		단위	sccm
		예시	100; 200; 250
P305	name (CVD)	정의	name of substrate
		동의어	
		타입	string
		단위	
		예시	SiO_{2}
P309	time (CVD)	정의	reaction time
		동의어	
		타입	number
		단위	min
		예시	20
P328	pressure (drying)	정의	pressure for drying
		동의어	
		타입	number
		단위	atm
		예시	1; 0.5
P329	casting temperature (casting)	정의	temperature of molten metal before cooling
		동의어	pouring temperature
		타입	number
		단위	K
		예시	1000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P330	mold temperature (casting)	정의	temperature maintained before pouring and during casting to ensure fluidity of molten metals
		동의어	pre-heating temperature
		타입	number
		단위	K
		예시	200
P331	pouring speed (casting)	정의	volumetric rate at which molten metal is delivered into the mold
		동의어	pouring rate, casting speed
		타입	number
		단위	m s^{-1}
		예시	0.1
P332	cooling rate (casting)	정의	rate at which temperature is lowered during solidification
		동의어	
		타입	number
		단위	K s^{-1}
		예시	10
P333	solidification time (casting)	정의	time required for the casting to solidify after pouring
		동의어	
		타입	number
		단위	s
		예시	1000
P334	method (heat treatment)	정의	types of heat treatment
		동의어	
		타입	string
		단위	
		예시	1. annealing 2. normalizing 3. tempering 4. hardening 5. sintering 6. calcination
P335	atmosphere (heat treatment)	정의	atmospheric environment to which a given heat treatment process is exposed • "Dry air" represent air with low relative humidity but where the relative humidity is not known • "Ambient" represent air where the relative humidity is not known. For ambient conditions where the relative humidity is known, state this as "Air" • "Vacuum" (of unspecified pressure) is for this purpose considered as an atmospheric gas.
		동의어	heat treatment environment
		타입	string
		단위	
		예시	Ar, N_{2}+5H_{2}, inert, reducing, oxidizing
P336	temperature (heat treatment / thermal history)	정의	soaking temperature maintained during the isothermal heat treatment
		동의어	
		타입	number
		단위	K
		예시	403
P337	time (heat treatment / thermal history)	정의	holding time for the heat treatment at the holding temperature
		동의어	
		타입	number
		단위	min
		예시	5; 240
P338	mixing type (mixing)	정의	materials mixing type
		동의어	
		타입	string
		단위	
		예시	magnetic stirrer

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P339	material mixed (mixing)	정의	information of material mixture
		동의어	
		타입	dictionary of {materials: quantity, ...
		단위	{materials name:g}
		예시	{Material_1:100, Material_2:50}; {carbon black:10, Material_2:30}; {PVDF:100}
P340	name (mixing / solvent)	정의	name of solvent
		동의어	
		타입	string
		단위	
		예시	toluene
P341	amount (mixing / solvent)	정의	amount of solvent
		동의어	
		타입	number
		단위	mL
		예시	200
P342	rotation speed (mixing)	정의	number of revolutions per unit time for mixing process
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	1500
P343	temperature (mixing)	정의	mixing temperature
		동의어	
		타입	number
		단위	K
		예시	300
P344	time (mixing)	정의	mixing time
		동의어	
		타입	number
		단위	h
		예시	0.5
P350	angle (pulsed laser deposition / laser)	정의	incident angle of a pulsed laser on a target material
		동의어	
		타입	number
		단위	degree
		예시	60
P351	rotation speed (pulsed laser deposition)	정의	rotation speed of the substrate
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	3000
P352	axis (pulsed laser deposition)	정의	a system configuration in terms of an axis type specifying the location of a substrate with respect to that of a target material
		동의어	
		타입	string
		단위	
		예시	1. On-axis 2. Off-axis
P353	working pressure (pulsed laser deposition)	정의	pressure in a processing chamber during the PLD process
		동의어	
		타입	number
		단위	Torr
		예시	0.03

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P354	base pressure (pulsed laser deposition)	정의	the lowest pressure of the processing chamber
		동의어	
		타입	number
		단위	Torr
		예시	0.000015
P355	name (pulsed laser deposition / atmosphere)	정의	name of the atmosphere
		동의어	
		타입	string
		단위	
		예시	Ar; vacuum; inert; air
P356	pressure (pulsed laser deposition / atmosphere)	정의	pressure of the atmosphere
		동의어	
		타입	number
		단위	Pa
		예시	1
P357	target-substrate distance (pulsed laser deposition)	정의	distance between a target material and a substrate
		동의어	
		타입	number
		단위	mm
		예시	200
P358	lens-target distance (pulsed laser deposition)	정의	distance between a laser focusing lens and a target material
		동의어	
		타입	number
		단위	mm
		예시	600
P359	load type (pressing)	정의	sample pressing type
		동의어	
		타입	string
		단위	
		예시	uniaxial pressing, isostatic pressing, injection molding, extrusion
P360	temperature (pressing)	정의	pressing temperature
		동의어	
		타입	number
		단위	K
		예시	298
P361	pressure (pressing)	정의	pressure for pressing
		동의어	
		타입	number
		단위	MPa
		예시	300
P362	time (pressing)	정의	time for pressing
		동의어	
		타입	number
		단위	min
		예시	2
P363	pressure (solvothermal synthesis)	정의	reaction pressure for synthesis
		동의어	
		타입	number
		단위	atm
		예시	1; 0.5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P364	name (sol-gel synthesis / precursor)	정의	name of precursor
		동의어	
		타입	string
		단위	
		예시	tetramethoxysilane
P365	solvent (sol-gel synthesis)	정의	solvent for sol-gel synthesis
		동의어	
		타입	string
		단위	
		예시	water
P366	power (sonication)	정의	amount of power delivered to samples
		동의어	
		타입	number
		단위	W
		예시	500
P367	name (spin coating / precursor)	정의	name of precursor
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P368	volume (spin coating / precursor)	정의	volume of precursor • The volumes refer the volumes used, not the volume of the stock solutions. Thus if 0.15 ml of a solution is used, the volume is 0.15 ml • When volumes are unknown, state that as ‘nan’
		동의어	amount
		타입	number
		단위	mL
		예시	50.2
P369	temperature (spin coating)	정의	process temperature
		동의어	
		타입	number
		단위	K
		예시	298
P370	composition (spin coating / atmosphere)	정의	composition of atmosphere in partial pressure
		동의어	environmental composition
		타입	dictionary of {value : {constituent : concentration, …}, unit : unit value}
		단위	bar
		예시	{value : {Ar : 0.5, N_2 : 0.5}, unit : bar}; {value : {vacuum : 0}, unit : bar}
P371	time (spin coating)	정의	time of the process
		동의어	
		타입	number
		단위	s
		예시	20
P372	rpm (spin coating)	정의	rotation speed of the substrate
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	3000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P373	name (spray pyrolysis coating / precursor)	정의	neme of precursor
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P374	volume (spray pyrolysis coating / precursor)	정의	volume of precursor • The volumes refer the volumes used, not the volume of the stock solutions. Thus if 0.15 ml of a solution is used, the volume is 0.15 ml • When volumes are unknown, state that as 'nan'
		동의어	amount
		타입	number
		단위	mL
		예시	50.2
P375	composition (spray pyrolysis coating / solvent)	정의	composition of solvent; volume ratio of each constituent of solvent
		동의어	
		타입	dictionary of {solvent_(n) : volume ratio in solvent mixture without dimension, ... }
		단위	none
		예시	{H2O : 9, acetonil : 0.6, methanol : 0.4}
P377	droplet size (spray pyrolysis coating)	정의	size of droplet
		동의어	
		타입	number
		단위	m
		예시	0.00001
P378	spray speed (spray pyrolysis coating)	정의	spray speed in volume per hour
		동의어	
		타입	number
		단위	m^3 h^{-1}
		예시	0.0034
P379	flame temperature (spray pyrolysis coating)	정의	flame temperature for material synthesis
		동의어	
		타입	number
		단위	K
		예시	160
P380	heat treatment time (spray pyrolysis coating)	정의	heat treatment time for material synthesis
		동의어	
		타입	number
		단위	h
		예시	12
P381	rotating speed (sputter deposition / jig)	정의	speed of rotation in rpm
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	10; 1
P385	substrate temperature (evaporation deposition)	정의	temperature of the substrate
		동의어	
		타입	number
		단위	K
		예시	300

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P386	deposition rate (evaporation deposition)	정의	material deposition rate
		동의어	
		타입	number
		단위	nm s^{-1}
		예시	1.5
P387	time (evaporation deposition)	정의	time of evaporation deposition process
		동의어	coating time; evaporation time
		타입	number
		단위	s
		예시	120
P388	power (sonochemical synthesis)	정의	amount of power delivered to samples
		동의어	
		타입	number
		단위	W
		예시	500
P389	solvent (wet etching / etching solution)	정의	solvent of an etching solution
		동의어	
		타입	string
		단위	
		예시	water
P390	additive (wet etching / etching solution)	정의	additional substances added to an etching solution that affect the etch rate, surface morphology and undercutting
		동의어	
		타입	string
		단위	
		예시	$NH_{2}OH$
P391	stirring rate (wet etching)	정의	solution stirring process during the wet etching
		동의어	
		타입	numeric
		단위	revolutions min^{-1}
		예시	100
P392	material (ball milling)	정의	description of material to be milled
		동의어	
		타입	string
		단위	
		예시	Co3O4 powder form ABC (d=30um)
P393	ball material (ball milling)	정의	materials of ball
		동의어	
		타입	string
		단위	
		예시	Stainless steel; Zirconium oxide; Silicon nitride
P395	ball size (ball milling)	정의	diameter of ball
		동의어	
		타입	number
		단위	mm
		예시	0.1 ; 2
P396	coating material (blade coating)	정의	coating materials by blade coating
		동의어	
		타입	string
		단위	
		예시	Material_3; LCO Slurry

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P397	substrate (blade coating)	정의	substrate on which a material is to be blade coated
		동의어	
		타입	string
		단위	
		예시	Material_3; Cu foil; Al foil
P398	gap (blade coating)	정의	distance between the die lip and the substrate
		동의어	blade gap
		타입	number
		단위	{\mu m}
		예시	200
P399	speed (blade coating)	정의	substrate speed
		동의어	
		타입	number
		단위	mm s^{-1}
		예시	1
P400	forming force (bulk metal forming)	정의	force applied to workpiece during which bulk metal forming process is performed
		동의어	applied force
		타입	number
		단위	N
		예시	10000
P401	cooling rate (bulk metal forming)	정의	rate at which temperature is lowered to the final temperature
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P402	tube material (centrifugation)	정의	materials of centrifuge tube
		동의어	
		타입	string
		단위	
		예시	polyethylene terephthalate (PET); polypropylene; PPCO (polypropylene copolymer), polycarbonate or polystyrene
P403	tube volume (centrifugation)	정의	volume of centrifuge tube
		동의어	
		타입	number
		단위	mL
		예시	15
P404	spin type (electrospinning)	정의	the way of electrospinning depending on the setup (especially for the number of nozzles)
		동의어	
		타입	string
		단위	
		예시	1. Uniaxial Electrospinning 2. Coaxial Electrospinning 3. Multi-axial Electrospinning
P405	name (electrospinning / precursor / solute)	정의	name of a solute material
		동의어	
		타입	string
		단위	
		예시	poly(vinylidene fluoride), poly(vinylidene fluoride-trifluoroethylene), polystyrene, polycaprolactone
P406	amount (electrospinning / precursor / solute)	정의	amount of a solute material
		동의어	
		타입	number
		단위	g
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P407	name (electrospinning / precursor / solvent)	정의	name of a solvent material
		동의어	
		타입	string
		단위	
		예시	acetone, N,N-dimethylformamide, chloroform, dichloromethane
P408	amount (electrospinning / precursor / solvent)	정의	amount of a solvent material
		동의어	
		타입	number
		단위	g
		예시	1
P409	concentration (electrospinning / precursor)	정의	concentration of the precursor solution
		동의어	
		타입	number
		단위	wt%
		예시	20
P410	inner diameter (electrospinning / tip size)	정의	inner diameter of needle tip
		동의어	
		타입	number
		단위	mm
		예시	0.26
P411	outer diameter (electrospinning / tip size)	정의	outer diameter of needle tip
		동의어	
		타입	number
		단위	mm
		예시	0.5144
P412	collector type (electrospinning / collector)	정의	type of collector
		동의어	
		타입	string
		단위	
		예시	1. Drum Collector 2. Plate Collector 3. Disk Collector 4. Customized Collector
P413	rotating speed (electrospinning / collector)	정의	rotating speed
		동의어	
		타입	numeric
		단위	revolutions min^{-1}
		예시	100
P414	diameter (electrospinning / collector)	정의	the diameter of drum or disk collector
		동의어	
		타입	number
		단위	cm
		예시	18
P415	tip to collector distance (electrospinning)	정의	the distance between where the fiber ejected and arrived
		동의어	
		타입	number
		단위	cm
		예시	15
P416	applied voltage (electrospinning)	정의	voltage applied to the syringe tip
		동의어	
		타입	number
		단위	kV
		예시	18

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P417	injection rate (electrospinning)	정의	speed of solution ejection
		동의어	
		타입	number
		단위	{\mu L} min^{-1}
		예시	20
P418	relative humidity (electrospinning)	정의	the physical quantity of water vapor in the air, particularly inside the electrospinning chamber
		동의어	
		타입	number
		단위	%
		예시	42; 45; 50
P419	temperature (electrospinning)	정의	temperature inside the electrospinning chamber
		동의어	
		타입	number
		단위	K
		예시	22
P420	time (corona poling)	정의	time applying the high voltage
		동의어	
		타입	number
		단위	min
		예시	30
P421	applied voltage (corona poling)	정의	value of voltage applied to the needle
		동의어	
		타입	number
		단위	kV
		예시	20
P422	atmosphere (corona poling)	정의	poling atmosphere
		동의어	process environment
		타입	string
		단위	
		예시	Air, Silicone oil, Ni_{2}
P423	temperature (corona poling)	정의	poling temperature
		동의어	
		타입	number
		단위	K
		예시	350
P424	corona needle (corona poling)	정의	part where high direct current potential is applied which has the effect of intensifying the electric field, causing ionization of the molecules in the medium
		동의어	
		타입	string
		단위	
		예시	tungsten needle
P425	needle to sample distance (corona poling)	정의	the distance between needle and sample
		동의어	
		타입	number
		단위	cm
		예시	15
P426	applied electric field (direct current poling)	정의	the process of aligning the direction of spontaneous polarization in one direction by applying a dc(direct current) electric field
		동의어	
		타입	number
		단위	kV m^{-1}
		예시	20

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P427	casting type (casting)	정의	casting type
		동의어	
		타입	string
		단위	
		예시	1. sand casting 2. die casting 3. permanent mold casting
P428	heating rate (heat treatment / thermal history)	정의	rate at which temperature is raised to holding(or target) temperature
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P429	cooling rate (heat treatment / thermal history)	정의	rate at which temperature is lowered to the final cold temperature
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P430	working temperature (hydrothermal reaction / working temperature / thermal cycle)	정의	temperature at which the hydrothermal reaction takes place
		동의어	
		타입	number
		단위	K
		예시	400
P431	ramping rate (hydrothermal reaction / working temperature / thermal cycle)	정의	rate at which the temperature is increased
		동의어	
		타입	number
		단위	K min^{-1}
		예시	30
P432	cooling rate (hydrothermal reaction / working temperature / thermal cycle)	정의	rate at which the temperature is decreased
		동의어	
		타입	number
		단위	K min^{-1}
		예시	30
P433	name (hydrothermal reaction / precursor / reactant)	정의	reactant for hydrothermal reaction
		동의어	
		타입	string
		단위	
		예시	CH3Cl
P434	amount (hydrothermal reaction / precursor / reactant)	정의	amount of reactant material_(m)
		동의어	
		타입	number
		단위	mg
		예시	64
P435	concentration (hydrothermal reaction / precursor / reactant)	정의	concentration of reactant
		동의어	
		타입	string
		단위	mol
		예시	0.1
P436	name (hydrothermal reaction / precursor / surfactant)	정의	name of surfactant material_(m)
		동의어	
		타입	string
		단위	
		예시	SDS

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P437	amount (hydrothermal reaction / precursor / surfactant)	정의	amount of surfactant material_(m)
		동의어	
		타입	number
		단위	mg
		예시	64
P438	concentration (hydrothermal reaction / precursor / surfactant)	정의	concentration of surfactant
		동의어	
		타입	string
		단위	mol
		예시	0.1
P439	solvent (hydrothermal reaction / precursor)	정의	solvent for hydrothermal reaction
		동의어	
		타입	string
		단위	
		예시	ethanol;
P440	ramping rate (hydrothermal reaction)	정의	rate at which the temperature is increased
		동의어	
		타입	number
		단위	K min^{-1}
		예시	30
P441	working pressure (hydrothermal reaction)	정의	pressure at which the hydrothermal reaction takes place
		동의어	
		타입	number
		단위	Torr
		예시	0.03
P442	substrate (hydrothermal reaction)	정의	substrate on which the hydrothermal reaction takes place
		동의어	
		타입	string
		단위	
		예시	SiO_{2}
P443	time (hydrothermal reaction)	정의	total duration of the hydrothermal reaction
		동의어	
		타입	number
		단위	h
		예시	0.75
P444	cooling rate (hydrothermal reaction)	정의	rate at which the temperature is decreased
		동의어	
		타입	number
		단위	K min^{-1}
		예시	30
P445	method (polishing)	정의	polishing method
		동의어	
		타입	string
		단위	
		예시	mechanical; electrolytic
P446	medium (polishing)	정의	medium used for polishing
		동의어	
		타입	string
		단위	
		예시	diamond embeded disk; Al2O3 solution

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P447	mixer type (slurry casting)	정의	slurry mixing type
		동의어	
		타입	string
		단위	
		예시	magnetic stirrer
P448	material mix (slurry casting)	정의	information of material mixture
		동의어	
		타입	dictionary of {material : name, amount : value}
		단위	mg
		예시	[{material : Material_01, amount : 10}, {material : PVDF, amount : 100}]
P449	name (slurry casting / solvent)	정의	name of solvent
		동의어	
		타입	string
		단위	
		예시	toluene
P450	amount (slurry casting / solvent)	정의	amount of solvent
		동의어	
		타입	number
		단위	mL
		예시	20
P451	rotation speed (slurry casting)	정의	number of rotations per unit time for mixing process
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	1500
P452	temperature (slurry casting)	정의	temperature of the process
		동의어	
		타입	number
		단위	K
		예시	300
P453	humidity (slurry casting)	정의	the concentration of water vapor present in the air
		동의어	
		타입	number
		단위	%
		예시	30
P454	time (slurry casting)	정의	time of the process
		동의어	
		타입	number
		단위	h
		예시	0.5
P455	pump rotation (slot coating)	정의	pump rotation
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	1000
P456	gap (slot coating)	정의	distance between the die lip and the substrate
		동의어	
		타입	number
		단위	{\mu m}
		예시	200

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P457	substrate speed (slot coating)	정의	substrate speed
		동의어	
		타입	number
		단위	m min^{-1}
		예시	1
P458	heating rate (solvothermal synthesis)	정의	temperature at which heat treatment process is performed
		동의어	
		타입	number
		단위	K min^{-1}
		예시	5
P459	cooling rate (solvothermal synthesis)	정의	temperature at which heat treatment process is performed
		동의어	
		타입	number
		단위	K min^{-1}
		예시	5
P460	sonication type (sonication)	정의	type of sonication
		동의어	
		타입	string
		단위	
		예시	tip; bath
P461	pump rotation (transporting)	정의	pump rotation
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	1650
P462	pipe material (transporting)	정의	wall material of pipes
		동의어	
		타입	string
		단위	
		예시	PFA
P463	pipe diameter (transporting)	정의	inner diameter of pipes
		동의어	
		타입	number
		단위	mm
		예시	8
P464	temperature (transporting)	정의	transporting temperature
		동의어	
		타입	number
		단위	K
		예시	300
P465	time (transporting)	정의	transporting time
		동의어	
		타입	number
		단위	h
		예시	0.5
P466	blade size (blade coating)	정의	size of blade
		동의어	
		타입	number
		단위	cm
		예시	30

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P467	feed (blade coating)	정의	total amount of coated materials
		동의어	
		타입	number
		단위	ml
		예시	30.5
P468	casting pressure (casting)	정의	pressurization level to feed the molten metal into the mold
		동의어	
		타입	number
		단위	bar
		예시	5
P469	name (chemical bath deposition / precursor)	정의	name of precursor
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P470	composition (chemical bath deposition / precursor)	정의	composition of precursor
		동의어	
		타입	dictionary of {value : {constituent : composition, …}, unit : unit value}
		단위	
		예시	{value : {Titanium diisopropoxide bis(acetylacetonate) : 56.0, C60 : 34.7}, unit : wt.%}
P471	excess (chemical bath deposition / precursor)	정의	Components that are in excess in the material synthesis. E.g. to form stoichiometric perovskite MAPbI3, PbI2 and MAI are mixed in the proportions 1:1. If one of them are in excess compared to the other, then that component is considered to be in excess. This information can be inferred from data entered on the concentration for all reaction solutions but this gives a convenient shorthand filtering option. • If more than one component is in excess, order them in alphabetic order and separate them by semicolons. • It there are no components that are in excess, write Stoichiometric
		동의어	
		타입	string
		단위	
		예시	PbI2; MAI
P472	purity (chemical bath deposition / precursor)	정의	purity of precursor
		동의어	
		타입	dictionary of {precursor : purity level in text string, ...
		단위	
		예시	{Titanium diisopropoxide bis(acetylacetonate) : Pro analysis, C60 : none}
P473	supplier (chemical bath deposition / precursor)	정의	supplier of precursor
		동의어	seller
		타입	dictionary of {precursor : supplier name in text string, ... }
		단위	
		예시	{Titanium diisopropoxide bis(acetylacetonate) : Sigma Aldrich, C60 : Acros}
P474	volume (chemical bath deposition / precursor)	정의	volume of precursor • The volumes refer the volumes used, not the volume of the stock solutions. Thus if 0.15 ml of a solution is used, the volume is 0.15 ml • When volumes are unknown, state that as 'nan'
		동의어	amount
		타입	number
		단위	mL
		예시	50.2

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P475	age (chemical bath deposition / precursor)	정의	The age of the solutions • As a general guideline, the age refers to the time from the preparation of the final precursor mixture to the reaction procedure. • When the age of a solution is not known, state that as 'nan' • For reaction steps where no solvents are involved, state this as 'nan' • For solutions that is stored a long time, an order of magnitude estimate is adequate.
		동의어	
		타입	number
		단위	h
		예시	3
P476	temperature (chemical bath deposition / precursor)	정의	precursor temperature
		동의어	
		타입	number
		단위	K
		예시	298
P477	state (chemical bath deposition / precursor)	정의	The physical state of the precursor • The three basic categories are Solid/Liquid/Gas • Most cases are clear cut, e.g. spin-coating involves species in solution and evaporation involves species in gas phase. For less clear-cut cases, consider where the reaction really is happening as in: o For a spray-coating procedure, it is droplets of liquid that enters the substrate (thus a liquid phase reaction) o For sputtering and thermal evaporation, it is species in gas phase that reaches the substrate (thus a gas phase reaction) • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	liquid; gas; solid
P478	molarity (chemical bath deposition / precursor)	정의	used quantity of precursor in mole
		동의어	
		타입	number
		단위	mol
		예시	2.0; 1.5
P479	composition (chemical bath deposition / solvent)	정의	composition of solvent; volume fraction of each constituent of solvent and its purity
		동의어	
		타입	array of {constituent: value, fraction: value, purity: value, supplier: value}
		단위	
		예시	[{constituent : H20, fraction : 0.9, purity : Pro analysis, supplier : Aldrich}, {consitituent : acetonil, fraction : 0.06, purity : puris, supplier : Aldrich}, {consitituent : methanol, fraction : 0.04}, purity : unknown, supplier : Fischer]
P480	purity (chemical bath deposition / solvent)	정의	note on purity
		동의어	
		타입	string
		단위	
		예시	Pro analysis; puris; unknown
P481	supplier (chemical bath deposition / solvent)	정의	name of supplier
		동의어	
		타입	string
		단위	
		예시	Sigma Aldrich; Acros; Fischer

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P482	temperature (chemical bath deposition)	정의	process temperature
		동의어	
		타입	number
		단위	K
		예시	298
P483	composition (chemical bath deposition / atmosphere)	정의	composition of atmosphere in partial pressure
		동의어	environmental composition
		타입	dictionary of {value : {constituent : concentration, … }, unit : unit value}
		단위	bar
		예시	{value : {Ar : 0.5, N_2 : 0.5}, unit : bar}; {value : {vacuum : 0}, unit : bar}
P484	total pressure (chemical bath deposition / atmosphere)	정의	total pressure of atmosphere
		동의어	
		타입	number
		단위	bar
		예시	1.0; 0.95
P485	relative humidity (chemical bath deposition / atmosphere)	정의	relative humidity of atmosphere
		동의어	
		타입	number
		단위	%
		예시	28.4
P486	time (chemical bath deposition)	정의	time of the process
		동의어	
		타입	number
		단위	s
		예시	20
P487	electrode distance (electrochemical deposition)	정의	distance between working electrode and reference electrode
		동의어	
		타입	number
		단위	cm
		예시	1
P488	encapsulating materials (encapsulation)	정의	The stack sequence of the encapsulation • Every layer should be separated by a space, a vertical bar, and a space, i.e. (' | ') • If two materials, e.g. A and B, are mixed in one layer, list the materials in alphabetic order and separate them with semicolons, as in (A; B) • Use common abbreviations when appropriate but spell it out if risk for confusion. • There are now separate filed for doping. Indicate doping with colons. E.g. wither aluminium doped NiO-np as Al:NiO-np
		동의어	
		타입	string
		단위	
		예시	SGL; Eposy | Cover glass; PMMA
P489	edge sealing materials (encapsulation)	정의	Edge sealing materials • If two materials, e.g. A and B, are mixed in one layer, list the materials in alphabetic order and separate them with semicolons, as in (A; B)
		동의어	
		타입	string
		단위	
		예시	Epoxy; Surlyn; UV-glue
P490	atmosphere (encapsulation)	정의	process atmosphere
		동의어	process environment
		타입	string
		단위	
		예시	air; O2; N2

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P491	pre-etching (pre-etching)	정의	For the most common substrates, i.e. FTO and ITO it is common that part of the conductive layer is removed before perovskite deposition. State the method by which it was removed in sequence (mapping: Perovskite DB 5.4)
		동의어	
		타입	string
		단위	
		예시	[Zn-powder]; [HCl, Mecanical scrubbing, Laser etching]
P492	method (cleaning)	정의	general description of this cleaning sequence or specific wet cleaning process name
		동의어	
		타입	string
		단위	
		예시	cleaning in ultrasonic bath; UV-ozone raidation; Wiping; SPM
P493	solute (direct ink writing)	정의	solutes for ink jet printing materials
		동의어	
		타입	string
		단위	
		예시	P3HT
P494	solvent (direct ink writing)	정의	solvent for ink jet printing materials
		동의어	
		타입	string
		단위	
		예시	toluene
P495	concentration (direct ink writing)	정의	concentration of solution
		동의어	
		타입	number
		단위	mg mL^{-1}
		예시	3
P496	temperature (direct ink writing)	정의	solution temperature
		동의어	
		타입	number
		단위	K
		예시	298
P497	atmosphere (direct ink writing)	정의	atmosphere gas during casting
		동의어	process environment
		타입	string
		단위	
		예시	Ar, N_{2}
P498	metal AM type (metal additive manufacturing)	정의	type of metal additive manufacturing processes which use either a laser or electron beam to melt and fuse material
		동의어	
		타입	string
		단위	
		예시	1. direct metal laser sintering (DMLS) 2. selective laser sintering (SLS) 3. selective laser melting (SLM) 4. electron beam melting (EBM) 5. electron beam freeform (EBF) 6. wire laser additive manufacturing (WLAM) 7. wire arc additive manufacturing (WAAM)
P499	beam wavelength (metal additive manufacturing)	정의	wavelength of laser or e-beam, which varies depending on the specific type of laser or e-beam being used
		동의어	
		타입	number
		단위	nm
		예시	1000

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P500	beam power (metal additive manufacturing)	정의	power at which energy is emitted from a laser or e-beam
		동의어	laser power
		타입	number
		단위	W
		예시	300
P501	beam diameter (metal additive manufacturing)	정의	beam diameter emitted from source
		동의어	laser beam spot size
		타입	number
		단위	mm
		예시	0.22
P502	beam current (metal additive manufacturing)	정의	electric current through the source used to generate the laser or e-beam
		동의어	
		타입	number
		단위	A
		예시	50
P503	focus offset (metal additive manufacturing)	정의	adjustment of the focal point of the laser beam in relation to the surface of the material being processed
		동의어	
		타입	number
		단위	mm
		예시	0.1
P504	build direction (metal additive manufacturing)	정의	orientation or the specific direction in which the part is being constructed layer by layer
		동의어	layer rotation angle
		타입	number
		단위	degree
		예시	45
P505	layer thickness (metal additive manufacturing)	정의	thickness of materials spread over the build platform
		동의어	
		타입	number
		단위	mm
		예시	0.1
P506	scan speed (metal additive manufacturing)	정의	rate at which energy source is moving
		동의어	travel speed; scanning velocity
		타입	number
		단위	mm s^{-1}
		예시	100
P507	working distance (metal additive manufacturing)	정의	distance between nozzle and substrate
		동의어	
		타입	number
		단위	mm
		예시	15
P508	overlap spacing (metal additive manufacturing)	정의	distance or gap between adjacent paths or tracks of deposited material
		동의어	
		타입	number
		단위	{\mu m}
		예시	10
P509	hatch spacing (metal additive manufacturing)	정의	distance between adjacent paths or lines within a single layer of material
		동의어	hatch distance; track spacing
		타입	number
		단위	{\mu m}
		예시	50

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P510	powder feed rate (metal additive manufacturing)	정의	rate at which powder material is delivered and deposited onto the substrate or workpiece
		동의어	
		타입	number
		단위	g s^{-1}
		예시	300
P511	wire diameter (metal additive manufacturing)	정의	diameter of feeding wire in wire laser or arc additive manufacturing
		동의어	
		타입	number
		단위	mm
		예시	2
P512	wire feed rate (metal additive manufacturing)	정의	rate at which wire material is delivered and deposited onto the substrate or workpiece
		동의어	
		타입	number
		단위	mm s^{-1}
		예시	1
P513	welding speed (metal additive manufacturing)	정의	rate at which the deposition head or welding tool travels along the substrate
		동의어	
		타입	number
		단위	mm s^{-1}
		예시	50
P514	welding voltage (metal additive manufacturing)	정의	electrical potential difference applied between the welding tool or deposition head and the workpiece or substrate
		동의어	
		타입	number
		단위	mV
		예시	300
P515	gas flow rate (metal additive manufacturing)	정의	flow rate of shielding or inert gases used during the welding or additive manufacturing process
		동의어	
		타입	number
		단위	L min^{-1}
		예시	300
P516	substrate temperature (metal additive manufacturing)	정의	substrate temperature maintained before or during additive manufacturing process
		동의어	
		타입	number
		단위	K
		예시	400
P517	preheating temperature (metal additive manufacturing)	정의	preheating temperature of materials (usually powder) to be deposited, which is maintained before or during additive manufacturing process
		동의어	
		타입	number
		단위	K
		예시	400
P518	heating rate (microwave-assisted synthesis)	정의	raising the temperature to a desired level at a constant rate per unit of time
		동의어	Heating speed, heating velocity, warm-up rate, temperature increase rate, ramp rate, reaction rate
		타입	number
		단위	K min^{-1}
		예시	1
P519	cooling rate (microwave-assisted synthesis)	정의	decreasing the temperature to room temperature at a constant rate per unit of time
		동의어	cooling speed, cooling velocity, cooling temperature, chill rate
		타입	number
		단위	K s^{-1}
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P520	microwave power (microwave-assisted synthesis)	정의	the amount of electromagnetic energy per unit time in a microwave
		동의어	
		타입	number
		단위	W
		예시	700
P521	microwave frequency (microwave-assisted synthesis)	정의	the number of waves in a certain period of time for a microwave,
		동의어	
		타입	number
		단위	GHz
		예시	2.45
P522	perovskite postprocess (perovskite postprocess)	정의	Any after treatment of the formed perovskite. Most possible reaction steps should have been entered before this point. This is an extra category for procedures that just does not fit into any of the other categories.
		동의어	
		타입	string
		단위	
		예시	Hot isostatic pressing, 150˚C, 15 min Magnetic field, UV radiation, 30 min
P523	media (quenching)	정의	quenching media used in the quenching process i.e. antisolvent treatment • If gas quenching was used, state the gas used • If the sample quickly after spin coating was subjected to a vacuum, state this as 'Vacuum' • If an antisolvent was used but it is unknown which one, stat this as "Antisolvent" • If no antisolvent was used, leave this field blank
		동의어	
		타입	dictionary of {composition : {constituent : concentration, … }, unit : unit value}
		단위	none
		예시	{composition : {Chlorobenzene : 0.8, Toluene : 0.2}, unit : none}
P524	media volume (quenching)	정의	The volume of the media • For gas and vacuum assisted quenching, stat the volume as 'nan' • If the sample is dipped or soaked in the media, state the volume of the entire solution
		동의어	
		타입	number
		단위	{mu L}
		예시	30
P525	media additive (quenching)	정의	list of the dopants and additives in the media
		동의어	
		타입	dictionary of {composition : {constituent : concentration, … }, unit : unit value}
		단위	1. wt.% 2. at.% 3. mg ml^{-1}
		예시	{composition : {toluene : 2.5, acetone : 3.5}, unit : wt.%}
P526	pH (sol-gel synthesis)	정의	quantitative measure of the acidity or basicity of aqueous or other liquid solutions
		동의어	
		타입	number
		단위	none
		예시	7
P527	coating type (spin coating)	정의	type of spin coating
		동의어	
		타입	string
		단위	
		예시	1. drop & run 2. dynamic

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P528	composition (spin coating / precursor / composition)	정의	composition of precursor; weight percent of each constituent of solvent
		동의어	
		타입	dictionary of {precursor : concentration in wt.%, ...
		단위	{precursor name : wt.%}
		예시	{Titanium diisopropoxide bis(acetylacetonate) : 15.5, C60 : 10}
P529	purity (spin coating / precursor / composition)	정의	purity of precursor
		동의어	
		타입	string
		단위	
		예시	Pro analysis; puris; unknown
P530	supplier (spin coating / precursor / composition)	정의	supplier of precursor constituent
		동의어	seller
		타입	string
		단위	
		예시	Sigma Aldrich; Acros
P531	age (spin coating / precursor)	정의	The age of the solutions • As a general guideline, the age refers to the time from the preparation of the final precursor mixture to the reaction procedure. • When the age of a solution is not known, state that as 'nan' • For reaction steps where no solvents are involved, state this as 'nan' • For solutions that is stored a long time, an order of magnitude estimate is adequate.
		동의어	
		타입	number
		단위	h
		예시	3
P532	temperature (spin coating / precursor)	정의	precursor temperature
		동의어	
		타입	number
		단위	K
		예시	298
P533	state (spin coating / precursor)	정의	The physical state of the precursor • The three basic categories are Solid/Liquid/Gas • Most cases are clear cut, e.g. spin-coating involves species in solution and evaporation involves species in gas phase. For less clear-cut cases, consider where the reaction really is happening as in: o For a spray-coating procedure, it is droplets of liquid that enters the substrate (thus a liquid phase reaction) o For sputtering and thermal evaporation, it is species in gas phase that reaches the substrate (thus a gas phase reaction) • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	liquid; gas; solid
P534	molarity (spin coating / precursor)	정의	used quantity of precursor in mole
		동의어	
		타입	number
		단위	mol
		예시	2.0; 1.5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P535	composition (spin coating / solvent)	정의	composition of solvent; volume fraction of each constituent of solvent and its purity
		동의어	
		타입	array of {constituent: value, fraction: value, purity: value, supplier: value}
		단위	
		예시	[{constituent : H2, fraction : 0.5, purity : Pro analysis, supplier : Sigma Aldrich}, {constituent : acetone, fraction : 0.5, purity : Pro analysis, supplier : Sigma Aldrich}]
P536	purity (spin coating / solvent)	정의	purity of solvent constituent
		동의어	
		타입	string
		단위	
		예시	Pro analysis; puris; unknown
P537	supplier (spin coating / solvent)	정의	supplier of solvent
		동의어	seller
		타입	string
		단위	
		예시	Sigma Aldrich; Acros
P538	total pressure (spin coating / atmosphere)	정의	total pressure of atmosphere; sum of partial pressures in composition value
		동의어	
		타입	number
		단위	bar
		예시	1.0; 0.95
P539	relative humidity (spin coating / atmosphere)	정의	relative humidity of atmosphere
		동의어	
		타입	number
		단위	%
		예시	28.4
P540	rpm ramping-up rate (spin coating)	정의	ramping rate to target rpm
		동의어	
		타입	number
		단위	revolutions min^{-1} (rpm) s^{-1}
		예시	100
P541	composition (spray pyrolysis coating / precursor / composition)	정의	composition of precursor; volume ratio of each constituent of solvent
		동의어	
		타입	dictionary of {precursor : concentration in wt.%, ...
		단위	{precursor name : wt.%}
		예시	{Titanium diisopropoxide bis(acetylacetonate) : 15.5, C60 : 10}
P542	purity (spray pyrolysis coating / precursor / composition)	정의	purity of precursor
		동의어	
		타입	string
		단위	
		예시	Pro analysis; puris; unknown
P543	supplier (spray pyrolysis coating / precursor / composition)	정의	supplier of precursor
		동의어	seller
		타입	string
		단위	
		예시	Sigma Aldrich; Acros; Fischer
P544	age (spray pyrolysis coating / precursor)	정의	The age of the solutions • As a general guideline, the age refers to the time from the preparation of the final precursor mixture to the reaction procedure. • When the age of a solution is not known, state that as ‘nan’ • For reaction steps where no solvents are involved, state this as ‘nan’ • For solutions that is stored a long time, an order of magnitude estimate is adequate.
		동의어	
		타입	number
		단위	h
		예시	3

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P545	temperature (spray pyrolysis coating / precursor)	정의	precursor temperature
		동의어	
		타입	number
		단위	K
		예시	298
P546	state (spray pyrolysis coating / precursor)	정의	The physical state of the precursor • The three basic categories are Solid/Liquid/Gas • Most cases are clear cut, e.g. spin-coating involves species in solution and evaporation involves species in gas phase. For less clear-cut cases, consider where the reaction really is happening as in: o For a spray-coating procedure, it is droplets of liquid that enters the substrate (thus a liquid phase reaction) o For sputtering and thermal evaporation, it is species in gas phase that reaches the substrate (thus a gas phase reaction) • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	
		타입	string
		단위	
		예시	liquid; gas; solid
P547	molarity (spray pyrolysis coating / precursor)	정의	used quantity of precursor in mole
		동의어	
		타입	number
		단위	mol
		예시	2.0; 1.5
P548	purity (spray pyrolysis coating / solvent)	정의	purity of solvent
		동의어	
		타입	string
		단위	
		예시	Pro analysis; puris; unknown
P549	supplier (spray pyrolysis coating / solvent)	정의	supplier of solvent
		동의어	seller
		타입	string
		단위	
		예시	Sigma Aldrich; Acros; Fischer
P550	composition (spray pyrolysis coating / atmosphere)	정의	composition of atmosphere
		동의어	environmental composition
		타입	dictionary of {gas_(n) : partial pressure, ... }
		단위	bar
		예시	{Ar:0.5, N_{2}:0.5}; {N_2:1}; {Vacuum}
P551	total pressure (spray pyrolysis coating / atmosphere)	정의	total pressure of atmosphere
		동의어	
		타입	number
		단위	bar
		예시	1.0; 0.95
P552	relative humidity (spray pyrolysis coating / atmosphere)	정의	relative humidity of atmosphere
		동의어	
		타입	number
		단위	%
		예시	28.4

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P553	substrate temperature (spray pyrolysis coating)	정의	process temperature
		동의어	
		타입	number
		단위	K
		예시	298
P554	solvent (solvent annealing)	정의	The solvent used in the annealing step • If the atmosphere is a mixture of different solvents and gases, e.g. A and B, list them in alphabetic order and separate them with semicolons: as in (A; B)
		동의어	
		타입	string
		단위	
		예시	DMSO, DMF; DMSO
P555	temperature (solvent annealing)	정의	process temperature
		동의어	
		타입	number
		단위	K
		예시	298
P556	time (solvent annealing)	정의	time of the process
		동의어	
		타입	number
		단위	s
		예시	20
P557	time (storage before process)	정의	The time between the additional layer front is finalised and the next layer is deposited • If there are uncertainties, only state the best estimate, e.g. write 35 and not 20-50.
		동의어	aging time
		타입	number
		단위	h
		예시	2.3
P558	atmosphere (storage before process)	정의	The atmosphere in which the sample with the finalised additional layer front is stored until the next deposition step.
		동의어	
		타입	string
		단위	
		예시	air; N2; vacuum
P559	relative humidity (storage before process)	정의	The relive humidity under which the sample with the finalised additional layer front is stored until next deposition step • If there are uncertainties, only state the best estimate, e.g. write 35 and not 20-50.
		동의어	
		타입	number
		단위	%
		예시	38
P560	surface pre-treatment (surface pre-treatment)	정의	Description of any type of surface treatment or other treatment the sample with the finalised additional layer front undergoes before the next deposition step. • If more than one treatment, list the treatments and separate them by a double forward angel bracket (' >> ') • If no special treatment, state that as 'none' • This category was included after the projects initial phase wherefor the list of reported categories is short. Thus, be prepared to expand the given list of alternatives in the data template.
		동의어	pre-treatment
		타입	string
		단위	
		예시	none; Ar plasma

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P561	name (evaporation deposition / evaporating material)	정의	name of materials
		동의어	
		타입	string
		단위	
		예시	Al; Cu-Al;
P562	composition (evaporation deposition / evaporating material)	정의	composition of evaporating materials; atomic ratio of each constituent of solvent
		동의어	
		타입	dictionary of {constituent : value, concentration : value, supplier : value, purity : value}
		단위	at.%
		예시	{constituent : Cu, concentration : 56.0, supplier : Aldrich, purity : 99.999}
P563	supplier (evaporation deposition / evaporating material)	정의	supplier of materials
		동의어	seller
		타입	string
		단위	
		예시	Sigma Aldrich; Acros
P564	purity (evaporation deposition / evaporating material)	정의	purity of evaporating material
		동의어	
		타입	number
		단위	at.%
		예시	99.999
P565	age (evaporation deposition / evaporating material)	정의	The age of the evaporating materials • As a general guideline, the age refers to the time from the production data of the evaporating material to the process date. • When the age of the material is not known, state that as ‘nan’
		동의어	
		타입	number
		단위	d
		예시	25
P566	state (evaporation deposition / evaporating material)	정의	The physical state of the evaporating materials; The two basic categories are Solid/Liquid
		동의어	
		타입	string
		단위	
		예시	liquid; gas; solid
P567	time (vacuum suction)	정의	suction time
		동의어	
		타입	number
		단위	s
		예시	10
P568	name (CVD / catalyst)	정의	name of catalyst
		동의어	
		타입	string
		단위	
		예시	Au
P569	amount (CVD / catalyst)	정의	amount of catalyst (number of catalytic particles per unit area)
		동의어	
		타입	number
		단위	cm^{-2}
		예시	100; 200; 250

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P570	name (CVD / promotor)	정의	name of promoter
		동의어	
		타입	string
		단위	
		예시	NaCl; KCl
P571	amount (CVD / promotor)	정의	amount of promoter
		동의어	
		타입	number
		단위	mg
		예시	100; 200; 250
P572	depth (DTM)	정의	depth of cut in diamond turning process
		동의어	
		타입	number
		단위	\mu m
		예시	10
P573	feed rate (DTM)	정의	feedrate in diamond turning process
		동의어	feed
		타입	number
		단위	\mu m
		예시	1
P574	rpm (DTM)	정의	revolutions per minute in diamond turning process
		동의어	
		타입	numeric
		단위	
		예시	100
P575	membrane type (dialysis)	정의	type of the membrane
		동의어	
		타입	string
		단위	
		예시	tubing; cassette
P576	membrane molecular weight cut-off (dialysis)	정의	molecule weight cut-off (MWCO) of the membrane
		동의어	
		타입	number
		단위	kDa (kilo Dalton)
		예시	10; 200
P577	dialysis buffer (dialysis)	정의	buffer solution used for the process
		동의어	
		타입	string
		단위	
		예시	bicarbonate
P578	temperature (dialysis)	정의	temperature of the process
		동의어	
		타입	number
		단위	K
		예시	300
P579	time (dialysis)	정의	time of each dialysis step
		동의어	
		타입	number
		단위	h
		예시	2

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P580	name (ion intercalation / intercalation reagent)	정의	name of intercalation reagent
		동의어	
		타입	string
		단위	
		예시	n-butyllithium
P581	amount (ion intercalation / intercalation reagent)	정의	amount of intercalation reagent
		동의어	
		타입	number
		단위	M; mol/L
		예시	2
P582	name (ion intercalation / matrix)	정의	name of matrix
		동의어	
		타입	string
		단위	
		예시	MoS_{2}
P583	amount (ion intercalation / matrix)	정의	amount of matrix (host material)
		동의어	
		타입	number
		단위	mg
		예시	64,
P584	temperature (ion intercalation)	정의	reaction temperature for ion intercalation
		동의어	
		타입	number
		단위	K
		예시	673
P585	time (ion intercalation)	정의	reaction time for the ion intercalation
		동의어	
		타입	number
		단위	h
		예시	0.75
P586	name (rinsing / solvent)	정의	name of solvent
		동의어	
		타입	string
		단위	
		예시	water
P587	amount (rinsing / solvent)	정의	amount of solvent
		동의어	
		타입	number
		단위	L
		예시	20
P588	hearth (arc melting)	정의	a water-cooled metal tray for melting and holding molten metal during the arc melting process. It provides a stable, non-contaminating surface for high-temperature metal processing, ensuring efficient heat dissipation and preventing unwanted reactions between the molten material and the container
		동의어	reaction chamber; reactor vessel
		타입	string
		단위	
		예시	Cu hearth

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P589	temperature (arc melting)	정의	processing temperature reached during the reaction process in the furnace
		동의어	
		타입	number
		단위	K
		예시	122
P590	heating rate (arc melting)	정의	the rate at which temperature increases to the processing temperature
		동의어	
		타입	number
		단위	K h^{-1}
		예시	12
P591	holding time (arc melting)	정의	duration for maintaining the sample at the processing temperature
		동의어	
		타입	number
		단위	h
		예시	3
P592	cooling rate (arc melting)	정의	rate to lower the temperature after melting
		동의어	
		타입	number
		단위	K h^{-1}
		예시	500
P593	equipment (ball milling)	정의	conventional name or model name of equipment used
		동의어	
		타입	string
		단위	
		예시	Pulverisette 6
P594	mold material (casting)	정의	substances used to create the mold cavity into which molten metal is poured
		동의어	die material; form material
		타입	string
		단위	
		예시	sand; SiO2
P595	coating material (dip coating)	정의	material to be dip coated
		동의어	
		타입	string
		단위	
		예시	polysulfone; SiO2
P596	coating solvent (dip coating)	정의	solvent to dissolve or disperse a coating material
		동의어	
		타입	string
		단위	
		예시	ethanol; water
P597	material (dip coating / substrate)	정의	substrate material
		동의어	
		타입	string
		단위	
		예시	glass; Al foil
P598	speed (dip coating)	정의	dipping or withdrawing speed
		동의어	
		타입	number
		단위	mm s^{-1}
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P599	temperature (dip coating)	정의	temperature of the process
		동의어	process environment
		타입	number
		단위	K
		예시	300
P600	temperature (direct current poling)	정의	poling temperature
		동의어	
		타입	number
		단위	K
		예시	350
P601	time (direct current poling)	정의	time applying the high voltage
		동의어	
		타입	number
		단위	min
		예시	30
P602	atmosphere (direct current poling)	정의	poling atmosphere
		동의어	process environment
		타입	string
		단위	
		예시	Air, Silicone oil, Ni_{2}
P603	inner diameter of the ampoule (encapsulated melting)	정의	the measurement of the internal width of the ampoule (quartz tube), typically expressed in millimeters, affecting the volume of the contained material and the ease of handling during experiments or storage
		동의어	
		타입	number
		단위	mm
		예시	10
P604	melting method (encapsulated melting)	정의	the technique used to transform solid materials into liquid form through the application of heat, including processes like induction melting, arc melting, and furnace melting
		동의어	
		타입	string
		단위	
		예시	furnace melting; induction melting
P605	heating rate (encapsulated melting / thermal cycle)	정의	the speed at which the temperature of a material increases over time, typically expressed in degrees per unit time, influencing the material's thermal behavior and phase transformations
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P606	melting temperature (encapsulated melting / thermal cycle)	정의	temperature at which a solid material transform to a liquid state, with both solid and liquid phases coexisting in equilibrium during this process
		동의어	
		타입	number
		단위	K
		예시	2000
P607	holding time (encapsulated melting / thermal cycle)	정의	duration at which the melted state of materials is maintained at the melting temperature
		동의어	
		타입	number
		단위	h
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P608	cooling rate (encapsulated melting / thermal cycle)	정의	the speed at which a material's temperature decreases over time, usually measured in degrees per unit time, affecting crystallization, microstructure, and material properties
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P609	cooling method (encapsulated melting)	정의	the technique used to reduce a material's temperature after heating, including methods such as air cooling, water quenching, oil quenching, and controlled cooling, impacting the material's microstructure and properties
		동의어	
		타입	string
		단위	
		예시	air cooling; water quenching; oil quenching
P610	post heat-treatment temperature (encapsulated melting)	정의	the specific temperature at which a material is processed after solidification, aimed at enhancing its mechanical properties, relieving stresses, or achieving desired microstructural changes
		동의어	
		타입	number
		단위	K
		예시	673
P611	post heat-treatment time (encapsulated melting)	정의	the duration for which a material is held at the post heat-treatment temperature, allowing for effective diffusion, phase transformations, and improvement of mechanical properties
		동의어	
		타입	number
		단위	h
		예시	12
P612	atmosphere (filtration)	정의	atmospheric environment to which a given filtration process is exposed
		동의어	
		타입	string
		단위	
		예시	air; Ar
P613	time (filtration)	정의	filtration processing time
		동의어	
		타입	number
		단위	h
		예시	60
P614	method (grinding)	정의	method or model name of grinder
		동의어	
		타입	string
		단위	
		예시	hand grinding; Beuhler grinder
P615	speed (grinding)	정의	rotating speed of rubbing materials by grinder
		동의어	
		타입	number
		단위	rpm (revolutions per minute)
		예시	1
P616	time (grinding)	정의	duration of grinding process
		동의어	
		타입	number
		단위	h
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P617	name (hot injection / precursor)	정의	name of precursor
		동의어	
		타입	string
		단위	
		예시	PtCl_{4}
P618	concentration (hot injection / precursor)	정의	concentration of precursor in solvent
		동의어	
		타입	number
		단위	mol L^{-1}
		예시	1.5
P619	amount (hot injection / precursor)	정의	amount of precursor
		동의어	amount
		타입	number
		단위	mg
		예시	10
P620	temperature (hot injection / precursor)	정의	precursor temperature
		동의어	
		타입	number
		단위	K
		예시	298
P621	composition (hot injection / solvent)	정의	composition of solvent
		동의어	
		타입	dictionary of {volume fraction : {constituent : concentration, … }, unit : none}
		단위	
		예시	{volume fraction : {H20 : 0.9, acetone : 0.1}, unit : none}
P622	volume (hot injection / solvent)	정의	volume of solvent
		동의어	amount
		타입	number
		단위	mL
		예시	10
P623	ligand composition (hot injection)	정의	chemical composition of the ligands
		동의어	
		타입	dictionary of {constituent : fraction, ...}
		단위	none
		예시	{Oleic acid : 0.7, Oleylamine : 0.3}
P624	reaction temperature (hot injection)	정의	temperature at which a reaction occurs
		동의어	
		타입	number
		단위	K
		예시	298
P625	reaction time (hot injection)	정의	time of the reaction
		동의어	
		타입	number
		단위	s
		예시	20
P626	quenching temperature (hot injection)	정의	temperature at which a quenching occurs
		동의어	
		타입	number
		단위	K
		예시	298

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P627	quenching time (hot injection)	정의	time of the quenching
		동의어	
		타입	number
		단위	s
		예시	20
P628	composition (hot injection / atmosphere)	정의	composition of atmosphere in partial pressure
		동의어	
		타입	dictionary of {value : {constituent : composition}, unit : unit value}
		단위	bar
		예시	{value : {Ar : 0.5, N_2 : 0.5}, unit : bar}; {value : {vacuum : 0}, unit : bar}
P629	total pressure (hot injection / atmosphere)	정의	total pressure of atmosphere; sum of partial pressures in composition value
		동의어	
		타입	number
		단위	bar
		예시	1.0; 0.95
P632	sintering pressure (hot pressing / sintering cycle)	정의	the force applied during the sintering process to compact powder materials, typically expressed in pascals (Pa), influencing the density, microstructure, and properties of the final sintered product
		동의어	
		타입	number
		단위	MPa
		예시	70
P633	sintering temperature (hot pressing / sintering cycle)	정의	the temperature at which the pressure is applied and sintering occurs, typically expressed in K
		동의어	
		타입	number
		단위	K
		예시	2000
P634	heating rate (hot pressing / sintering cycle)	정의	the speed at which the temperature of a material increases over time, typically expressed in degrees per unit time, influencing the material's thermal behavior and phase transformations
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P635	sintering time (hot pressing / sintering cycle)	정의	duration for applying sintering pressure at sintering temperature
		동의어	
		타입	number
		단위	h
		예시	10
P636	sintering atmosphere (hot pressing / sintering cycle)	정의	the gas environment present during the sintering process, which can include inert gases, reducing or oxidizing conditions, and influences the material's properties, reactions, and overall sintering behavior
		동의어	
		타입	string
		단위	
		예시	inert gases; vacuum
P637	traveling speed (direct ink writing)	정의	the velocity at which the print head (or the substrate) moves during the deposition process
		동의어	feed rate; printing speed; scan rate
		타입	number
		단위	mm s^{-1}
		예시	1

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P638	dispensing pressure (direct ink writing)	정의	the applied pressure (in Pa) controlling ink flow through the nozzle, affecting droplet size, line thickness, and consistency of the deposited material during printing.
		동의어	ink delivery pressure; flow rate control; nozzle pressure
		타입	number
		단위	Pa
		예시	1
P639	nozzle diameter (direct ink writing)	정의	the internal width of the nozzle (measured in millimeter) that determines the resolution, ink flow rate, and minimal feature size possible during the ink printing process
		동의어	nozzle width; printing head size
		타입	number
		단위	mm
		예시	0.5
P640	3D modelling SW (direct ink writing)	정의	software tools like SolidWorks, AutoCAD, or Cura used to design 3D models and simulate printing paths, ensuring precise control of material deposition and structural accuracy.
		동의어	CAD tools; design software; modeling program
		타입	string
		단위	
		예시	TinkerCAD with Ultimaker Cura
P641	reaction temperature (interfacial reaction)	정의	temperature at which the interfacial reaction takes place
		동의어	
		타입	number
		단위	K
		예시	293
P642	name (interfacial reaction / composition phase 1 / reactant)	정의	name of reactant material
		동의어	
		타입	string
		단위	
		예시	m-phenylenediamine
P643	concentration (interfacial reaction / composition phase 1 / reactant)	정의	concentration of reactant material
		동의어	
		타입	number
		단위	mol L^{-1}
		예시	1
P644	name (interfacial reaction / composition phase 1 / additive)	정의	name of additive material
		동의어	
		타입	string
		단위	
		예시	sodium chloride
P645	concentration (interfacial reaction / composition phase 1 / additive)	정의	concentration of additive material
		동의어	
		타입	number
		단위	mol L^{-1}
		예시	1
P646	name (interfacial reaction / composition phase 1 / surfactant)	정의	name of surfactant material
		동의어	
		타입	string
		단위	
		예시	SDS

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P647	concentration (interfacial reaction / composition phase 1 / surfactant)	정의	concentration of surfactant material
		동의어	
		타입	number
		단위	mol L^{-1}
		예시	0.1
P648	solvent (interfacial reaction / composition phase 1)	정의	solvent of phase 1 for interfacial reaction
		동의어	
		타입	string
		단위	
		예시	water; ethanol
P649	name (interfacial reaction / composition phase 2 / reactant)	정의	name of reactant material
		동의어	
		타입	string
		단위	
		예시	m-phenylenediamine
P650	concentration (interfacial reaction / composition phase 2 / reactant)	정의	concentration of reactant material
		동의어	
		타입	number
		단위	mol L^{-1}
		예시	1
P651	name (interfacial reaction / composition phase 2 / additive)	정의	name of additive material
		동의어	
		타입	string
		단위	
		예시	sodium chloride
P652	concentration (interfacial reaction / composition phase 2 / additive)	정의	concentration of additive material
		동의어	
		타입	number
		단위	mol L^{-1}
		예시	1
P653	name (interfacial reaction / composition phase 2 / surfactant)	정의	name of surfactant material
		동의어	
		타입	string
		단위	
		예시	SDS
P654	concentration (interfacial reaction / composition phase 2 / surfactant)	정의	concentration of surfactant material
		동의어	
		타입	number
		단위	mol L^{-1}
		예시	0.1
P655	solvent (interfacial reaction / composition phase 2)	정의	solvent of phase 2 for interfacial reaction
		동의어	
		타입	string
		단위	
		예시	water; ethanol
P656	substrate (interfacial reaction)	정의	substrate on which the interfacial reaction takes place
		동의어	
		타입	string
		단위	
		예시	polysulfone; polyacrylonitrile

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P657	reaction time (interfacial reaction)	정의	total duration of the interfacial reaction
		동의어	
		타입	number
		단위	min
		예시	5
P658	milling method (mechanical alloying)	정의	the method or equipment used for the mechanical process of mechanical alloying, including techniques like ball milling, rod milling, or jet milling, each with distinct operational principles and applications
		동의어	
		타입	string
		단위	
		예시	ball milling; rod milling
P659	milling ball material (mechanical alloying)	정의	the composition of the balls used in milling processes, which can include materials like stainless steel, tungsten carbide, or ceramic, chosen based on their hardness, density, and suitability for the specific milling application
		동의어	
		타입	string
		단위	
		예시	stainless steel; tungsten carbide; zirconia
P660	milling ball diameter (mechanical alloying)	정의	the diameter of the milling balls used in a milling process, usually expressed in millimeters, which affects the efficiency of particle size reduction and the energy transferred during milling
		동의어	
		타입	number
		단위	mm
		예시	5; 10
P661	milling speed (mechanical alloying)	정의	the rotational speed of the milling ball container at which the milling equipment operates, usually measured in revolutions per minute (RPM), affecting the effectiveness of the milling process and the energy imparted to the materials
		동의어	
		타입	number
		단위	rpm
		예시	300
P662	milling time (mechanical alloying)	정의	the duration for which materials are subjected to the milling process, influencing the particle size reduction, homogeneity, and overall effectiveness of the milling operation
		동의어	
		타입	number
		단위	h
		예시	12
P663	milling container material (mechanical alloying)	정의	the type of material used to construct the container in which milling takes place, such as steel, ceramic, or plastic, affecting the process's durability, contamination risk, and material compatibility
		동의어	
		타입	string
		단위	
		예시	stainless steel; zirconia
P664	milling container volume (mechanical alloying)	정의	the volume of the container used in the milling process, which affects the amount of material that can be processed, the efficiency of size reduction, and the dynamics of the milling action
		동의어	
		타입	number
		단위	cm^{3}
		예시	500

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P665	milling atmosphere (mechanical alloying)	정의	the environment in which the milling process occurs, including factors like the presence of gases (e.g., inert gases, air, or vacuum), which can influence oxidation, contamination, and material properties during milling
		동의어	
		타입	string
		단위	
		예시	inert gases; air; vacuum
P666	milling agent (mechanical alloying)	정의	the substance added during the milling process to enhance the efficiency of size reduction, improve material flow, reduce agglomeration, or prevent oxidation, thereby optimizing the milling outcome
		동의어	
		타입	string
		단위	
		예시	surfactant; solvent; antioxidant; dispersant
P667	substrate (metal additive manufacturing)	정의	base material or platform onto which the first layers of metal are deposited during the building process
		동의어	
		타입	string
		단위	
		예시	304 stainless steel
P668	tilt angle (metal additive manufacturing)	정의	orientation of the beam nozzle axis measured in the plane perpendicular to the deposition direction
		동의어	lean angle
		타입	number
		단위	degree
		예시	45
P669	welding current (metal additive manufacturing)	정의	electrical current used to generate the arc between the welding wire and the workpiece
		동의어	
		타입	number
		단위	A
		예시	10
P670	equipment (mixing)	정의	conventional name of model name of the mixing equipment
		동의어	
		타입	string
		단위	
		예시	AR-100
P671	metal (molten metal-flux melting)	정의	types of metals and metalloids used in their molten state to form a liquid medium at high temperatures, facilitating the melting and reaction of materials
		동의어	
		타입	string
		단위	
		예시	Pb; Sn; Sb
P672	composition of metals (molten metal-flux melting)	정의	exact molar ratio of each metal used in the reaction
		동의어	
		타입	dictionary of {metal : metal molar ratio, … }
		단위	none
		예시	[{metal : Pb, molar ratio : 10}; [{metal : Ca, molar ratio : 1}, {metal : Zn, molar ratio : 2}, {metal : Sb, molar ratio : 2}, {metal : Pb, molar ratio : 10}]

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P673	reaction container (molten metal-flux melting)	정의	high-temperature resistant crucible, such as alumina or zirconia, sealed in a fused-silica jacket under vacuum to prevent oxidation of the reaction mixtures
		동의어	
		타입	string
		단위	
		예시	Alumina crucible; zirconia crucible
P674	heating rate (molten metal-flux melting)	정의	rate at which the temperature increases during the heating of the reaction mixture
		동의어	
		타입	number
		단위	K h^{-1}
		예시	120
P675	temperature (molten metal-flux melting)	정의	highest temperature reached during the reaction
		동의어	
		타입	number
		단위	K
		예시	1223
P676	holding time (molten metal-flux melting)	정의	duration for maintaining the reaction at the maximum temperature
		동의어	
		타입	number
		단위	h
		예시	24
P677	cooling temperature (molten metal-flux melting)	정의	target temperature to be reached during the cooling process
		동의어	
		타입	number
		단위	K
		예시	823
P678	cooling time (molten metal-flux melting)	정의	time required for the cooling process
		동의어	
		타입	number
		단위	h
		예시	92
P679	holding time at cooled temperature (molten metal-flux melting)	정의	duration for maintaining the reaction at a specific temperature after cooling
		동의어	
		타입	number
		단위	h
		예시	72
P680	centrifuge speed (molten metal-flux melting)	정의	rotation speed of the centrifuge
		동의어	
		타입	numeric
		단위	revolutions min^{-1}
		예시	2400
P681	centrifuge time (molten metal-flux melting)	정의	duration of the centrifuge operation
		동의어	
		타입	number
		단위	min
		예시	3
P682	name (NIPS / polymer solution composition / polymer)	정의	name of polymer material
		동의어	
		타입	string
		단위	
		예시	polysulfone; polyimide

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P683	concentration (NIPS / polymer solution composition / polymer)	정의	concentration of polymer material
		동의어	
		타입	number
		단위	g L^{-1}
		예시	10
P684	name (NIPS / polymer solution composition / additive)	정의	name of additive material
		동의어	
		타입	string
		단위	
		예시	lithium chloride
P685	concentration (NIPS / polymer solution composition / additive)	정의	concentration of additive material
		동의어	
		타입	number
		단위	g L^{-1}
		예시	1
P686	solvent (NIPS / polymer solution composition)	정의	solvent of the polymer solution for non-solvent induced phase separation
		동의어	
		타입	string
		단위	
		예시	dimethyl sulfoxide; water
P687	polymer solution temperature (NIPS)	정의	temperature of the polymer solution
		동의어	
		타입	number
		단위	K
		예시	293
P688	name (NIPS / non-solvent composition / additive)	정의	name of additive material
		동의어	
		타입	string
		단위	
		예시	lithium chloride
P689	concentration (NIPS / non-solvent composition / additive)	정의	concentration of additive material
		동의어	
		타입	number
		단위	g L^{-1}
		예시	0.1
P690	solvent (NIPS / non-solvent composition)	정의	solvent of non-solvent for non-solvent induced phase separation
		동의어	
		타입	string
		단위	
		예시	water; ethanol
P691	non-solvent temperature (NIPS)	정의	temperature at which non-solvent induced phase separation takes place
		동의어	
		타입	number
		단위	K
		예시	293
P692	phase separation time (NIPS)	정의	total duration of non-solvent induced phase separation
		동의어	
		타입	number
		단위	min
		예시	5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P693	plasma gas (plasma treatment)	정의	flow rate of reaction gas in the reactive ion plating; to be used in the case of reactive ion plating
		동의어	process gas
		타입	dictionary of {value : {constituent : concentration, … }, unit : unit value}
		단위	cm^{3} min^{-1}
		예시	{value : {C2H6 : 50, Ar : 100}, unit : SCCM}
P694	plasma power (plasma treatment)	정의	plasma power used in plasma treatment
		동의어	
		타입	number
		단위	W
		예시	10
P695	plasma frequency (plasma treatment)	정의	frequency of pulsed plasma
		동의어	
		타입	number
		단위	Hz
		예시	30
P696	temperature (plasma treatment)	정의	temperature of plasma treatment
		동의어	
		타입	number
		단위	K
		예시	300
P697	pressure (plasma treatment)	정의	pressure of plasma treatment
		동의어	
		타입	number
		단위	kPa
		예시	0.09
P698	time (plasma treatment)	정의	duration of plasma treatment
		동의어	
		타입	number
		단위	s
		예시	10
P699	temperature (quenching)	정의	temperature of quenching media
		동의어	
		타입	number
		단위	K
		예시	30
P700	time (quenching)	정의	duration time of the sample in the quenching media
		동의어	
		타입	number
		단위	min
		예시	1
P701	ingot diameter (Bridgman growth method)	정의	the diameter of the cylindrical ingot produced in the Bridgman method, typically controlled to achieve desired material properties and uniformity.
		동의어	
		타입	number
		단위	mm
		예시	20
P702	ingot length (Bridgman growth method)	정의	the length of the ingot produced in the Bridgman method, typically controlled to achieve desired material properties and uniformity.
		동의어	
		타입	number
		단위	mm
		예시	50

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P703	temperature (Bridgman growth method)	정의	the process temperature precisely controlled to ensure proper melting and solidification of the material, which is crucial for forming high-quality ingots.
		동의어	
		타입	number
		단위	K
		예시	200
P704	holding time (Bridgman growth method)	정의	the duration for which the material is maintained at a specific temperature, allowing uniform melting before controlled solidification.
		동의어	
		타입	number
		단위	h
		예시	24
P705	heating rate (Bridgman growth method)	정의	the rate at which the furnace temperature is increased, affecting the material's melting behavior and reducing the formation of defects.
		동의어	
		타입	number
		단위	K min^{-1}
		예시	10
P706	down rate (Bridgman growth method)	정의	the rate at which the crucible, typically a vacuum-sealed glass or metal tube, is lowered in the furnace, determining the quality of ingot by controlling the crystal growth behavior.
		동의어	
		타입	number
		단위	mm h^{-1}
		예시	0.2
P707	temperature (solid state reaction)	정의	process temperature; reaction temperature
		동의어	
		타입	number
		단위	K
		예시	50
P708	time (solid state reaction)	정의	process time
		동의어	
		타입	number
		단위	h
		예시	10
P709	temperature (spark plasma sintering)	정의	target temperature of the process
		동의어	
		타입	number
		단위	K
		예시	1300
P710	time (spark plasma sintering)	정의	duration for applying uniaxial pressure and pulsed direct current
		동의어	
		타입	number
		단위	min
		예시	30
P711	heating rate (spark plasma sintering)	정의	rate at which temperature is raised
		동의어	temperature raising rate
		타입	number
		단위	K min^{-1}
		예시	500

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P712	average powder size (spark plasma sintering)	정의	size of the used powder
		동의어	
		타입	number
		단위	{\mu m}
		예시	20
P713	weight (sol-gel synthesis / precursor)	정의	weight of precursor
		동의어	
		타입	number
		단위	mg
		예시	64
P714	chemical information (spin coating / anti-solvent)	정의	composition of anti-precursor
		동의어	
		타입	dictionary of {value : {constituent : composition, …}, unit : unit value}
		단위	none
		예시	{value : {chlorobenzene : 0.5, acetonitrile : 0.5}, unit : none}
P715	dropping time (spin coating / anti-solvent)	정의	time of the process
		동의어	
		타입	number
		단위	s
		예시	20
P716	dropping volume (spin coating / anti-solvent)	정의	volume of anti-solvent
		동의어	
		타입	number
		단위	mL
		예시	20
P717	thickness (evaporation deposition)	정의	thickness of the deposited layer
		동의어	
		타입	number
		단위	nm
		예시	100
P718	name (washing / washing solvent)	정의	name of solvent
		동의어	
		타입	string
		단위	
		예시	MDF; DI water; EtOH
P719	time (washing)	정의	washing time
		동의어	
		타입	number
		단위	min
		예시	2
P720	temperature (e-beam lithography / develop)	정의	developer temperature
		동의어	
		타입	number
		단위	K
		예시	298
P721	material (electrospinning / collector)	정의	the substrate material onto which the electro spun fibers are deposited
		동의어	support, underlying material
		타입	string
		단위	
		예시	aluminum foil; copper; stainless steel glass; PP

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P722	crucible (heat treatment)	정의	a container used to hold the sample undergoing heat treatment
		동의어	dish, container, vessel
		타입	string
		단위	
		예시	alumina crucible; quartz plate
P723	layer thickness (additive manufacturing)	정의	the thickness of each printed layer
		동의어	
		타입	number
		단위	mm
		예시	0.2
P724	print speed (additive manufacturing)	정의	the speed at which the material is deposited during printing
		동의어	
		타입	number
		단위	mm s^{-1}
		예시	50
P725	nozzle temperature (additive manufacturing)	정의	the temperature of the printing nozzle
		동의어	
		타입	number
		단위	K
		예시	500
P726	bed temperature (additive manufacturing)	정의	the temperature of the print bed during printing
		동의어	
		타입	number
		단위	K
		예시	300
P727	infusion pattern (additive manufacturing)	정의	the pattern of resin infusion in printed composites
		동의어	
		타입	string
		단위	
		예시	
P728	electric capacity (arc melting)	정의	quantity of electric charge stored or released by a capacitor per unit voltage
		동의어	
		타입	number
		단위	W
		예시	400
P729	pressure (autoclave molding)	정의	the applied pressure during the autoclave molding process
		동의어	
		타입	number
		단위	MPa
		예시	0.6
P730	temperature cycle (autoclave molding)	정의	the cycle of heating, holding, and cooling during the process
		동의어	
		타입	string
		단위	
		예시	120 °C heating, 30 min hold, cooling to 60 °C
P731	vacuum level (autoclave molding)	정의	the level of vacuum applied to the mold during processing
		동의어	
		타입	number
		단위	Pa
		예시	500

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P732	curing time (autoclave molding)	정의	the total time required for curing in the autoclave
		동의어	
		타입	number
		단위	min
		예시	150
P733	tool material (autoclave molding)	정의	the material used for the mold tooling
		동의어	
		타입	string
		단위	
		예시	Al; Stainless steel; carbon composite; FRP; ceramic
P734	fiber tow width (AFP)	정의	the nominal width of each fiber tow supplied to the AFP head, specifying the range of continuous tows deposited in a single pass
		동의어	
		타입	string
		단위	mm
		예시	6-12
P735	placement speed (AFP)	정의	the linear speed at which the AFP head applies fiber tows onto the tool, balancing throughput with consolidation quality
		동의어	
		타입	number
		단위	m min^{-1}
		예시	300
P736	consolidation method (AFP)	정의	the technique used to bond and compact the laid-up tows during AFP, typically involving a heated roller applying pressure to achieve proper adhesion and void reduction
		동의어	
		타입	string
		단위	
		예시	heat and pressure roller
P737	tape width (ATP)	정의	the width of the prepreg tape used in ATP, defining the coverage per layup pass and influencing the resolution and efficiency of the build
		동의어	
		타입	number
		단위	mm
		예시	25-65
P738	placement speed (ATP)	정의	the rate at which prepreg tape is laid onto the mold surface in ATP, affecting both production speed and laminate quality
		동의어	
		타입	number
		단위	m min^{-1}
		예시	150
P739	temperature (ATP)	정의	the processing temperature at which the prepreg tape is heated during ATP to activate its resin system and ensure proper flow and bonding
		동의어	
		타입	number
		단위	K
		예시	500
P740	consolidation pressure (ATP)	정의	the pressure applied to the tape against the mold in ATP to compact the layers and eliminate trapped air
		동의어	
		타입	number
		단위	bar
		예시	5

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P741	material type (ATP)	정의	the specific prepreg material processed by ATP defining its resin chemistry and handling characteristics
		동의어	
		타입	string
		단위	
		예시	thermoplastic prepreg
P742	materials type (blow (bottle))	정의	type of materials to be processed
		동의어	
		타입	string
		단위	
		예시	PE; PP; PET; PS
P743	material loading (blow (bottle))	정의	the quantity of material used to create the parison
		동의어	
		타입	number
		단위	g
		예시	
P744	temperature (blow (bottle))	정의	temperature of the polymer during processing
		동의어	
		타입	number
		단위	K
		예시	
P745	length (blow (bottle) / parison)	정의	length of parison
		동의어	
		타입	number
		단위	cm
		예시	
P746	diameter (blow (bottle) / parison)	정의	diameter of parison
		동의어	
		타입	number
		단위	cm
		예시	
P747	blow pressure (blow (bottle))	정의	pressure applied to the air or gas that inflates the parison whthin the mold
		동의어	
		타입	number
		단위	MPa
		예시	
P748	inflation rate (blow (bottle))	정의	speed at which the air or gas is introduced into the parison
		동의어	
		타입	number
		단위	cm^{3} min^{-1} (sccm)
		예시	
P749	mold temperature (blow (bottle))	정의	temperature of the mold during the blowing process
		동의어	
		타입	number
		단위	K
		예시	
P750	cooling time (blow (bottle))	정의	total time allowed for the blown part to cool
		동의어	
		타입	number
		단위	min
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P751	cycle time (blow (bottle))	정의	total time required to complete one blowing cycle, including heating, blowing, cooling and ejection
		동의어	
		타입	number
		단위	min
		예시	
P752	materials type (blown film extrusion)	정의	type of materials to be processed
		동의어	
		타입	string
		단위	
		예시	PE; PP; PET; PS
P753	melt temperature (blown film extrusion)	정의	temperature of the polymer as it is extruded
		동의어	
		타입	number
		단위	K
		예시	
P754	die temperature (blown film extrusion)	정의	the temperature of the die through which the polymer is extruded
		동의어	
		타입	number
		단위	K
		예시	
P755	blow-up ratio (blown film extrusion)	정의	the ratio of the diameter of the inflated film to the diameter of the die
		동의어	
		타입	number
		단위	none
		예시	2; 6
P756	extruder speed (blown film extrusion)	정의	the speed at which the polymer is fed through the extruder
		동의어	
		타입	number
		단위	kg h^{-1}
		예시	
P757	inflation pressure (blown film extrusion)	정의	the pressure applied to inflate the tube
		동의어	
		타입	number
		단위	MPa
		예시	
P758	cooling method (blown film extrusion)	정의	the technique used to cool the film after it exits the die
		동의어	
		타입	string
		단위	
		예시	air cooling; water cooling
P759	take-up speed (blown film extrusion)	정의	the speed at which the finished film is collected onto rolls after it has been cooled and solidified.
		동의어	
		타입	number
		단위	m h^{-1}
		예시	
P760	film thickness (blown film extrusion)	정의	the final thickness of the extruded film
		동의어	
		타입	number
		단위	{\mu m}
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P761	additives (blown film extrusion)	정의	chemicals added to the polymer before or during the blown film process to enhance specific properties
		동의어	
		타입	string
		단위	
		예시	ethoxylated alcohols
P762	reaction container (Bridgman growth method)	정의	A reaction tube in the Bridgman process refers to a specialized containment vessel designed to hold and transport molten materials during directional solidification. It provides a controlled environment for phase transitions, ensuring uniform heat distribution, minimizing contamination, and facilitating the formation of high-quality single crystals or polycrystalline structures. Reaction tubes are commonly made from quartz, alumina, or refractory metals to withstand high temperatures and prevent unwanted interactions with the molten material,
		동의어	
		타입	string
		단위	
		예시	quartz; alumina; refactory metals
P763	materials type (calendering)	정의	type of materials to be processed
		동의어	
		타입	string
		단위	
		예시	PE; PP; PET; PS
P764	roller temperature (calendering)	정의	temperature of the calender rolls
		동의어	
		타입	number
		단위	K
		예시	
P765	roller speed (calendering)	정의	rotational speed of the calender rolls
		동의어	
		타입	number
		단위	m min^{-1}
		예시	
P766	roller gap (calendering)	정의	distance between the calender rolls
		동의어	
		타입	number
		단위	mm
		예시	
P767	configuration (calendering)	정의	configuration of the calender
		동의어	
		타입	string
		단위	
		예시	two-roll calender; three-roll calender
P768	feed rate (calendering)	정의	the rate at which the polymer material is fed into the calender machine
		동의어	
		타입	number
		단위	g min^{-1}
		예시	
P769	cooling method (calendering)	정의	technique used to cool the film or sheet after it is formed
		동의어	
		타입	string
		단위	
		예시	air cooling; water cooling

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P770	slip agent (calendering)	정의	additives used to enhance processability, improve surface quality and reduce friction between the polymer and rollers
		동의어	
		타입	string
		단위	
		예시	paraffin; silicone oil; PEG
P771	CVD type (CVD)	정의	type of energy supplied for chemical vapor deposition
		동의어	
		타입	string
		단위	
		예시	
P772	applied voltage (CVD / power supply)	정의	electric voltage applied to the CVD system
		동의어	
		타입	number
		단위	V
		예시	50; 1000; 5000
P773	applied current (CVD / power supply)	정의	electric current apply to the CVD system
		동의어	
		타입	number
		단위	A
		예시	50; 0.1; 3e-2
P774	applied power (CVD / power supply)	정의	electric power used for the CVD system
		동의어	
		타입	number
		단위	W
		예시	100; 300
P775	pulse width (CVD / power supply)	정의	turn-on period of pulsed DC power supply
		동의어	
		타입	number
		단위	ms
		예시	250
P776	frequency (CVD / power supply)	정의	frequency of AC or RF power used for the CVD system
		동의어	
		타입	number
		단위	Hz
		예시	250e3; 13.56e6
P777	laser (CVD)	정의	laser used for laser assisted CVD
		동의어	
		타입	string
		단위	
		예시	CO_{2} laser 100W; Xe layer 80W pulse
P778	microwave source (CVD)	정의	microwave power source for microwave CVD
		동의어	
		타입	string
		단위	
		예시	ESI micros wave plasma source; AETHALON 500W; APEX 1.5KW
P779	filament (CVD)	정의	filament for hot filament CVD
		동의어	
		타입	string
		단위	
		예시	W-Ta 1mm dia;

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P780	jig type (CVD / jig)	정의	type of jig
		동의어	
		타입	string
		단위	
		예시	columnar structure; simple plate; cubic columnar where specimen can be fixed vertically
P781	rotating speed (CVD / jig)	정의	speed of rotation in rpm
		동의어	
		타입	number
		단위	revolution min^{-1}
		예시	10; 1
P782	solvent (cleaning)	정의	solvent used in the cleaning sequence
		동의어	
		타입	string
		단위	
		예시	Ethanol; TCE; Soap; Acetone; DI water
P783	time (cleaning)	정의	cleaning time of the sequence
		동의어	
		타입	number
		단위	min
		예시	30; 60
P784	materials type (compression molding)	정의	type of polymer or elastomer used
		동의어	
		타입	string
		단위	
		예시	thermoset polymers; elastomers; thermoplastic polymers
P785	material loading (compression molding)	정의	the quantity of material placed in the mold cavity before compression
		동의어	
		타입	number
		단위	g
		예시	
P786	material flow characteristics (compression molding)	정의	the flow behavior of the material under heat and pressure during molding
		동의어	
		타입	string
		단위	
		예시	Viscous and homogeneous under molding conditions; High melt viscosity required cyclic pressurization to ensure uniform flow and fiber impregnation
P787	compression pressure (compression molding)	정의	the pressure applied to the material during molding
		동의어	
		타입	number
		단위	MPa
		예시	4
P788	molding temperature (compression molding)	정의	the temperature applied during the compression molding process
		동의어	
		타입	number
		단위	K
		예시	435
P789	molding time (compression molding)	정의	the total time required for the compression molding process
		동의어	
		타입	number
		단위	min
		예시	20

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P790	heating method (compression molding)	정의	method of heating
		동의어	
		타입	string
		단위	
		예시	electric heating; hot water; steam
P791	heating rate (compression molding)	정의	the rate at which temperature is increased during molding
		동의어	
		타입	number
		단위	K min^{-1}
		예시	5
P792	cooling method (compression molding)	정의	method to remove heat from the mold
		동의어	
		타입	string
		단위	
		예시	air cooling; water cooling
P793	cooling rate (compression molding)	정의	the rate at which temperature is decreased during cooling
		동의어	
		타입	number
		단위	K min^{-1}
		예시	3
P794	temperature (conditioning)	정의	the specific temperature at which the conditioning process occurs, influencing the polymer's molecular behavior and performance
		동의어	
		타입	number
		단위	K
		예시	
P795	humidity (conditioning)	정의	the level of moisture in the air surrounding the polymer during the conditioning
		동의어	
		타입	number
		단위	%
		예시	
P796	time (conditioning)	정의	the duration for which the polymer is subjected to the conditioning conditions
		동의어	
		타입	number
		단위	h
		예시	
P797	airflow (conditioning)	정의	the movement of air around the polymer during conditioning, which can enhance moisture absorption or evaporation depending on the process design
		동의어	
		타입	number
		단위	cm^{3} min^{-1}
		예시	
P798	initial moisture content (conditioning)	정의	the starting moisture level of the polymer prior to conditioning, which influences how much additional moisture can be absorbed during the process
		동의어	
		타입	number
		단위	%
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P799	process equipment (conditioning)	정의	the equipment used for conditioning
		동의어	
		타입	string
		단위	
		예시	oven; humidity-controlled chamber; vacuum dryer
P800	chemical additives (conditioning)	정의	additives may be incorporated during conditioning to facilitate moisture absorption or to enhance specific properties
		동의어	
		타입	string
		단위	
		예시	diethyl phthalates; dibutyl phthalate; citrate esters; natural oil; butylated hydroxytoluene (BHT); benzophenones
P801	cooling rate (conditioning)	정의	rate to cool down
		동의어	
		타입	number
		단위	K min^{-1}
		예시	
P803	temperature (crystallization)	정의	the temperature at which crystallization occurs, which can greatly influence the rate and extent of crystallization
		동의어	
		타입	number
		단위	K
		예시	
P804	cooling rate (crystallization)	정의	the rate at which the polymer is cooled after processing, affecting the degree of crystallization
		동의어	
		타입	number
		단위	K min^{-1}
		예시	
P805	time (crystallization)	정의	the duration for which the polymer is held at a specific temperature during crystallization
		동의어	
		타입	number
		단위	min
		예시	
P806	external stress (crystallization)	정의	stress applied during crystallization, which can affect the structure and orientation of the resulting polymer
		동의어	
		타입	number
		단위	MPa
		예시	
P807	polymer molecular weight (crystallization)	정의	molar mass of the polymer chains
		동의어	
		타입	number
		단위	g mol^{-1}
		예시	
P808	additive (crystallization)	정의	chemicals added to modify crystallization behavior and improve the properties of the polymer
		동의어	
		타입	string
		단위	
		예시	talc; sodium benzoate; phthalates
P809	shear rate (crystallization)	정의	the rate of applied shear during processing can affect the crystallization process
		동의어	
		타입	number
		단위	min^{-1}
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P810	cooling medium (crystallization)	정의	the medium used to cool the polymer during crystallization, which can affect cooling rates and crystallization kinetics
		동의어	
		타입	string
		단위	
		예시	air; water; oil baths
P811	humidity (crystallization)	정의	relative humidity in the process
		동의어	
		타입	number
		단위	%
		예시	
P812	atmosphere (crystallization)	정의	atmosphere of the process
		동의어	
		타입	string
		단위	
		예시	Air; O2; N2
P813	orientation (dip coating / substrate)	정의	orientation of substrate
		동의어	
		타입	string
		단위	
		예시	vertical; horizontal
P814	relative humidity (dip coating)	정의	water vapor pressure in the process environment
		동의어	
		타입	number
		단위	%
		예시	
P815	method (direct ink writing)	정의	printing method categorized by the driving mechanism, such as thermal, mechanical or chemical stimuli; method determines how materials is deposited, cured or solidified during the printing
		동의어	deposition process; printing method; fabrication technique
		타입	string
		단위	
		예시	Pneumatic controlled extrusion; electrohydrodynamic jet; ink jet
P816	ampoule material (encapsulated melting)	정의	material of ampoule containing substance to be melted
		동의어	
		타입	string
		단위	
		예시	quartz; stainless steel
P817	ampoule environment (encapsulated melting)	정의	environment inside of the ampoule
		동의어	
		타입	string
		단위	
		예시	vacuum; Ar; H2
P818	melted material (encapsulated melting)	정의	material to be melted
		동의어	
		타입	string
		단위	
		예시	Ga; Mg; Al
P819	atmosphere (evaporation deposition)	정의	atmosphere in the evaporation chamber
		동의어	
		타입	string
		단위	
		예시	vacuum; N2; O2; CH4

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P820	base pressure (evaporation deposition)	정의	the lowest pressure of the processing chamber to pump down without any gas flows
		동의어	
		타입	number
		단위	Torr
		예시	0.000001
P821	evaporation method (evaporation deposition)	정의	type of energy source for evaporating materials to be deposited on substrate
		동의어	
		타입	string
		단위	
		예시	1. electric heating 2. e-beam 3. laser ablation 4. induction heating
P822	source power (evaporation deposition)	정의	power supplied to evaporating materials
		동의어	
		타입	number
		단위	W
		예시	60
P823	evaporation type (evaporation deposition)	정의	type of vaporized species
		동의어	
		타입	string
		단위	
		예시	1. atomic evaporation 2. molecular evaporation 3. reactive evaporation
P824	substrate distance (evaporation deposition)	정의	the distance between the evaporation source and the substrate
		동의어	working distance
		타입	number
		단위	mm
		예시	60
P825	substrate rotation (evaporation deposition)	정의	rotation of substrate for uniform deposition
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	20
P826	materials type (extrusion)	정의	type of materials to be processed
		동의어	
		타입	string
		단위	
		예시	PE; PP; PET; PS
P827	material feed rate (extrusion)	정의	the rate at which polymer pellets or granules are fed into the extruder
		동의어	
		타입	number
		단위	g min^{-1}
		예시	
P828	T feed zone (extrusion / temperature profile)	정의	temperature at feed zone
		동의어	
		타입	number
		단위	K
		예시	
P829	T compression zone (extrusion / temperature profile)	정의	temperature at compression zone
		동의어	
		타입	number
		단위	K
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P830	T metering zone (extrusion / temperature profile)	정의	temperature at metering zone
		동의어	
		타입	number
		단위	K
		예시	
P831	extruder speed (extrusion)	정의	rotational speed of the screw
		동의어	
		타입	number
		단위	rpm
		예시	
P832	line speed (extrusion)	정의	the speed at which the extruded material is pulled away from the die
		동의어	
		타입	number
		단위	cm min^{-1}
		예시	
P833	pressure (extrusion)	정의	pressure applied within the extruder at the metering section
		동의어	
		타입	number
		단위	MPa
		예시	
P834	residence time (extrusion)	정의	time the polymer spends in the extruder
		동의어	
		타입	number
		단위	min
		예시	
P835	cooling method (extrusion)	정의	method used to cool and solidify the extrudate after leaving the die
		동의어	
		타입	string
		단위	
		예시	air cooling; water baths
P836	fiber tension (filament winding)	정의	the tension of the fiber during the winding process
		동의어	
		타입	number
		단위	N
		예시	40
P837	winding angle (filament winding)	정의	the angle at which fibers are wound around the mandrel
		동의어	
		타입	number
		단위	degree
		예시	45
P838	winding speed (filament winding)	정의	the speed of the winding process
		동의어	
		타입	number
		단위	m min^{-1}
		예시	15
P839	mandrel material (filament winding)	정의	the material used for the mandrel around which fibers are wound
		동의어	
		타입	string
		단위	
		예시	Al; steel; PVC, glass fiber composite

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P840	number of layers (filament winding)	정의	the number of fiber layers wound around the mandrel
		동의어	
		타입	number
		단위	none
		예시	
P841	resin injection rate (filament winding)	정의	supplying rate of the resin on the wired fiber
		동의어	
		타입	number
		단위	l min^{-1}
		예시	
P842	humidity (filament winding / winding atmosphere)	정의	relative humidity during process
		동의어	
		타입	number
		단위	%
		예시	
P843	temperature (filament winding / winding atmosphere)	정의	temperature during winding process
		동의어	
		타입	number
		단위	K
		예시	
P844	time (filament winding / curing)	정의	curing time
		동의어	
		타입	number
		단위	h
		예시	
P845	temperature (filament winding / curing)	정의	curing temperature
		동의어	
		타입	number
		단위	K
		예시	
P846	pressure (filament winding / curing)	정의	curing pressure
		동의어	
		타입	number
		단위	bar
		예시	
P847	flow rate (heat treatment)	정의	volumetric flow rate of the atmospheric gas during heat treatment
		동의어	
		타입	number
		단위	cm^{3} min^{-1}
		예시	100
P848	injection pressure (hot injection)	정의	the force applied by an injection molding machine to push molten material into a mold cavity
		동의어	
		타입	number
		단위	bar
		예시	20
P849	cooling time (injection molding)	정의	the duration the part spends in the mold after the material has been filled and packed
		동의어	
		타입	number
		단위	min
		예시	10

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P850	injection pressure (injection molding)	정의	the force applied to push molten plastic into the mold cavity
		동의어	
		타입	number
		단위	MPa
		예시	100
P851	injection speed (injection molding)	정의	the speed at which molten plastic is injected into a mold
		동의어	
		타입	number
		단위	mm s^{-1}
		예시	320
P852	instrument (injection molding)	정의	instrument used to do injection molding
		동의어	
		타입	string
		단위	
		예시	
P853	molded material (injection molding)	정의	the material used in the injection molding
		동의어	
		타입	string
		단위	
		예시	PP; PE; ABS
P854	melt temperature (injection molding)	정의	the temperature of molten plastic
		동의어	
		타입	number
		단위	K
		예시	550
P855	mold temperature (injection molding)	정의	the temperature of the mold cavity surface
		동의어	
		타입	number
		단위	K
		예시	375
P856	source material (ion beam deposition)	정의	source materials for ion beam generation
		동의어	
		타입	dictionary of {composition : {constituent : concentration, … }, unit : unit value}
		단위	none
		예시	{composition : {C2H6 : 0.5, Ar : 0.5}, unit : none}
P857	model (ion beam deposition / ion source)	정의	name of the manufacturer and the model of the ion source
		동의어	
		타입	string
		단위	
		예시	Thermo Fisher Scientific ISQ QD; AJA A300
P858	mechanism (ion beam deposition / ion source)	정의	method to ionize the vapor of the source material
		동의어	
		타입	string
		단위	
		예시	DC; RF; Microwave; pulsed; hollow cathode
P859	power supplied (ion beam deposition / ion source)	정의	supplied power to ionization mechanism
		동의어	
		타입	number
		단위	W
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P860	base pressure (ion beam deposition)	정의	the lowest pressure of the processing chamber to pump down without any gas flows
		동의어	
		타입	number
		단위	Torr
		예시	0.000001
P861	working pressure (ion beam deposition)	정의	the pressure in the processing chamber during the ion plating process
		동의어	processing pressure
		타입	number
		단위	Torr
		예시	0.001
P862	reaction gas (ion beam deposition)	정의	flow rate of reaction gas in the reactive ion plating; in standard cubic centimeter per minute (sccm); to be used in the case of reactive ion plating
		동의어	
		타입	dictionary of {gas name : value, flow rate : value}
		단위	cm^{3} min^{-1}
		예시	{gas name : N2, flow rate : 50}
P863	substrate distance (ion beam deposition)	정의	the distance between the ion plating source and the substrate
		동의어	working distance
		타입	number
		단위	mm
		예시	60
P864	bias type (ion beam deposition / substrate bias)	정의	type of electric bias applied on the substrate
		동의어	
		타입	string
		단위	
		예시	
P865	applied power (ion beam deposition / substrate bias)	정의	bias power applied on the substrate
		동의어	
		타입	number
		단위	W
		예시	100
P866	bias voltage (ion beam deposition / substrate bias)	정의	bias voltage effectively applied on the substrate
		동의어	
		타입	number
		단위	V
		예시	-300
P867	substrate temperature (ion beam deposition)	정의	temperature of the substrate
		동의어	
		타입	number
		단위	K
		예시	300
P868	substrate rotation (ion beam deposition)	정의	rotation of substrate for uniform deposition
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	20
P869	time (ion beam deposition)	정의	time of sputter deposition process
		동의어	coating time; sputtering time
		타입	number
		단위	s
		예시	120

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P870	composition (ion plating / source material)	정의	composition of source materials
		동의어	
		타입	dictionary of {value : {constituent : concentration, … }, unit : unit value}
		단위	none
		예시	{value : {Au : 0.5, Al : 0.5}, unit : none}
P871	generation method (ion plating / source material)	정의	method to generate the vapor phase of the source material
		동의어	
		타입	string
		단위	
		예시	thermal evaporation; sputtering; arc; laser evaporation
P872	ion source type (ion plating / ionization)	정의	method to ionize the vapor of the source material
		동의어	
		타입	string
		단위	
		예시	DC; RF; Microwave; pulsed; hollow cathode
P873	power supply (ion plating / ionization)	정의	supplied power to ionization mechanism
		동의어	
		타입	number
		단위	W
		예시	100
P874	base pressure (ion plating)	정의	the lowest pressure of the processing chamber to pump down without any gas flows
		동의어	
		타입	number
		단위	Torr
		예시	0.000001
P875	working pressure (ion plating)	정의	the pressure in the processing chamber during the ion plating process
		동의어	processing pressure
		타입	number
		단위	Torr
		예시	0.001
P876	reaction gas (ion plating)	정의	flow rate of reaction gas in the reactive ion plating; to be used in the case of reactive ion plating
		동의어	
		타입	dictionary of {value : {constituent : concentration, … }, unit : unit value}
		단위	cm^{3} min^{-1}
		예시	{value : {C2H6 : 50, Ar : 100}, unit : SCCM}
P877	substrate distance (ion plating)	정의	the distance between the ion plating source and the substrate
		동의어	working distance
		타입	number
		단위	mm
		예시	60
P878	bias type (ion plating / substrate bias)	정의	type of electric bias applied on the substrate
		동의어	
		타입	string
		단위	
		예시	DC; RF; pulse
P879	applied power (ion plating / substrate bias)	정의	bias power applied on the substrate
		동의어	
		타입	number
		단위	W
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P880	bias voltage (ion plating / substrate bias)	정의	bias voltage effectively applied on the substrate
		동의어	
		타입	number
		단위	V
		예시	-300
P881	substrate temperature (ion plating)	정의	temperature of the substrate
		동의어	
		타입	number
		단위	K
		예시	300
P882	substrate rotation speed (ion plating)	정의	rotation speed of substrate for uniform deposition
		동의어	
		타입	number
		단위	revolutions min^{-1}
		예시	20
P883	time (ion plating)	정의	time of sputter deposition process
		동의어	coating time; sputtering time
		타입	number
		단위	s
		예시	120
P884	atmosphere (mixing)	정의	process environment within the mixing chamber
		동의어	
		타입	string
		단위	
		예시	air; Ar; N2; vacuum
P885	encapsulation (molten metal-flux melting)	정의	Encapsulation refers to the process of enclosing or surrounding a material, component, or structure within a protective tube or coating to prevent contamination, enhance durability, and regulate thermal or chemical exposure. In molten metal processing, encapsulation using quartz or metal tubes ensures controlled melting conditions and minimizes unwanted reactions with the surrounding environment
		동의어	
		타입	string
		단위	
		예시	quartz; metallic tube
P886	non-solvent state (NIPS)	정의	the physical statoe of the nonsolvent used in phase separation
		동의어	
		타입	string
		단위	
		예시	liquid; vapor
P887	pressure (out of autoclave system)	정의	the process pressure maintained during out-of-autoclave curing, ranging from vacuum up to 2.5 bar, used to compact the composite and remove entrapped air
		동의어	
		타입	number
		단위	bar
		예시	2.5
P888	cure cycle time (out of autoclave system)	정의	total duration of the out-of-autoclave curing process required to fully polymerize the composite resin at the specified temperature and pressure
		동의어	
		타입	number
		단위	min
		예시	60

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P889	fluid type (out of autoclave system)	정의	the specific heat-transfer fluid circulated through the system—chosen for thermal stability and compatibility with the composite (e.g., silicone oil)
		동의어	
		타입	string
		단위	
		예시	silicone oil
P890	fluid flow rate (out of autoclave system)	정의	the volumetric flow rate of the heat-transfer fluid through the out-of-autoclave system to ensure uniform heating and cooling
		동의어	
		타입	number
		단위	L min^{-1}
		예시	10
P891	fluid temperature (out of autoclave system)	정의	the temperature of the circulating heat-transfer fluid during the curing cycle, controlling the resin cure kinetics and overall process efficiency
		동의어	
		타입	number
		단위	K
		예시	375
P892	power type (plasma treatment)	정의	type of power to generate plasma
		동의어	
		타입	string
		단위	
		예시	DC; RF; MW
P895	fiber volume fraction (prepreg production)	정의	the percentage of fiber volume in the prepreg material
		동의어	
		타입	number
		단위	%
		예시	55
P896	resin content (prepreg production)	정의	the amount of resin within the prepreg material by percentage
		동의어	
		타입	number
		단위	%
		예시	45
P897	prepreg width (prepreg production)	정의	the width of the prepreg sheet
		동의어	
		타입	number
		단위	mm
		예시	500
P898	impregnation method (prepreg production)	정의	the method used for impregnating resin
		동의어	
		타입	string
		단위	
		예시	hot melt; solution
P899	curing temperature (prepreg production)	정의	the temperature required to cure the prepreg
		동의어	
		타입	number
		단위	K
		예시	400
P900	curing time (prepreg production)	정의	the time required to cure the prepreg
		동의어	
		타입	number
		단위	min
		예시	90

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P901	average (prepreg production / storage conditions / temperature)	정의	average temperature
		동의어	
		타입	number
		단위	K
		예시	
P902	variation (prepreg production / storage conditions / temperature)	정의	temperature variation
		동의어	
		타입	number
		단위	K
		예시	
P903	humidity (prepreg production / storage conditions)	정의	relative humidity in storage
		동의어	
		타입	number
		단위	%
		예시	
P904	duration (prepreg production / storage conditions)	정의	duration of storage
		동의어	
		타입	number
		단위	d
		예시	
P905	vacuum sealed (prepreg production / storage conditions)	정의	stored in vacuum sealed condition
		동의어	
		타입	boolean
		단위	
		예시	
P906	pulling speed (pultrusion)	정의	the speed at which the composite is pulled through the die
		동의어	
		타입	number
		단위	m min^{-1}
		예시	1.5
P907	die temperature (pultrusion)	정의	the temperature of the die during the pultrusion process
		동의어	
		타입	number
		단위	K
		예시	473
P908	resin injection pressure (pultrusion)	정의	the pressure applied to inject resin into the fibers
		동의어	
		타입	number
		단위	MPa
		예시	2
P909	fiber orientation (pultrusion)	정의	the orientation of fibers in the pultruded composite profile
		동의어	
		타입	number
		단위	degree
		예시	10
P910	profile shape (pultrusion)	정의	the shape of the composite profile produced by pultrusion
		동의어	
		타입	string
		단위	
		예시	solid shape; hollow shape; complex shape

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P911	automation level (resin spray transfer)	정의	degree of automation of the Resin Spray Transfer system, indicating the extent of robotic control and minimal human intervention
		동의어	
		타입	string
		단위	
		예시	fully automated
P912	cycle time (resin spray transfer)	정의	time required to complete one full production cycle for a single carbon-fiber composite panel using the Resin Spray Transfer system
		동의어	
		타입	number
		단위	min
		예시	5
P913	injection pressure (RTM)	정의	the pressure at which resin is injected into the mold
		동의어	
		타입	number
		단위	MPa
		예시	1.8
P914	injection time (RTM)	정의	the time required to inject the resin fully
		동의어	
		타입	number
		단위	s
		예시	90
P915	mold temperature (RTM)	정의	the temperature of the mold during resin injection
		동의어	
		타입	number
		단위	K
		예시	435
P916	resin viscosity (RTM)	정의	the viscosity of the resin used in the RTM process
		동의어	
		타입	number
		단위	Pa s
		예시	0.4
P917	fiber preform orientation (RTM)	정의	the orientation angle of fibers in the preform
		동의어	
		타입	number
		단위	degree
		예시	10
P918	post-curing conditions (RTM)	정의	additional curing conditions after the initial molding
		동의어	
		타입	string
		단위	
		예시	2 h at 393 K; 4 h at 453 K
P919	materials (solid state reaction)	정의	materials used for the reaction. Each material plays a distinct role in determining mechanism, kinetics, and final product properties
		동의어	
		타입	string
		단위	
		예시	metal oxide; metallic powder; ceramic

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P920	milling and mixing (solid state reaction)	정의	method of milling and mixing to ensure the uniform distribution of reactants, enhancing diffusion kinetics, and optimizing reaction efficiency
		동의어	
		타입	string
		단위	
		예시	wet/dry mixing; ball milling; planetary milling
P921	pressure (spark plasma sintering)	정의	uniaxial mechanical force per unit area
		동의어	
		타입	number
		단위	MPa
		예시	10
P922	voltage (spark plasma sintering)	정의	applied voltage
		동의어	
		타입	number
		단위	V
		예시	1000
P923	plasma power (spark plasma sintering)	정의	plasma power used in SPS
		동의어	
		타입	number
		단위	W
		예시	
P924	solvent (solvent casting)	정의	solvent for dissolving polymer
		동의어	
		타입	string
		단위	
		예시	MDF; DI water; EtOH
P925	materials (solvent casting / mold)	정의	mold for casting
		동의어	
		타입	string
		단위	
		예시	glass; metal; Teflon
P926	shape (solvent casting / mold)	정의	shape of the mold
		동의어	
		타입	string
		단위	
		예시	flat;
P927	casting time (solvent casting)	정의	time for casting of polymer films or membrane before peeling from the mold
		동의어	
		타입	number
		단위	h
		예시	2
P928	temperature (solvent casting)	정의	casting temperature
		동의어	
		타입	number
		단위	K
		예시	300; 400
P929	rpm ramping-down rate (spin coating)	정의	the rate at which the rotational speed (RPM) decreases during the final stage of spin-coating
		동의어	
		타입	number
		단위	revolutions min^{-1} (rpm) s^{-1}
		예시	100

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P930	materials type (spinning)	정의	type of materials to be processed
		동의어	
		타입	string
		단위	
		예시	PE; PP; PET; PS
P931	temperature (spinning)	정의	temperature of the polymer melt or solution during the process
		동의어	
		타입	number
		단위	K
		예시	
P932	solution viscosity (spinning)	정의	viscosity of the polymer solution or melt
		동의어	
		타입	number
		단위	Pa s
		예시	
P933	spinning speed (spinning)	정의	the rate at which the extruded polymer is drawn away from the spinneret
		동의어	
		타입	number
		단위	m min^{-1}
		예시	
P934	spinneret design (spinning)	정의	configuration of holes in the spinneret
		동의어	
		타입	string
		단위	
		예시	description of design
P935	cooling method (spinning)	정의	technique used to solidify the extruded fibers
		동의어	
		타입	string
		단위	
		예시	air cooling; bath cooling; evaporation cooling
P936	drawing ratio (spinning)	정의	the ratio of the length of the fiber after drawing to the length of the fiber before drawing
		동의어	
		타입	number
		단위	none
		예시	
P937	solvent removal ratio (spinning)	정의	the proportion of solvent removed during the spinning process
		동의어	
		타입	number
		단위	%
		예시	
P938	coagulation bath composition (spinning)	정의	the chemical composition of the bath that the spun fibers pass through immediately after extrusion
		동의어	
		타입	dictionary of {value : {constituent : concentration, … }, unit : unit value}
		단위	at.%
		예시	{value : {Pt : 56.0, Ni : 34.7}, unit : at.%}; {value : {Fe2O3 : 97.8, Y2O3 : 2.2}, unit : at.%}
P939	sputter type (sputter deposition)	정의	type of sputtering process
		동의어	
		타입	string
		단위	
		예시	1. diode sputteirng 2. planar magnetron sputtering 3. circular magnetron sputtering 4. rotating target sputtering

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P940	reaction gas (sputter deposition)	정의	flow rate of reaction gas in the reactive sputter deposition
		동의어	
		타입	dictionary of {composition : {constituent : concentration, … }, unit : unit value}
		단위	cm^{3} min^{-1}
		예시	{composition : {C2H6 : 50, Ar : 100}, unit : SCCM}
P941	jig type (sputter deposition / jig)	정의	type of jig
		동의어	
		타입	string
		단위	
		예시	columnar structure; simple plate; cubic columnar where specimen can be fixed vertically
P942	layer thickness (stereolithography)	정의	the vertical distance between each layer during printing
		동의어	
		타입	number
		단위	{\mu m}
		예시	
P943	build speed (stereolithography)	정의	the rate at which the platform moves to create a layer
		동의어	
		타입	number
		단위	{\mu m} s^{-1}
		예시	
P944	exposure time (stereolithography)	정의	the duration that the UV light is applied to cure each layer
		동의어	
		타입	number
		단위	s
		예시	
P945	beam power (stereolithography)	정의	the intensity of the UV beam used to cure the resin
		동의어	
		타입	number
		단위	W
		예시	
P946	resin properties (stereolithography)	정의	characteristics of the photopolymer used, including viscosity, curing speed, and mechanical properties
		동의어	
		타입	string
		단위	
		예시	epoxy acrylates; methyl methacrylate; polyurethane acrylates
P947	platform temperature (stereolithography)	정의	the temperature of the build platform during the printing process
		동의어	
		타입	number
		단위	K
		예시	
P948	post-processing (stereolithography)	정의	steps taken after printing, such as rinsing, curing, and finishing
		동의어	
		타입	string
		단위	
		예시	rinsing; curing; polishing
P949	stretching type (stretching)	정의	type of stretching
		동의어	
		타입	string
		단위	
		예시	uniaxial; biaxial

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P950	materials type (stretching)	정의	type of materials to be processed
		동의어	
		타입	string
		단위	
		예시	PE; PP; PET; PS
P951	temperature (stretching)	정의	the temperature at which the stretching is performed, which affects the elasticity and strength of the polymer
		동의어	
		타입	number
		단위	K
		예시	
P952	stretching ratio (stretching)	정의	the ratio of the length after stretching to the length before stretching, which significantly affects the properties of the final product
		동의어	
		타입	number
		단위	none
		예시	
P953	stretching speed (stretching)	정의	the speed at which the polymer is stretched, which can influence the uniformity and properties of the film or fiber
		동의어	
		타입	number
		단위	mm min^{-1}
		예시	
P954	cooling rate (stretching)	정의	the rate at which the stretched material is cooled after orientation, which affects crystallization and dimensional stability
		동의어	
		타입	number
		단위	K min^{-1}
		예시	
P955	dwell time (stretching)	정의	the time the polymer is held under stretched conditions before cooling or solidifying
		동의어	
		타입	number
		단위	min
		예시	
P956	humidity (stretching)	정의	relative humidity in the process
		동의어	
		타입	number
		단위	%
		예시	
P957	atmosphere (stretching)	정의	atmosphere of the process
		동의어	
		타입	string
		단위	
		예시	Air; O2; N2
P958	additives (stretching)	정의	chemicals added to improve processing or final material properties
		동의어	
		타입	string
		단위	
		예시	diethyl phthalates; dibutyl phthalate; citrate esters; natural oil
P959	base material (tailored fiber placement)	정의	the backing substrate onto which fibrous tows are stitched in TFP to provide support and shape
		동의어	
		타입	string
		단위	
		예시	fabric backing or non-crimp fabric or nonwoven fabric

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P960	stitch pattern (tailored fiber placement)	정의	the layout and orientation of stitching paths used to secure fibrous tows to the base material, adapted to optimize composite performance
		동의어	
		타입	string
		단위	
		예시	curvilinear, stress-adapted
P961	stitch threads (tailored fiber placement)	정의	the upper and lower sewing threads employed in TFP to fix continuous fibrous tows to the base material, selected for strength and compatibility with the composite
		동의어	
		타입	string
		단위	
		예시	upper and lower sewing threads
P962	material placement (tailored fiber placement)	정의	the arrangement and orientation of continuous fibrous tows on the base material before stitching in TFP, defining the reinforcement architecture
		동의어	
		타입	string
		단위	
		예시	continuous fibrous tows
P963	heating method (zone melting)	정의	the method of heating used in zone melting
		동의어	
		타입	string
		단위	
		예시	electric resistance heating; laser heating; inductive heating
P964	temperature (zone melting)	정의	temperature of the molten zone
		동의어	
		타입	number
		단위	K
		예시	1500
P965	length of heated zone (zone melting)	정의	length of the heated zone
		동의어	
		타입	number
		단위	mm
		예시	50
P966	zone movement speed (zone melting)	정의	speed at which the molten zone passed through the material
		동의어	
		타입	number
		단위	mm min^{-1}
		예시	1
P967	atmosphere (zone melting)	정의	environment during the process
		동의어	
		타입	string
		단위	
		예시	N2; vacuum; H2
P968	electrode (arc melting)	정의	information of the electrode that generates and maintains the electric arc; typically made of graphite or tungsten
		동의어	
		타입	string
		단위	
		예시	graphite cylinder d=10 mm
P974	chill (casting)	정의	a heat-conducting material placed in the mold to promote rapid local solidification of molten metal, improving the casting's microstructure and reducing defects
		동의어	
		타입	string
		단위	
		예시	

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P976	wavelength (digital light processing)	정의	wavelength of the light used in DLP printing to initiate resin polymerization; typical range is 365 ~ 405 nm
		동의어	curing wavelength; light wavelength
		타입	number
		단위	nm
		예시	
P977	intensity (digital light processing)	정의	amount of light energy delivered to the resin surface, affecting polymerization rate and curing depth
		동의어	light intensity; irradiance; exposure power
		타입	number
		단위	mW cm{-2}
		예시	
P978	vat temperature (digital light processing)	정의	resin tank's ambient temperature; influencing resin flow, reactivity, and print reliability
		동의어	resin bath temperature; tray temperature
		타입	number
		단위	K
		예시	
P979	exposure time (digital light processing)	정의	duration of light exposure per layer required to adequately cure the resin
		동의어	curing time; exposure duration
		타입	number
		단위	s
		예시	
P980	layer thickness (digital light processing)	정의	vertical height of each cured layer, affecting print quality and build speed in DLP processes
		동의어	layer height; slice thickness
		타입	number
		단위	{\mu m}
		예시	
P981	platform temperature (digital light processing)	정의	temperature of the build platform, which affects adhesion and curing consistency during printing
		동의어	build plate temperature; bed temperature
		타입	number
		단위	K
		예시	
P982	resolution (digital light processing)	정의	minimum feature size that a DLP printer can reproduce, determined by the projector's pixel size and optics; higher resolution allows for finer details in printed parts
		동의어	XY resolution; pixel resolution; projected resolution
		타입	number
		단위	{\mu m}
		예시	
P983	build speed (digital light processing)	정의	vertical speed at which the printer constructs the object, typically in mm/h
		동의어	print speed; vertical build rate
		타입	number
		단위	mm h^{-1}
		예시	
P984	cooling method (heat treatment / thermal history)	정의	method of cooling without specified cooling rate
		동의어	
		타입	string
		단위	
		예시	1. air cooling (AC) 2. furnace cooling (FC) 3. oil quenching (OQ) 4. water quenching (WQ)
P985	finishing temperature (heat treatment / thermal history)	정의	temperature at which the thermal cycle finishes
		동의어	
		타입	number
		단위	K
		예시	298

어휘번호	표준어휘 (상위어휘체계)	구분	상세 설명
P1018	feed rate (slot coating)	정의	ink supply rate with pump
		동의어	pumping rate
		타입	number
		단위	ml min^{-1}
		예시	
P1019	polymer concentration (solvent annealing)	정의	amount of polymer dissolved in a given volume of solvent
		동의어	
		타입	number
		단위	mg mL^{-1}
		예시	
P1020	mode (sonication)	정의	operational scheme describing how ultrasonic energy is applied
		동의어	
		타입	string
		단위	
		예시	continuous; pulsed
P1021	frequency (sonication)	정의	oscillation rate of ultrasound waves applied to a sample during sonication
		동의어	
		타입	number
		단위	Hz
		예시	40000
P1022	composition (spinning / atmosphere)	정의	composition of atmosphere in partial pressure
		동의어	environmental composition
		타입	dictionary of {constituent : partial pressure, ...}
		단위	bar
		예시	{value: {Ar:0.5, N_2:0.5}, unit:bar}
P1023	total pressure (spinning / atmosphere)	정의	total pressure of atmosphere; sum of partial pressures in composition value
		동의어	
		타입	number
		단위	bar
		예시	1.0; 0.95
P1024	relative humidity (spinning)	정의	the ratio of the actual water-vapor content of air to the maximum possible at the same temperature
		동의어	
		타입	number
		단위	%
		예시	40
P1026	amount (washing / washing solvent)	정의	amount of solvent
		동의어	
		타입	number
		단위	ml
		예시	
P1027	drying (conditioning / post-conditioning)	정의	solvent evaporation with heat and air flow
		동의어	solvent evaporation
		타입	string
		단위	
		예시	drying at 200 K in oven
P1028	annealing (conditioning / post-conditioning)	정의	heat treatment with thermal energy
		동의어	heat treatment
		타입	string
		단위	
		예시	annealing at 300 K for 3h

소재 연구데이터 표준화위원회

	성 명	근 무 처	직 위
(위원장)	송 재 용	포항공과대학교	교 수
(위원)	김 선 태	전북대학교	교 수
	나 종 주	한국재료연구원	박 사
	안 재 평	한국과학기술연구원	박 사
	이 관 영	한국과학기술연구원	박 사
	이 병 주	포항공과대학교	교 수
	이 봉 상	한국원자력연구원	박 사
	차 필 령	국민대학교	교 수
	최 우 진	한국화학연구원	박 사

소재 연구데이터 표준화 전문위원회

	성 명	근 무 처	직 위
(위원장)	이 광 렬	한국과학기술연구원	박 사
(위원)	변 영 운	한국과학기술연구원	박 사
	신 정 호	한국화학연구원	박 사
	이 우	한국표준과학연구원	박 사
	전 종 배	동아대학교	교 수
	홍 현 욱	창원대학교	교 수

소재 연구데이터 표준화 실무위원 (표준어휘:2025-2)

성 명	근 무 처	직 위
방 준 하	고려대학교	교 수
신 세 은	국립순천대학교	교 수
신 호 선	한국표준과학연구원	박 사
오 창 석	한국재료연구원	박 사
이 정 현	고려대학교	교 수
최 윤 석	부산대학교	교 수
한 상 수	한국과학기술연구원	박 사

제 정 : 2026년 1월 29일
심 의 : 2026년 제1차 소재 연구데이터 표준화위원회

한국표준과학연구원 국가소재연구데이터센터
(34113) 대전광역시 유성구 가정로 267(한국표준과학연구원)
Tel. 042-868-5104 Fax. 042-604-1125